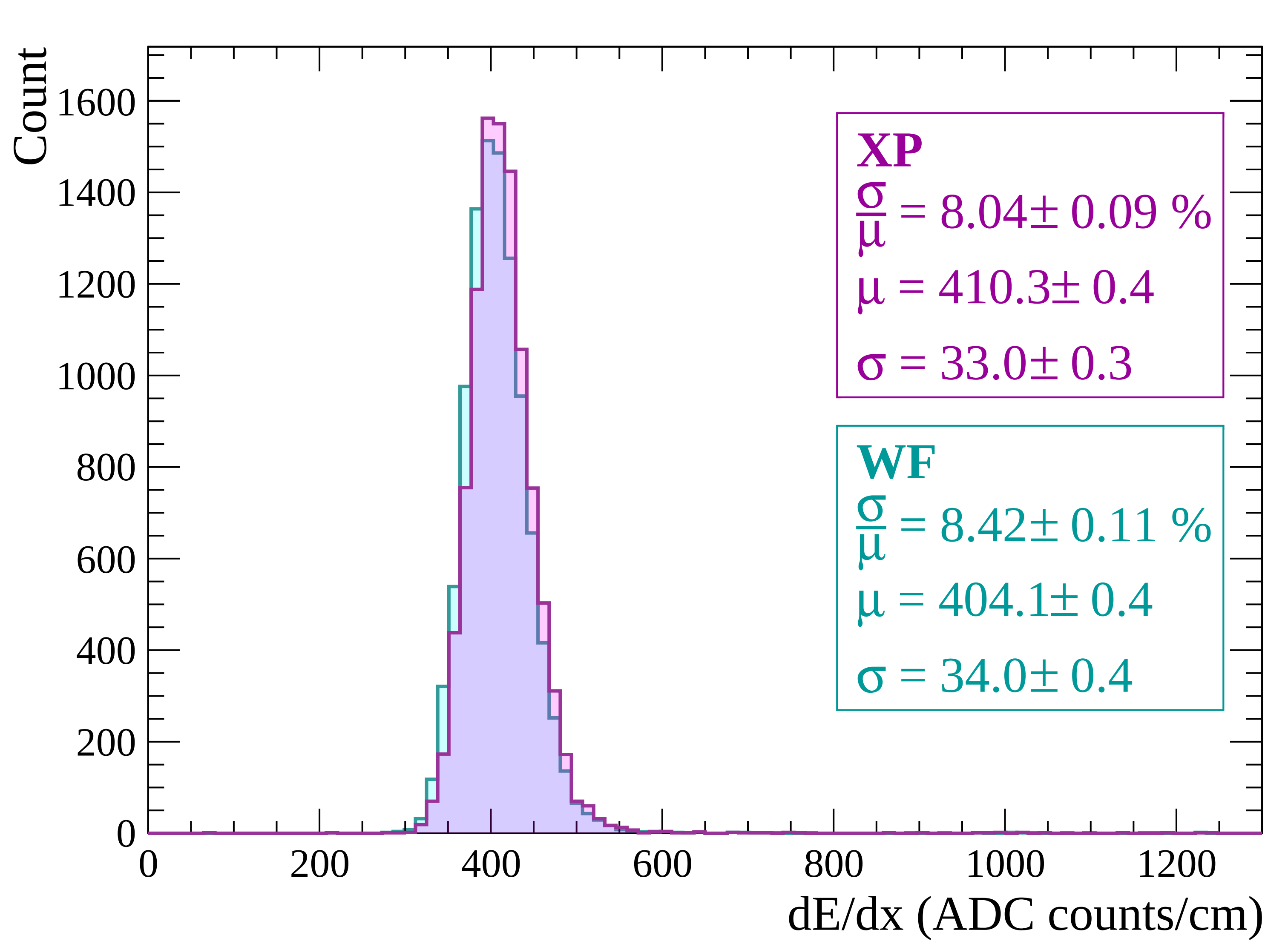
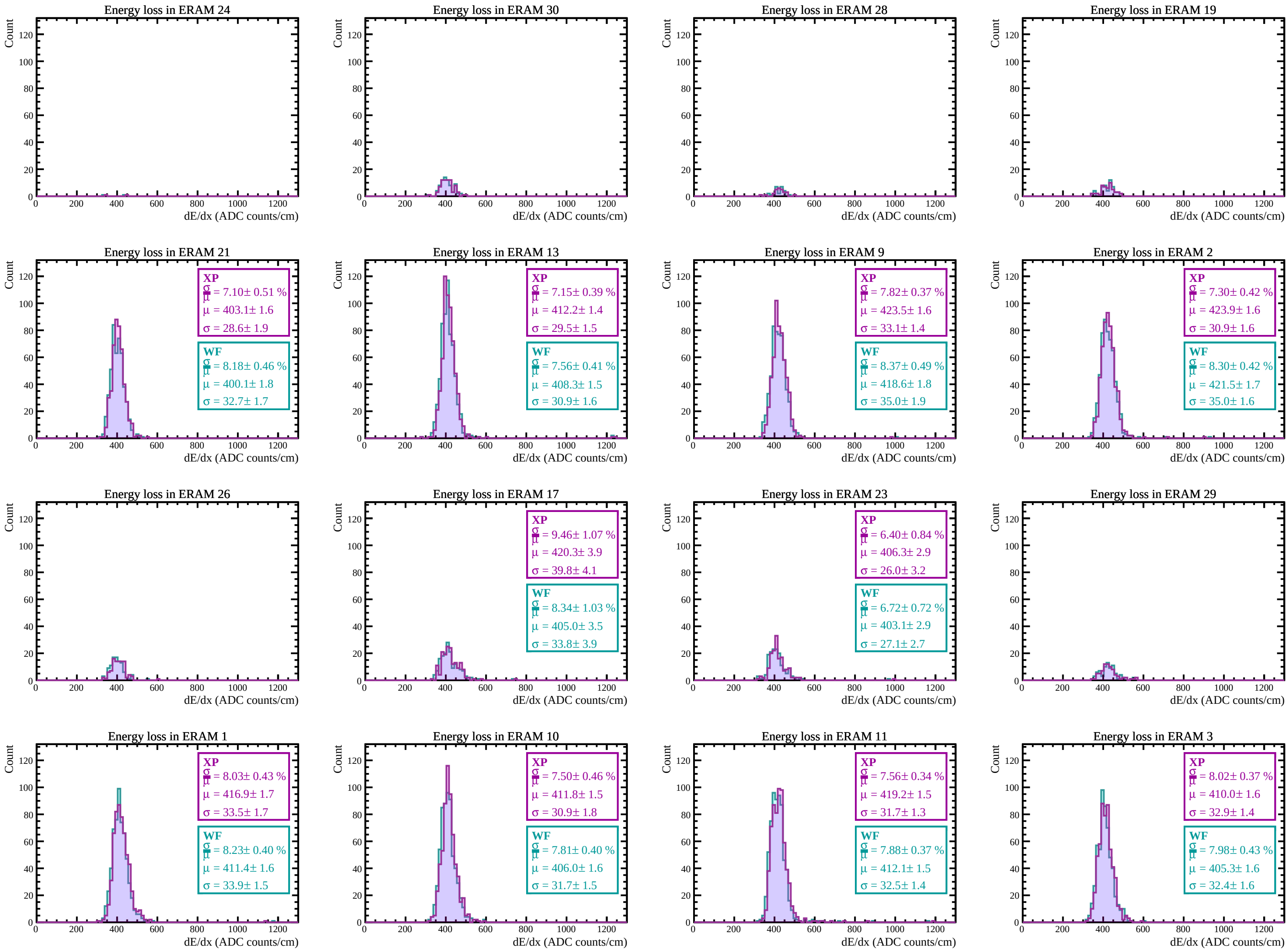
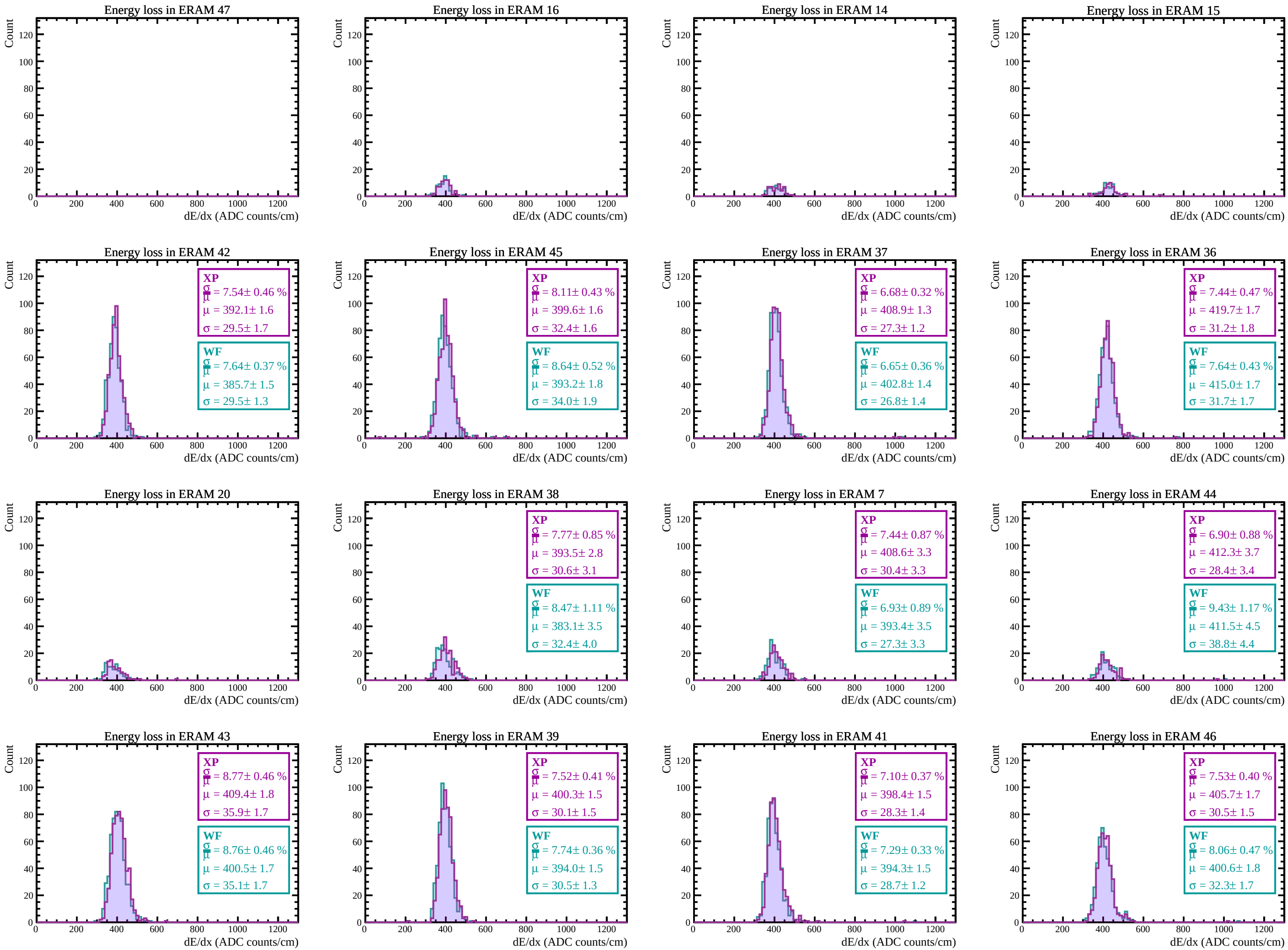
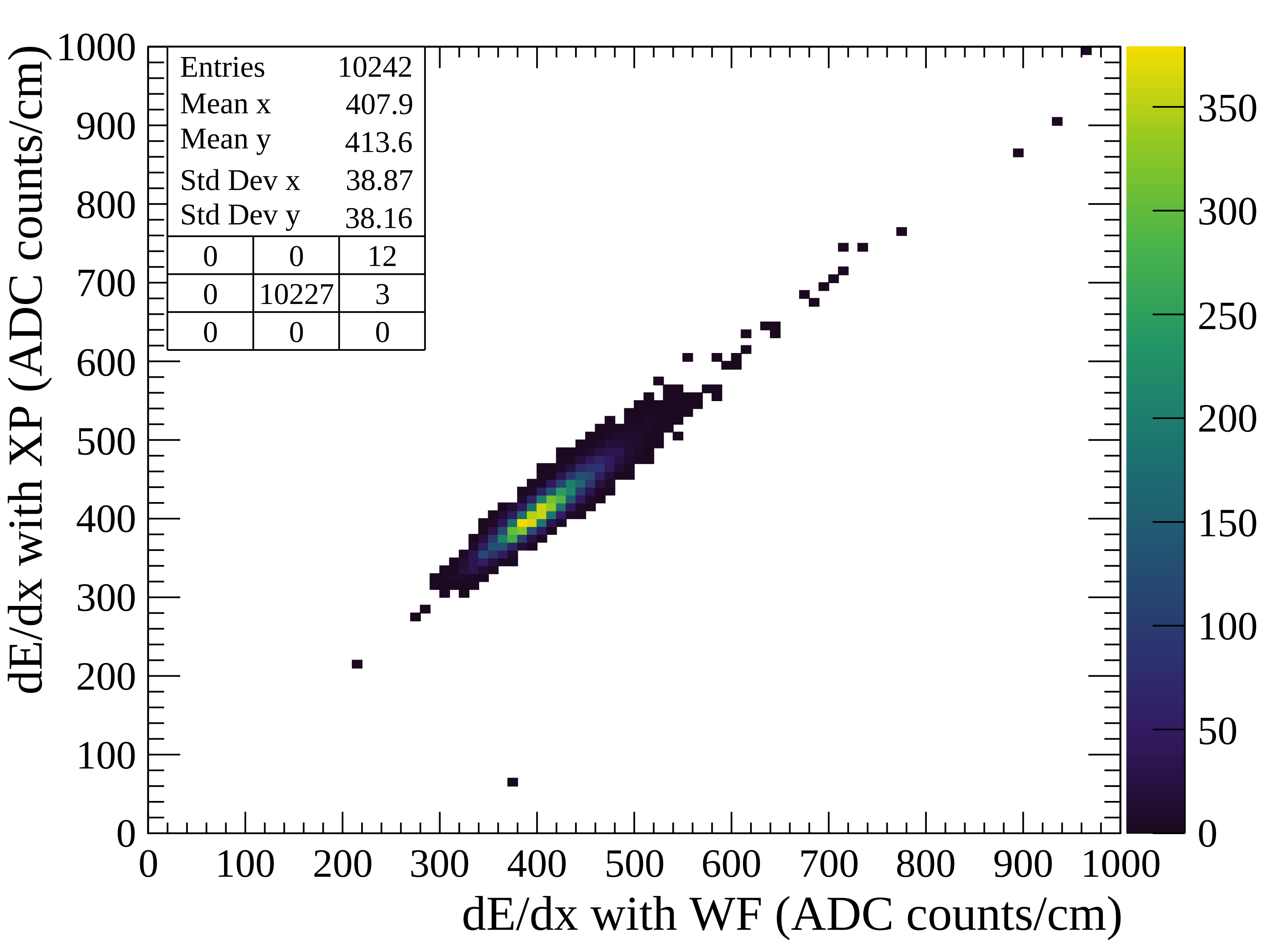

Count
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.04± 0.09 %
μ = 410.3± 0.4
σ = 33.0± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.42± 0.11 %
μ = 404.1± 0.4
σ = 34.0± 0.4



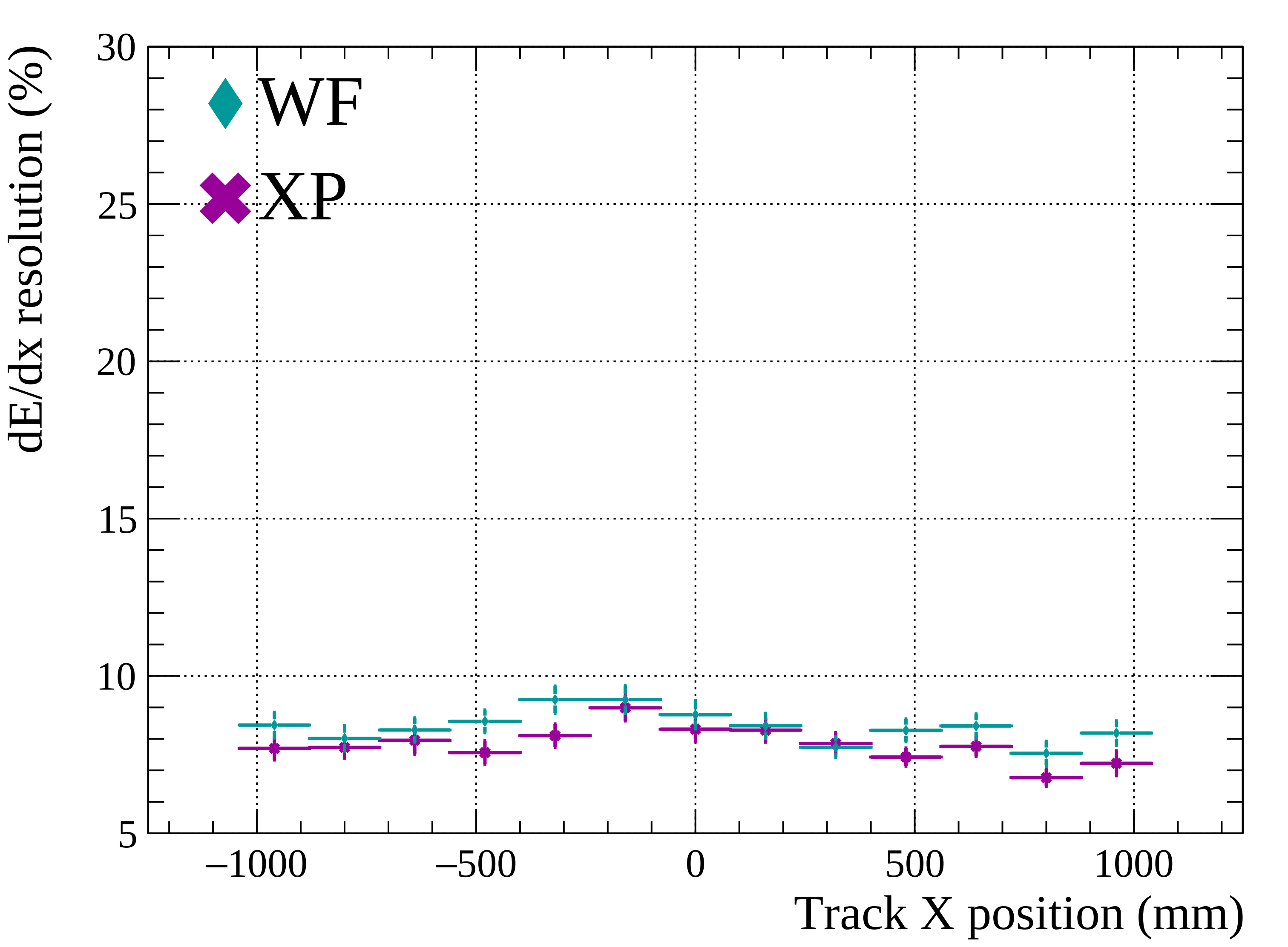
Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
XP σ/μ = 7.10± 0.51 % μ = 403.1± 1.6 σ = 28.6± 1.9
WF σ/μ = 8.18± 0.46 % μ = 400.1± 1.8 σ = 32.7± 1.7
Energy loss in ERAM 13
XP σ/μ = 7.15± 0.39 % μ = 412.2± 1.4 σ = 29.5± 1.5
WF σ/μ = 7.56± 0.41 % μ = 408.3± 1.5 σ = 30.9± 1.6
Energy loss in ERAM 9
XP σ/μ = 7.82± 0.37 % μ = 423.5± 1.6 σ = 33.1± 1.4
WF σ/μ = 8.37± 0.49 % μ = 418.6± 1.8 σ = 35.0± 1.9
Energy loss in ERAM 2
XP σ/μ = 7.30± 0.42 % μ = 423.9± 1.6 σ = 30.9± 1.6
WF σ/μ = 8.30± 0.42 % μ = 421.5± 1.7 σ = 35.0± 1.6
Energy loss in ERAM 26
Energy loss in ERAM 17
XP σ/μ = 9.46± 1.07 % μ = 420.3± 3.9 σ = 39.8± 4.1
WF σ/μ = 8.34± 1.03 % μ = 405.0± 3.5 σ = 33.8± 3.9
Energy loss in ERAM 23
XP σ/μ = 6.40± 0.84 % μ = 406.3± 2.9 σ = 26.0± 3.2
WF σ/μ = 6.72± 0.72 % μ = 403.1± 2.9 σ = 27.1± 2.7
Energy loss in ERAM 29
Energy loss in ERAM 1
XP σ/μ = 8.03± 0.43 % μ = 416.9± 1.7 σ = 33.5± 1.7
WF σ/μ = 8.23± 0.40 % μ = 411.4± 1.6 σ = 33.9± 1.5
Energy loss in ERAM 10
XP σ/μ = 7.50± 0.46 % μ = 411.8± 1.5 σ = 30.9± 1.8
WF σ/μ = 7.81± 0.40 % μ = 406.0± 1.6 σ = 31.7± 1.5
Energy loss in ERAM 11
XP σ/μ = 7.56± 0.34 % μ = 419.2± 1.5 σ = 31.7± 1.3
WF σ/μ = 7.88± 0.37 % μ = 412.1± 1.5 σ = 32.5± 1.4
Energy loss in ERAM 3
XP σ/μ = 8.02± 0.37 % μ = 410.0± 1.6 σ = 32.9± 1.4
WF σ/μ = 7.98± 0.43 % μ = 405.3± 1.6 σ = 32.4± 1.6
Count
dE/dx (ADC counts/cm)



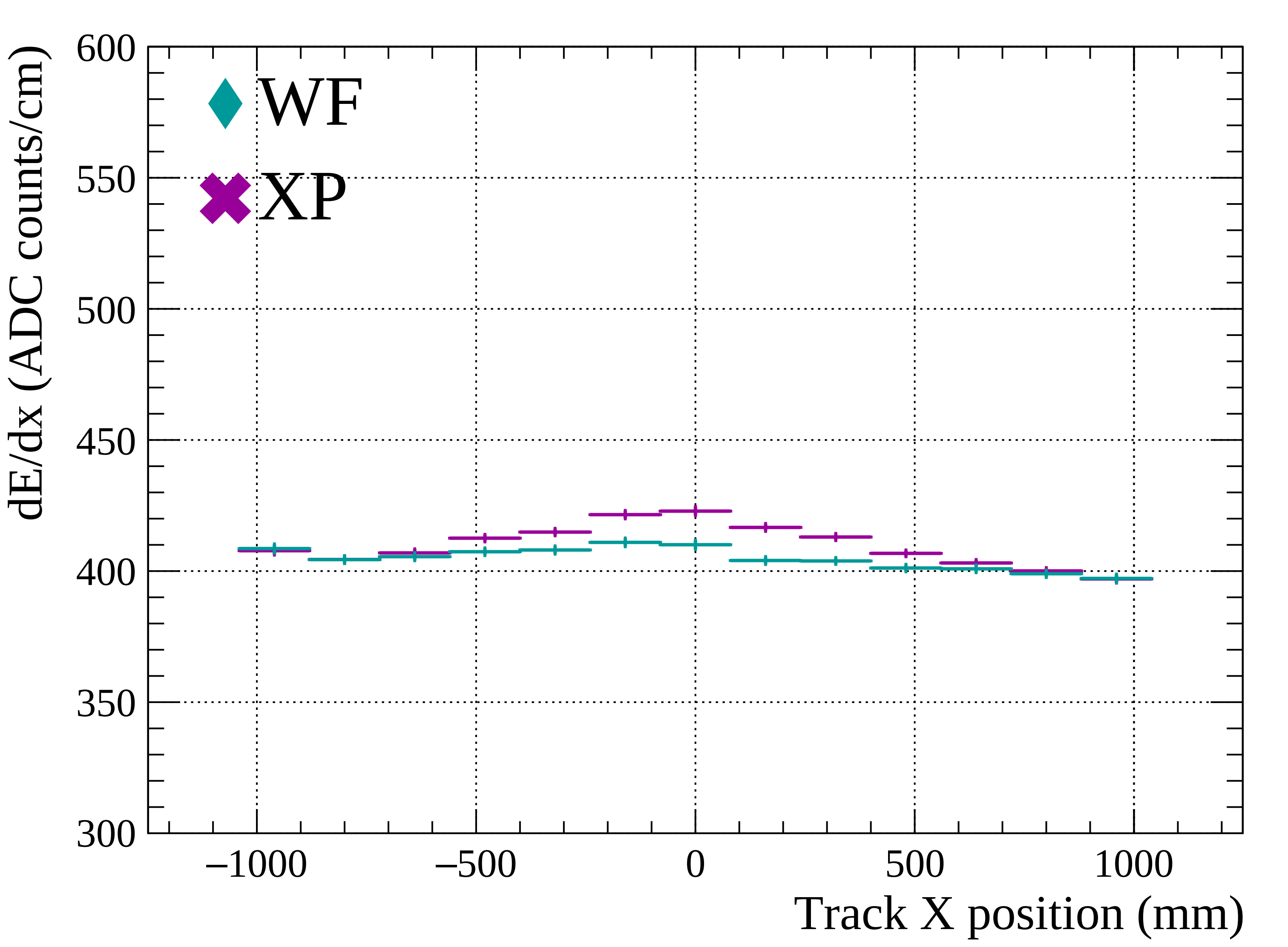
Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
Energy loss in ERAM 42
XP
σ/μ = 7.54± 0.46 %
μ = 392.1± 1.6
σ = 29.5± 1.7
WF
σ/μ = 7.64± 0.37 %
μ = 385.7± 1.5
σ = 29.5± 1.3
Energy loss in ERAM 45
XP
σ/μ = 8.11± 0.43 %
μ = 399.6± 1.6
σ = 32.4± 1.6
WF
σ/μ = 8.64± 0.52 %
μ = 393.2± 1.8
σ = 34.0± 1.9
Energy loss in ERAM 37
XP
σ/μ = 6.68± 0.32 %
μ = 408.9± 1.3
σ = 27.3± 1.2
WF
σ/μ = 6.65± 0.36 %
μ = 402.8± 1.4
σ = 26.8± 1.4
Energy loss in ERAM 36
XP
σ/μ = 7.44± 0.47 %
μ = 419.7± 1.7
σ = 31.2± 1.8
WF
σ/μ = 7.64± 0.43 %
μ = 415.0± 1.7
σ = 31.7± 1.7
Energy loss in ERAM 20
Energy loss in ERAM 38
XP
σ/μ = 7.77± 0.85 %
μ = 393.5± 2.8
σ = 30.6± 3.1
WF
σ/μ = 8.47± 1.11 %
μ = 383.1± 3.5
σ = 32.4± 4.0
Energy loss in ERAM 7
XP
σ/μ = 7.44± 0.87 %
μ = 408.6± 3.3
σ = 30.4± 3.3
WF
σ/μ = 6.93± 0.89 %
μ = 393.4± 3.5
σ = 27.3± 3.3
Energy loss in ERAM 44
XP
σ/μ = 6.90± 0.88 %
μ = 412.3± 3.7
σ = 28.4± 3.4
WF
σ/μ = 9.43± 1.17 %
μ = 411.5± 4.5
σ = 38.8± 4.4
Energy loss in ERAM 43
XP
σ/μ = 8.77± 0.46 %
μ = 409.4± 1.8
σ = 35.9± 1.7
WF
σ/μ = 8.76± 0.46 %
μ = 400.5± 1.7
σ = 35.1± 1.7
Energy loss in ERAM 39
XP
σ/μ = 7.52± 0.41 %
μ = 400.3± 1.5
σ = 30.1± 1.5
WF
σ/μ = 7.74± 0.36 %
μ = 394.0± 1.5
σ = 30.5± 1.3
Energy loss in ERAM 41
XP
σ/μ = 7.10± 0.37 %
μ = 398.4± 1.5
σ = 28.3± 1.4
WF
σ/μ = 7.29± 0.33 %
μ = 394.3± 1.5
σ = 28.7± 1.2
Energy loss in ERAM 46
XP
σ/μ = 7.53± 0.40 %
μ = 405.7± 1.7
σ = 30.5± 1.5
WF
σ/μ = 8.06± 0.47 %
μ = 400.6± 1.8
σ = 32.3± 1.7
Count
dE/dx (ADC counts/cm)



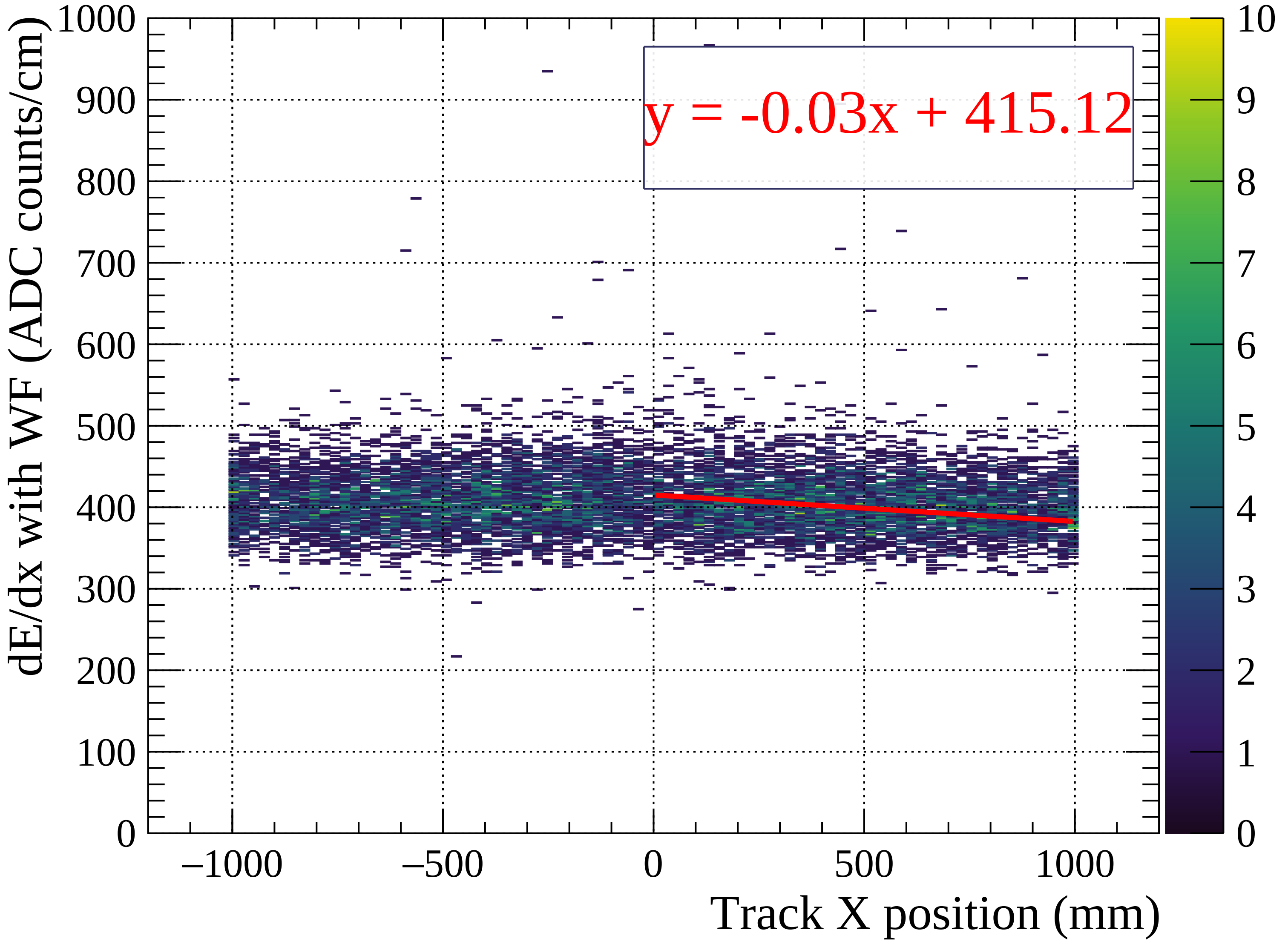
Entries 10242
Mean x 407.9
Mean y 413.6
Std Dev x 38.87
Std Dev y 38.16
0 0 12
0 10227 3
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
50
150
250
350



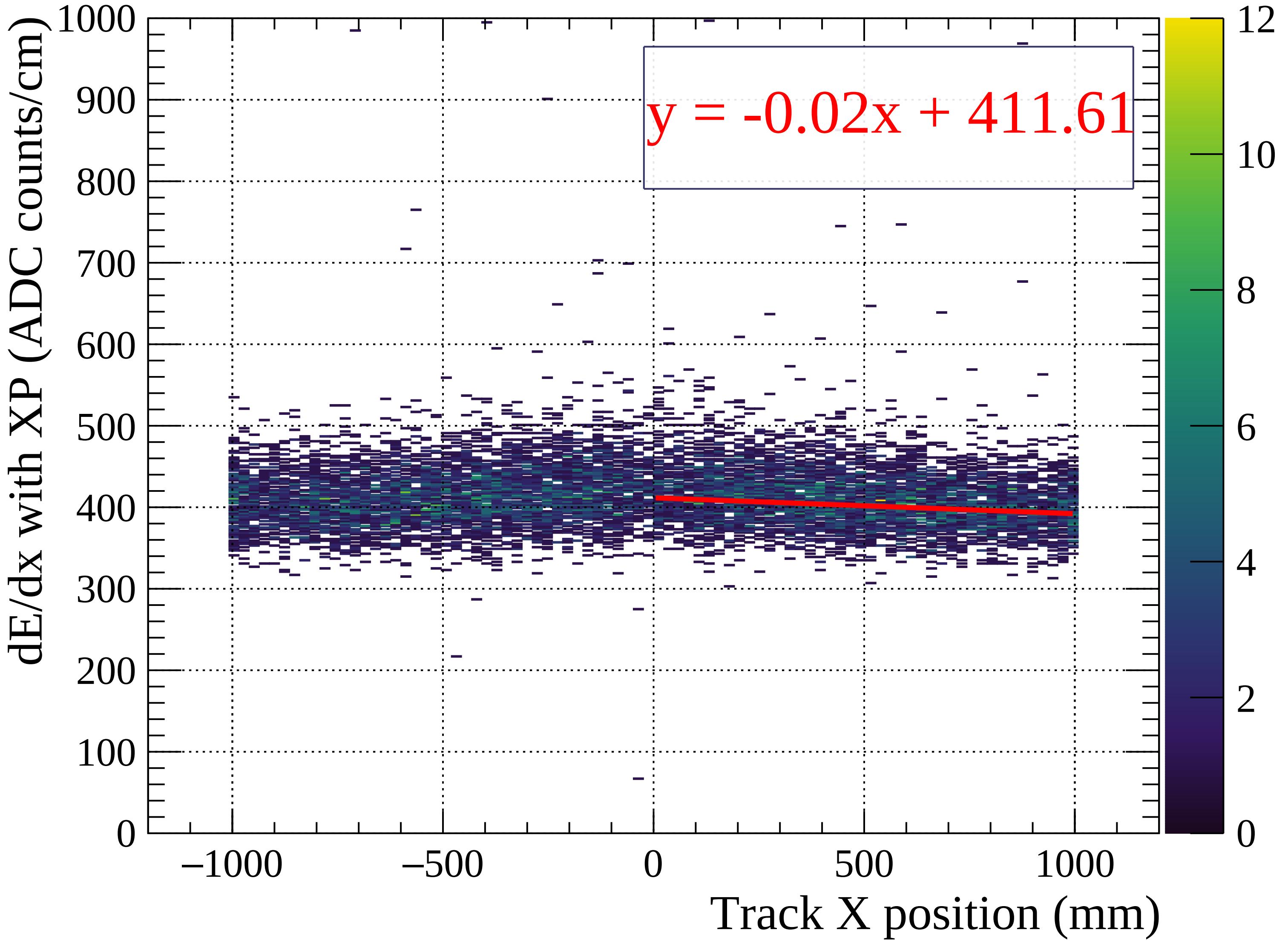
dE/dx resolution (%)
Track X position (mm)
WF
XP
30
25
20
15
10
5
−1000
−500
0
500
1000

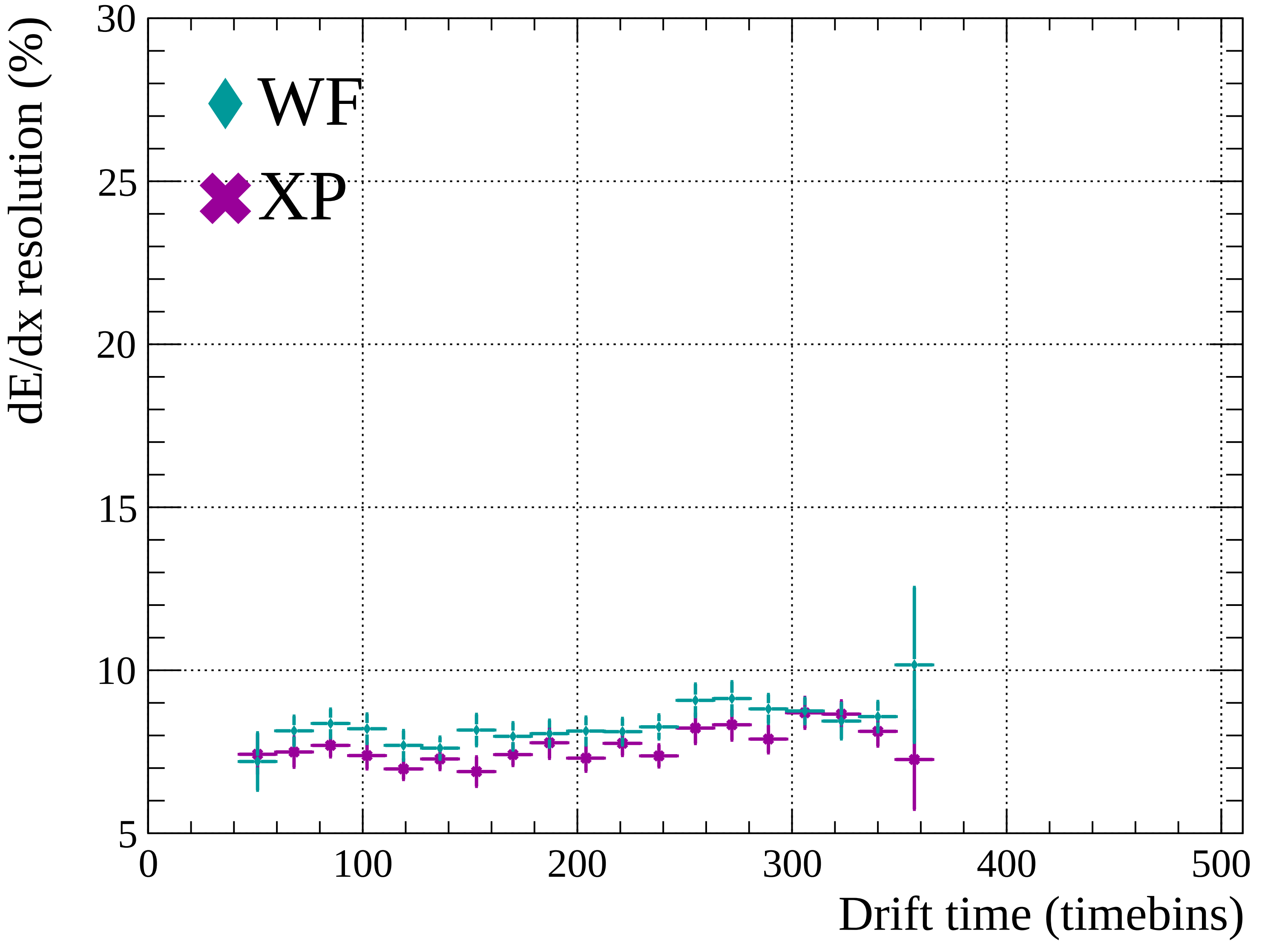


WF
XP
600
550
500
450
400
350
300
dE/dx (ADC counts/cm)
−1000
−500
0
500
1000
Track X position (mm)

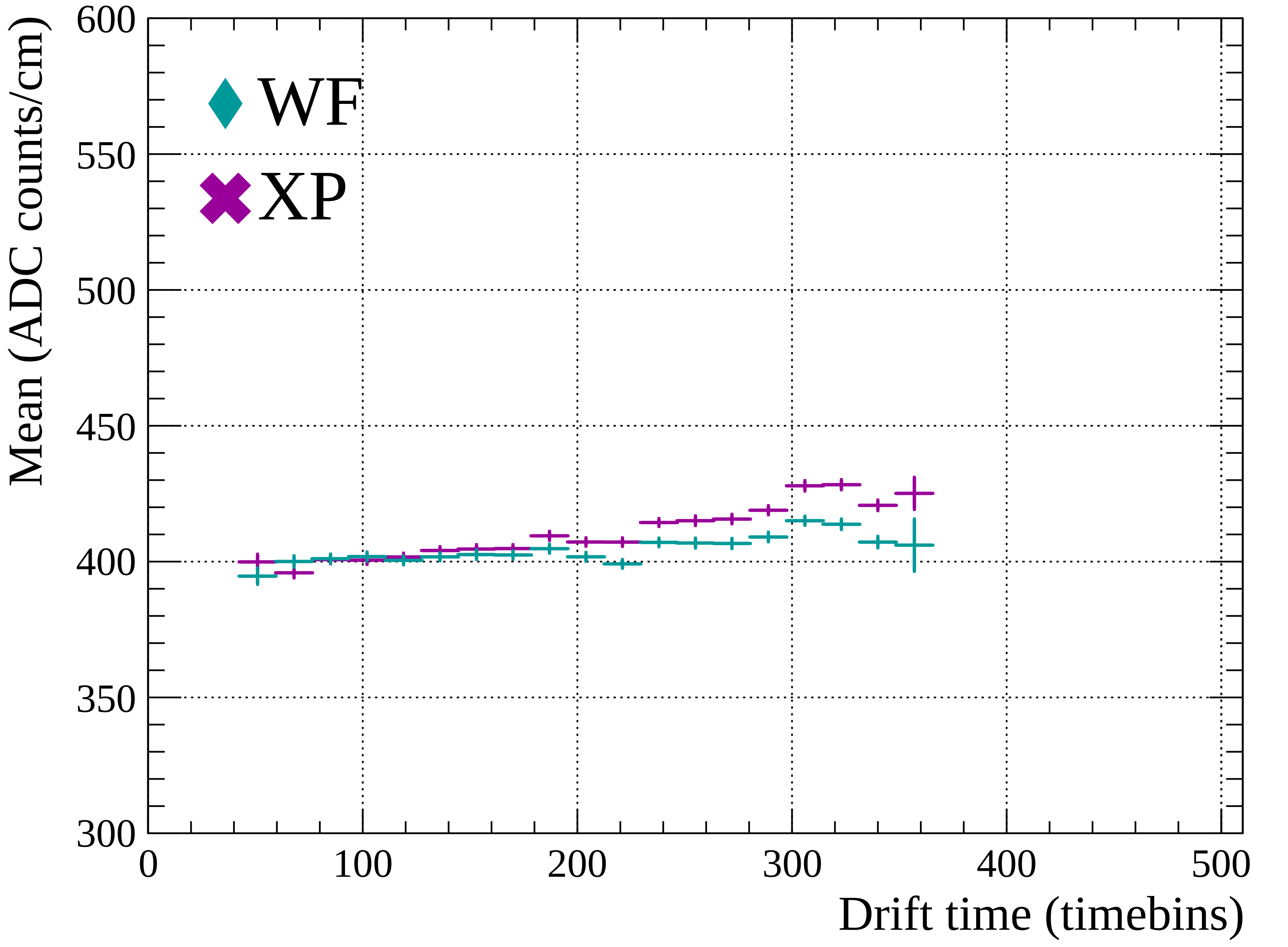


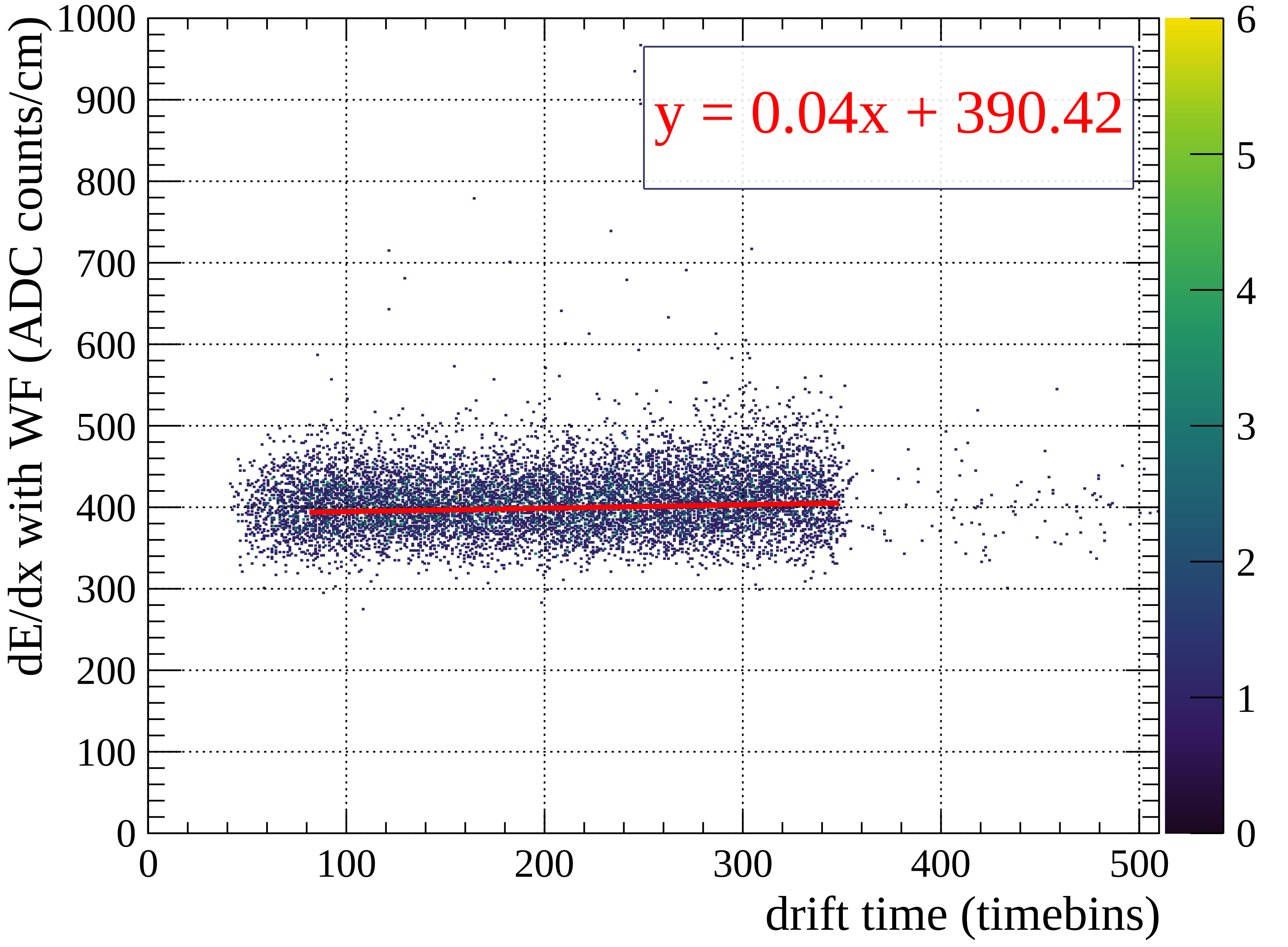
y = -0.03x + 415.12
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
1
2
3
4
5
6
7
8
9
10


y = -0.02x + 411.61
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
2
4
6
8
10
12

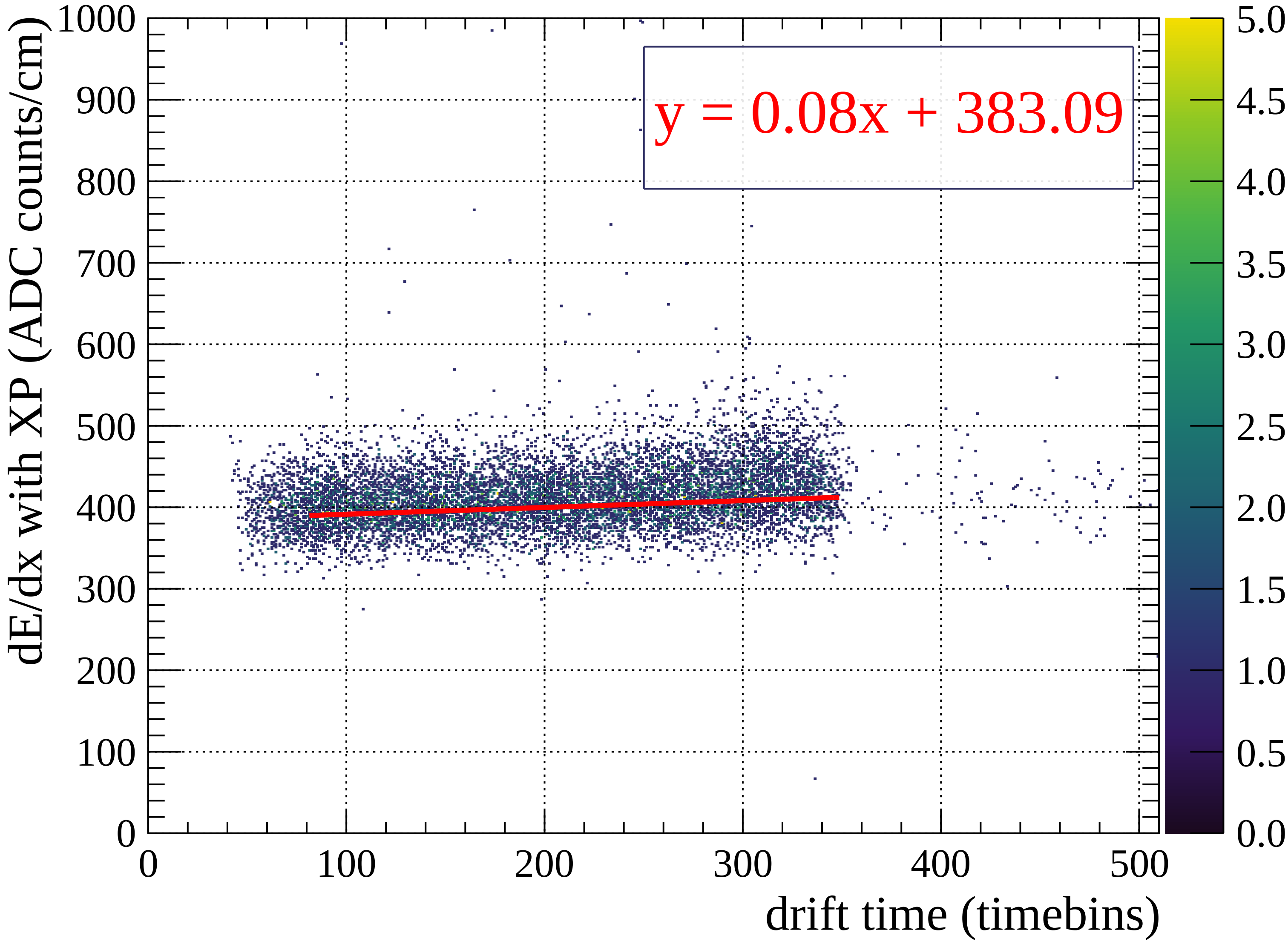


dE/dx resolution (%)
30
25
20
15
10
5
0
100
200
300
400
500
Drift time (timebins)
WF
XP

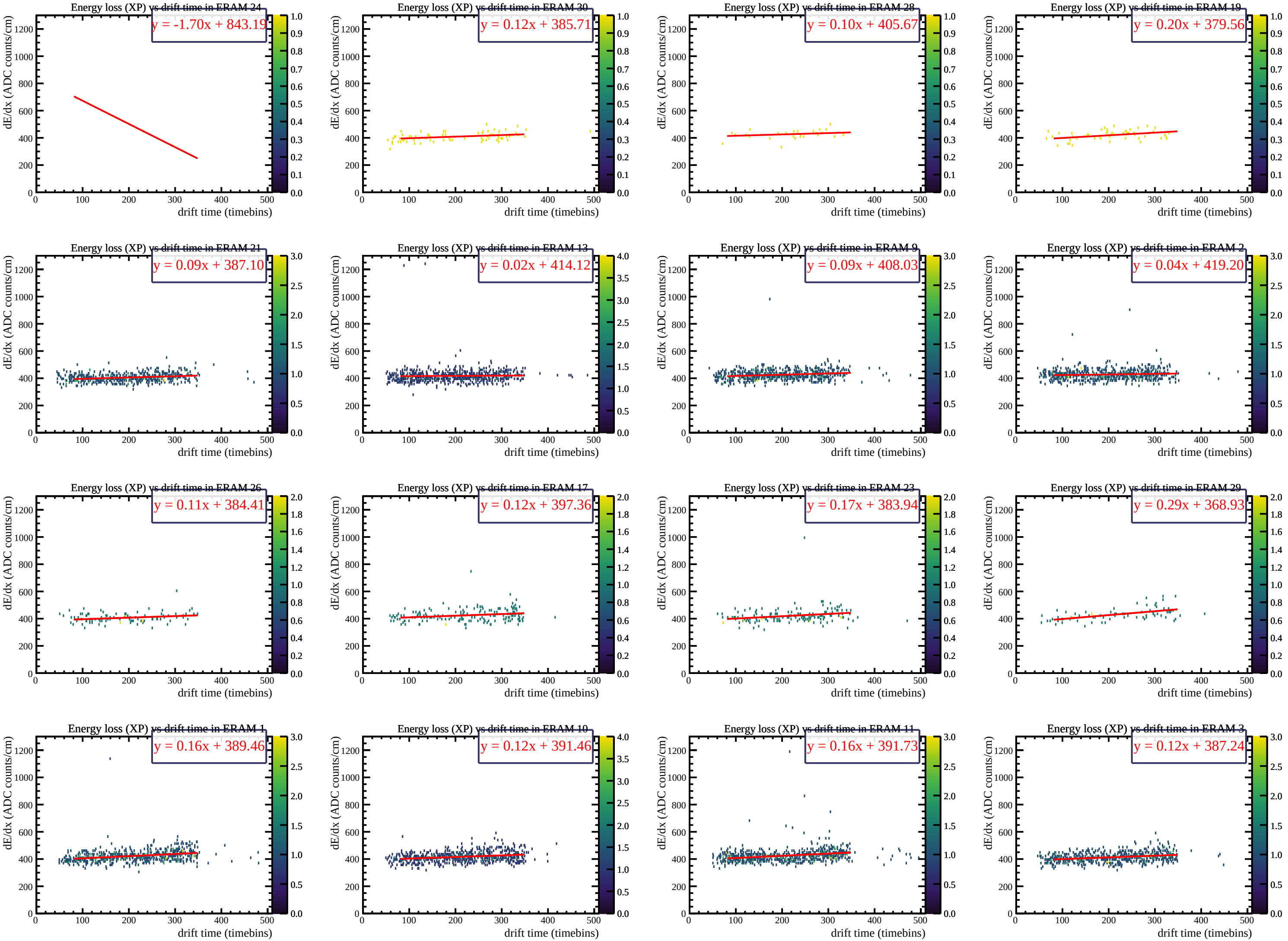

WF
XP
Mean (ADC counts/cm)
Drift time (timebins)
600
550
500
450
400
350
300
0
100
200
300
400
500

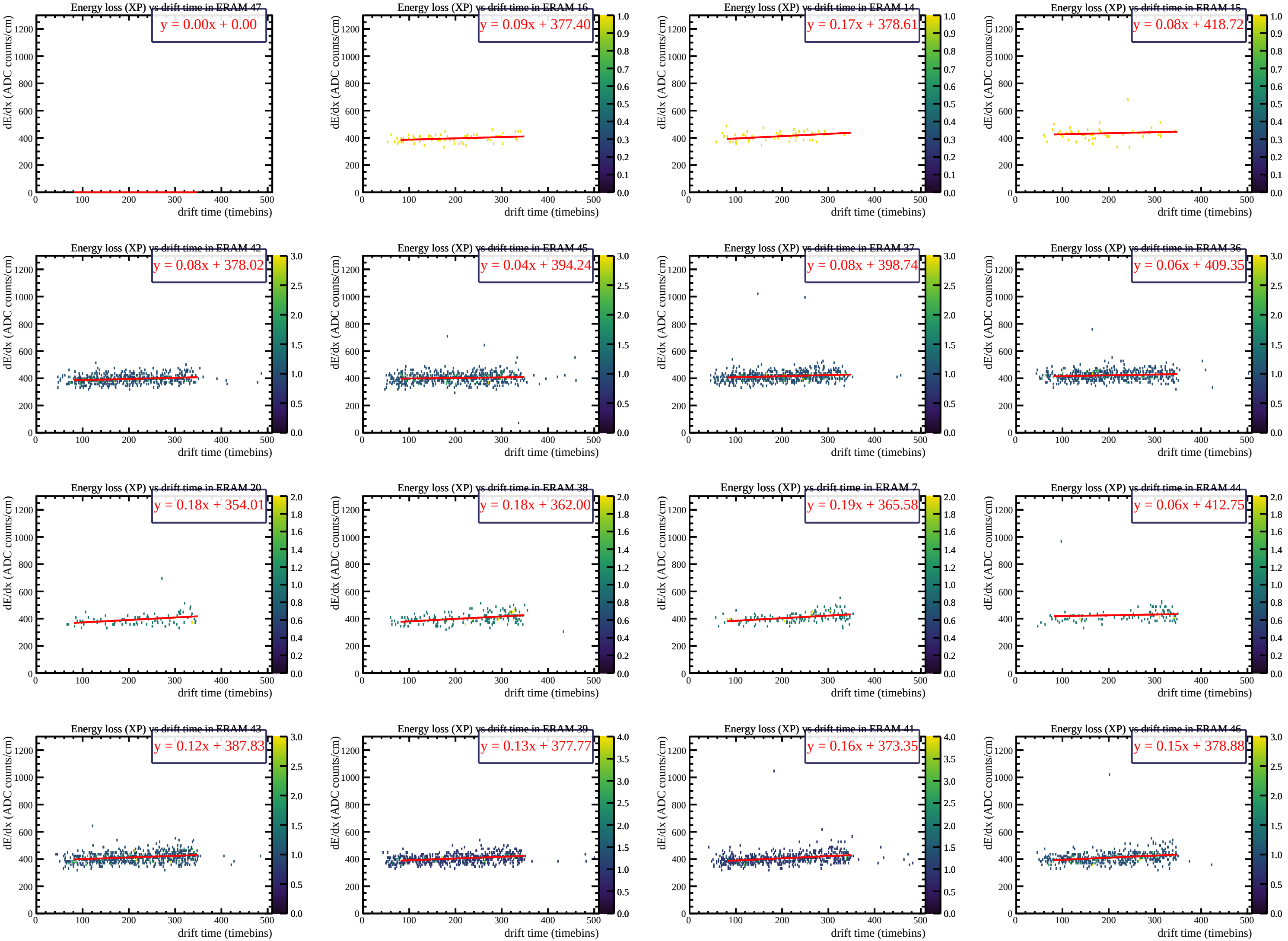


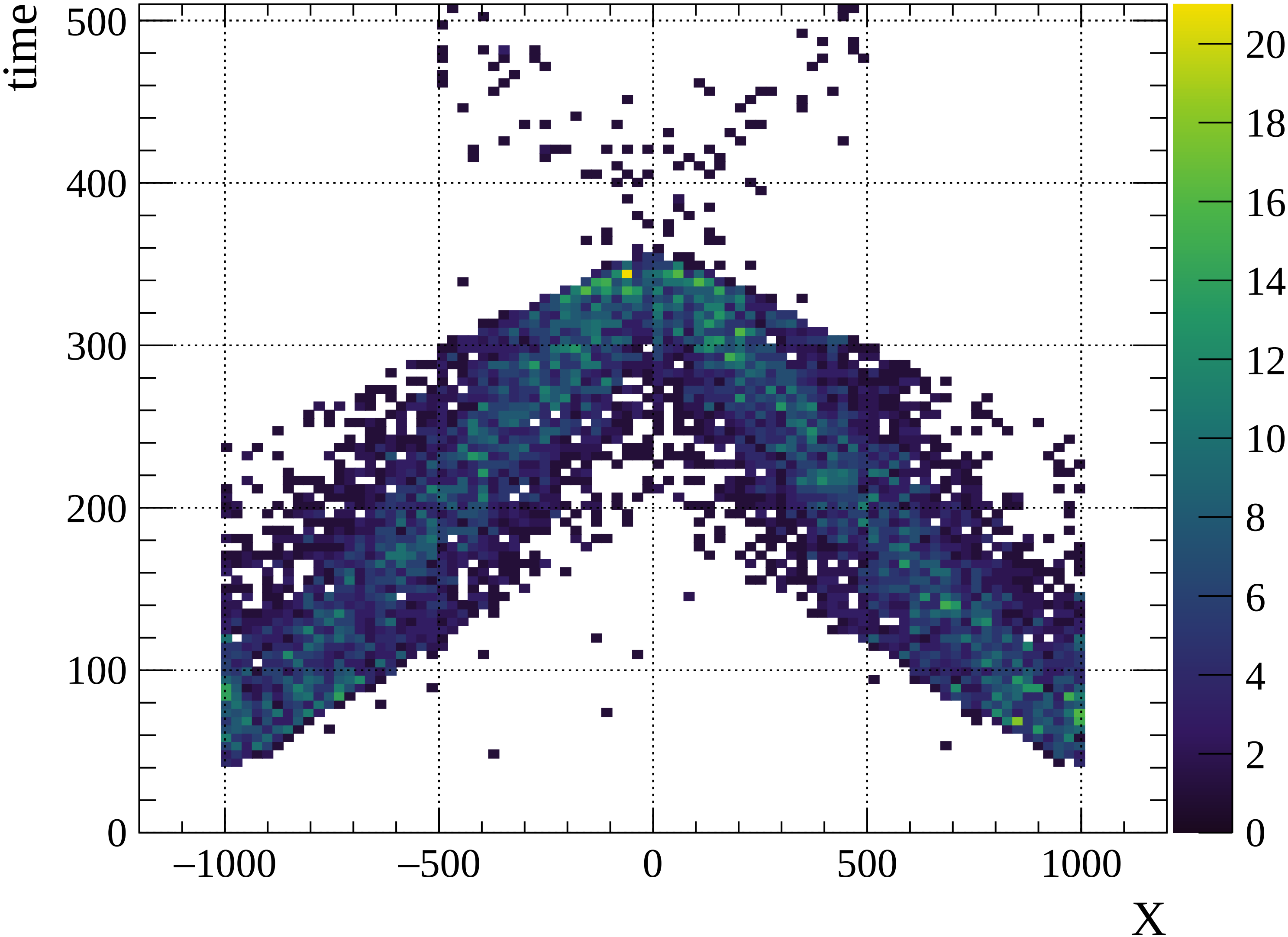
y = 0.04x + 390.42
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6



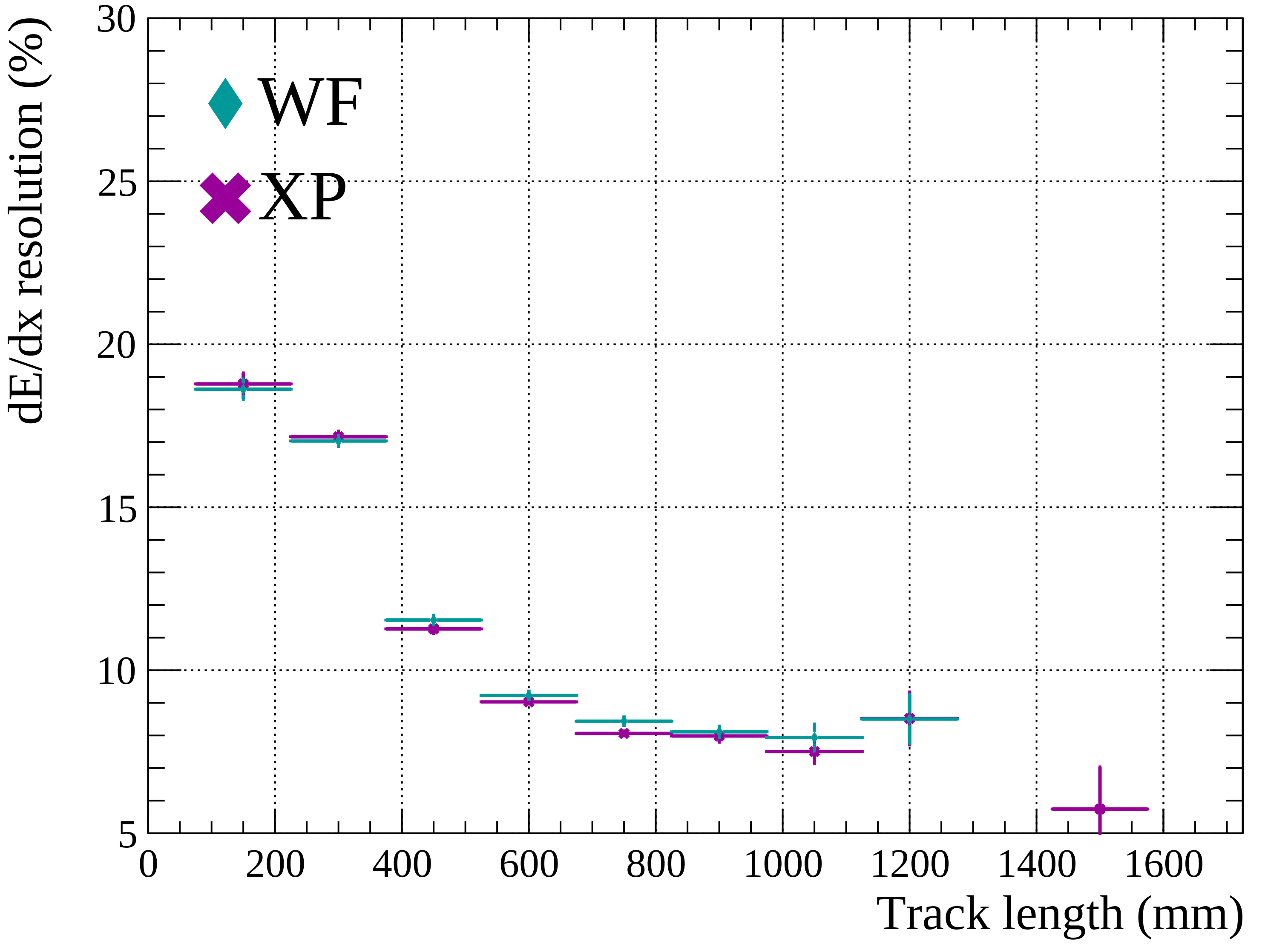
y = 0.08x + 383.09
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0


Energy loss (XP) vs drift time in ERAM 24
y = -1.70x + 843.19
Energy loss (XP) vs drift time in ERAM 30
y = 0.12x + 385.71
Energy loss (XP) vs drift time in ERAM 28
y = 0.10x + 405.67
Energy loss (XP) vs drift time in ERAM 19
y = 0.20x + 379.56
Energy loss (XP) vs drift time in ERAM 21
y = 0.09x + 387.10
Energy loss (XP) vs drift time in ERAM 13
y = 0.02x + 414.12
Energy loss (XP) vs drift time in ERAM 9
y = 0.09x + 408.03
Energy loss (XP) vs drift time in ERAM 2
y = 0.04x + 419.20
Energy loss (XP) vs drift time in ERAM 26
y = 0.11x + 384.41
Energy loss (XP) vs drift time in ERAM 17
y = 0.12x + 397.36
Energy loss (XP) vs drift time in ERAM 23
y = 0.17x + 383.94
Energy loss (XP) vs drift time in ERAM 29
y = 0.29x + 368.93
Energy loss (XP) vs drift time in ERAM 1
y = 0.16x + 389.46
Energy loss (XP) vs drift time in ERAM 10
y = 0.12x + 391.46
Energy loss (XP) vs drift time in ERAM 11
y = 0.16x + 391.73
Energy loss (XP) vs drift time in ERAM 3
y = 0.12x + 387.24
dE/dx (ADC counts/cm)
drift time (timebins)

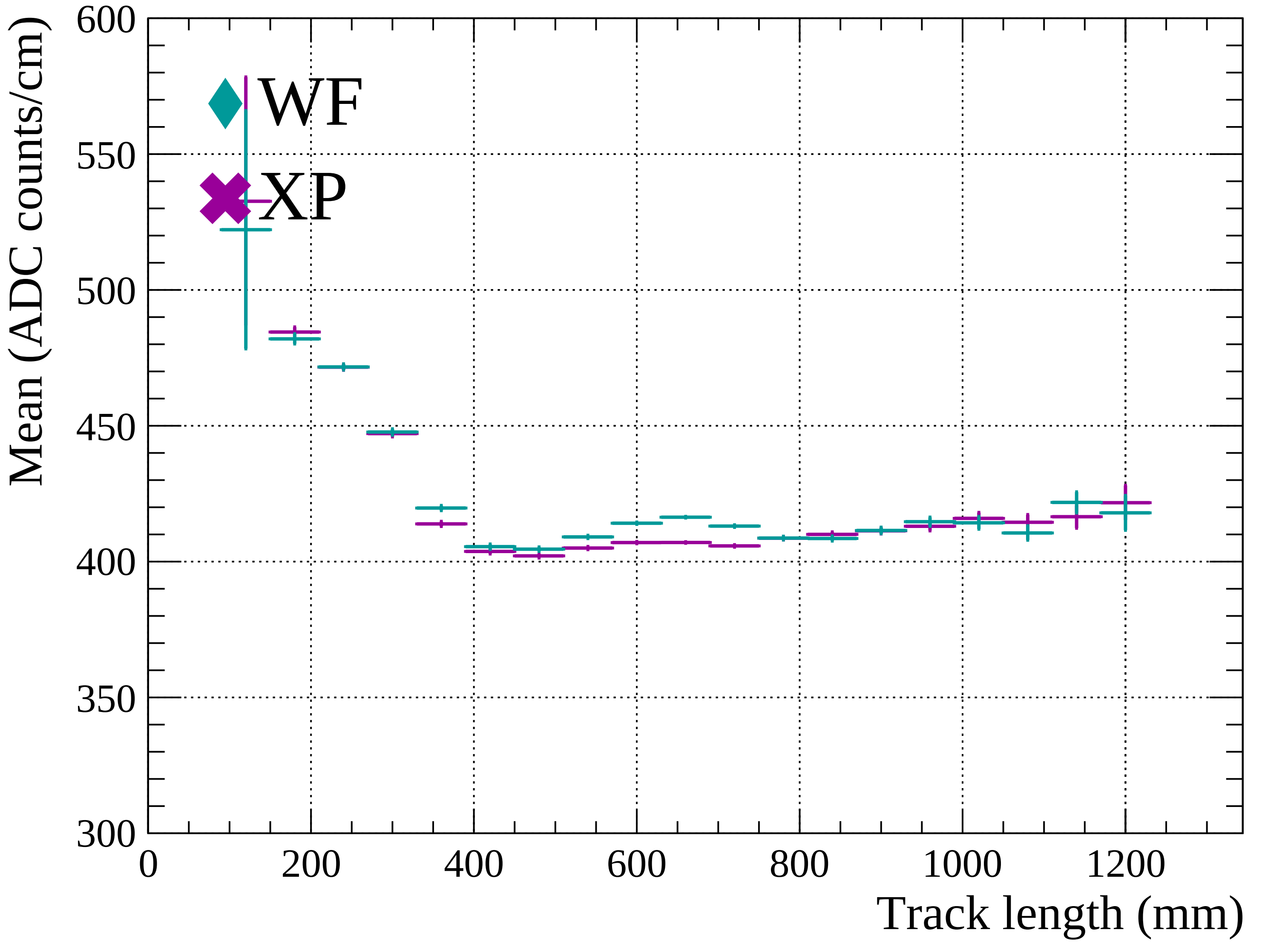

Energy loss (XP) vs drift time in ERAM 47
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 16
y = 0.09x + 377.40
Energy loss (XP) vs drift time in ERAM 14
y = 0.17x + 378.61
Energy loss (XP) vs drift time in ERAM 15
y = 0.08x + 418.72
Energy loss (XP) vs drift time in ERAM 42
y = 0.08x + 378.02
Energy loss (XP) vs drift time in ERAM 45
y = 0.04x + 394.24
Energy loss (XP) vs drift time in ERAM 37
y = 0.08x + 398.74
Energy loss (XP) vs drift time in ERAM 36
y = 0.06x + 409.35
Energy loss (XP) vs drift time in ERAM 20
y = 0.18x + 354.01
Energy loss (XP) vs drift time in ERAM 38
y = 0.18x + 362.00
Energy loss (XP) vs drift time in ERAM 7
y = 0.19x + 365.58
Energy loss (XP) vs drift time in ERAM 44
y = 0.06x + 412.75
Energy loss (XP) vs drift time in ERAM 43
y = 0.12x + 387.83
Energy loss (XP) vs drift time in ERAM 39
y = 0.13x + 377.77
Energy loss (XP) vs drift time in ERAM 41
y = 0.16x + 373.35
Energy loss (XP) vs drift time in ERAM 46
y = 0.15x + 378.88
dE/dx (ADC counts/cm)
drift time (timebins)



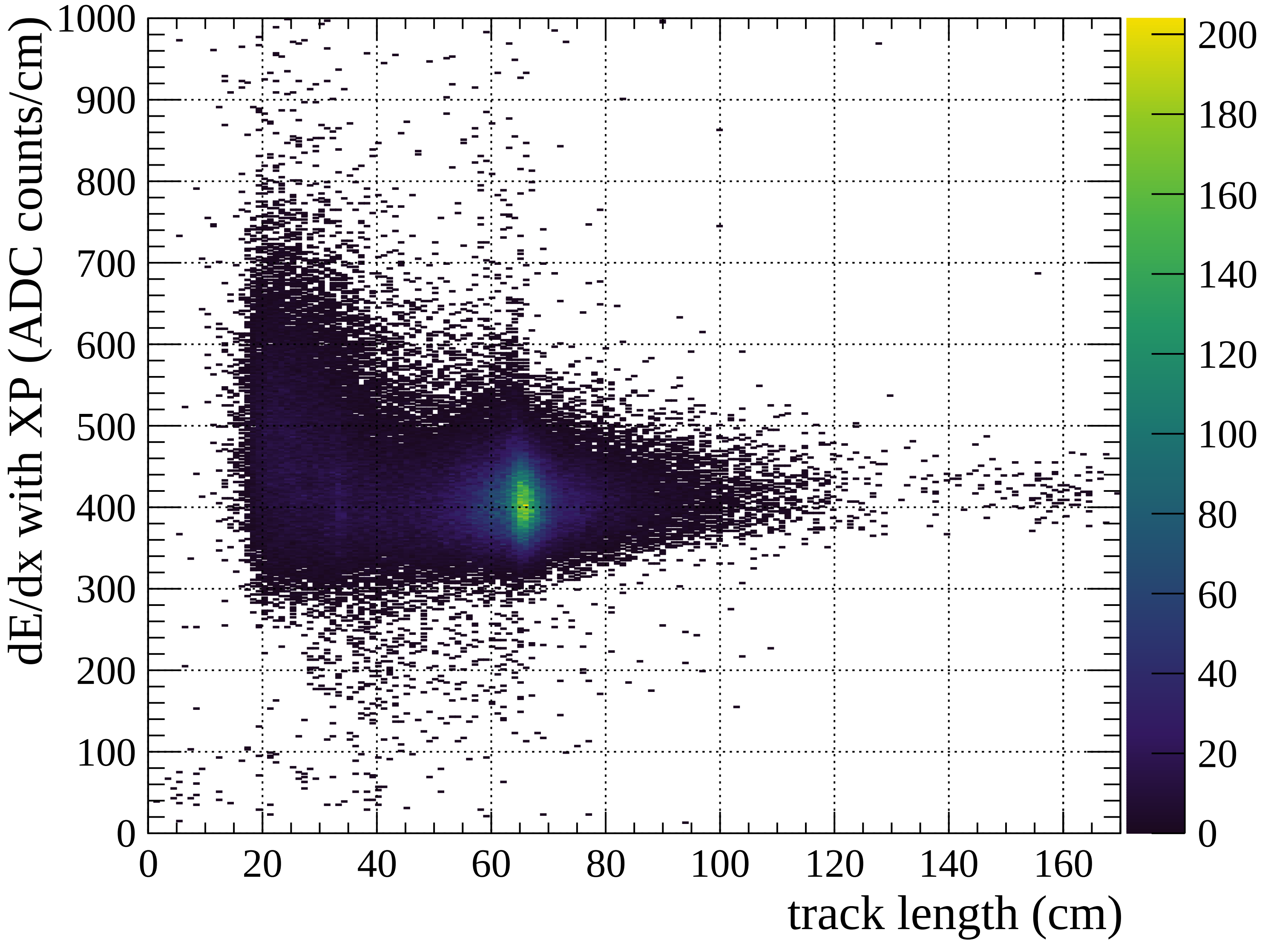
time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
20
18
16
14
12
10
8
6
4
2
0



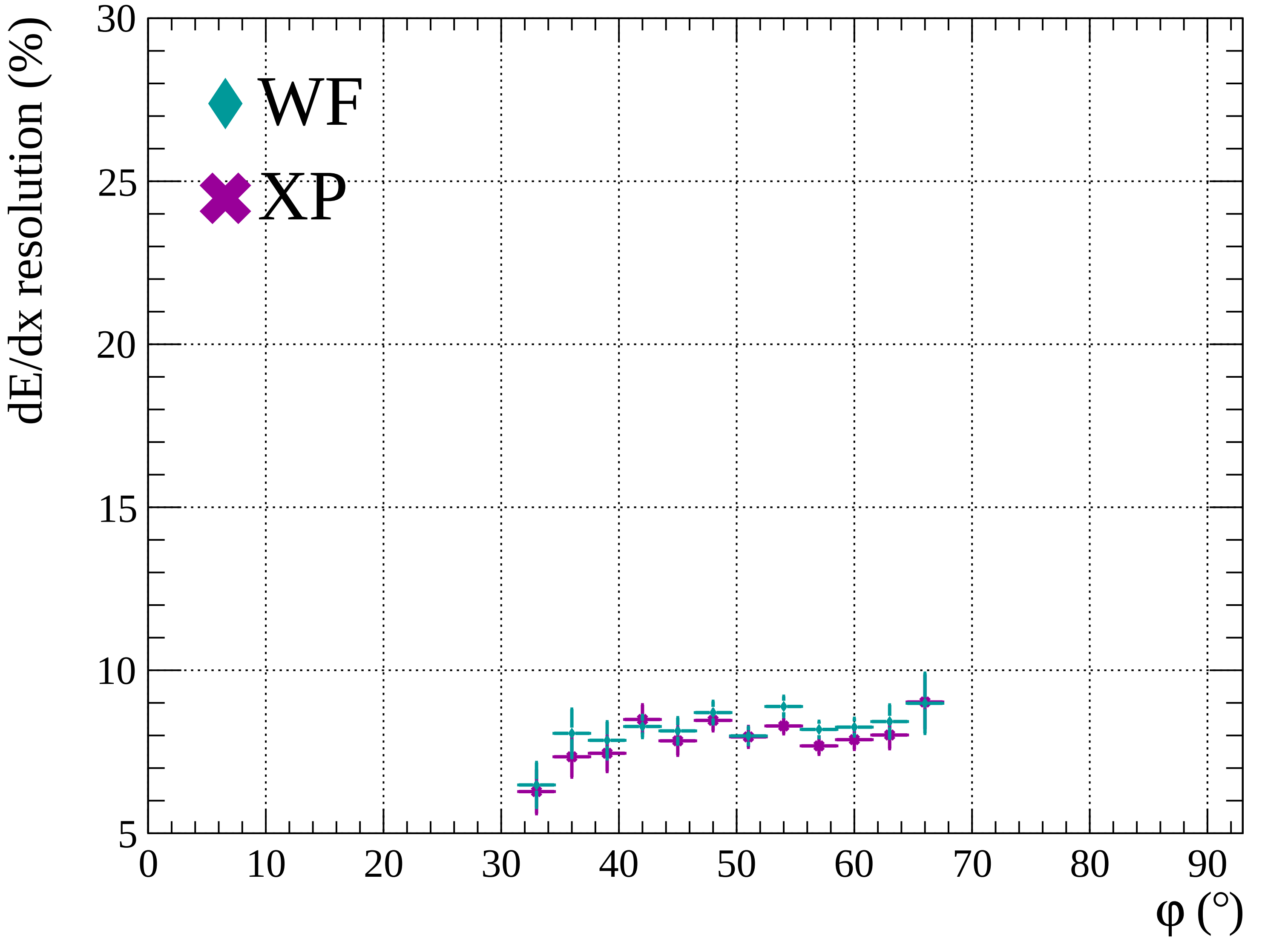
dE/dx resolution (%)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
Track length (mm)
WF
XP



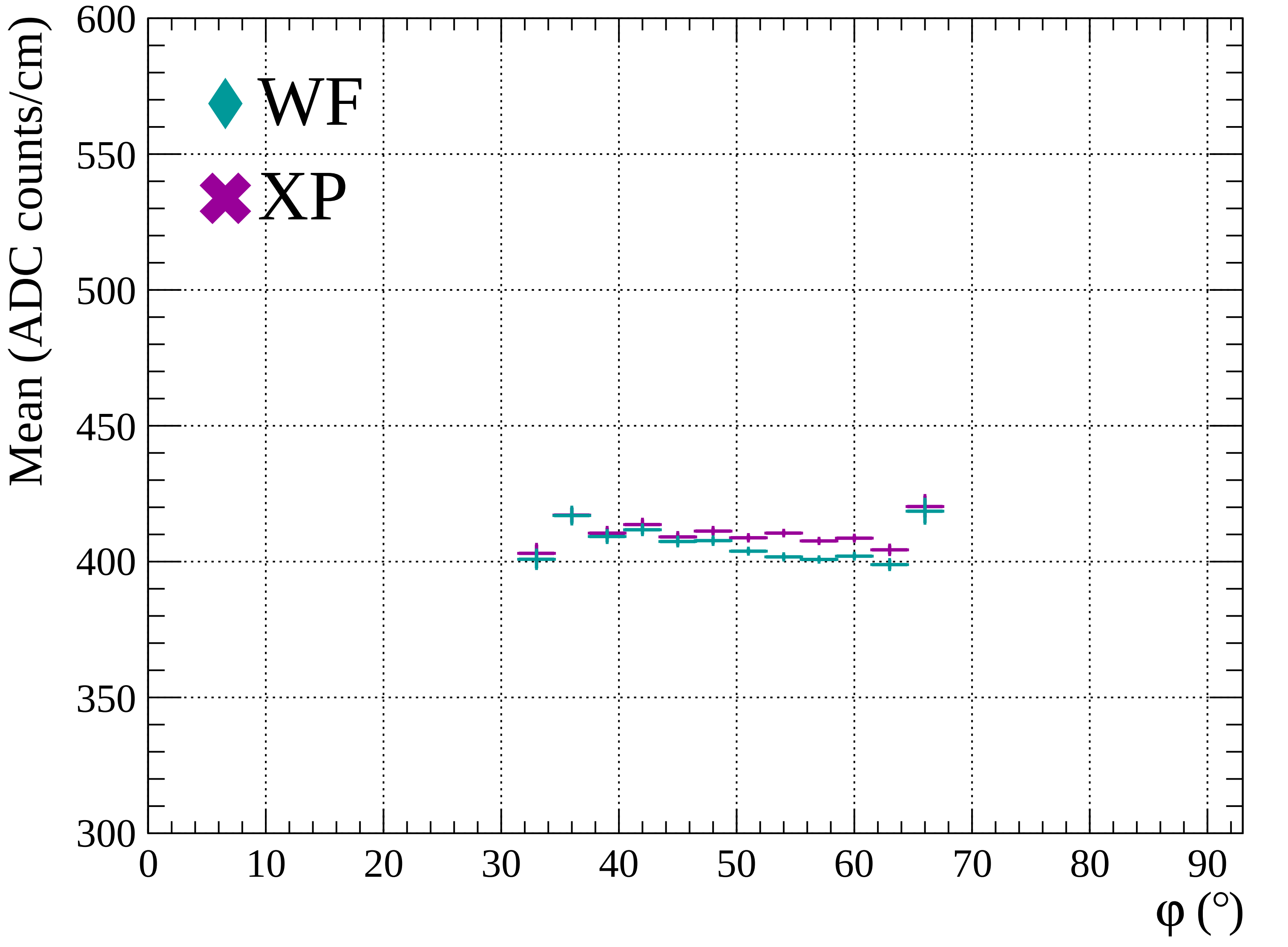
Mean (ADC counts/cm)
Track length (mm)
WF
XP
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200



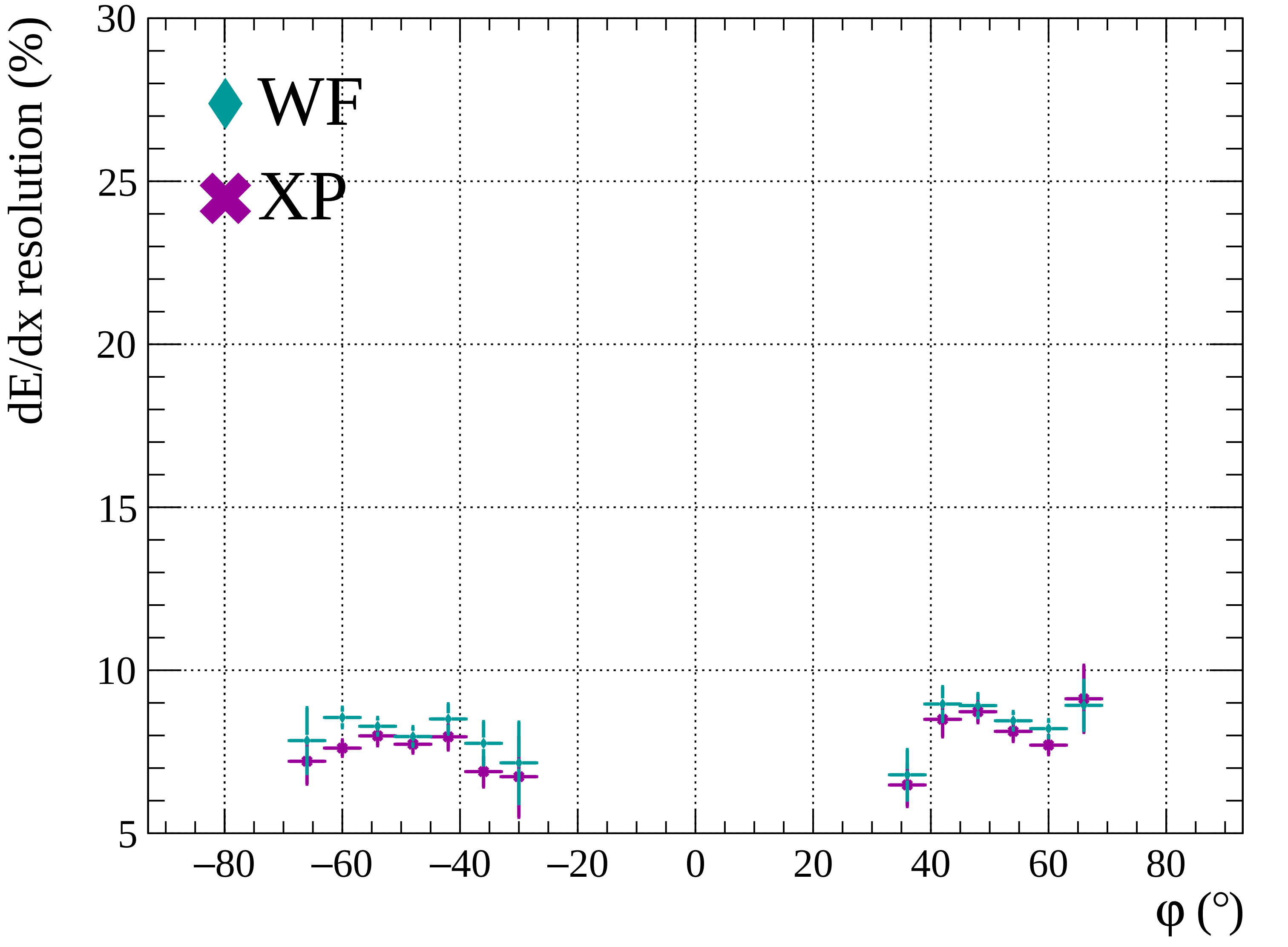
dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
20
40
60
80
100
120
140
160
180
200

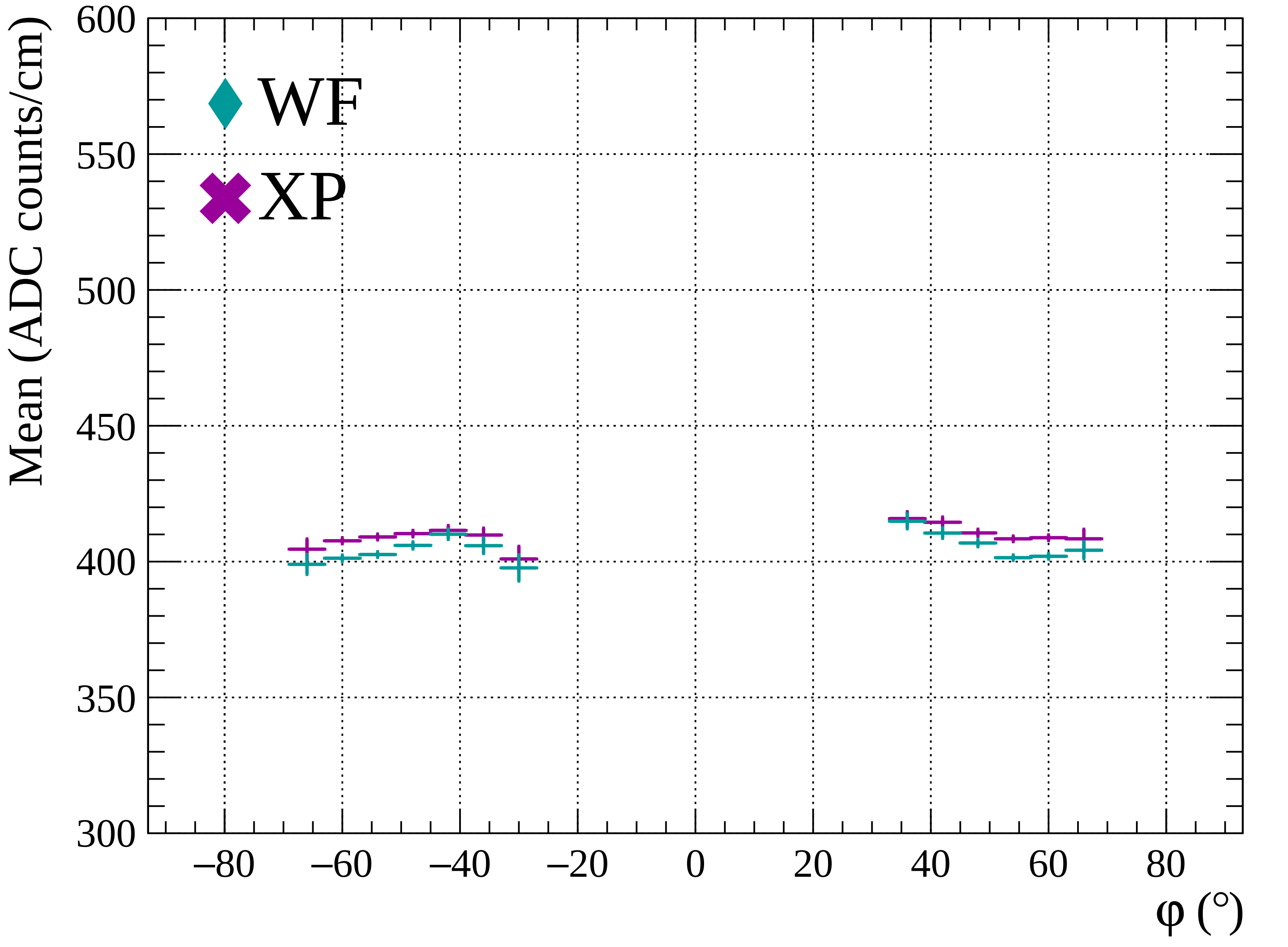


dE/dx resolution (%)
WF
XP
φ (°)
0
10
20
30
40
50
60
70
80
90
5
10
15
20
25
30

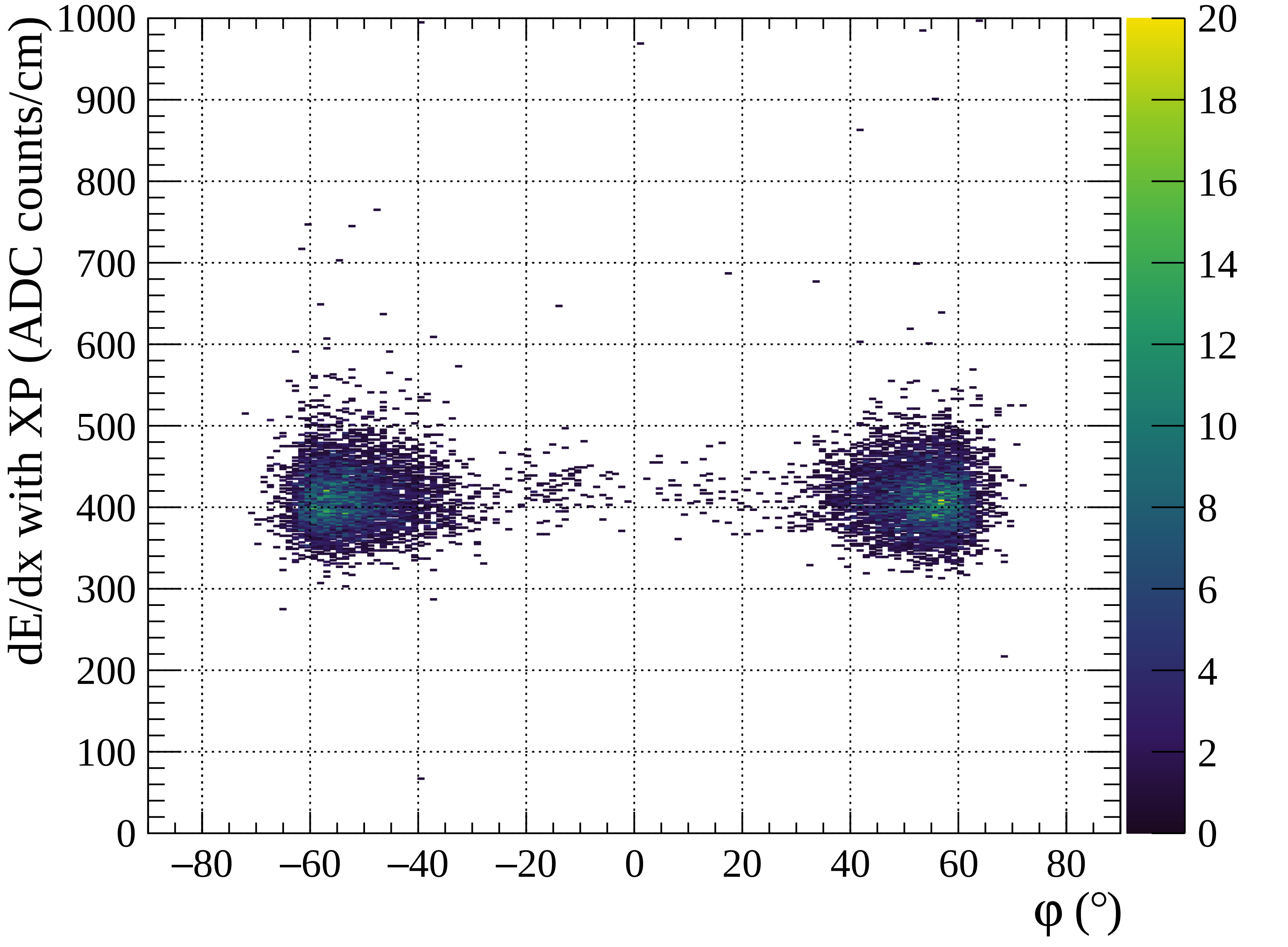


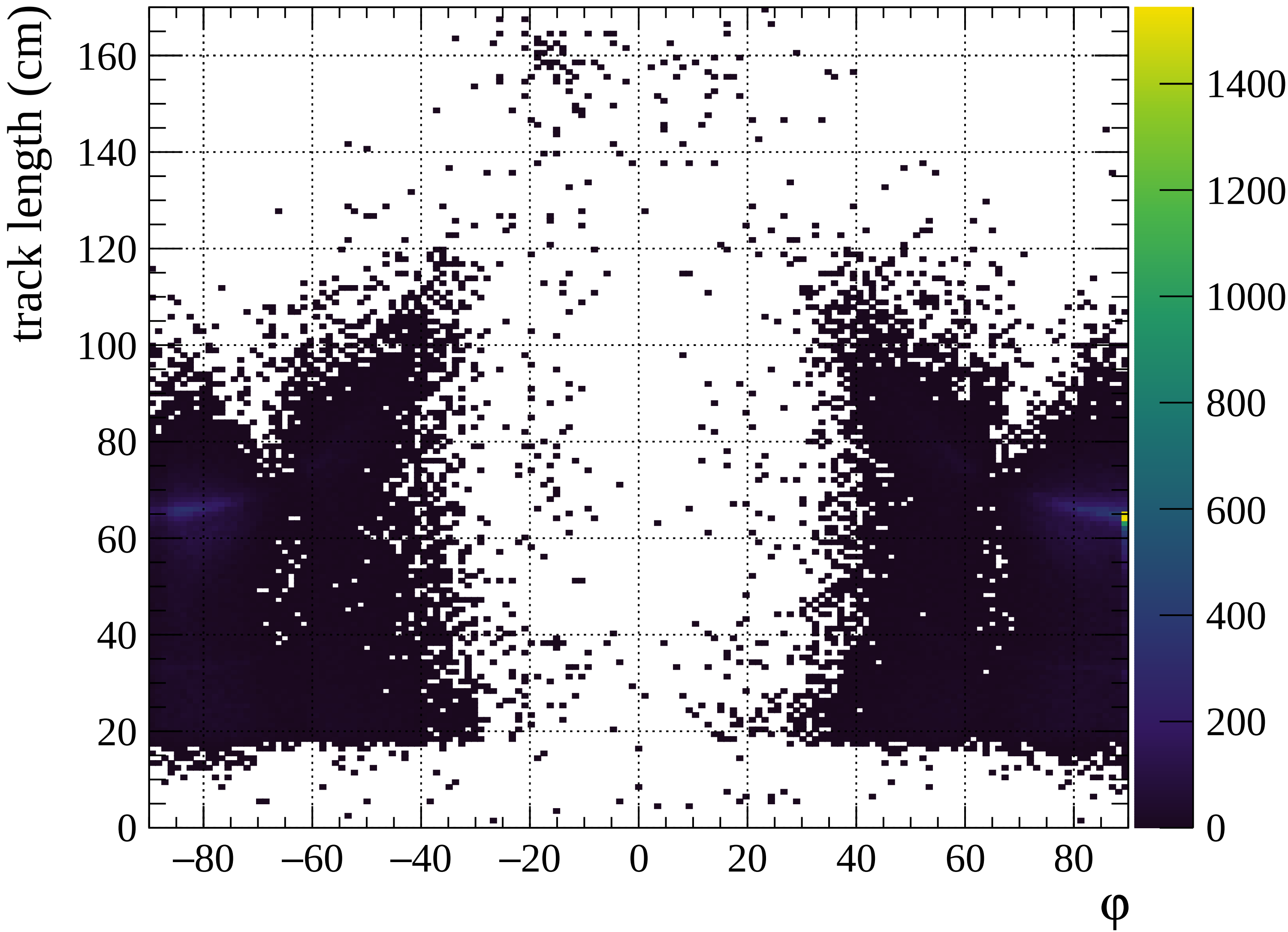
WF
XP
Mean (ADC counts/cm)
φ (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90


dE/dx resolution (%)
WF
XP
φ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80

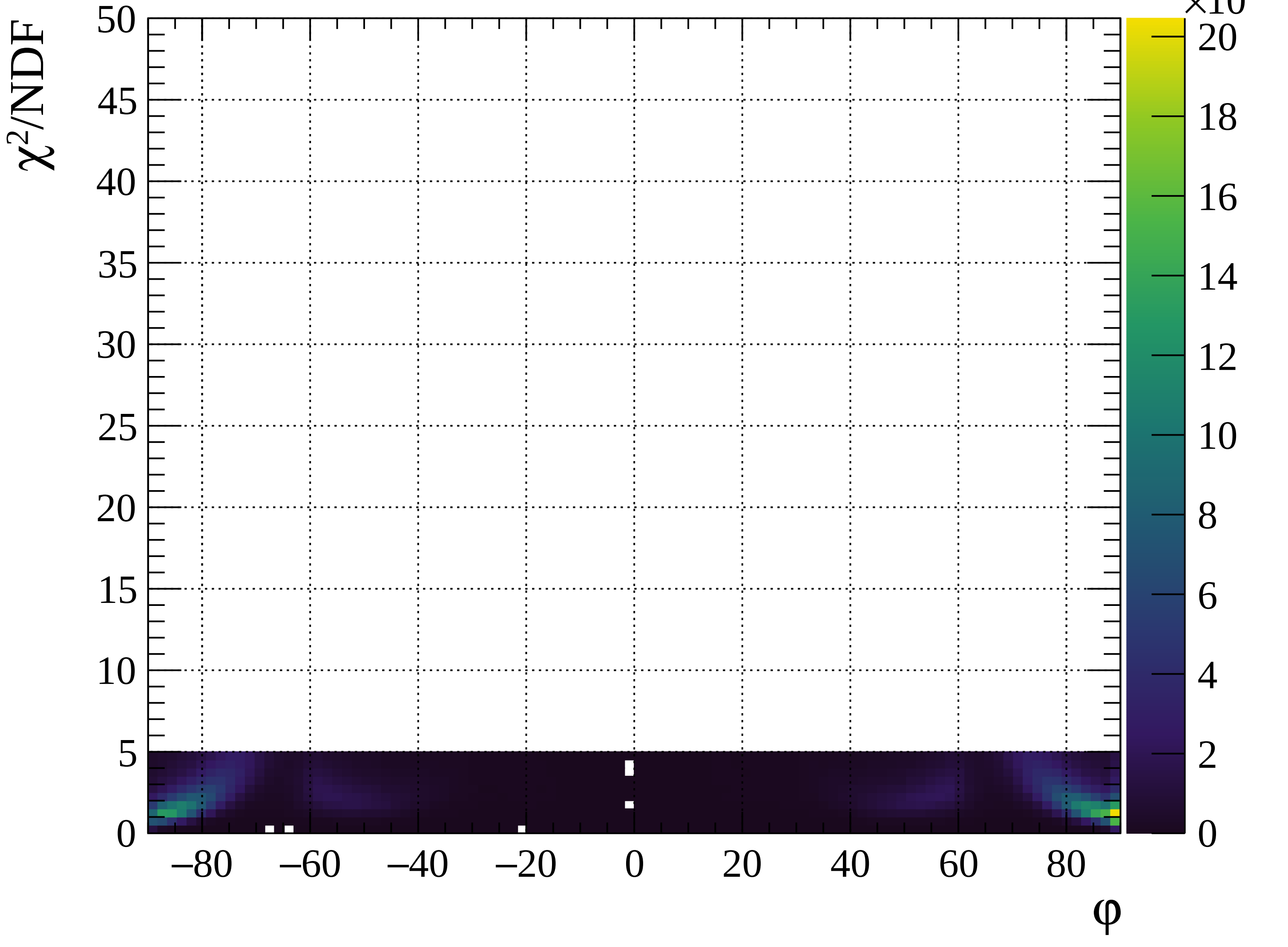


Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)

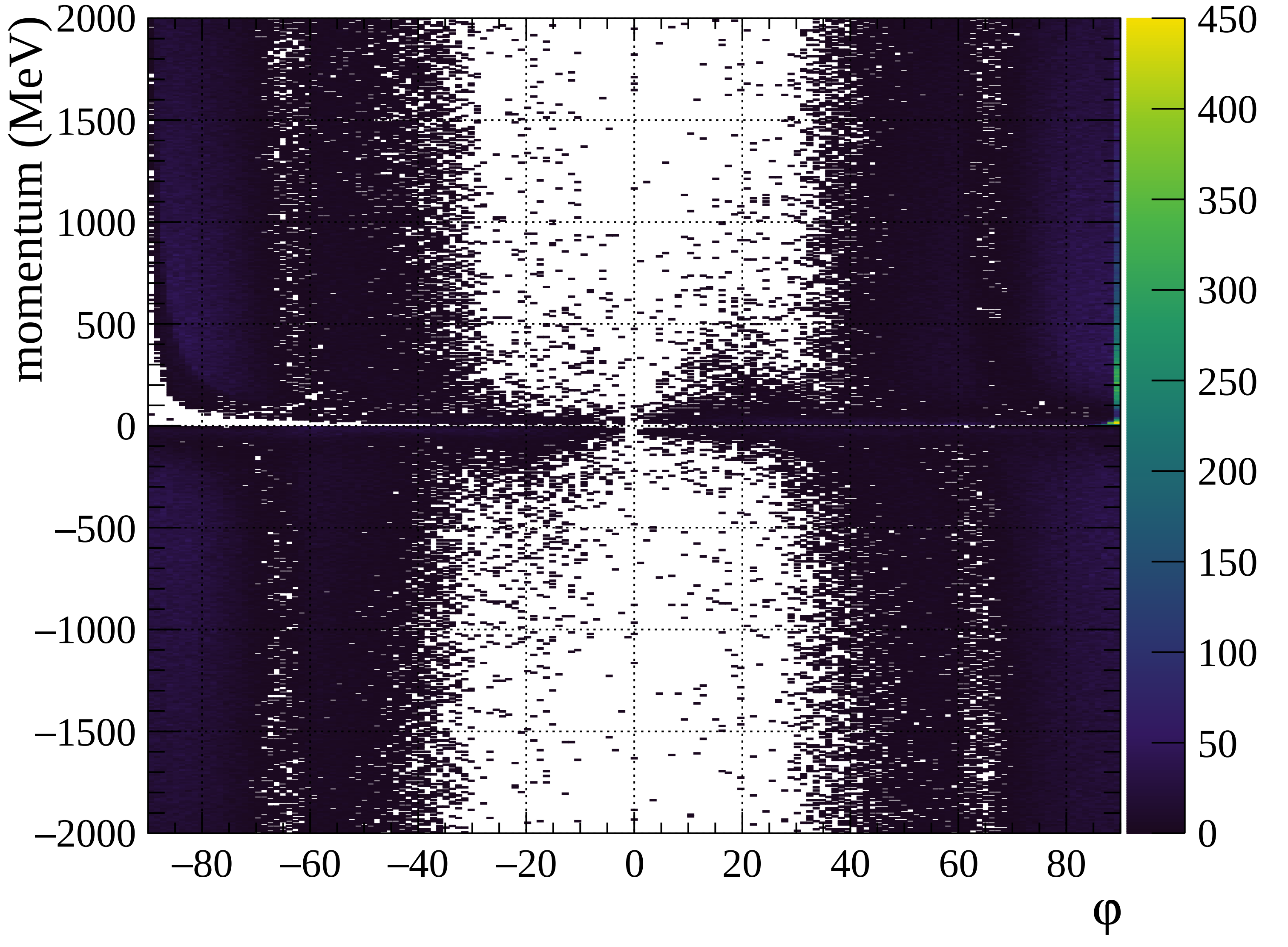

dE/dx with XP (ADC counts/cm)
φ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
20
18
16
14
12
10
8
6
4
2
0



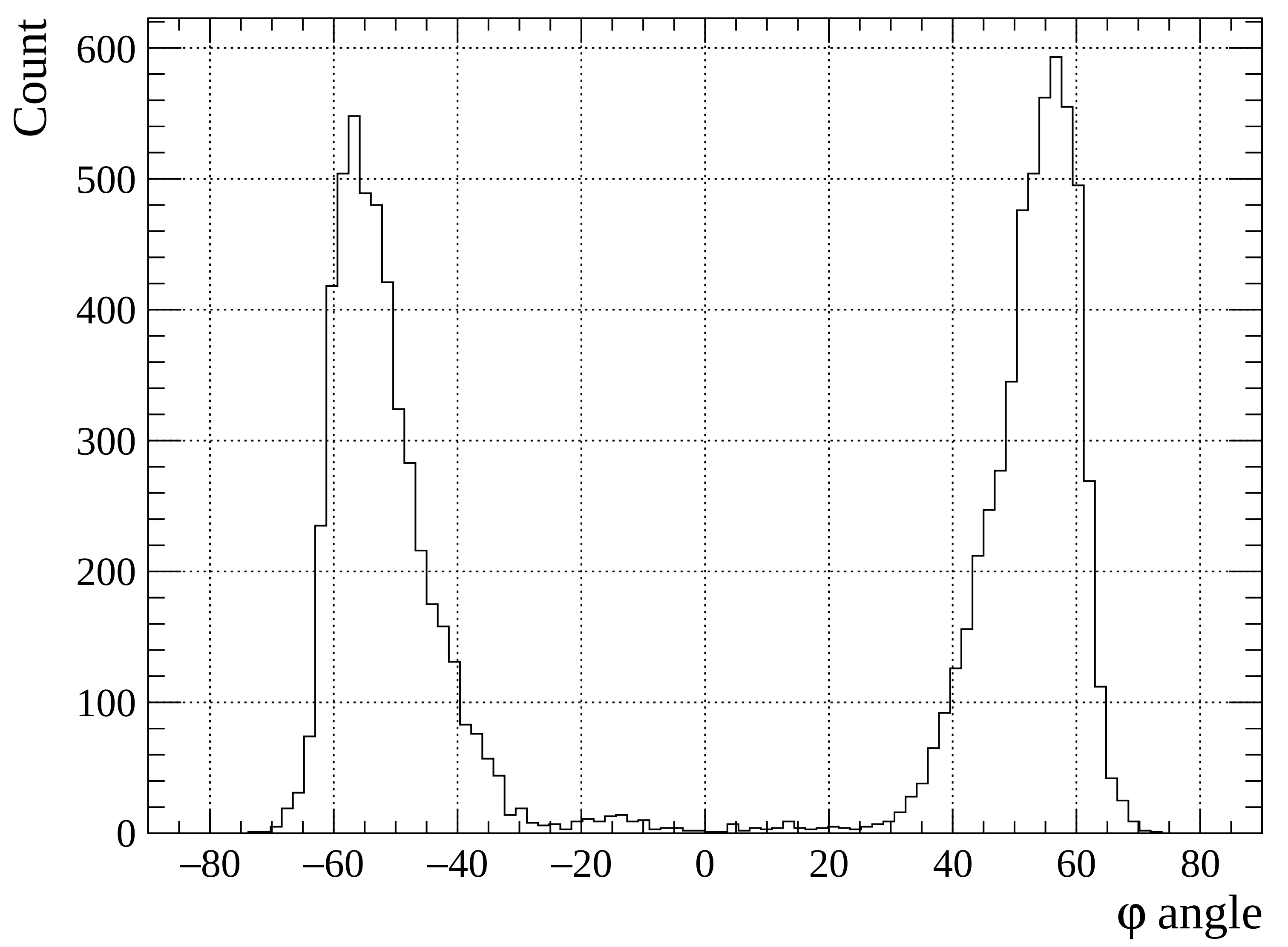
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
200
400
600
800
1000
1200
1400

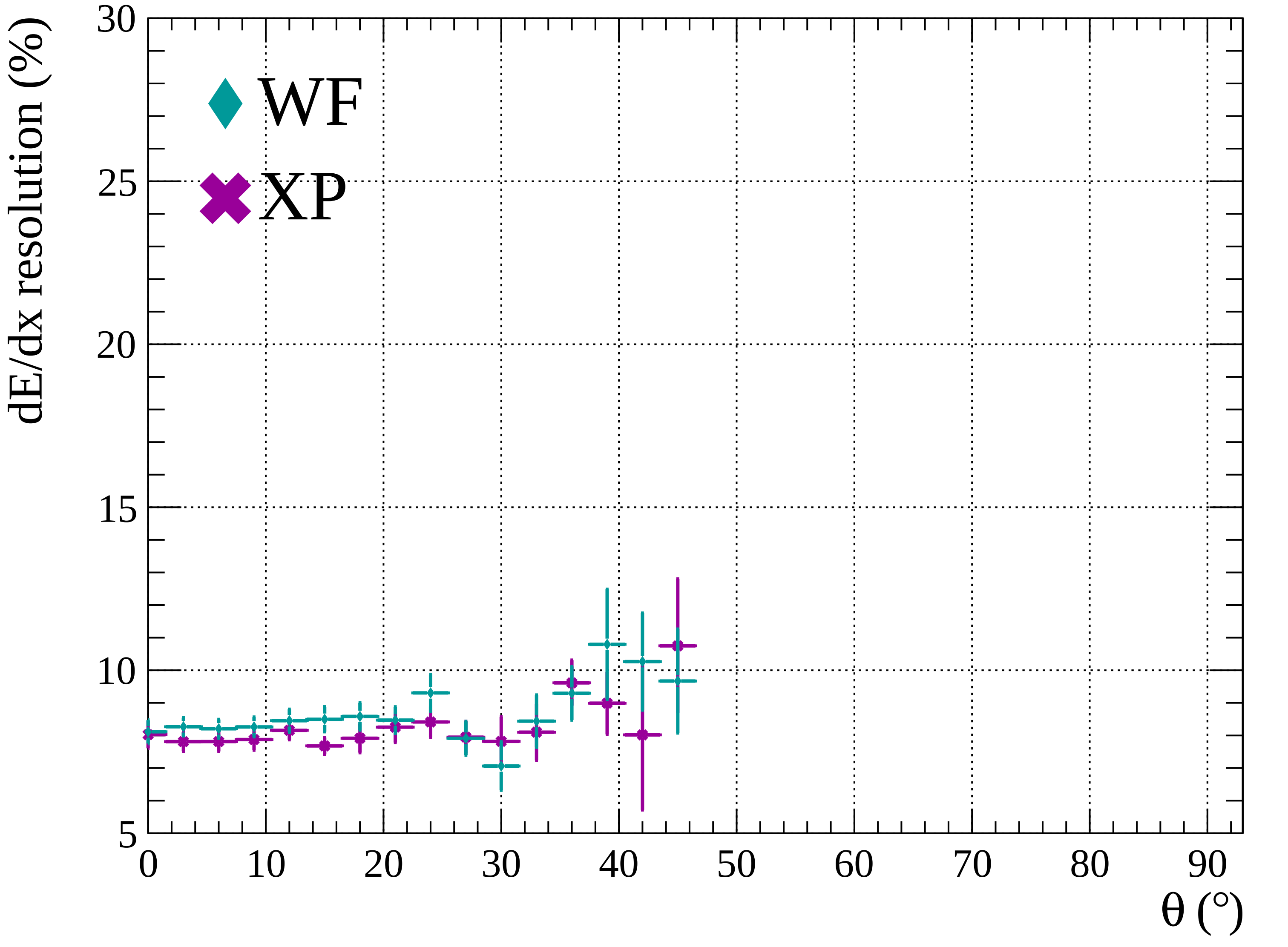


χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
×10
20
18
16
14
12
10
8
6
4
2
0

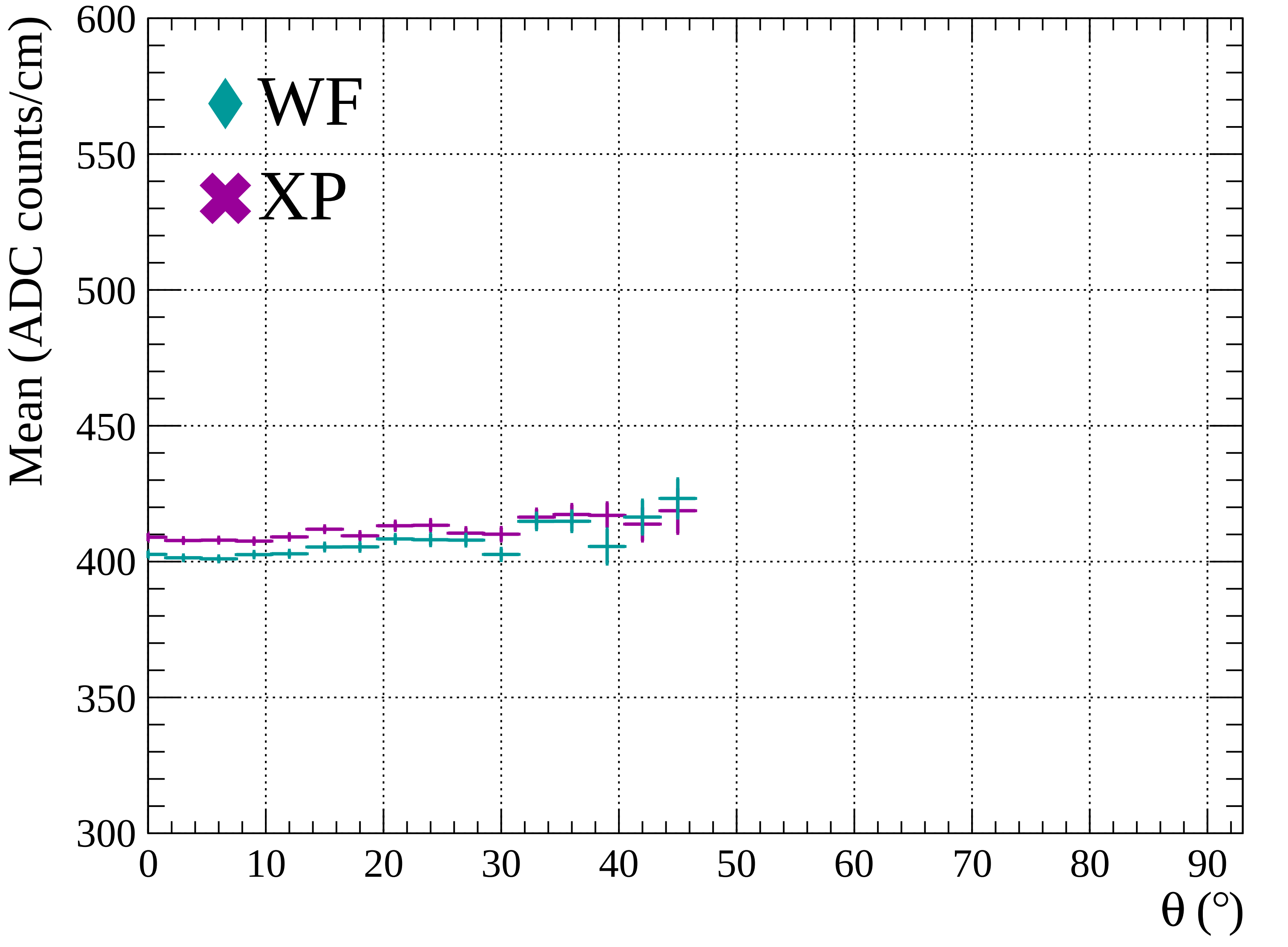


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
450
400
350
300
250
200
150
100
50
0

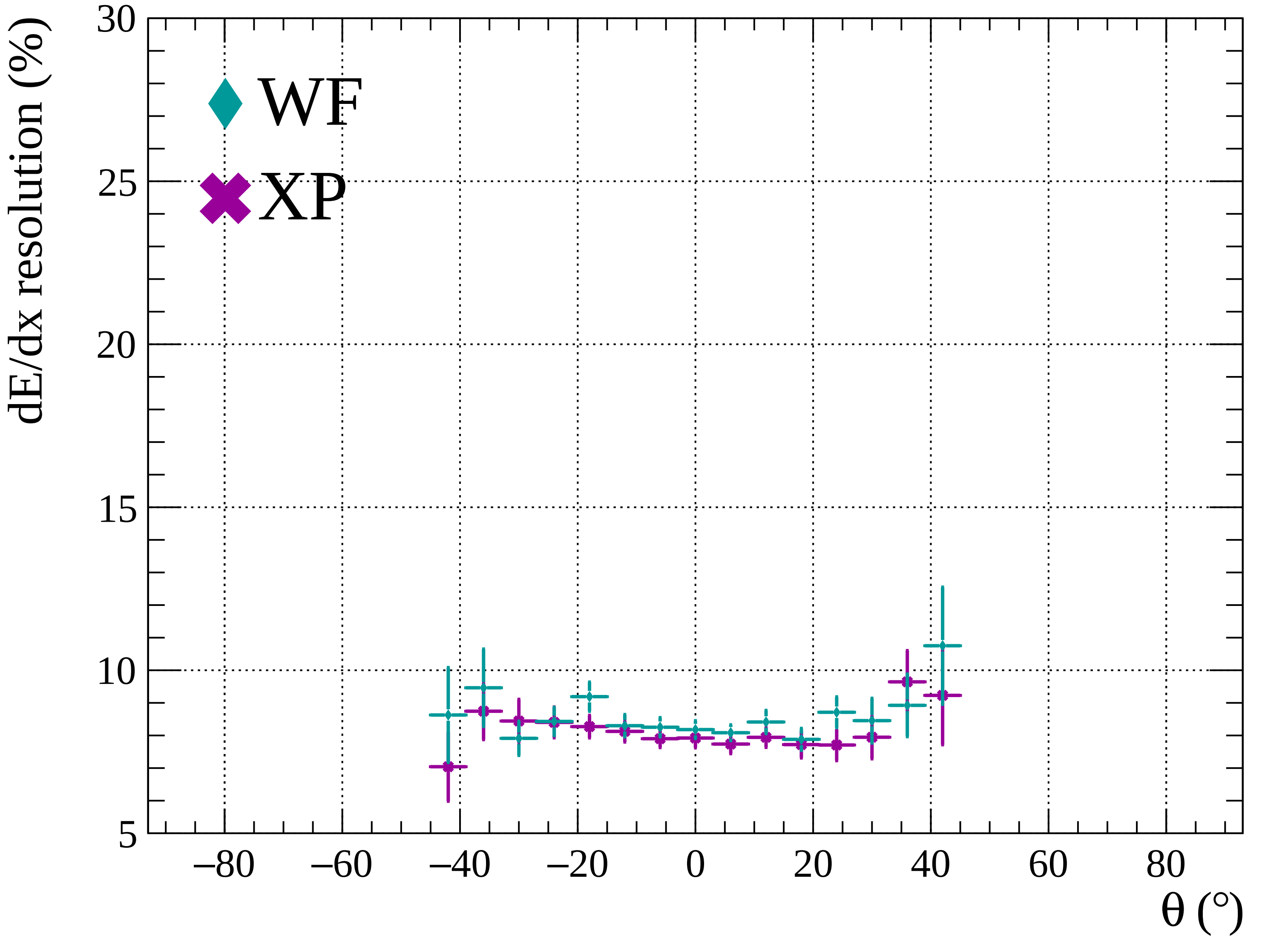

Count
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

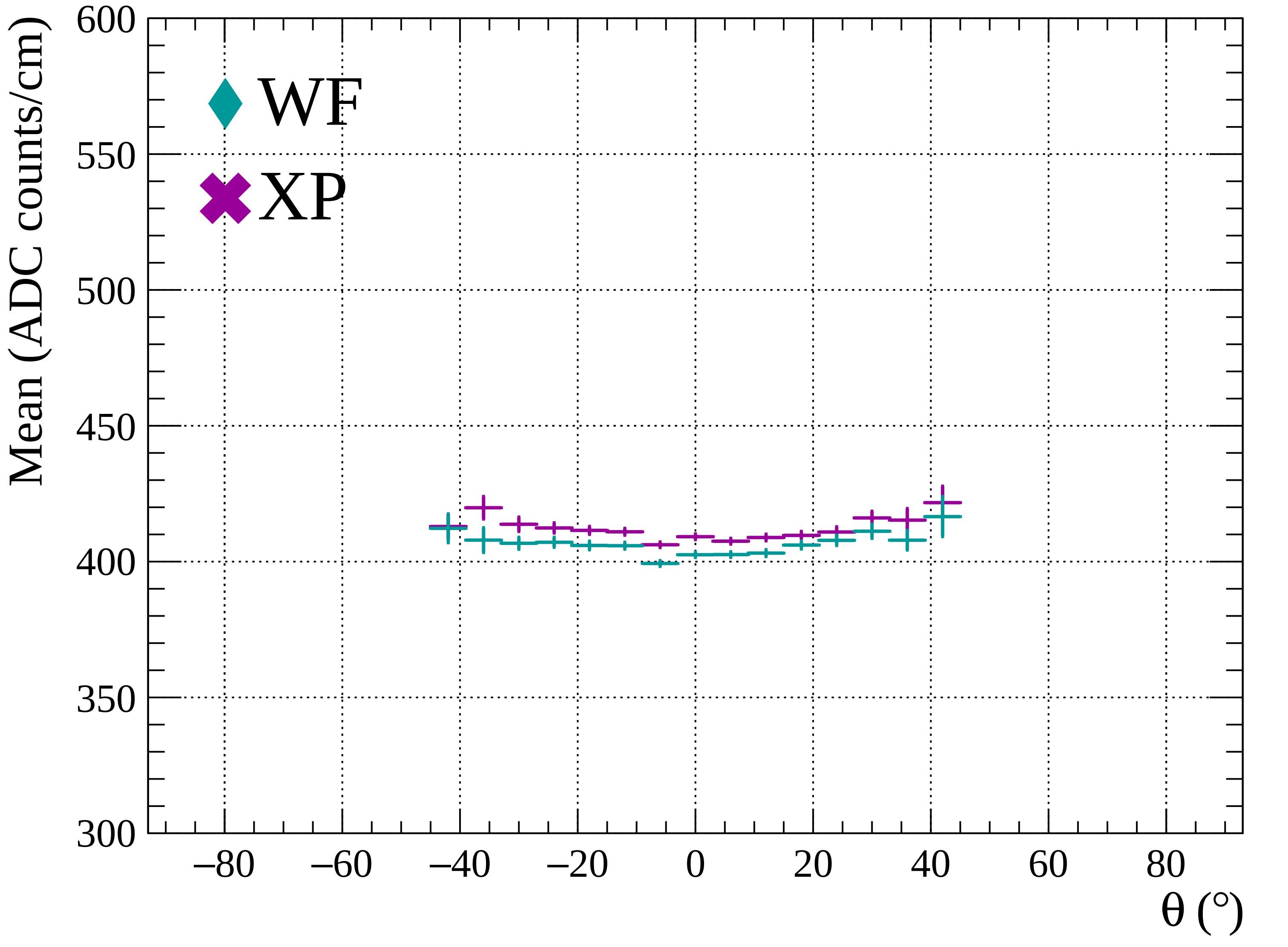


dE/dx resolution (%)
θ (°)
WF
XP
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90

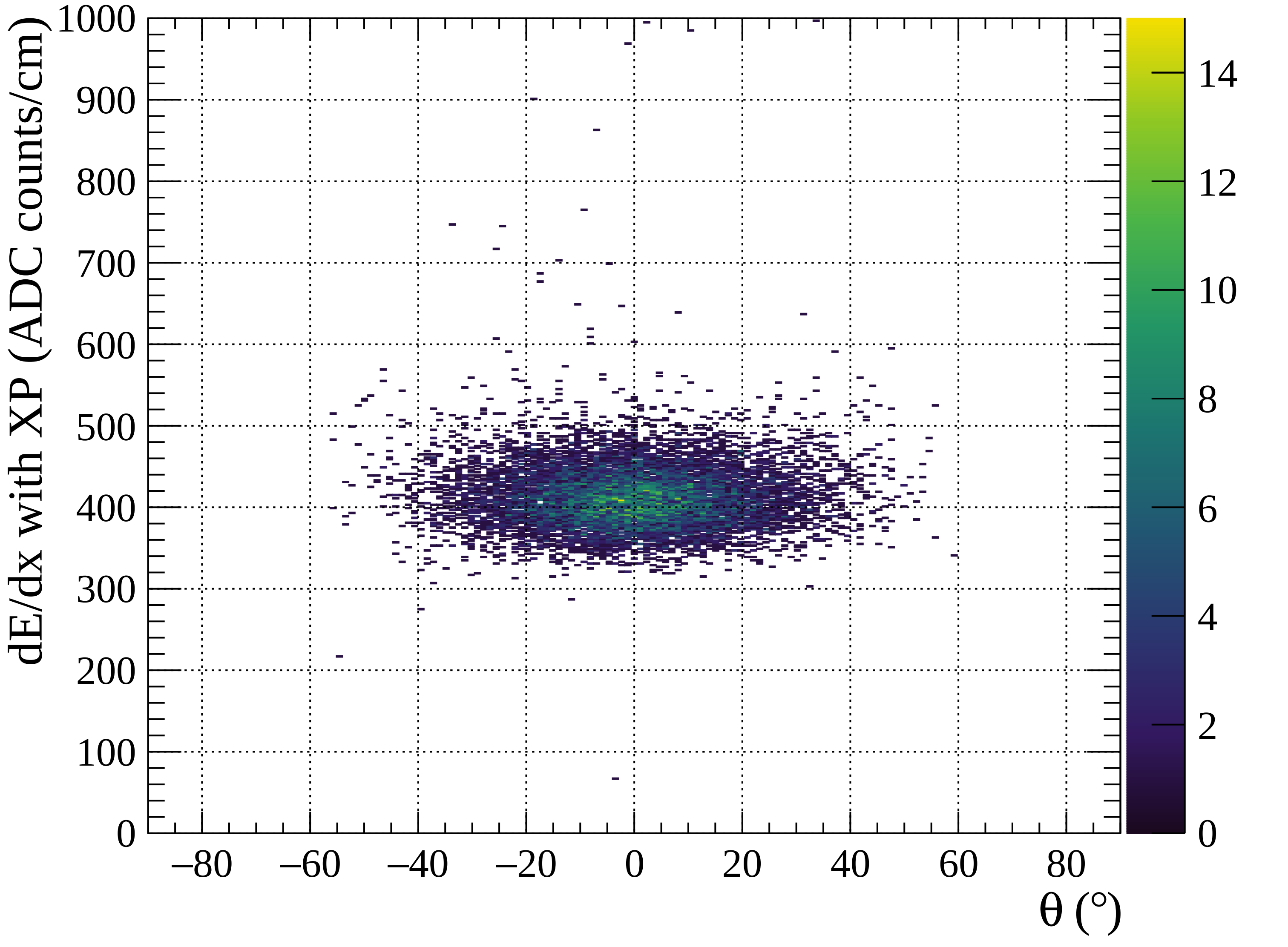


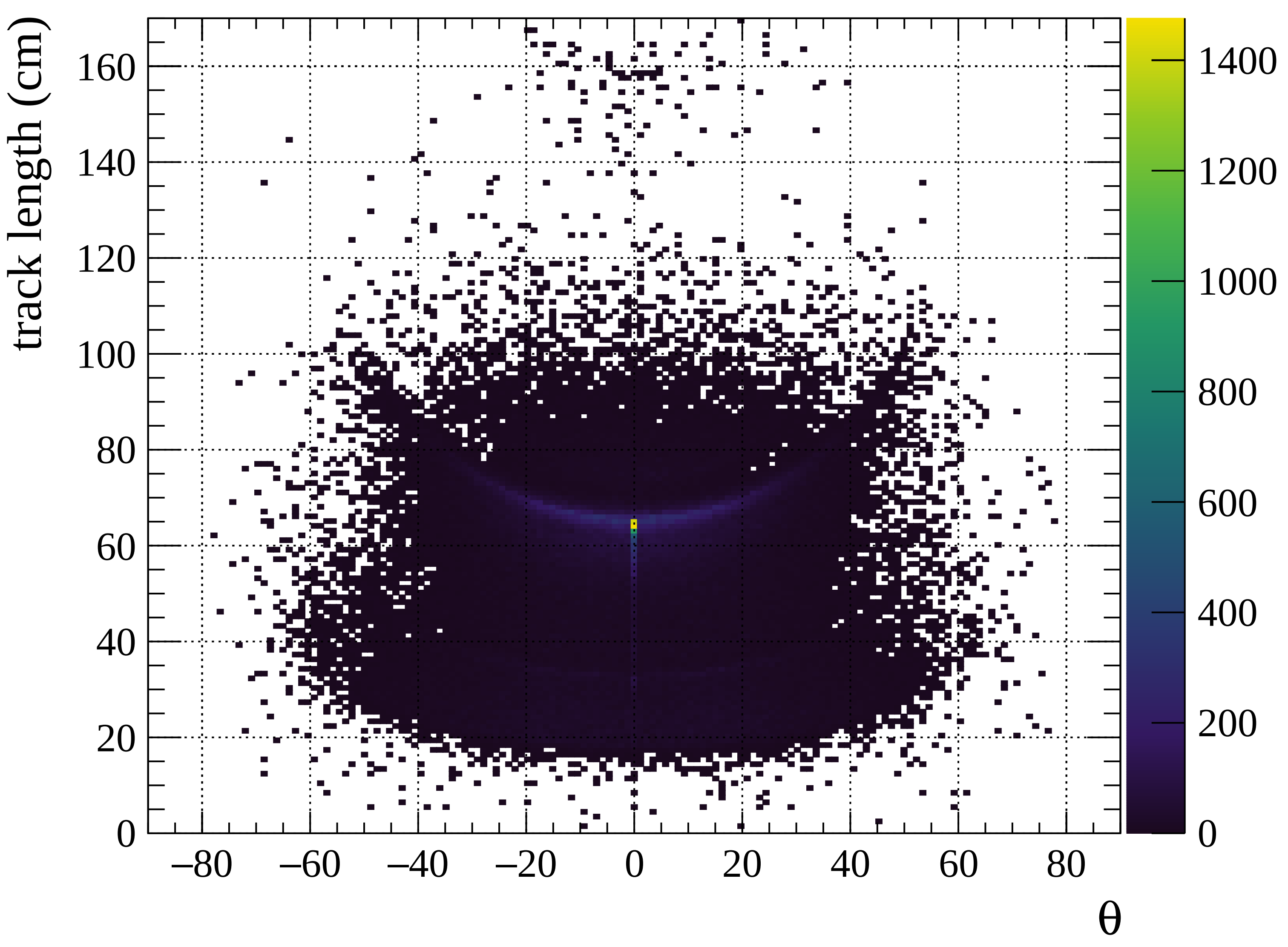
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
10
20
30
40
50
60
70
80
90
θ (°)

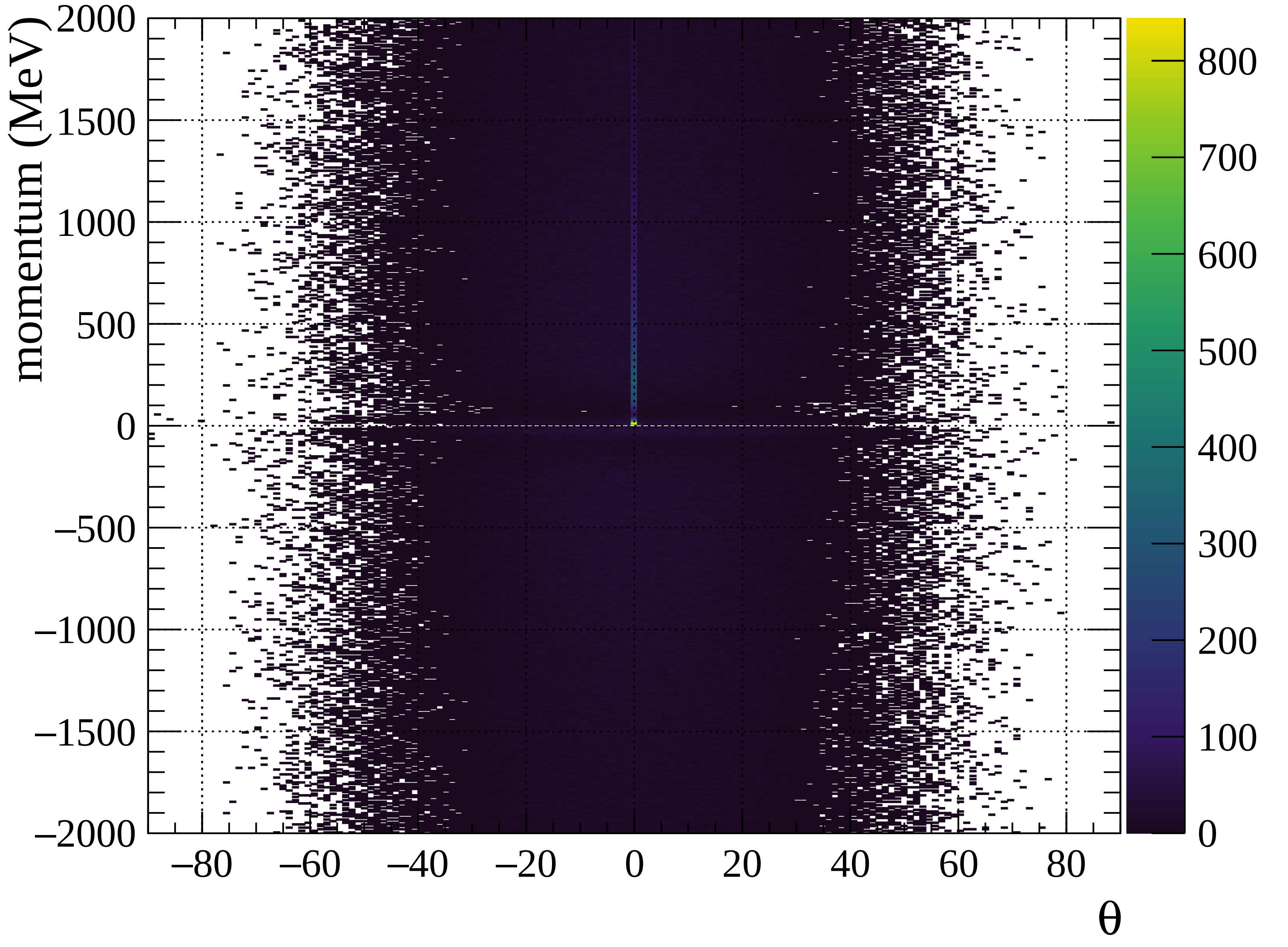

WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80

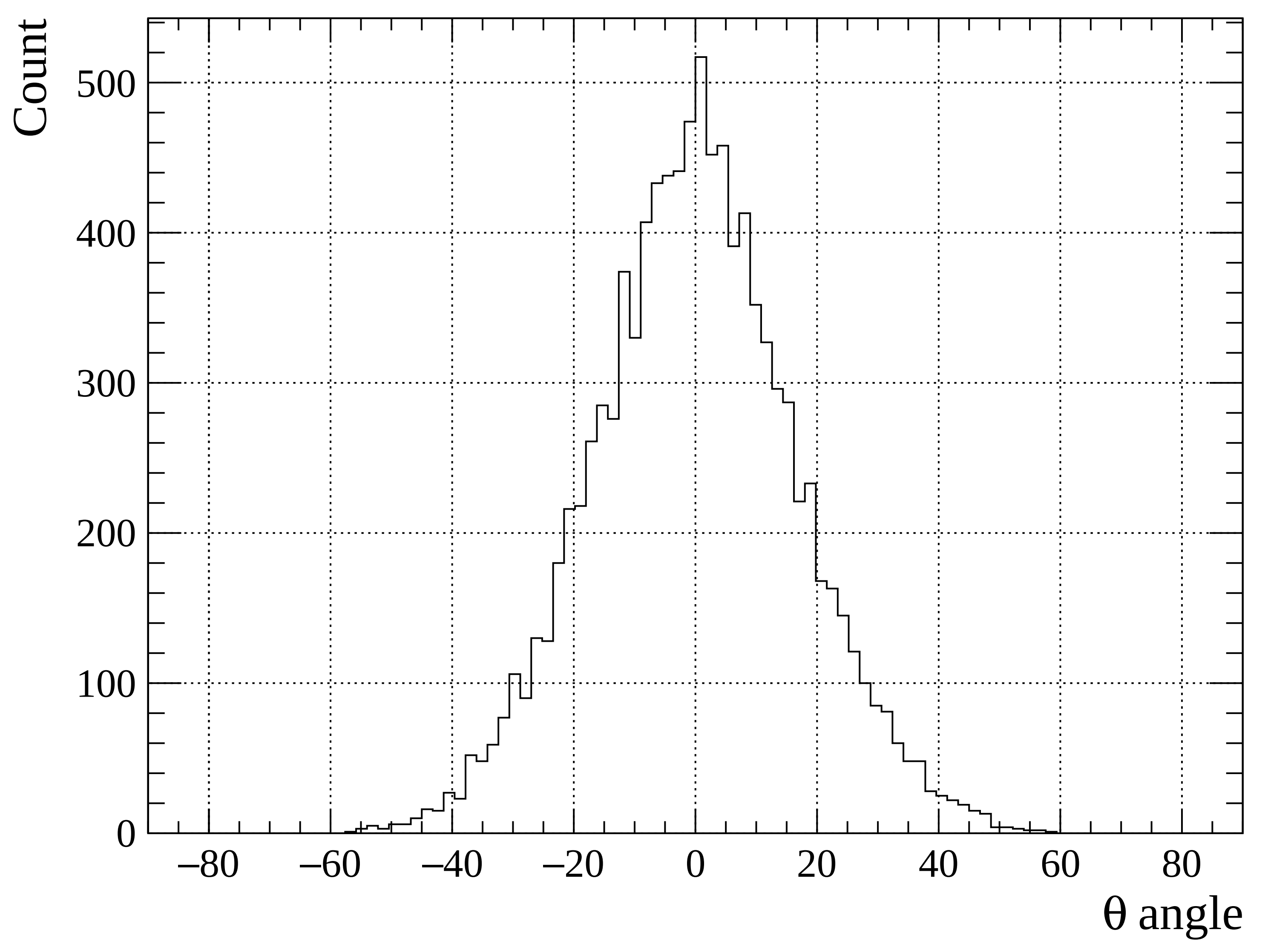


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
θ (°)

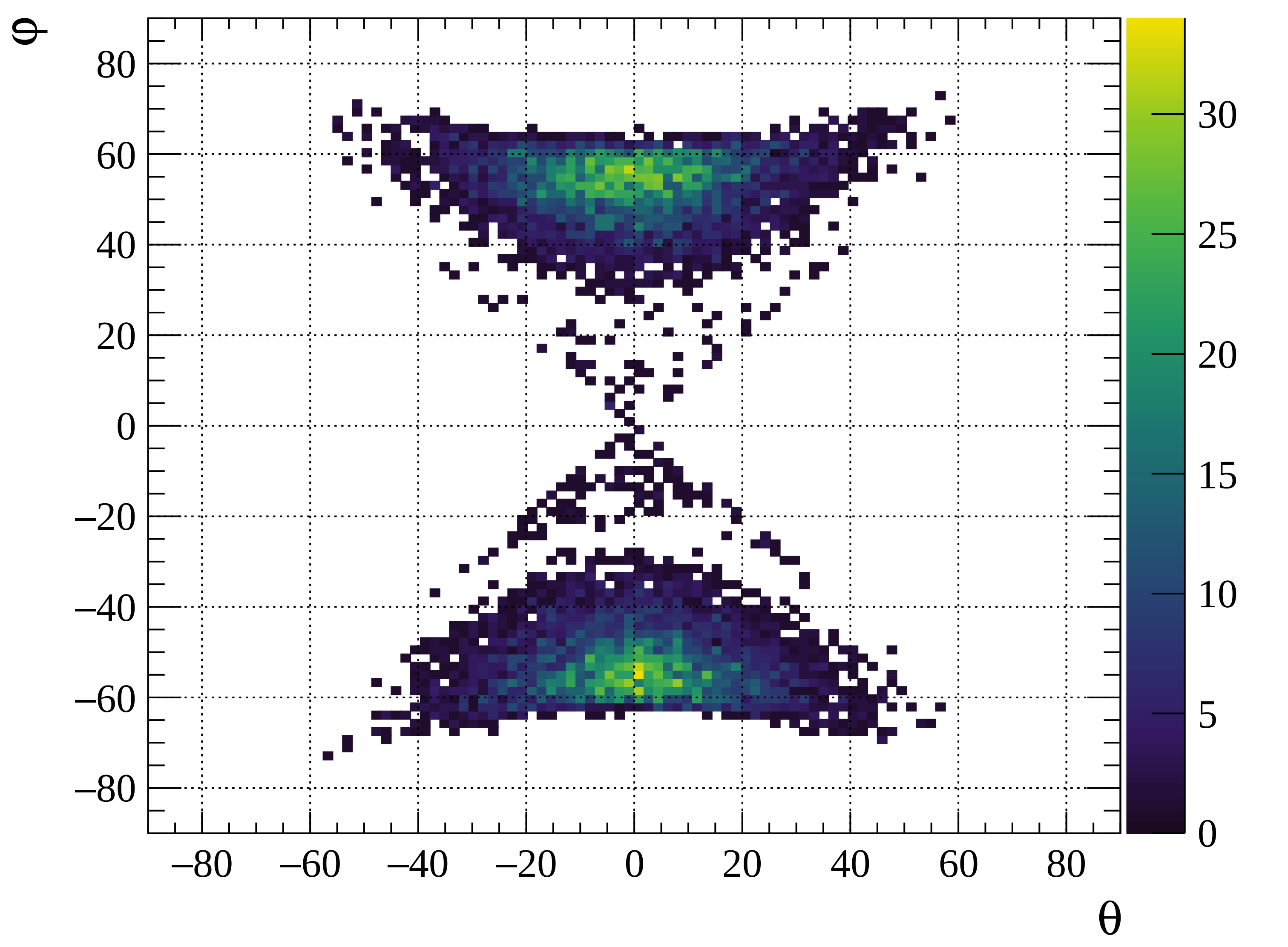

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
14
12
10
8
6
4
2
0


track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
1400
1200
1000
800
600
400
200
0

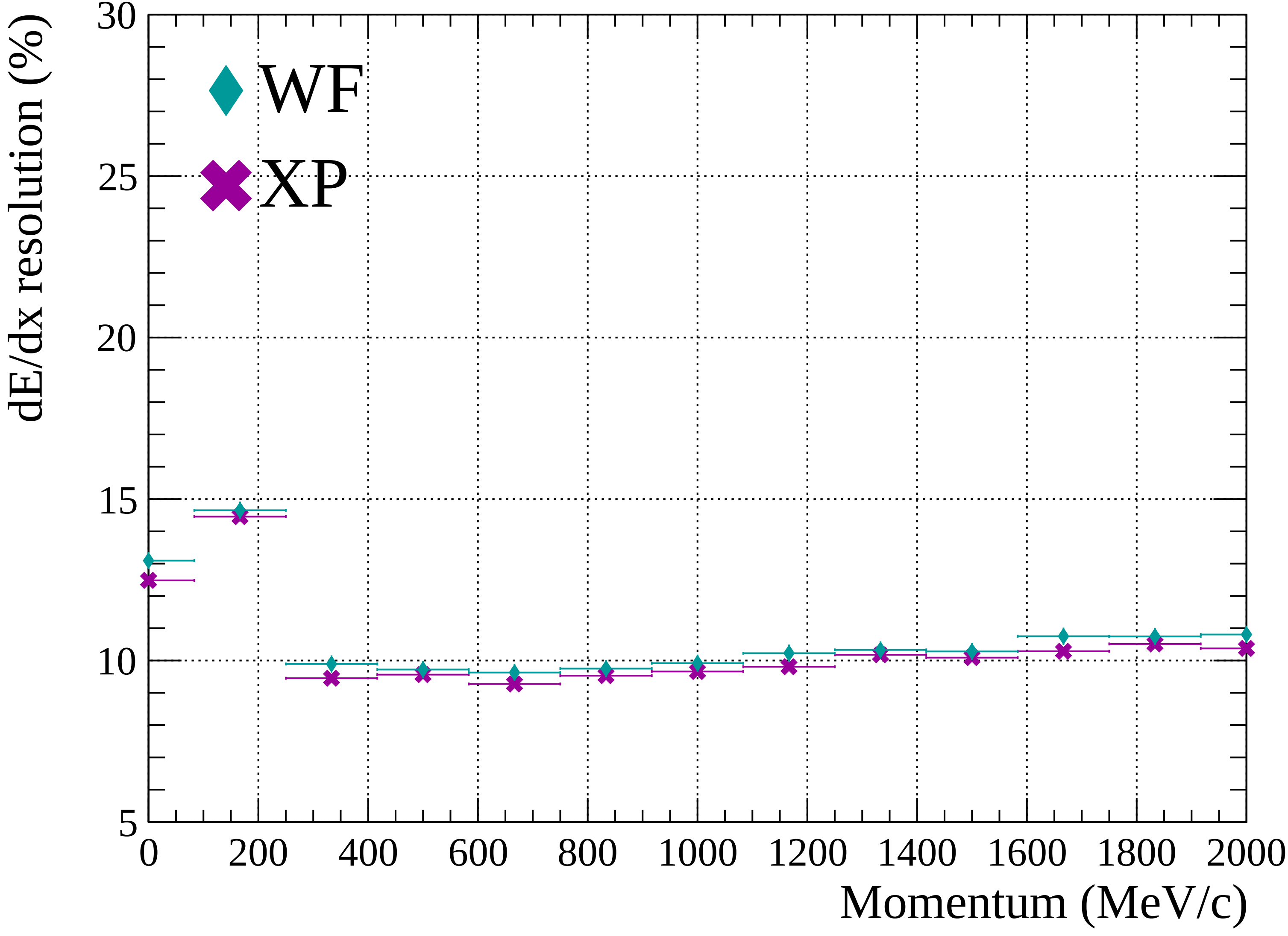

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
800
700
600
500
400
300
200
100
0



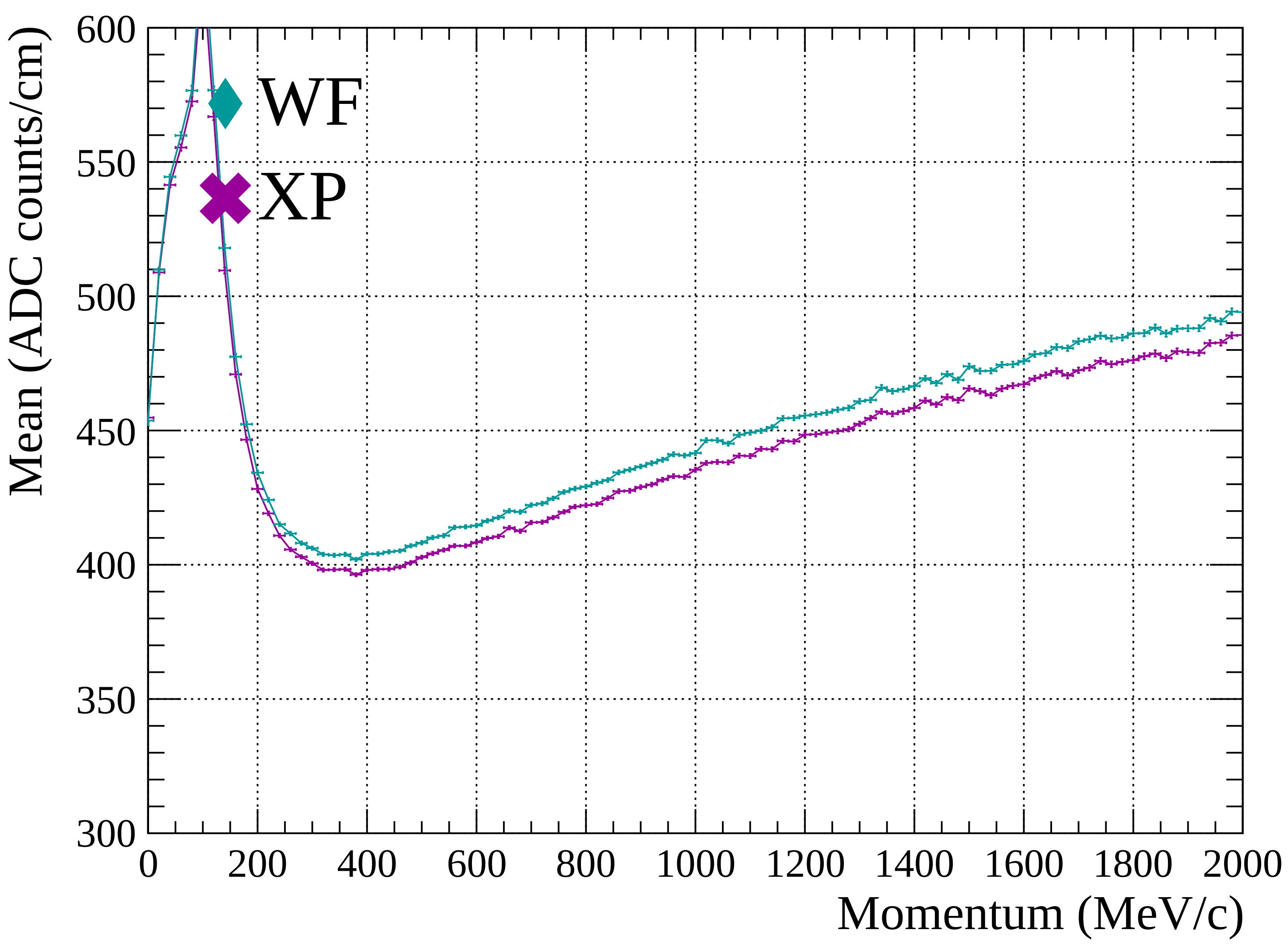
Count
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ angle



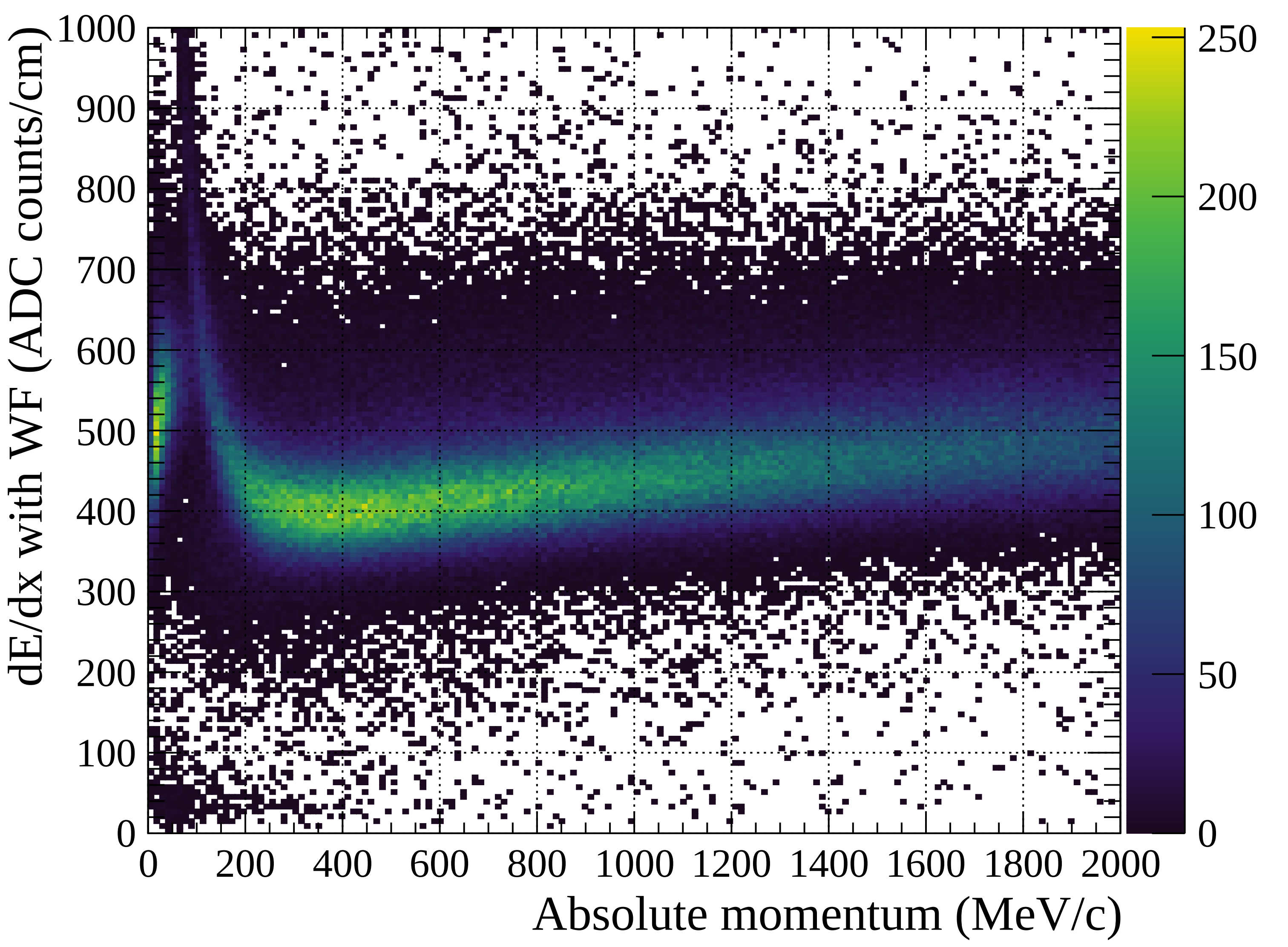
φ
80
60
40
20
0
−20
−40
−60
−80
θ
30
25
20
15
10
5
0



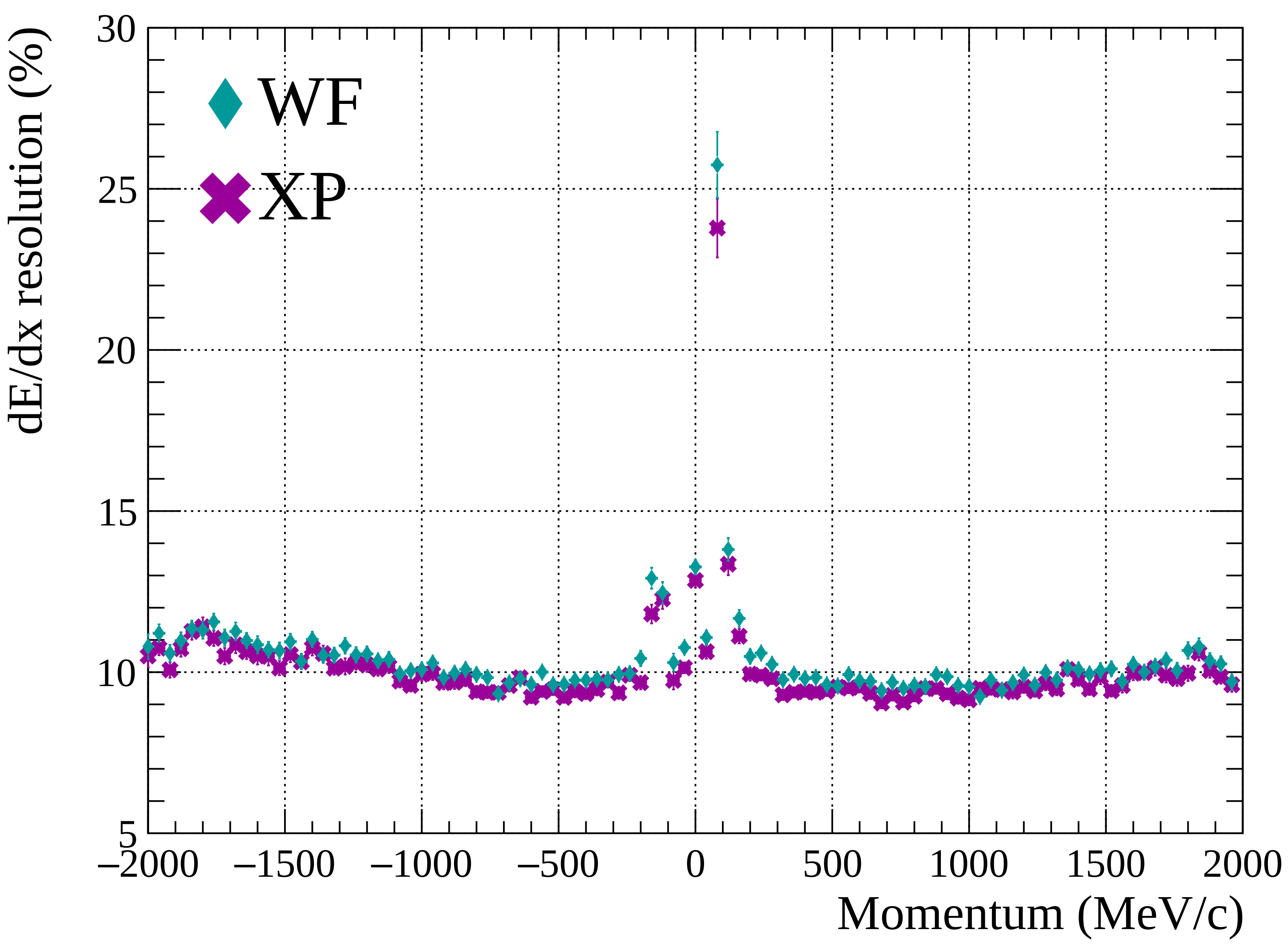
dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000



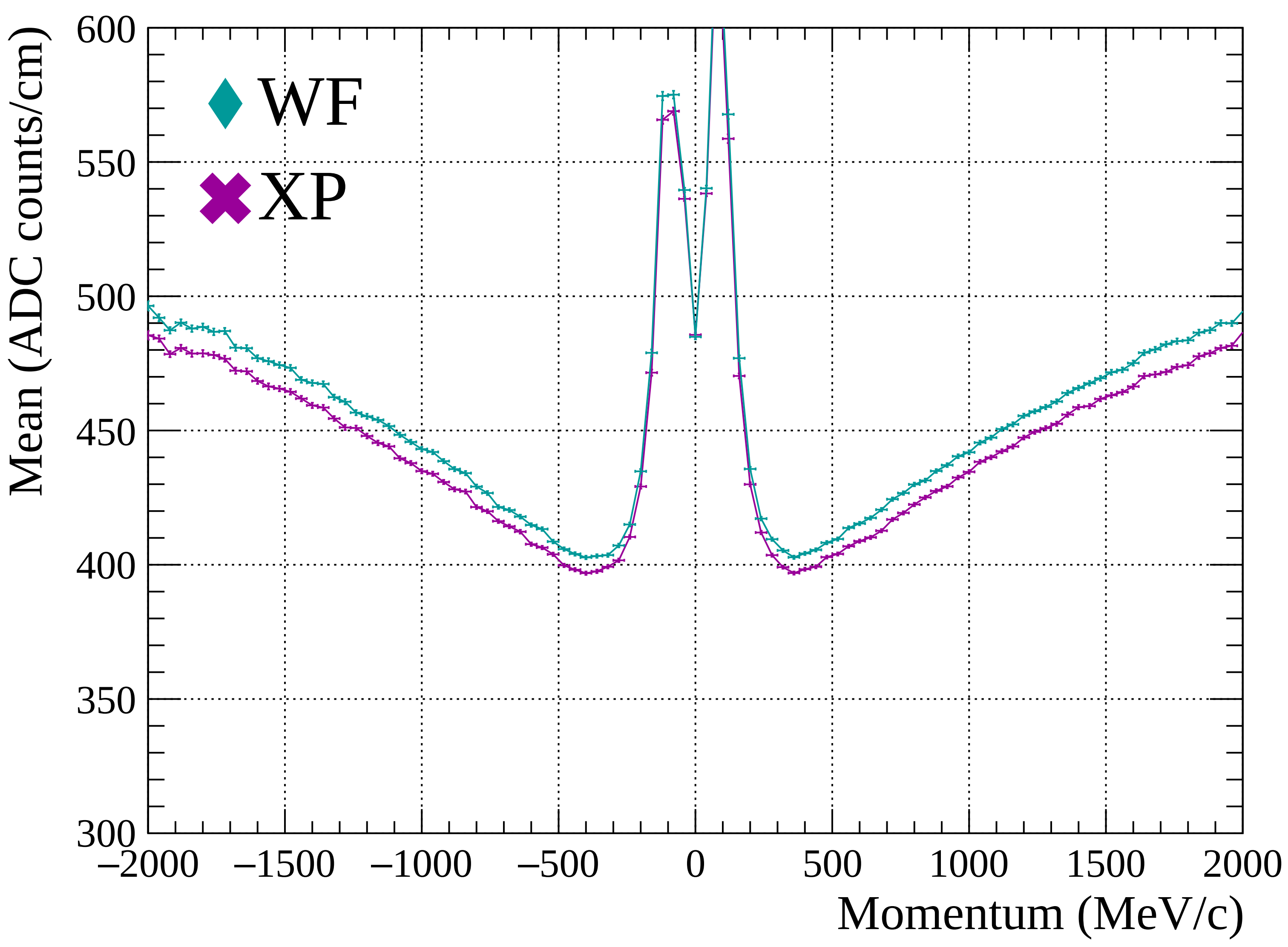
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000



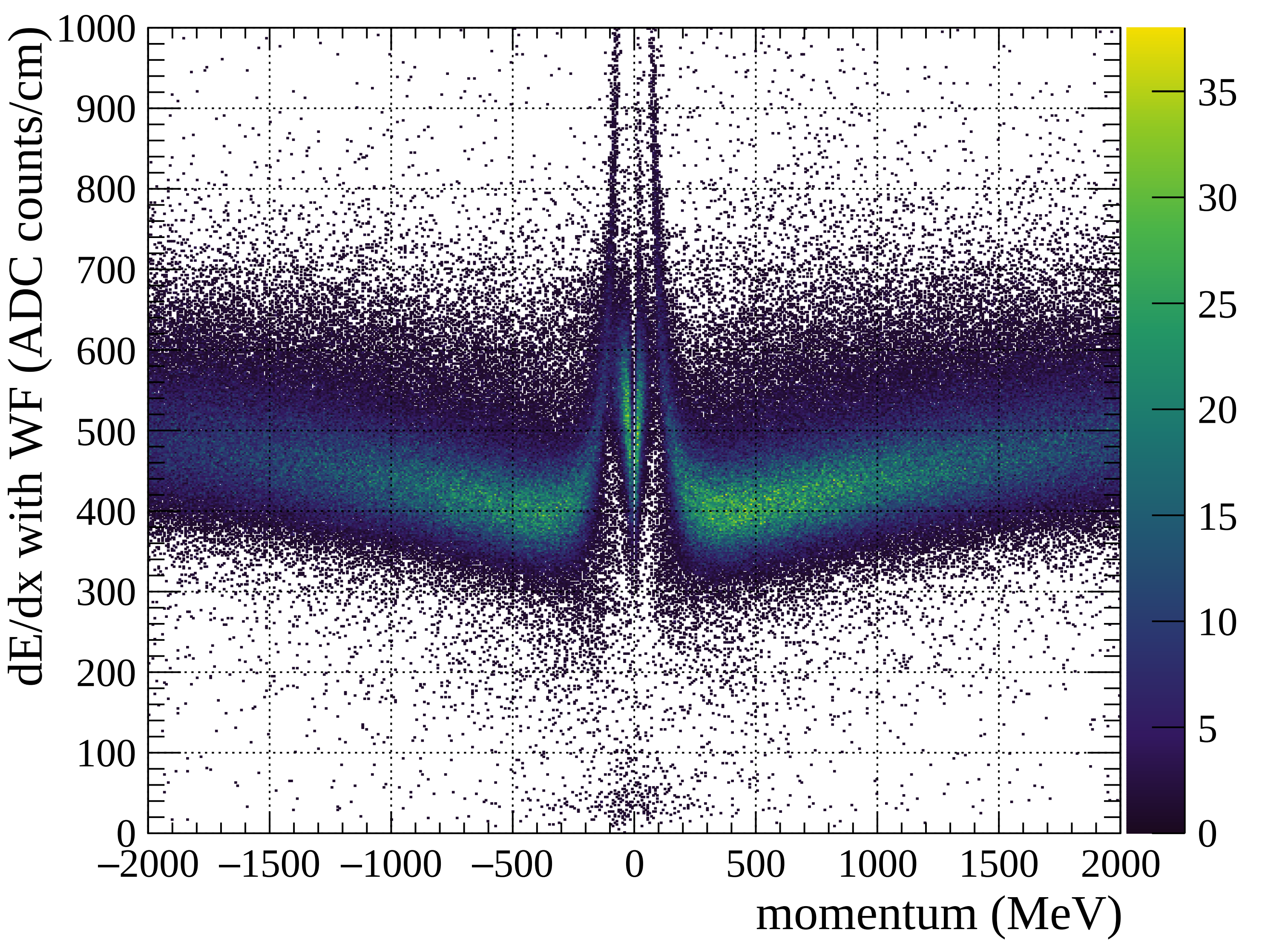
dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
50
100
150
200
250



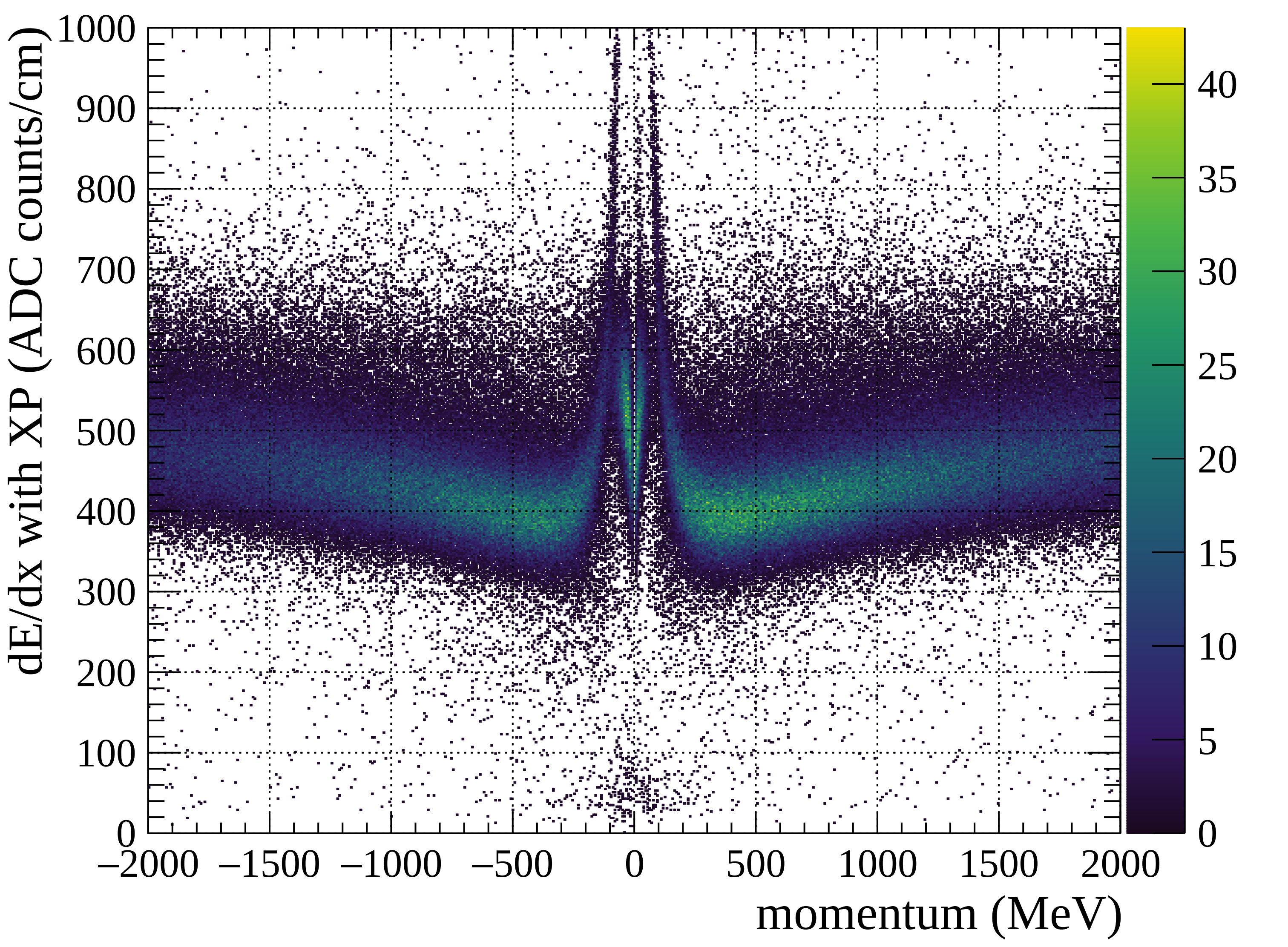
WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000



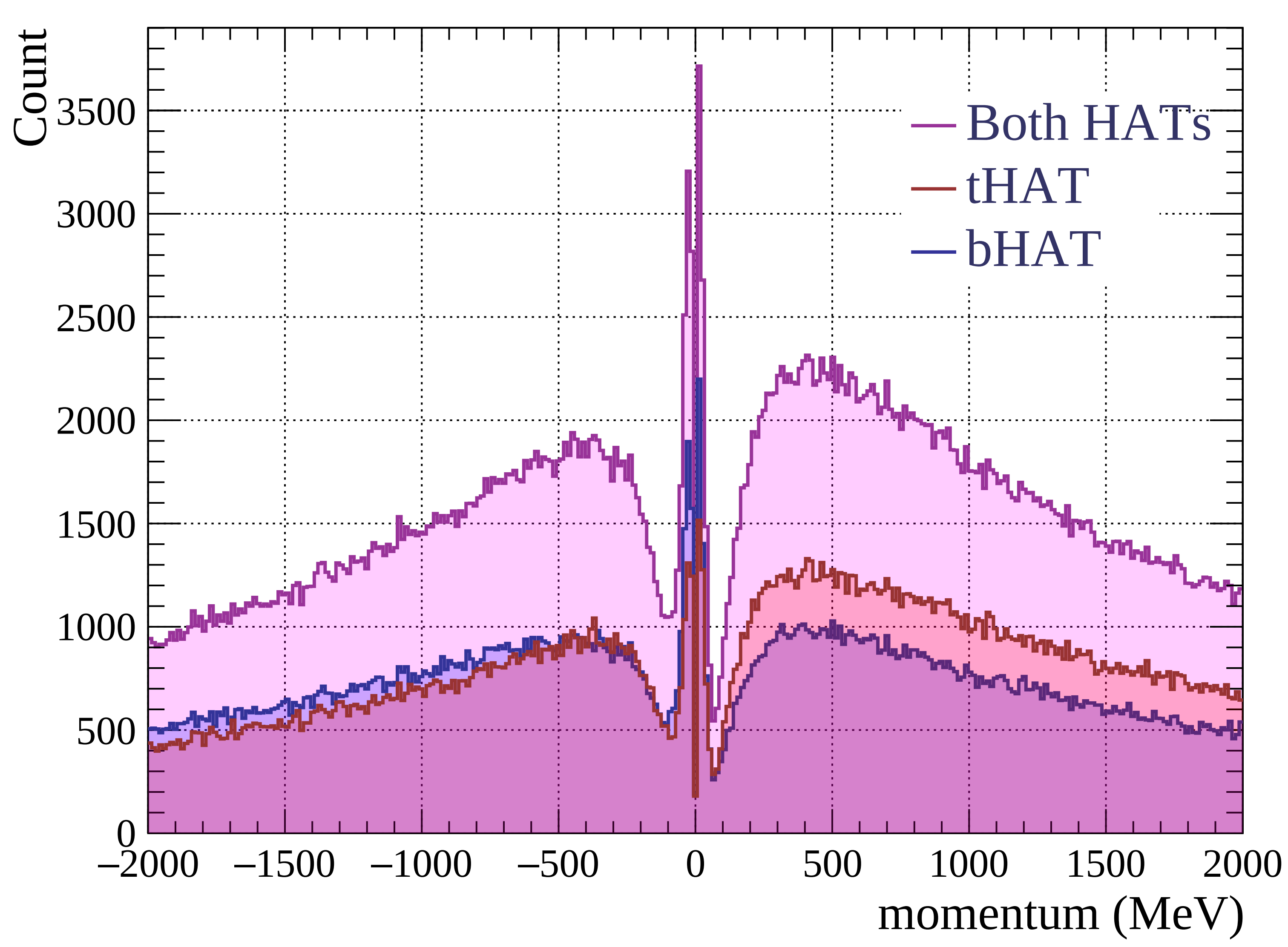
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000

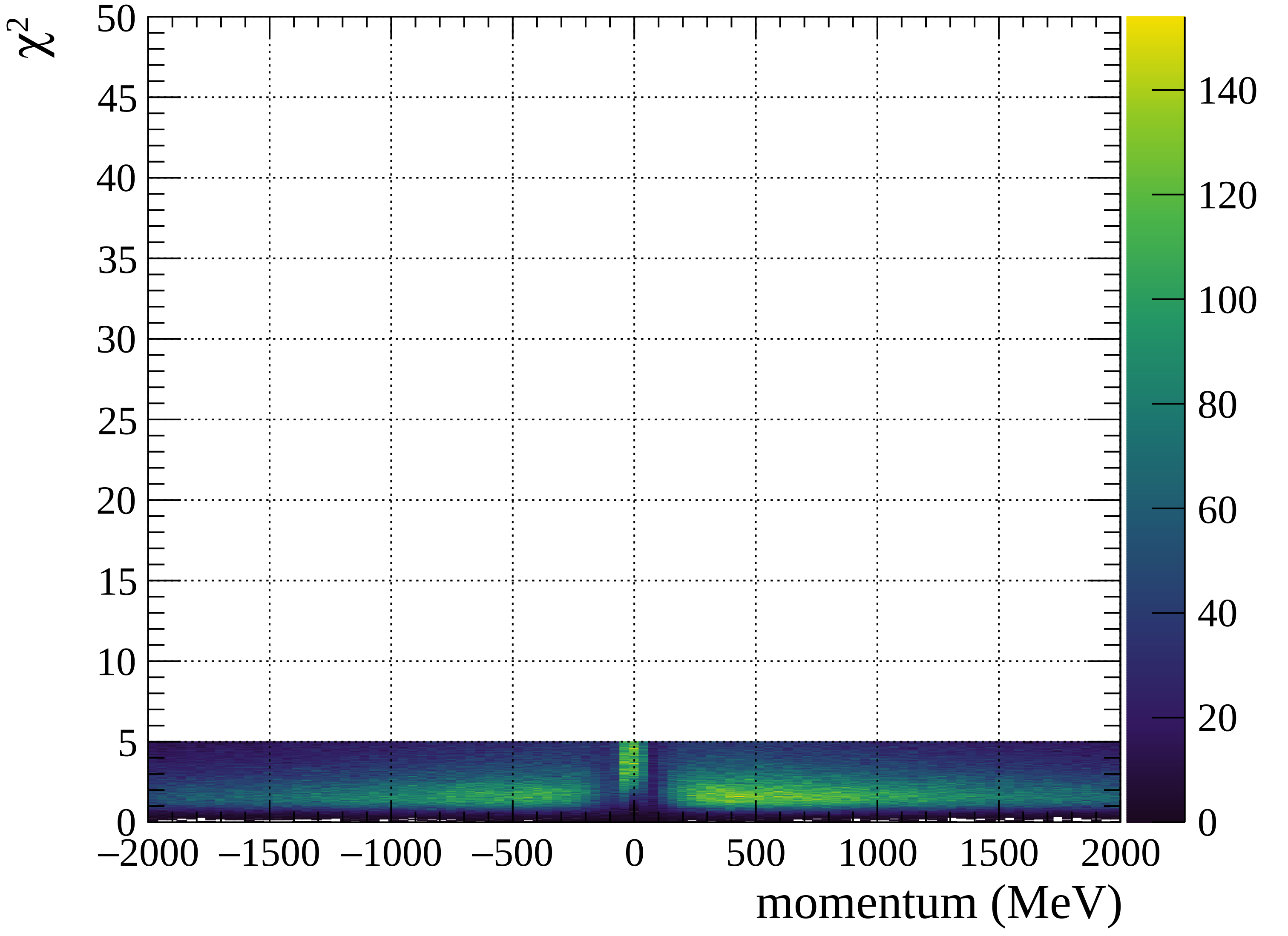


dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
5
10
15
20
25
30
35

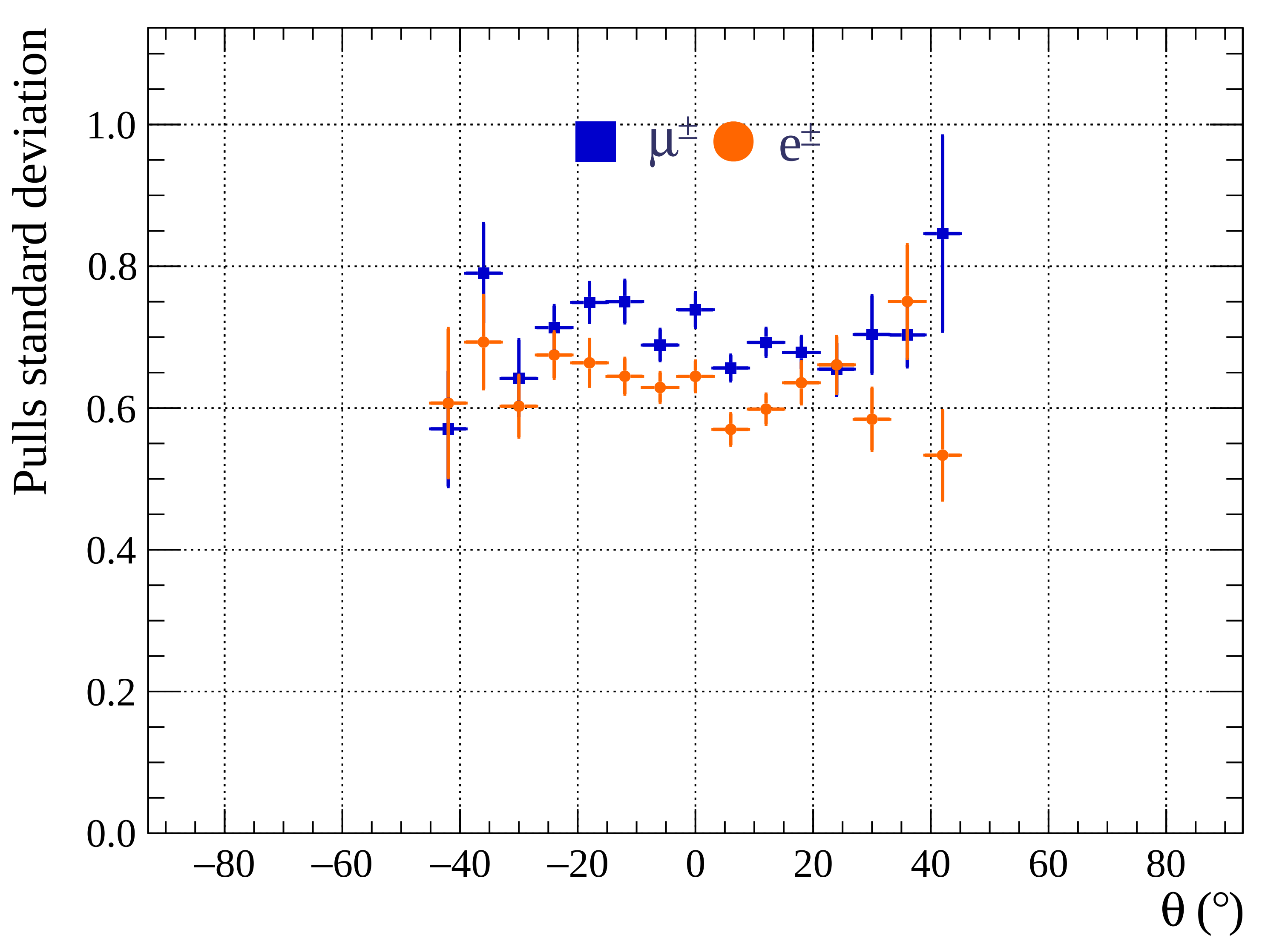


dE/dx with XP (ADC counts/cm)
momentum (MeV)

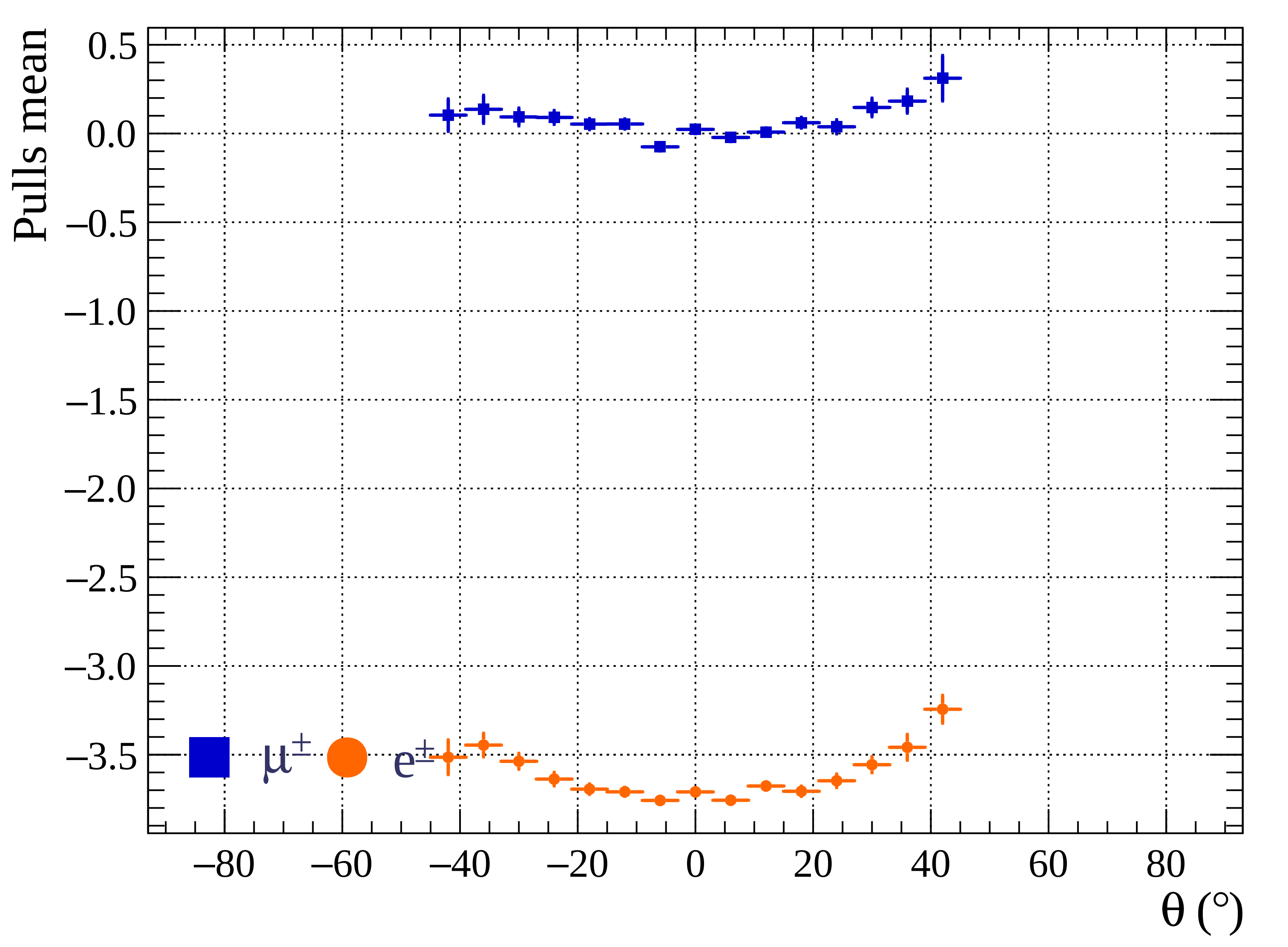

Count
3500
3000
2500
2000
1500
1000
500
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT

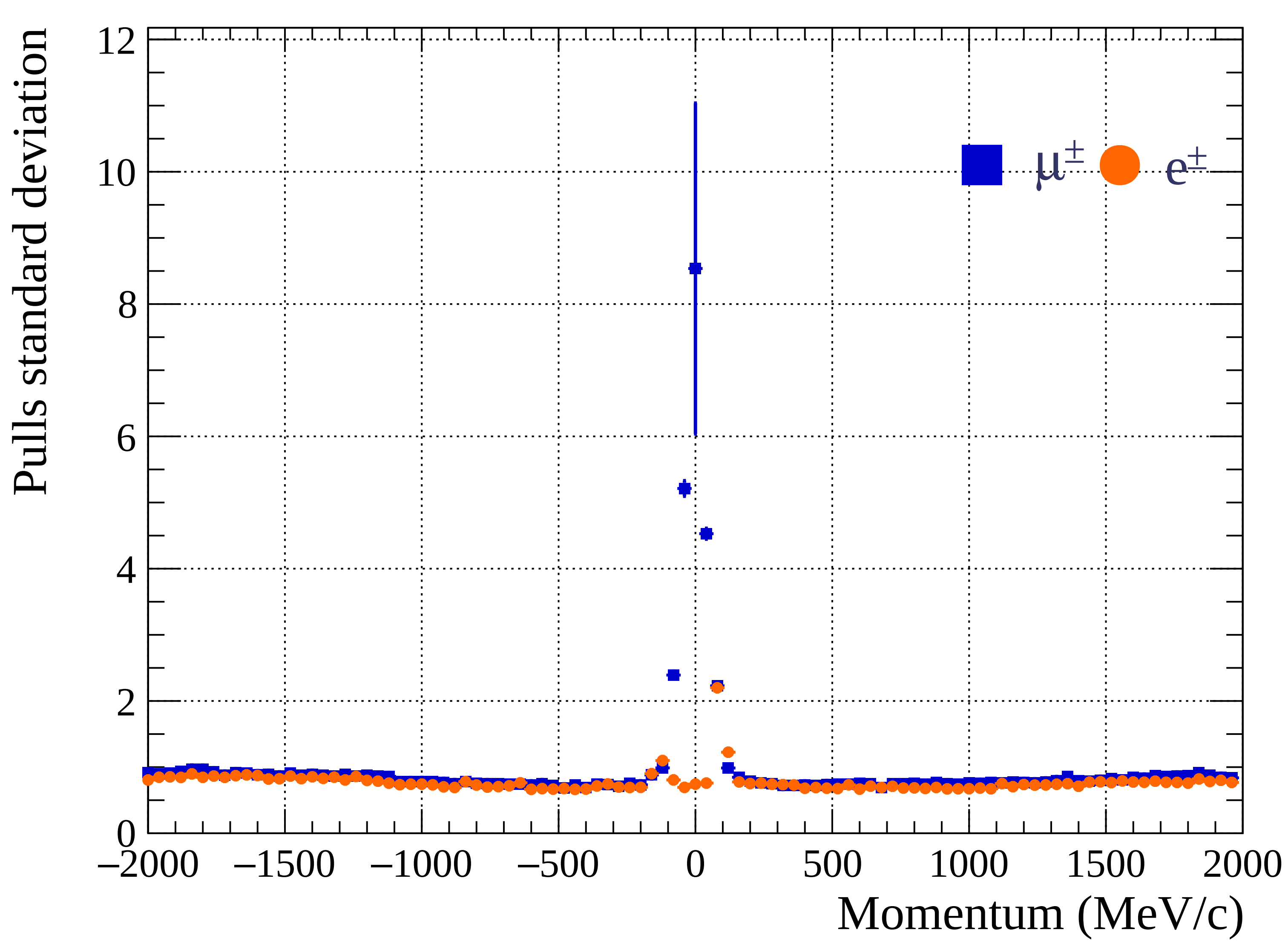


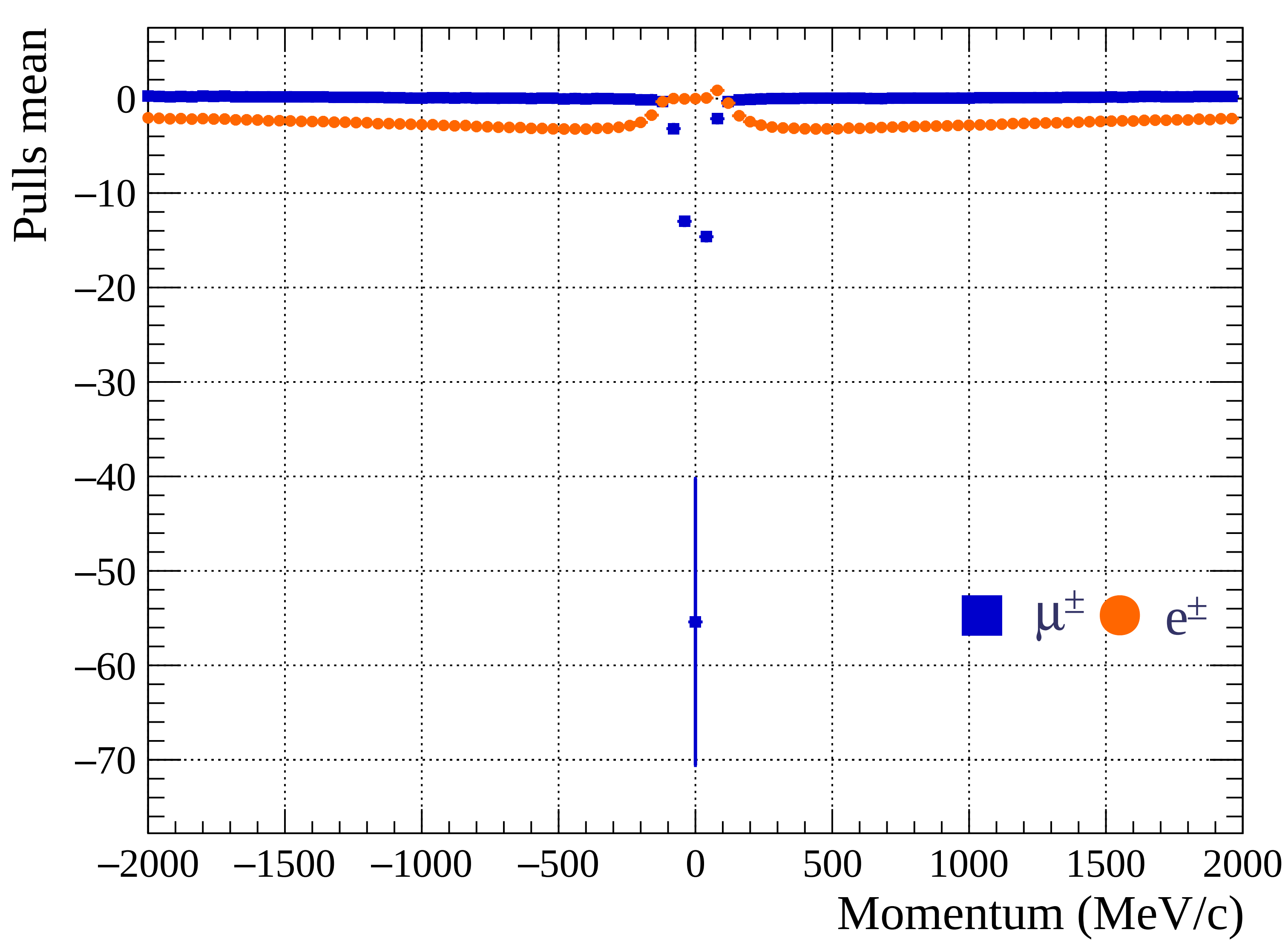
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
140
120
100
80
60
40
20
0



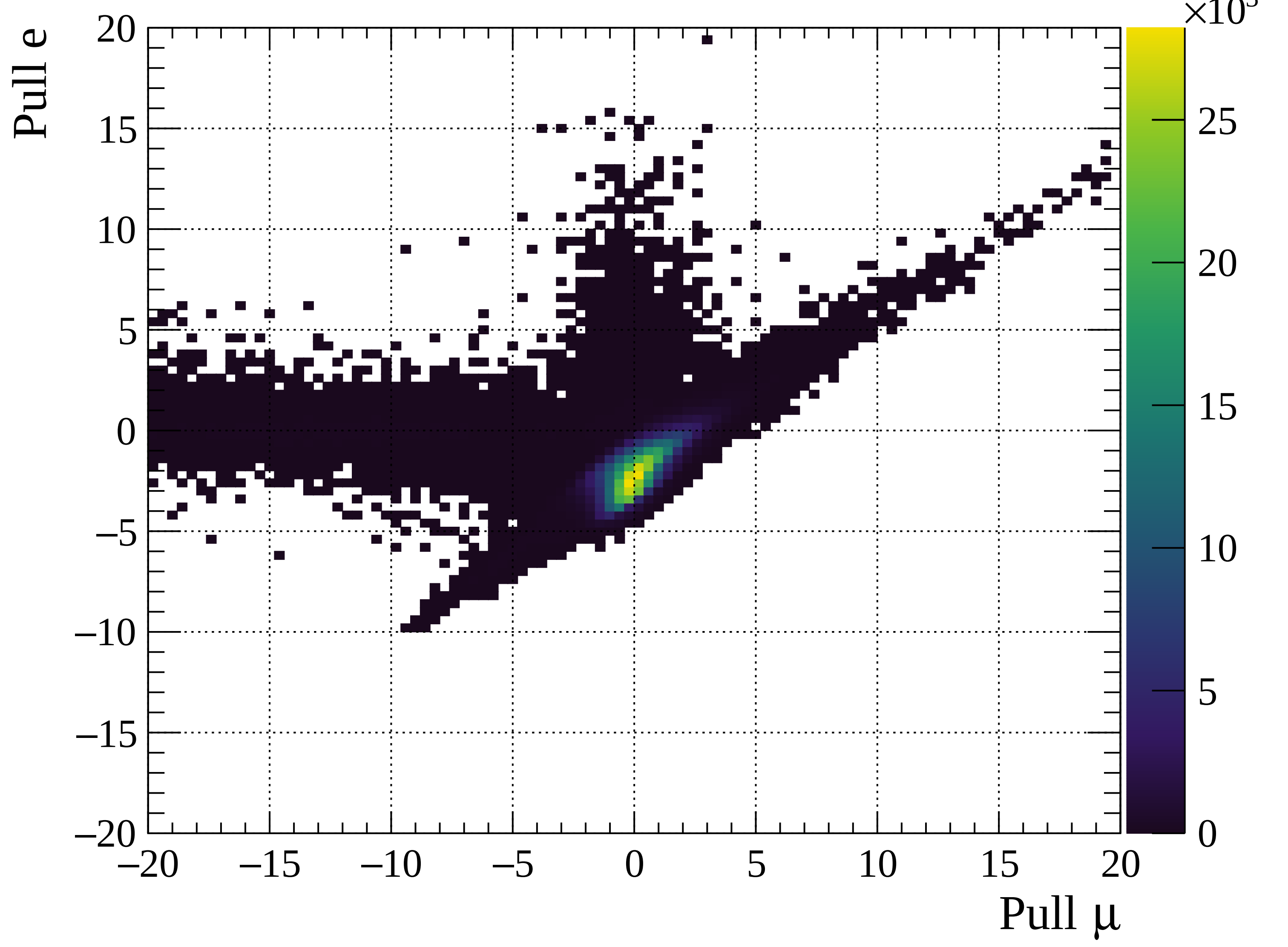
Pulls standard deviation
θ (°)
μ±
e±
0.0
0.2
0.4
0.6
0.8
1.0
−80
−60
−40
−20
0
20
40
60
80


Pulls mean
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

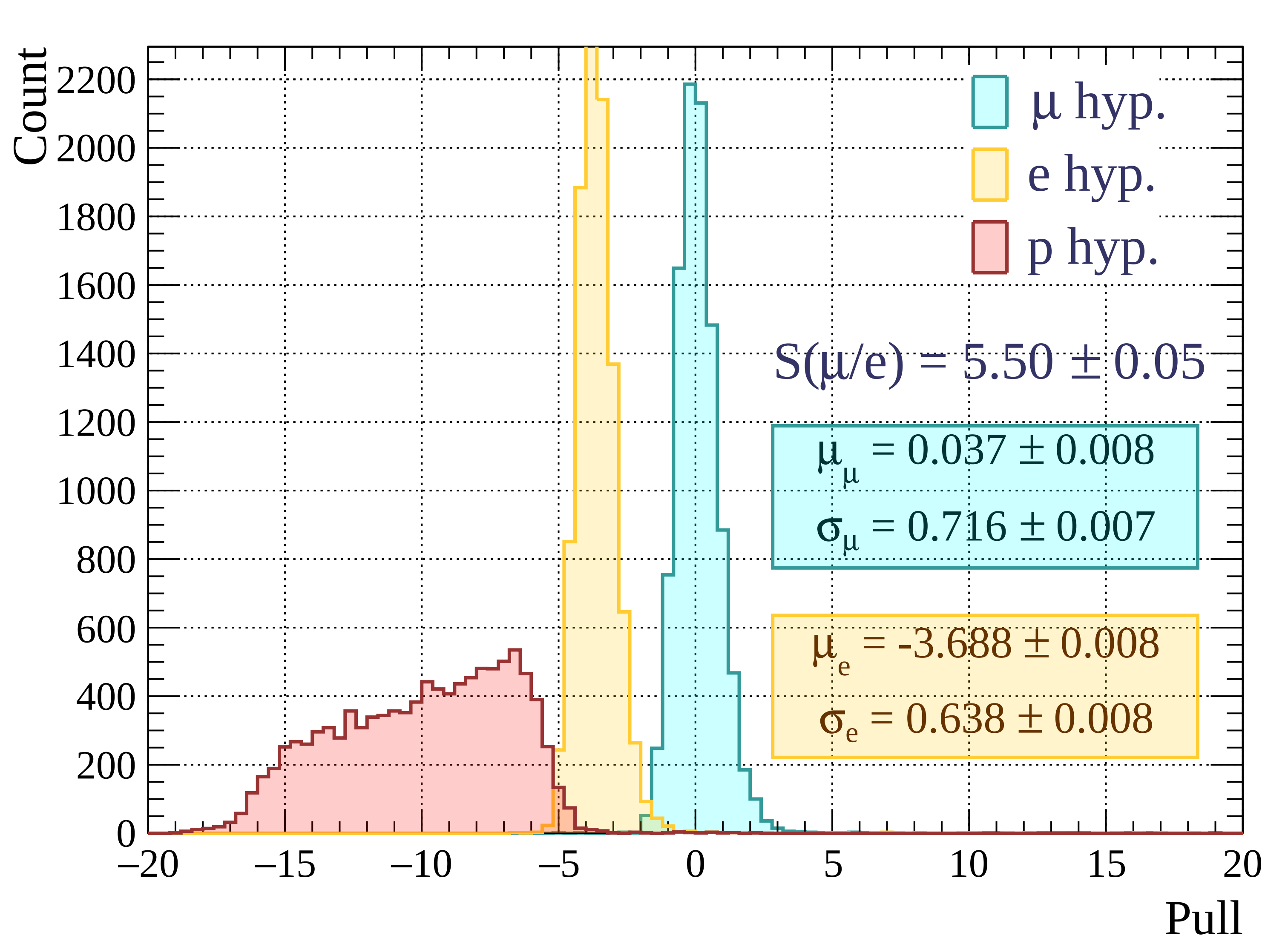

Pulls standard deviation
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$
0
2
4
6
8
10
12
−2000
−1500
−1000
−500
0
500
1000
1500
2000



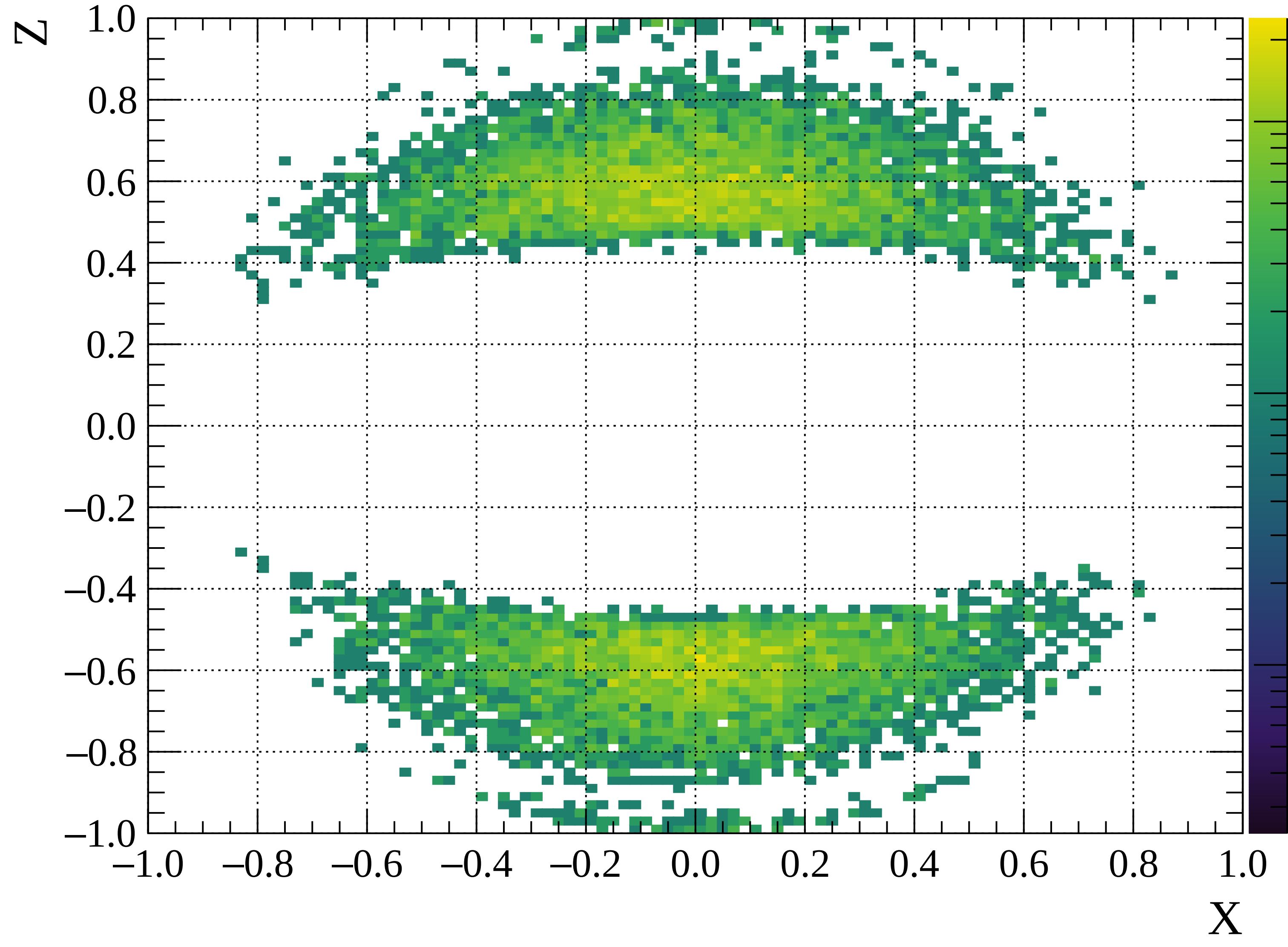
Pulls mean
0
−10
−20
−30
−40
−50
−60
−70
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±

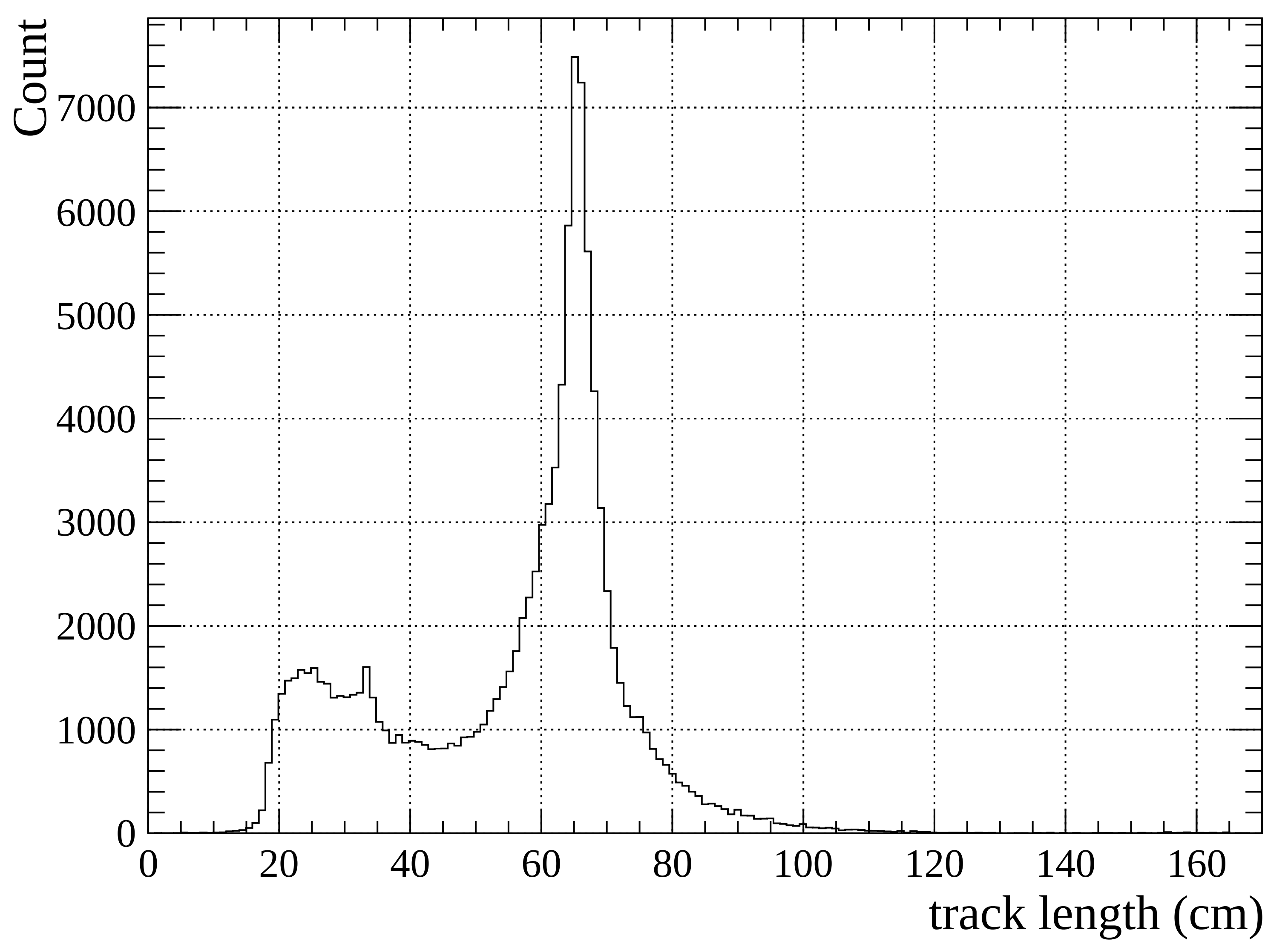


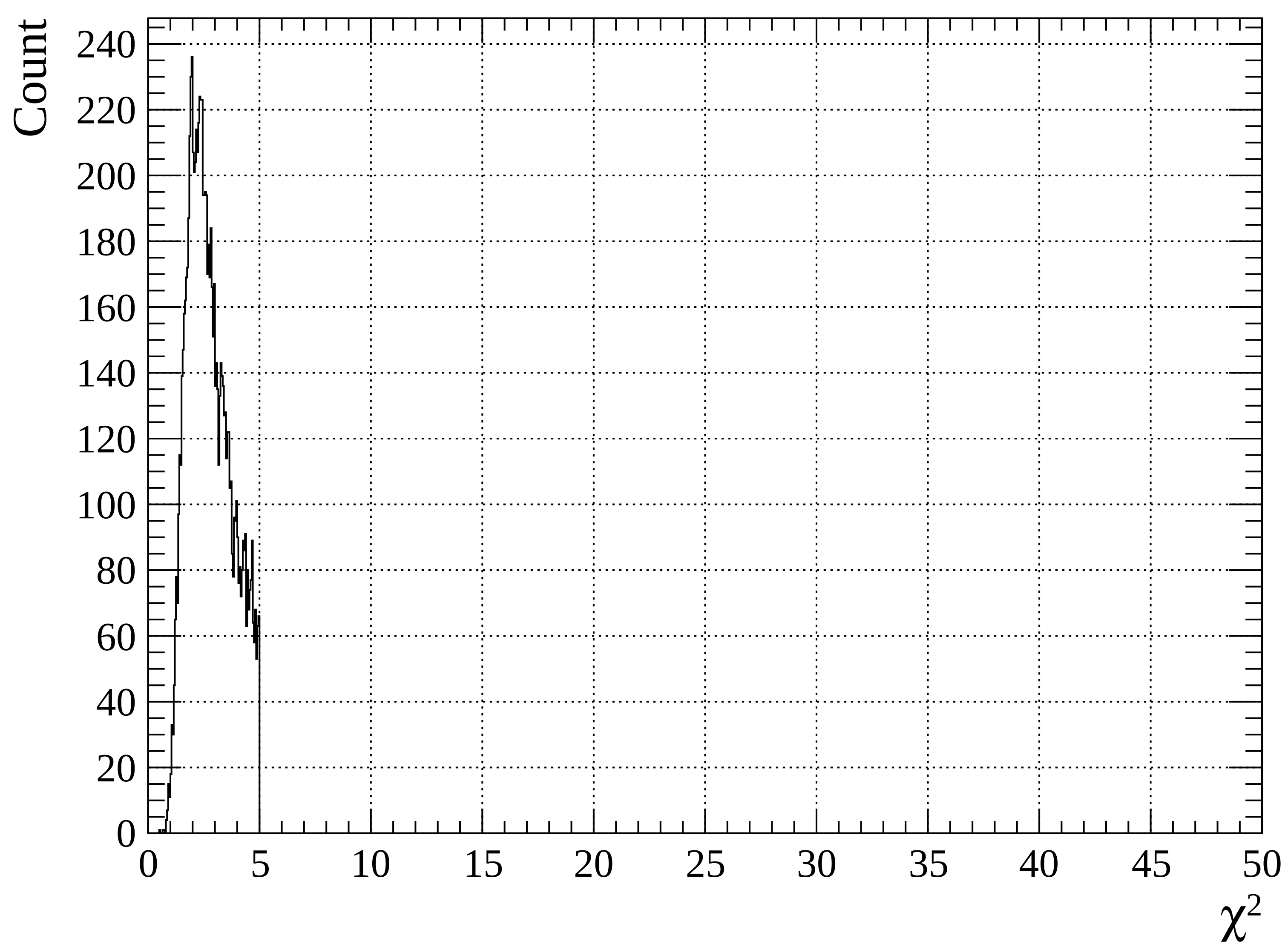
Pull e
Pull μ
×10³
20
15
10
5
0
−5
−10
−15
−20
25



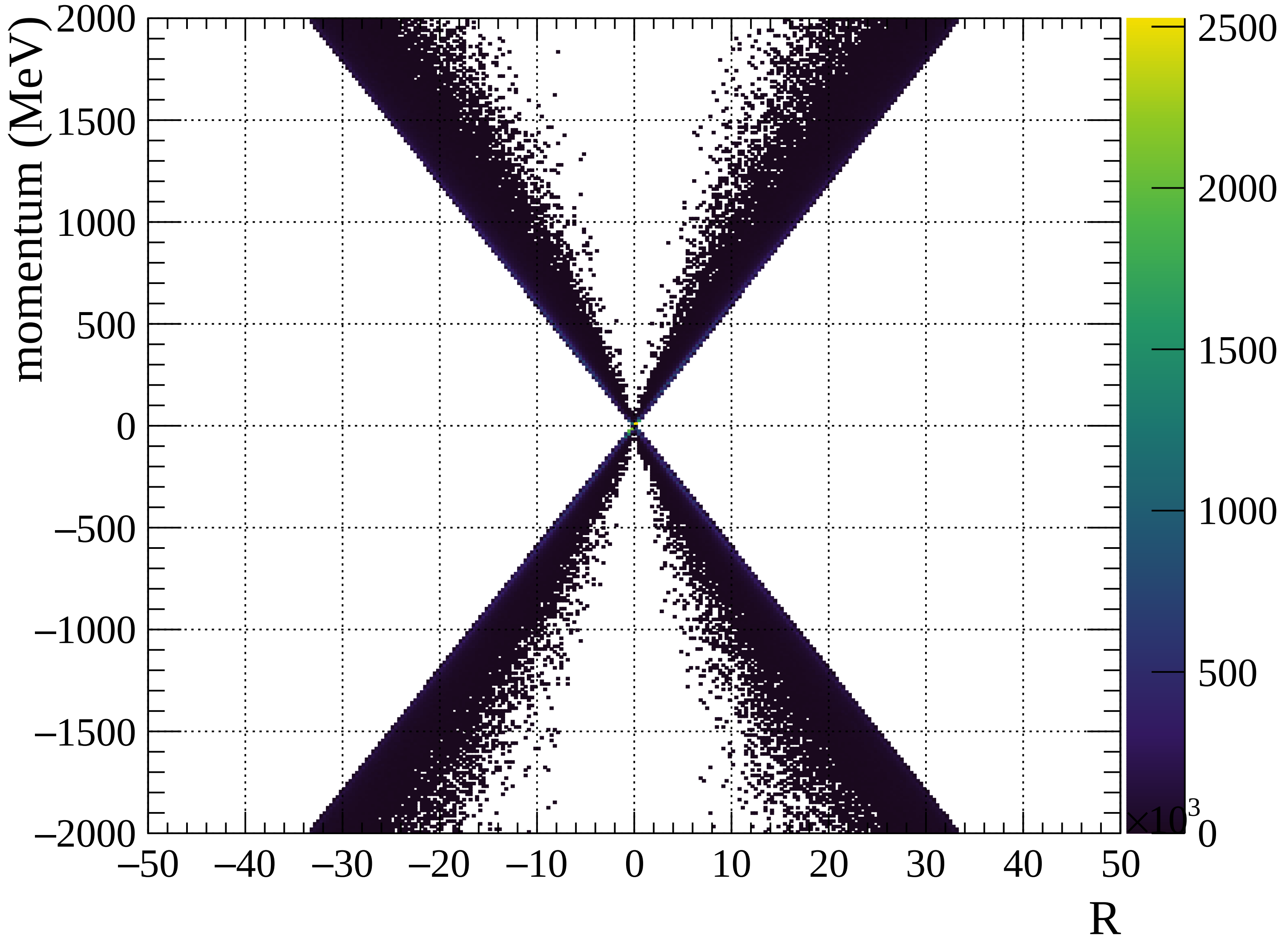
Count
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 5.50 ± 0.05
μ_μ = 0.037 ± 0.008
σ_μ = 0.716 ± 0.007
μ_e = -3.688 ± 0.008
σ_e = 0.638 ± 0.008


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

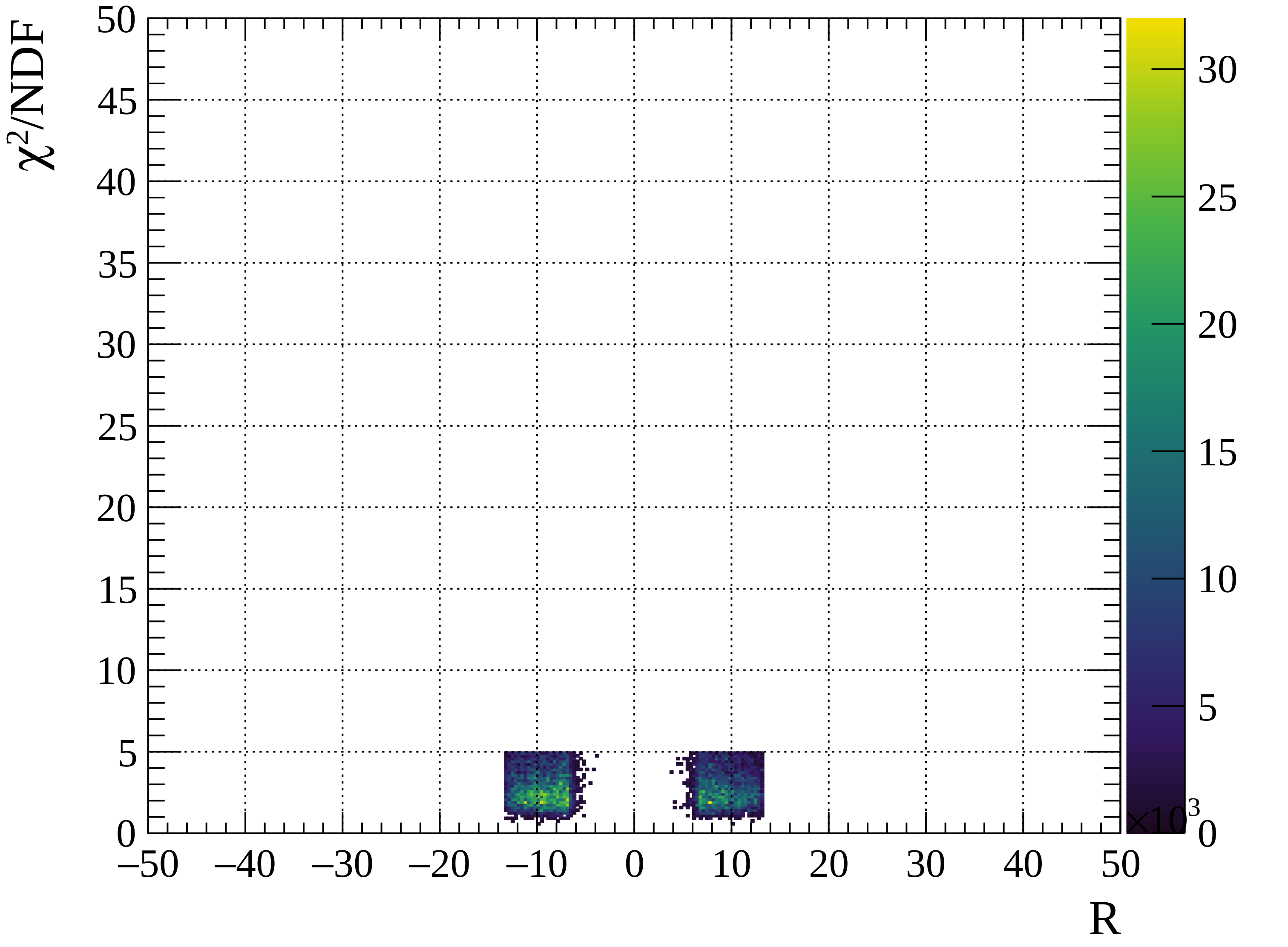

Count
7000
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
160
track length (cm)



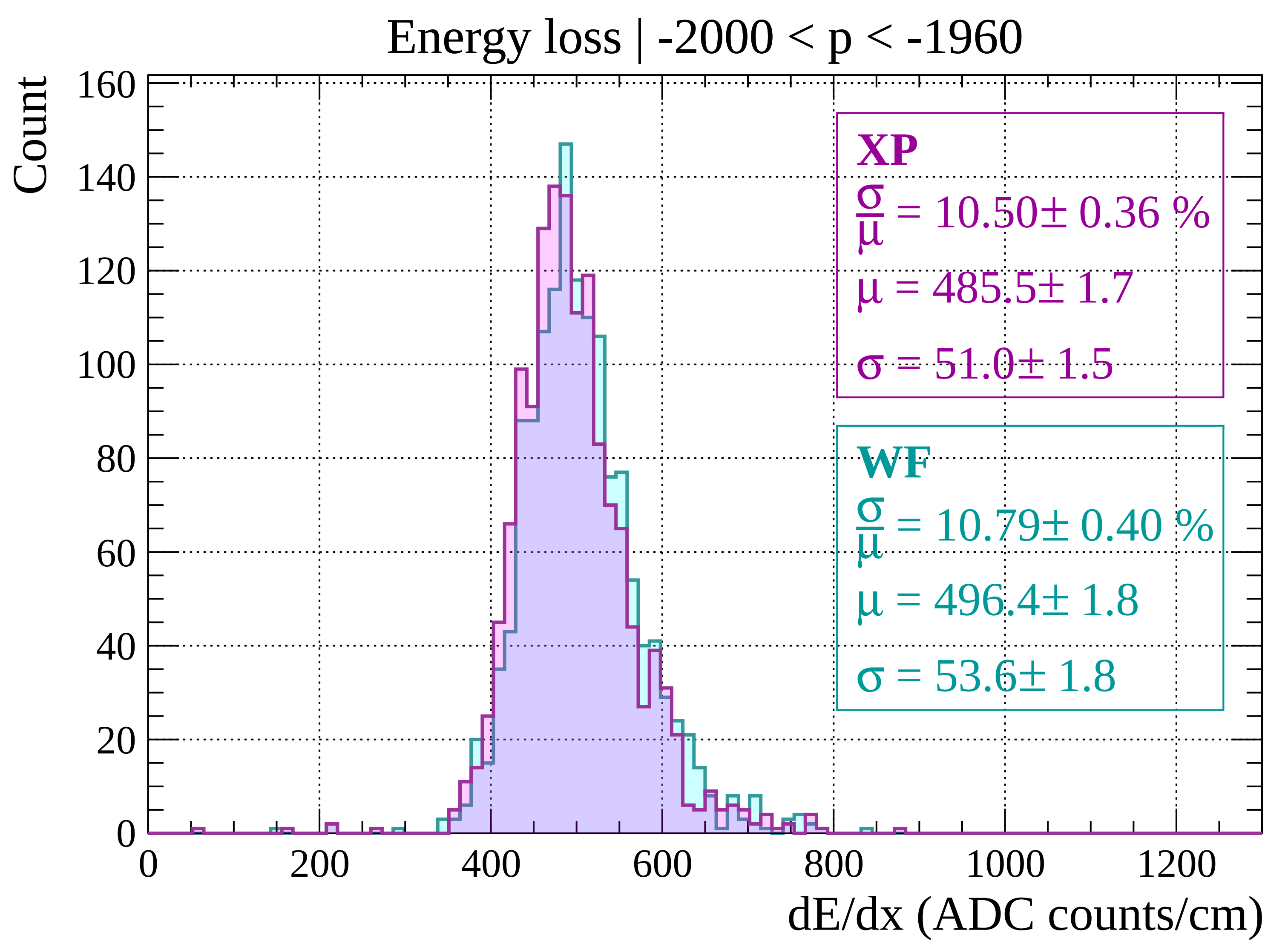
Count
0
20
40
60
80
100
120
140
160
180
200
220
240
0
5
10
15
20
25
30
35
40
45
50
χ^2

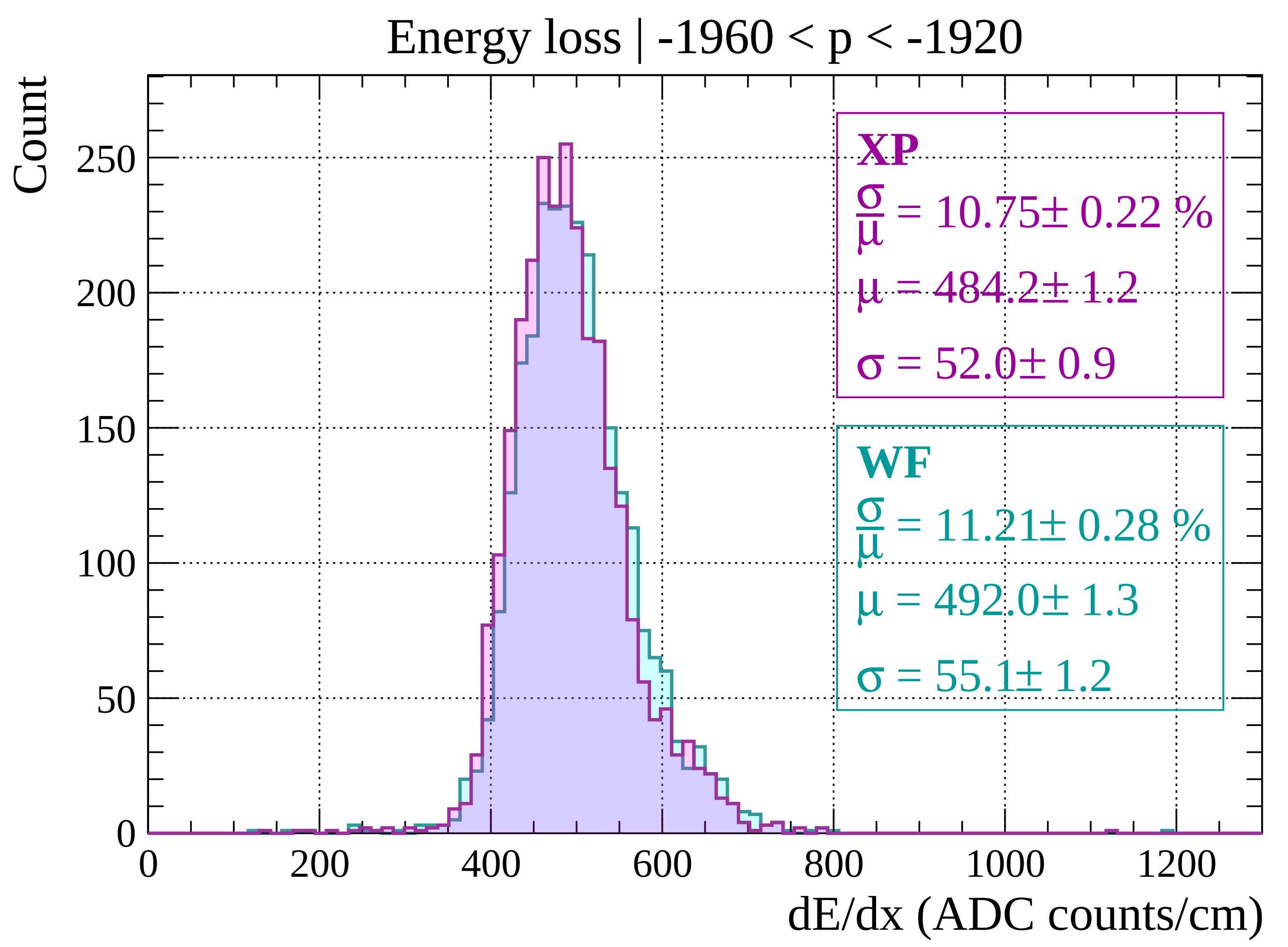


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
2500
2000
1500
1000
500
0
×10^3

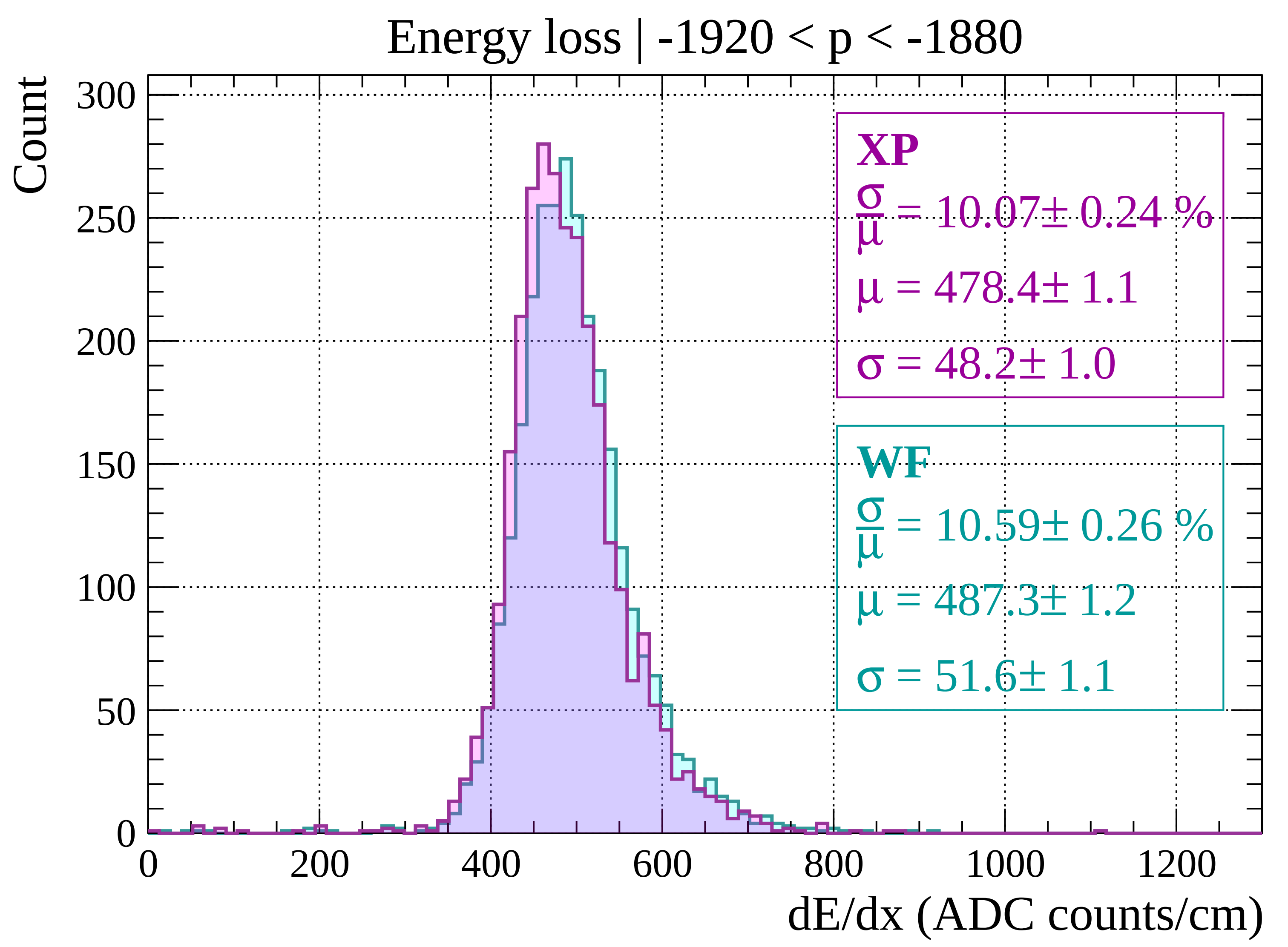


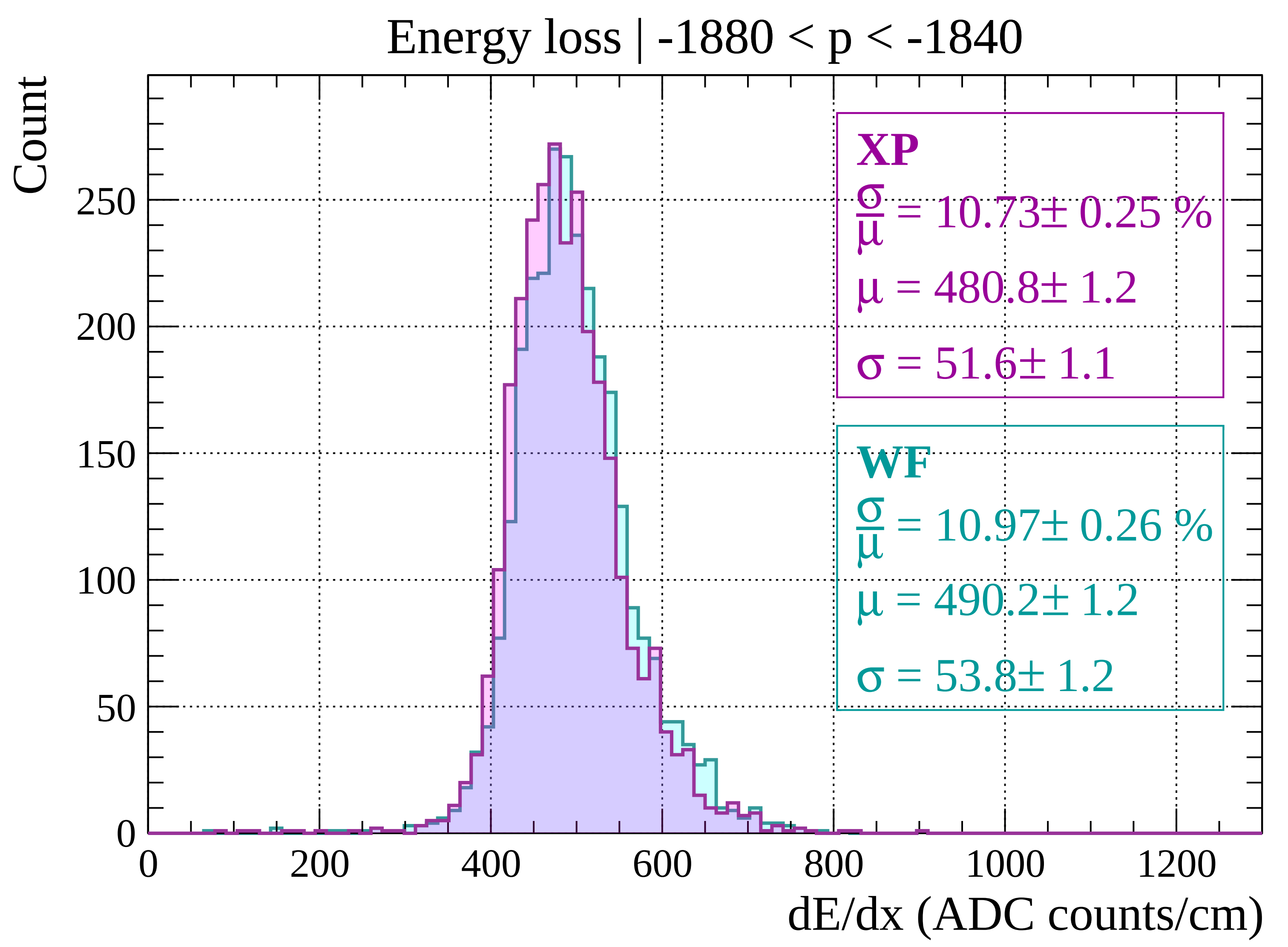
χ^2/NDF
R
−50
−40
−30
−20
−10
0
10
20
30
40
50
0
5
10
15
20
25
30
35
40
45
50
$\times 10^3$


Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.50± 0.36 %
μ = 485.5± 1.7
σ = 51.0± 1.5
WF
$\frac{\sigma}{\mu}$ = 10.79± 0.40 %
μ = 496.4± 1.8
σ = 53.6± 1.8
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160

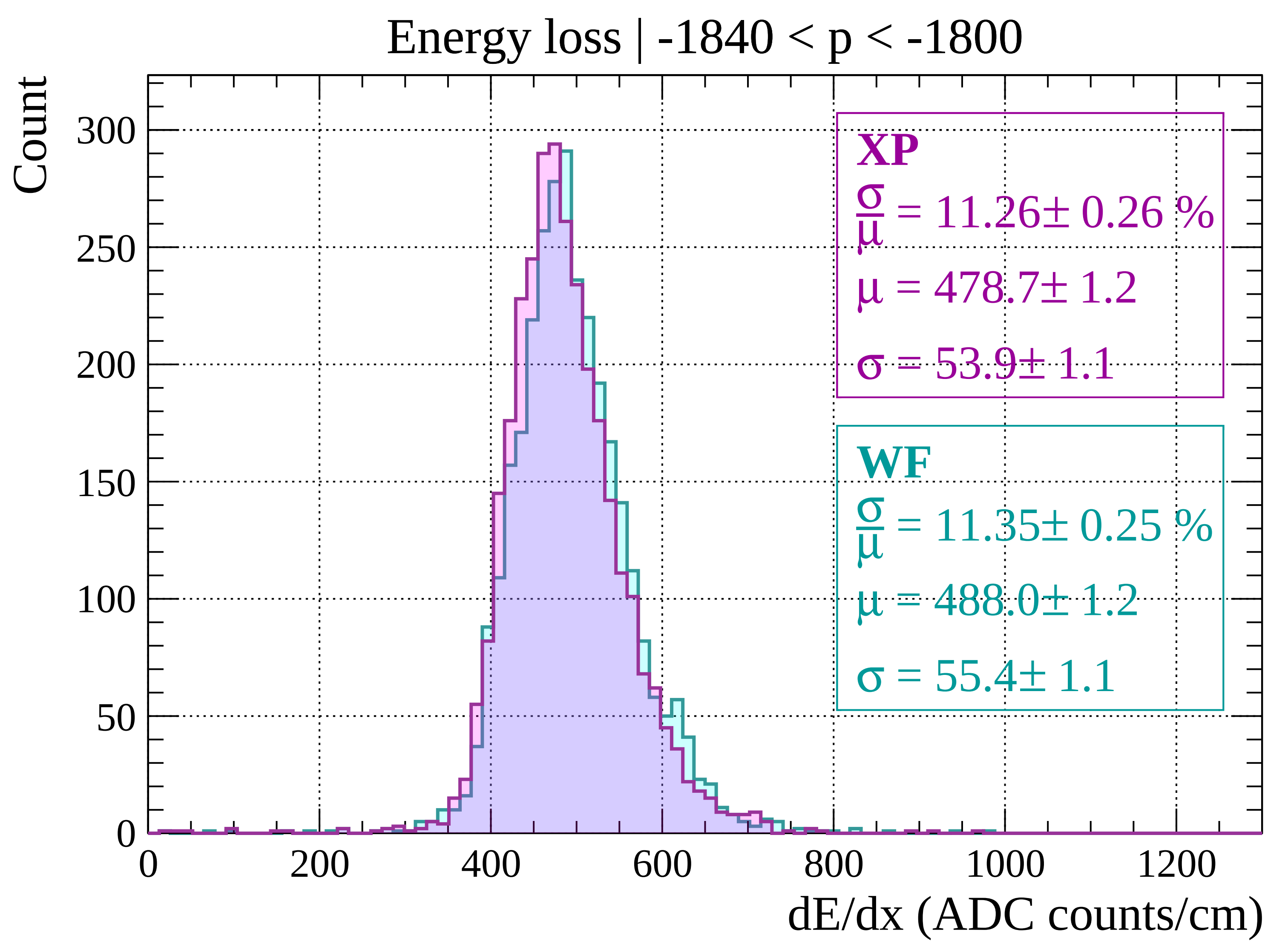


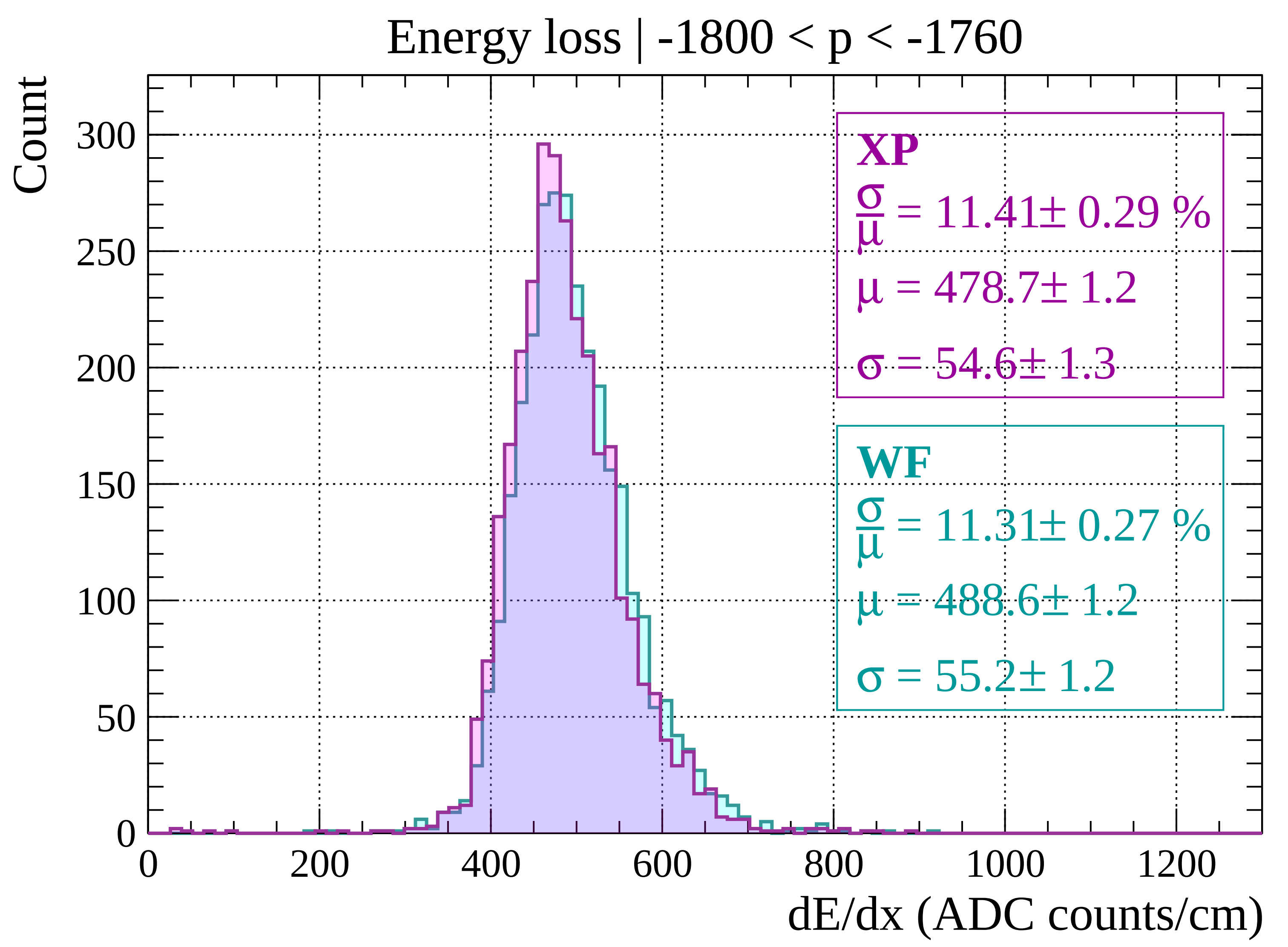
Energy loss | -1960 < p < -1920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP
$\frac{\sigma}{\mu}$ = 10.75± 0.22 %
μ = 484.2± 1.2
σ = 52.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 11.21± 0.28 %
μ = 492.0± 1.3
σ = 55.1± 1.2


Energy loss | -1920 < p < -1880
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.07± 0.24 %
μ = 478.4± 1.1
σ = 48.2± 1.0
WF
σ/μ = 10.59± 0.26 %
μ = 487.3± 1.2
σ = 51.6± 1.1

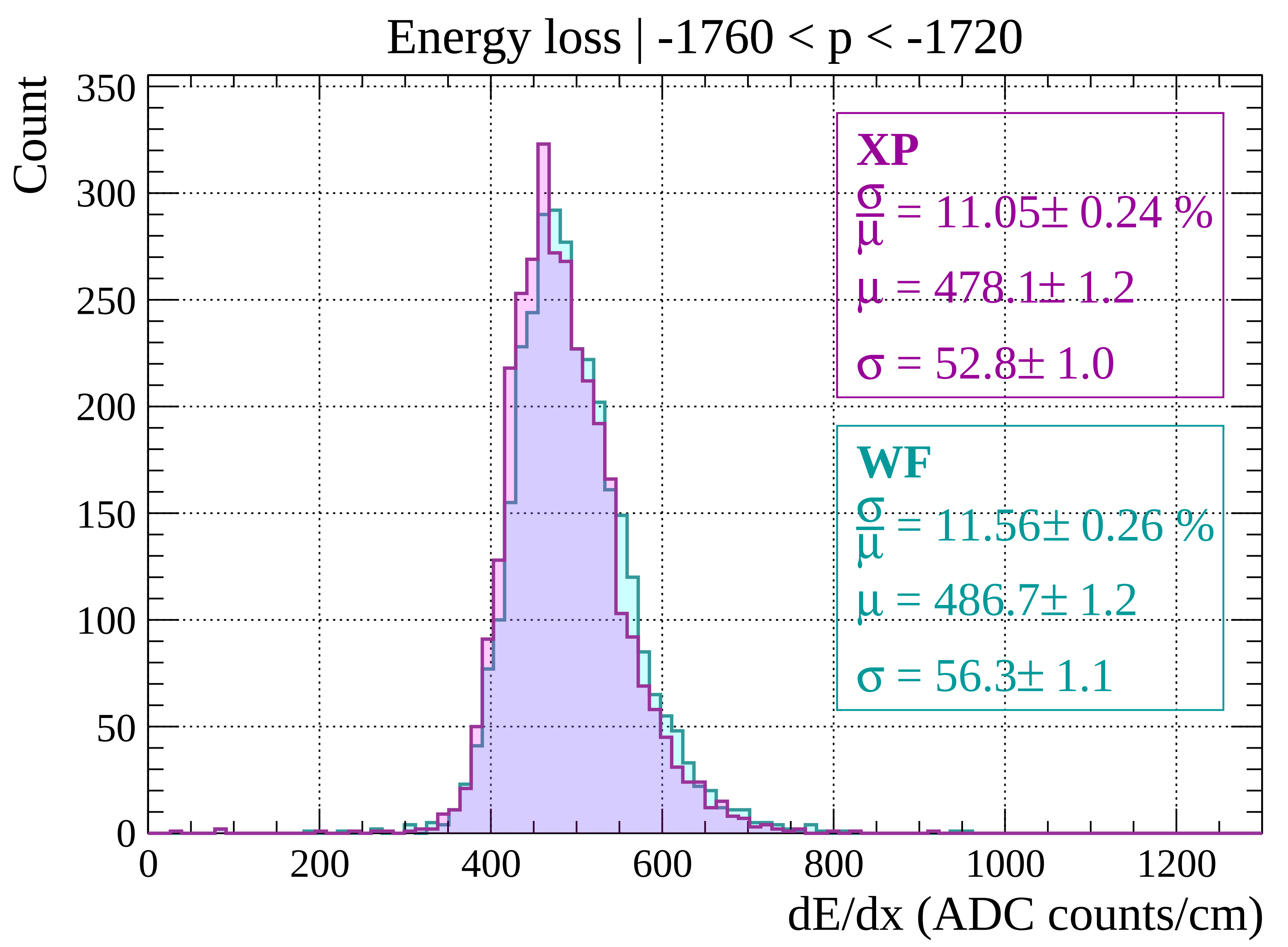


Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
σ/μ = 10.73± 0.25 %
μ = 480.8± 1.2
σ = 51.6± 1.1
WF
σ/μ = 10.97± 0.26 %
μ = 490.2± 1.2
σ = 53.8± 1.2

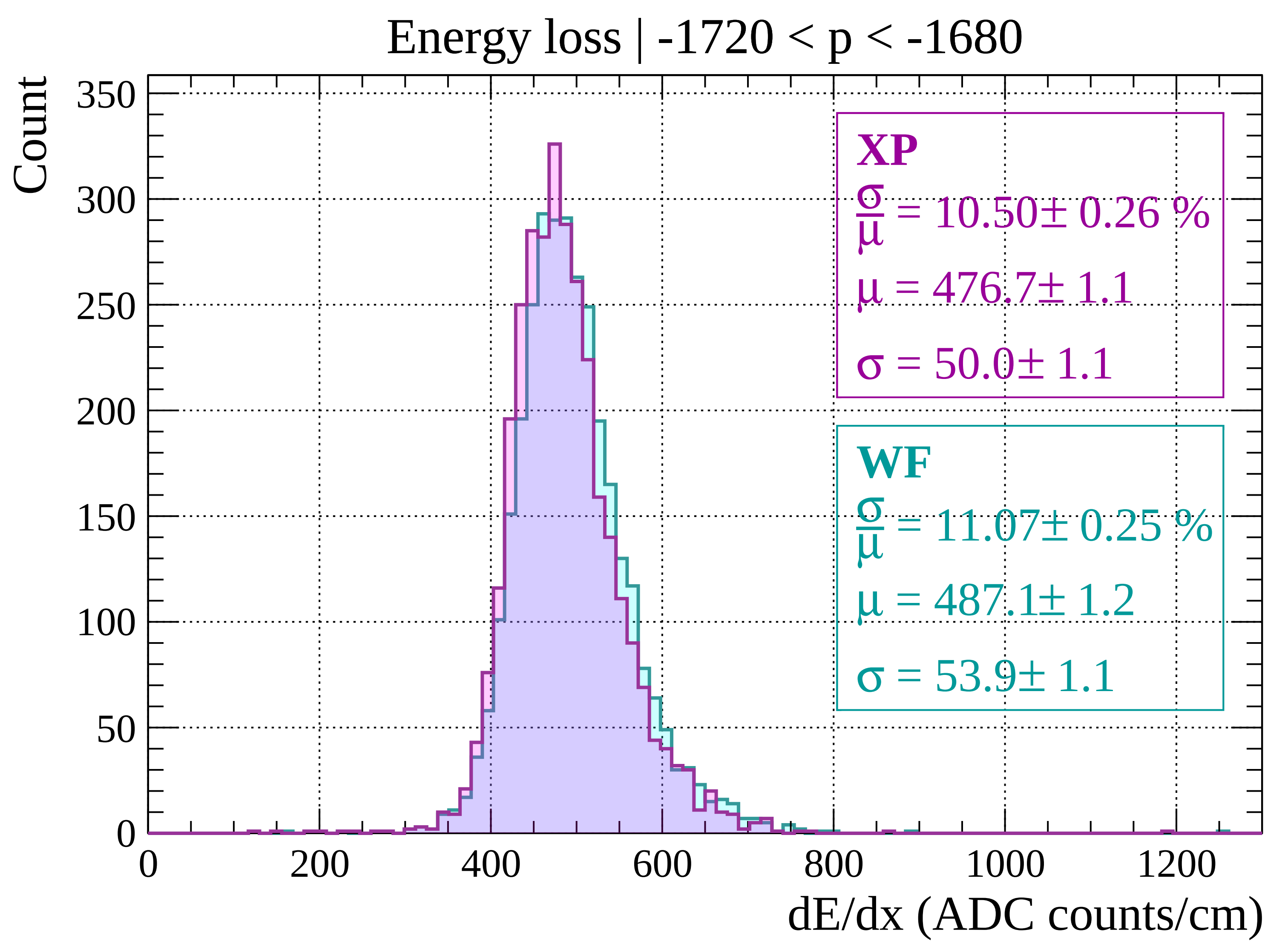

Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.26± 0.26 %
μ = 478.7± 1.2
σ = 53.9± 1.1
WF
σ/μ = 11.35± 0.25 %
μ = 488.0± 1.2
σ = 55.4± 1.1
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

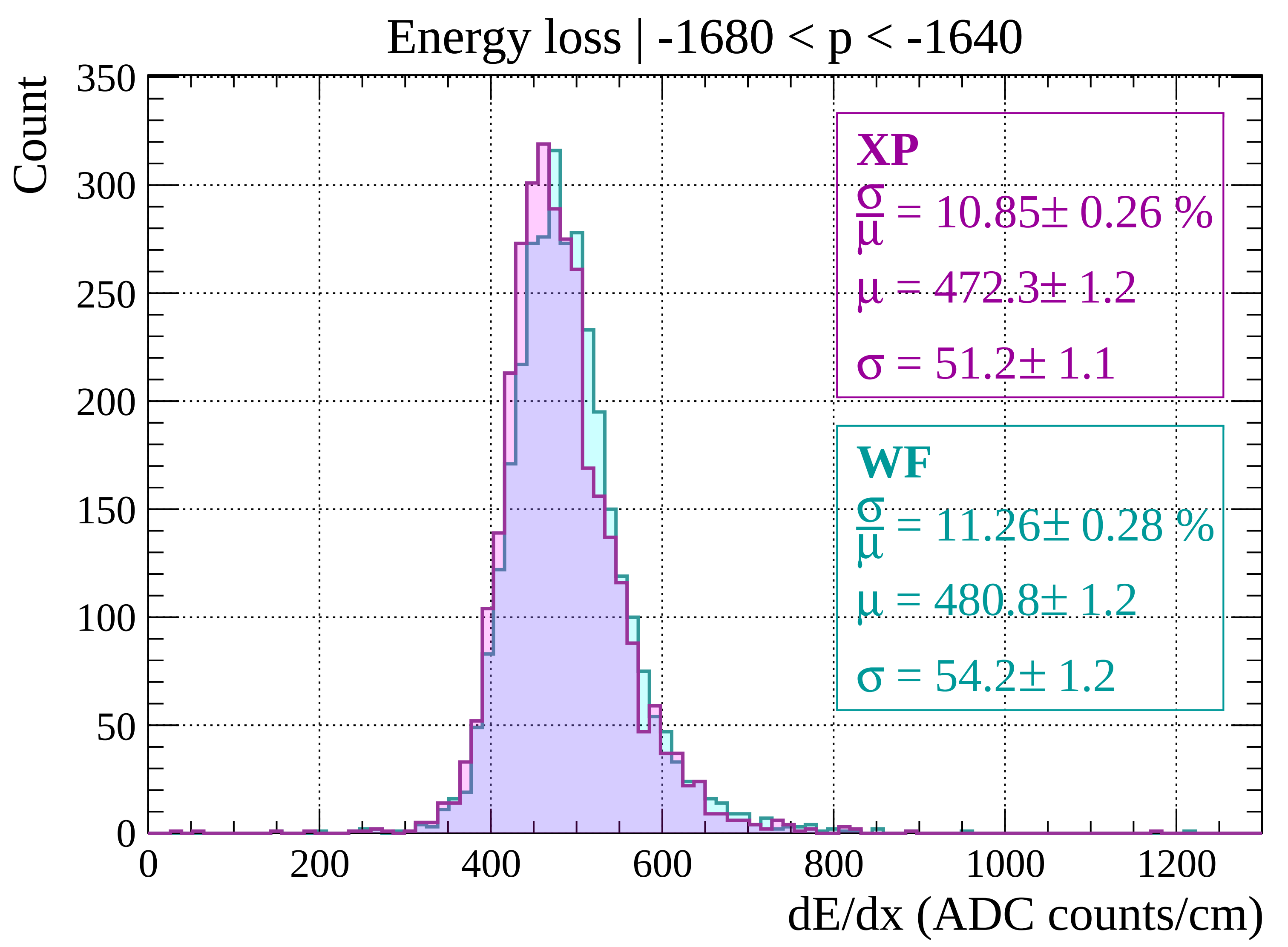


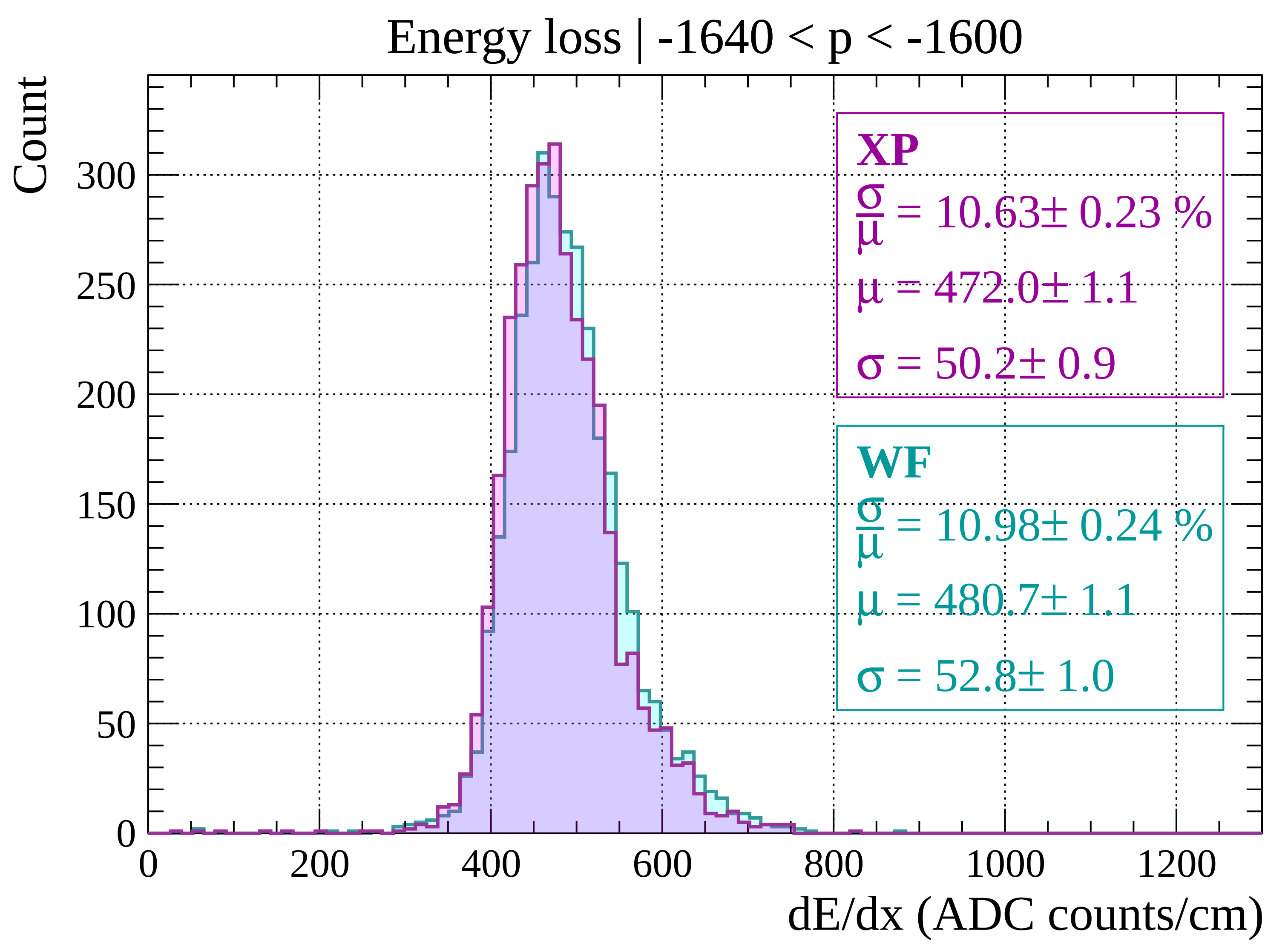
Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 11.41± 0.29 %
μ = 478.7± 1.2
σ = 54.6± 1.3
WF
σ/μ = 11.31± 0.27 %
μ = 488.6± 1.2
σ = 55.2± 1.2



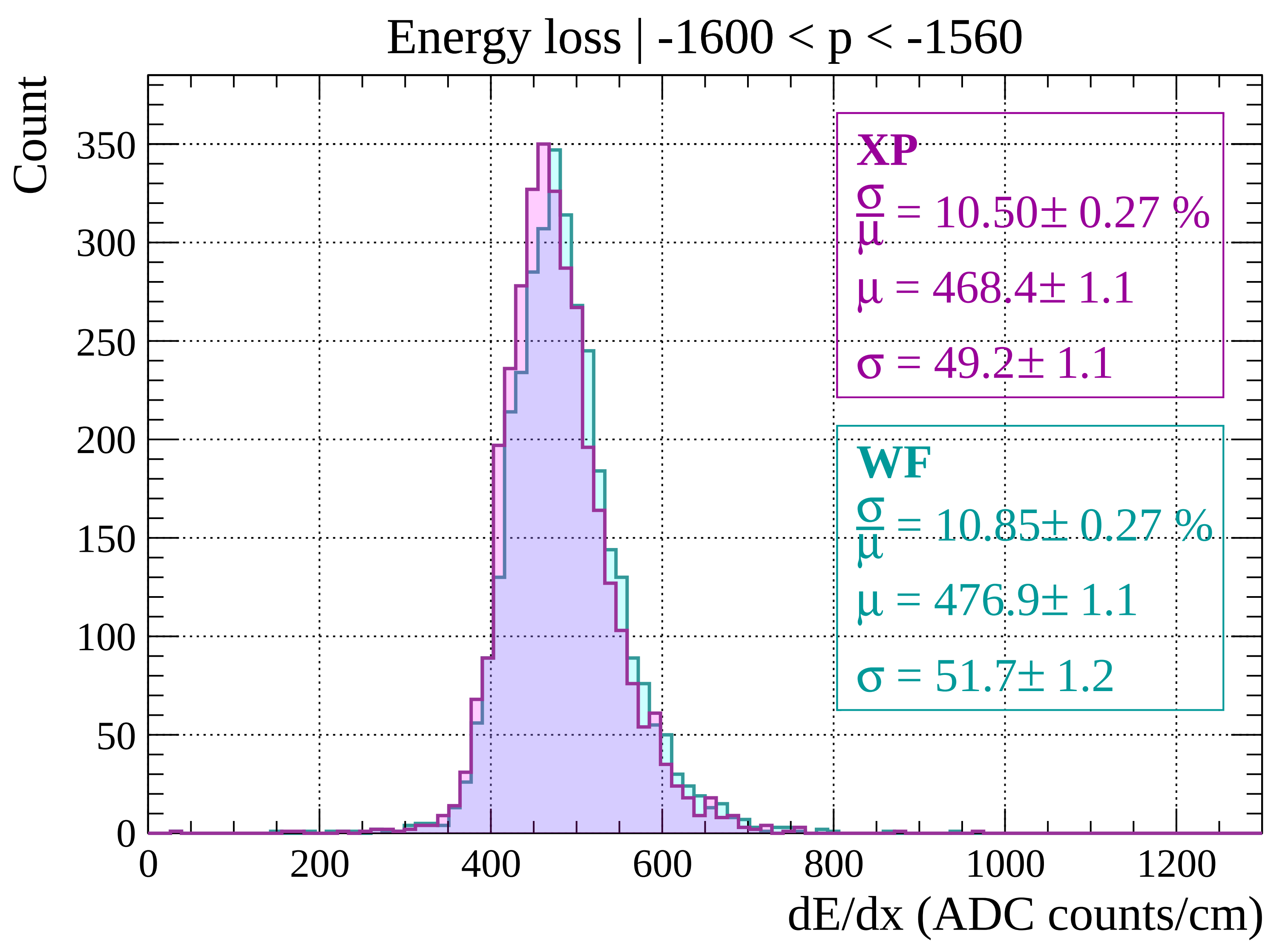
Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.05± 0.24 %
μ = 478.1± 1.2
σ = 52.8± 1.0
WF
$\frac{\sigma}{\mu}$ = 11.56± 0.26 %
μ = 486.7± 1.2
σ = 56.3± 1.1

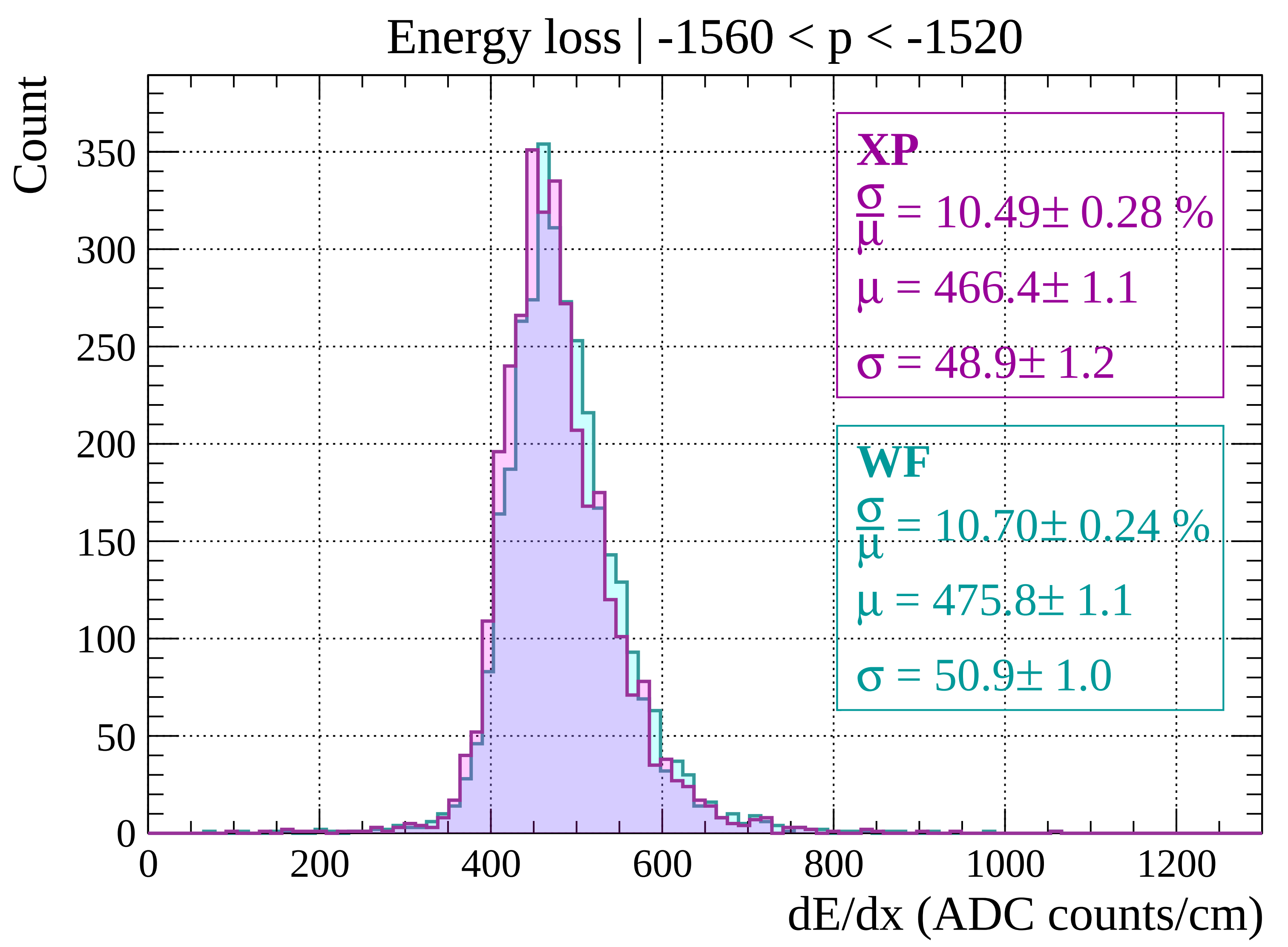

Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.50± 0.26 %
μ = 476.7± 1.1
σ = 50.0± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.07± 0.25 %
μ = 487.1± 1.2
σ = 53.9± 1.1


Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.85± 0.26 %
μ = 472.3± 1.2
σ = 51.2± 1.1
WF
σ/μ = 11.26± 0.28 %
μ = 480.8± 1.2
σ = 54.2± 1.2
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

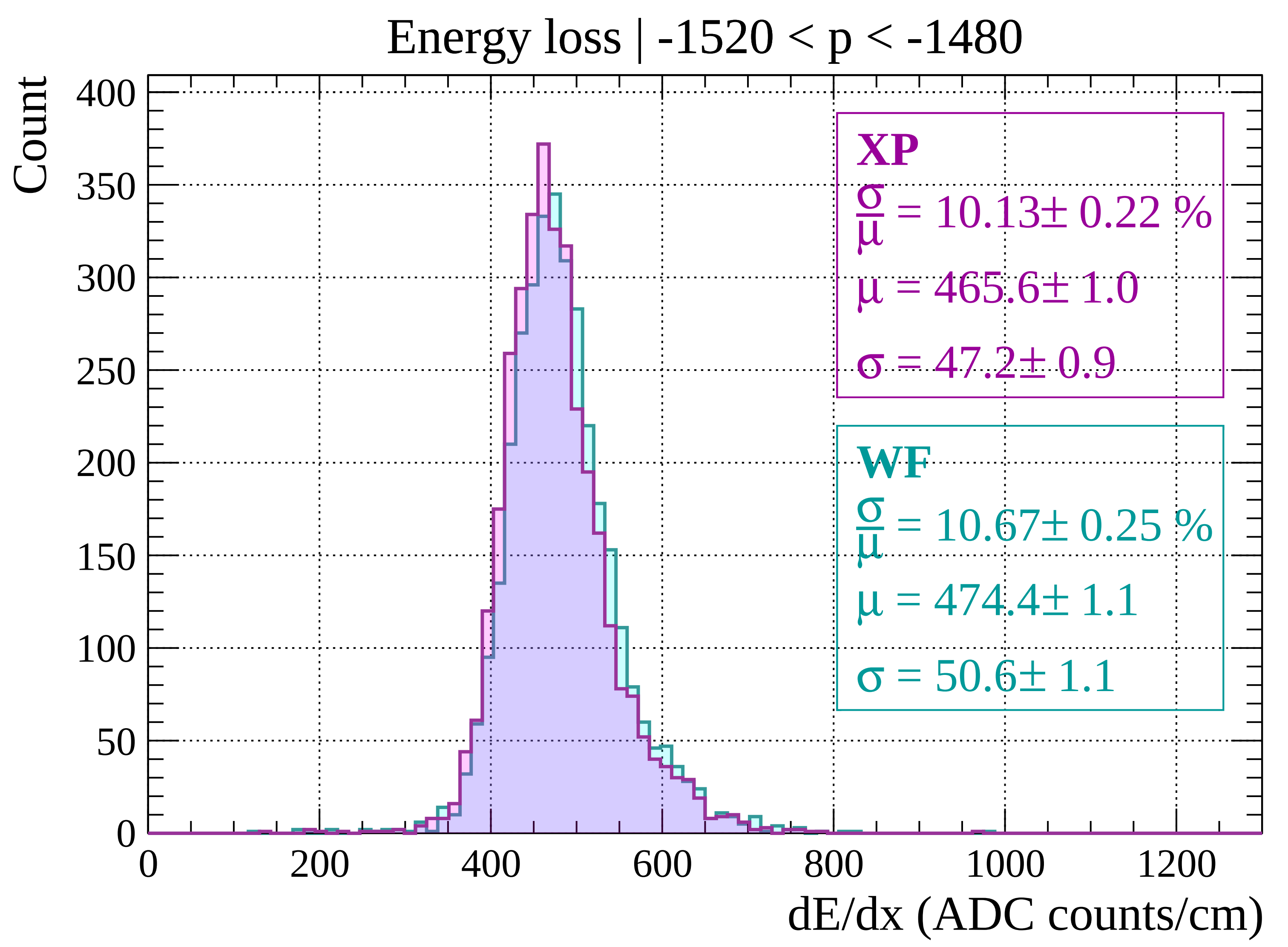


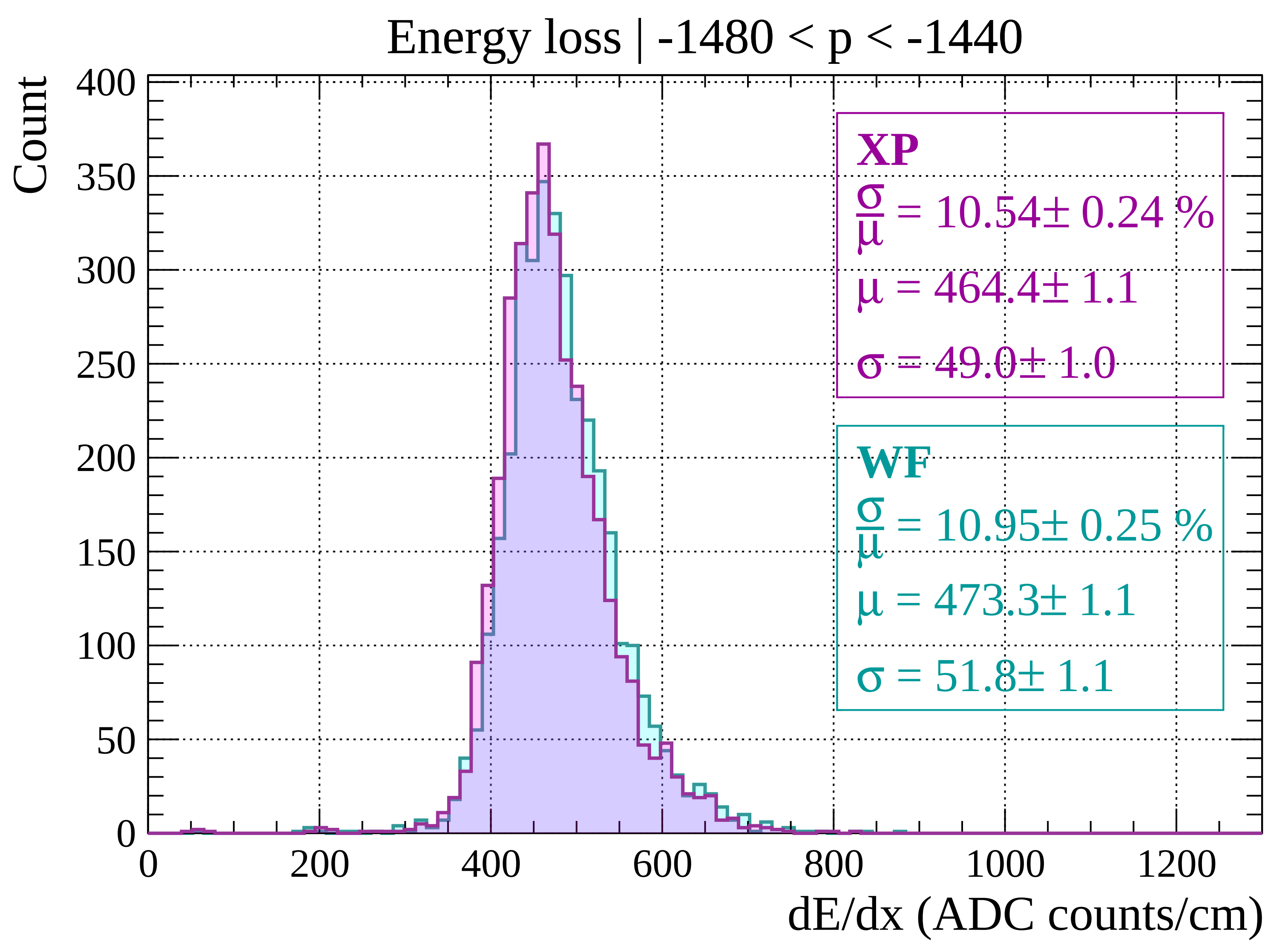
Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
σ/μ = 10.63± 0.23 %
μ = 472.0± 1.1
σ = 50.2± 0.9
WF
σ/μ = 10.98± 0.24 %
μ = 480.7± 1.1
σ = 52.8± 1.0


Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 10.50± 0.27 %
μ = 468.4± 1.1
σ = 49.2± 1.1
WF
σ/μ = 10.85± 0.27 %
μ = 476.9± 1.1
σ = 51.7± 1.2

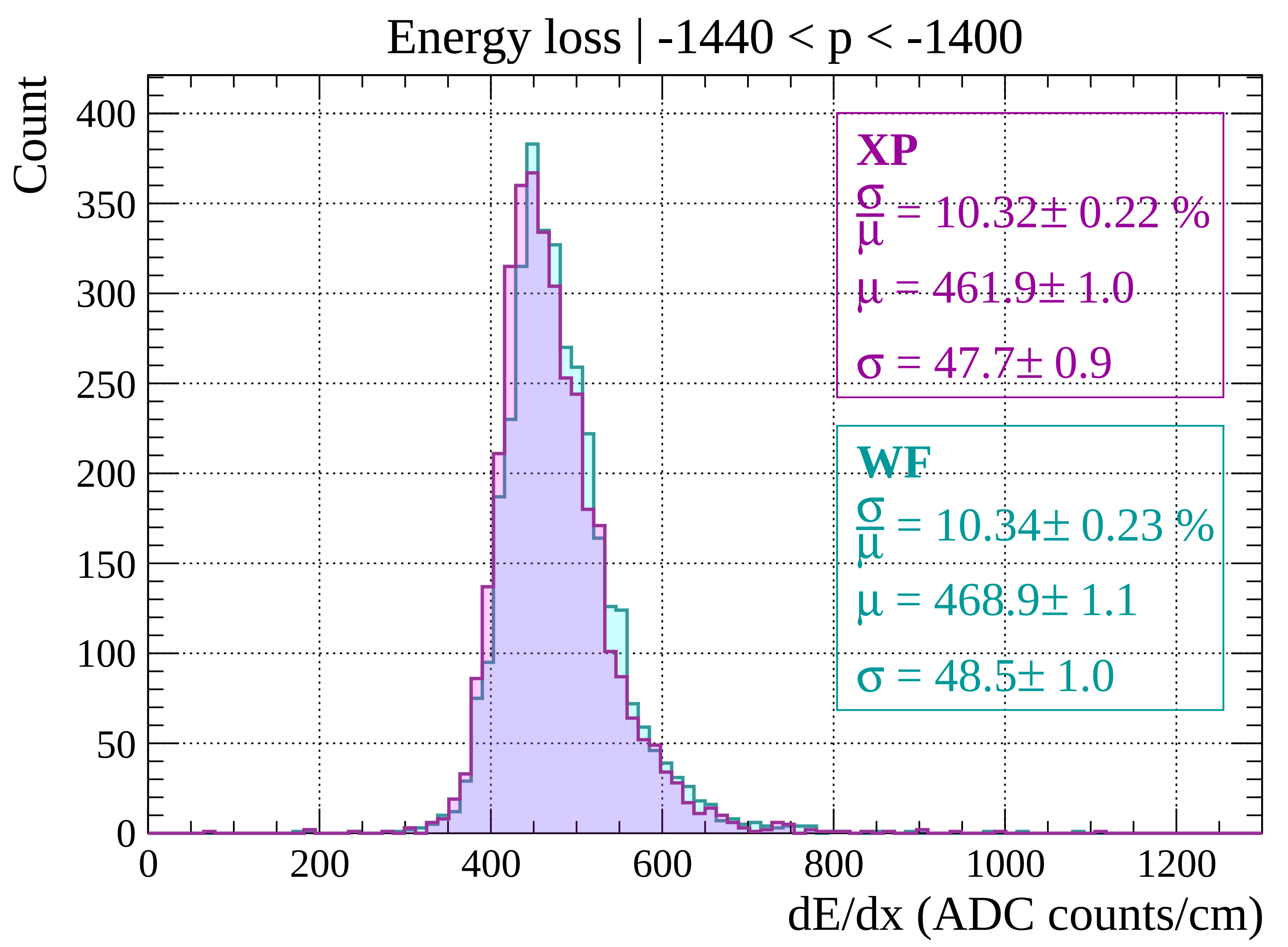


Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.49± 0.28 %
μ = 466.4± 1.1
σ = 48.9± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.70± 0.24 %
μ = 475.8± 1.1
σ = 50.9± 1.0

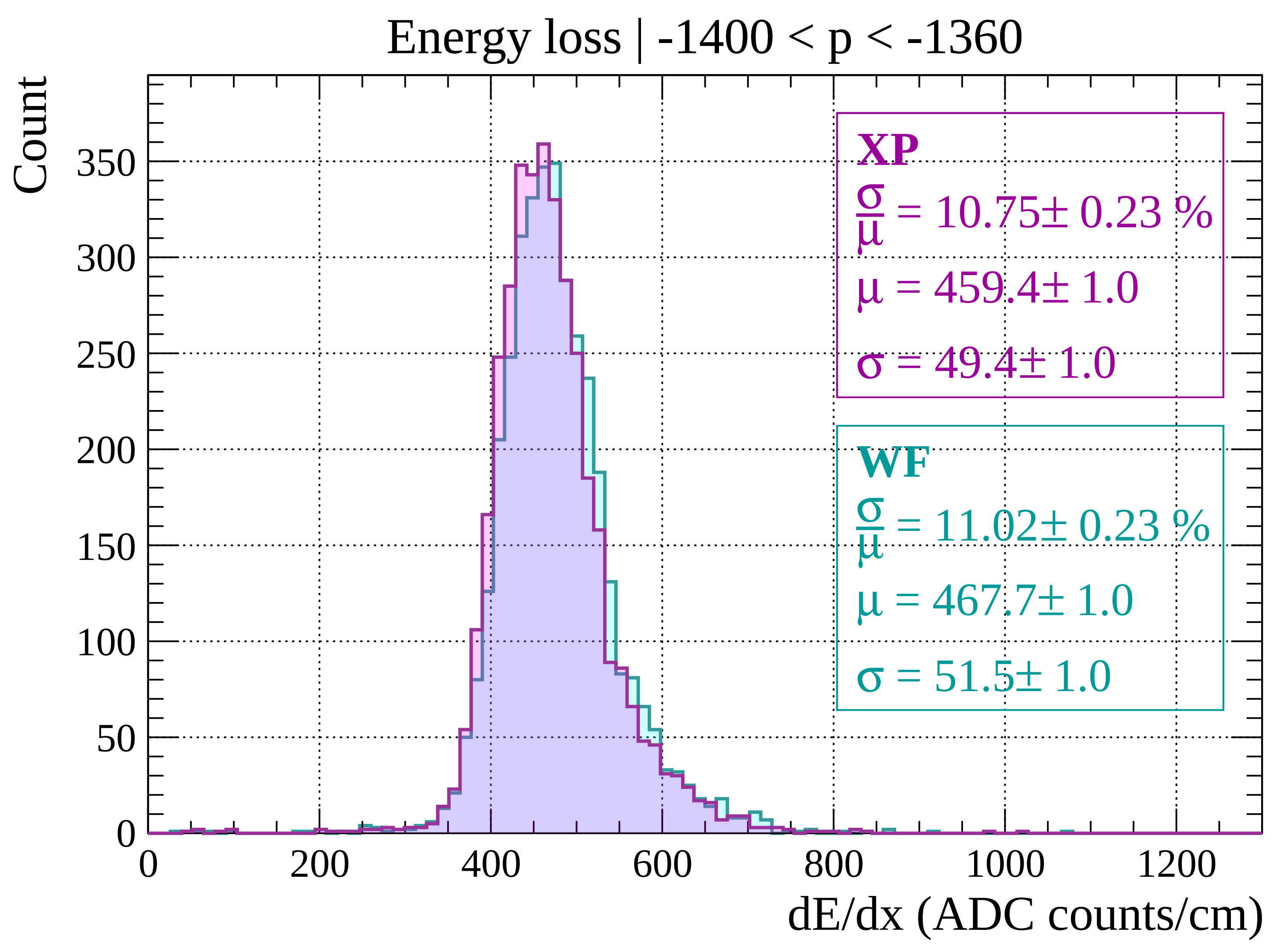

Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.13± 0.22 %
μ = 465.6± 1.0
σ = 47.2± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.67± 0.25 %
μ = 474.4± 1.1
σ = 50.6± 1.1



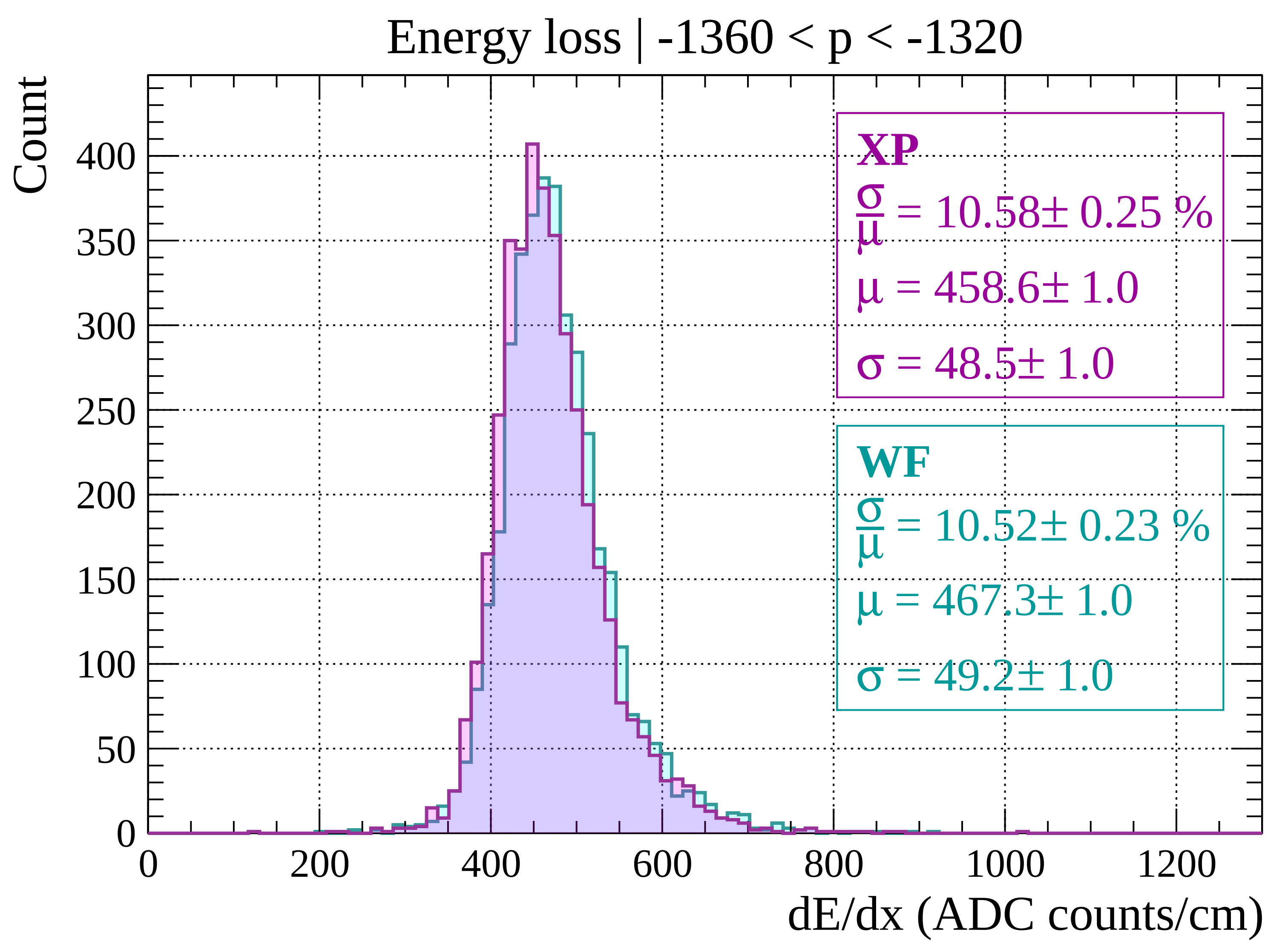
Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
σ/μ = 10.54± 0.24 %
μ = 464.4± 1.1
σ = 49.0± 1.0
WF
σ/μ = 10.95± 0.25 %
μ = 473.3± 1.1
σ = 51.8± 1.1



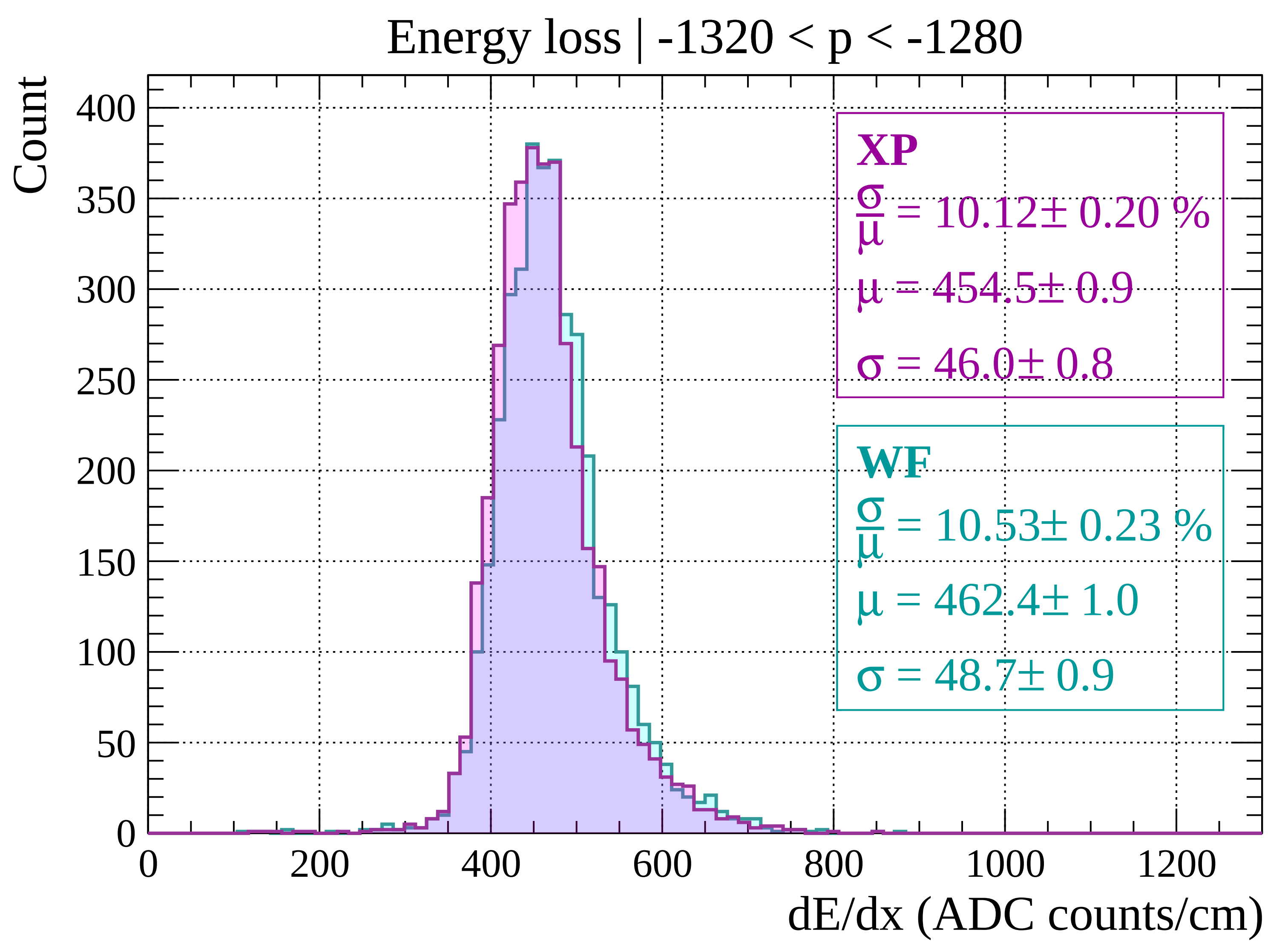
Energy loss | -1440 < p < -1400
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.32± 0.22 %
μ = 461.9± 1.0
σ = 47.7± 0.9
WF
σ/μ = 10.34± 0.23 %
μ = 468.9± 1.1
σ = 48.5± 1.0

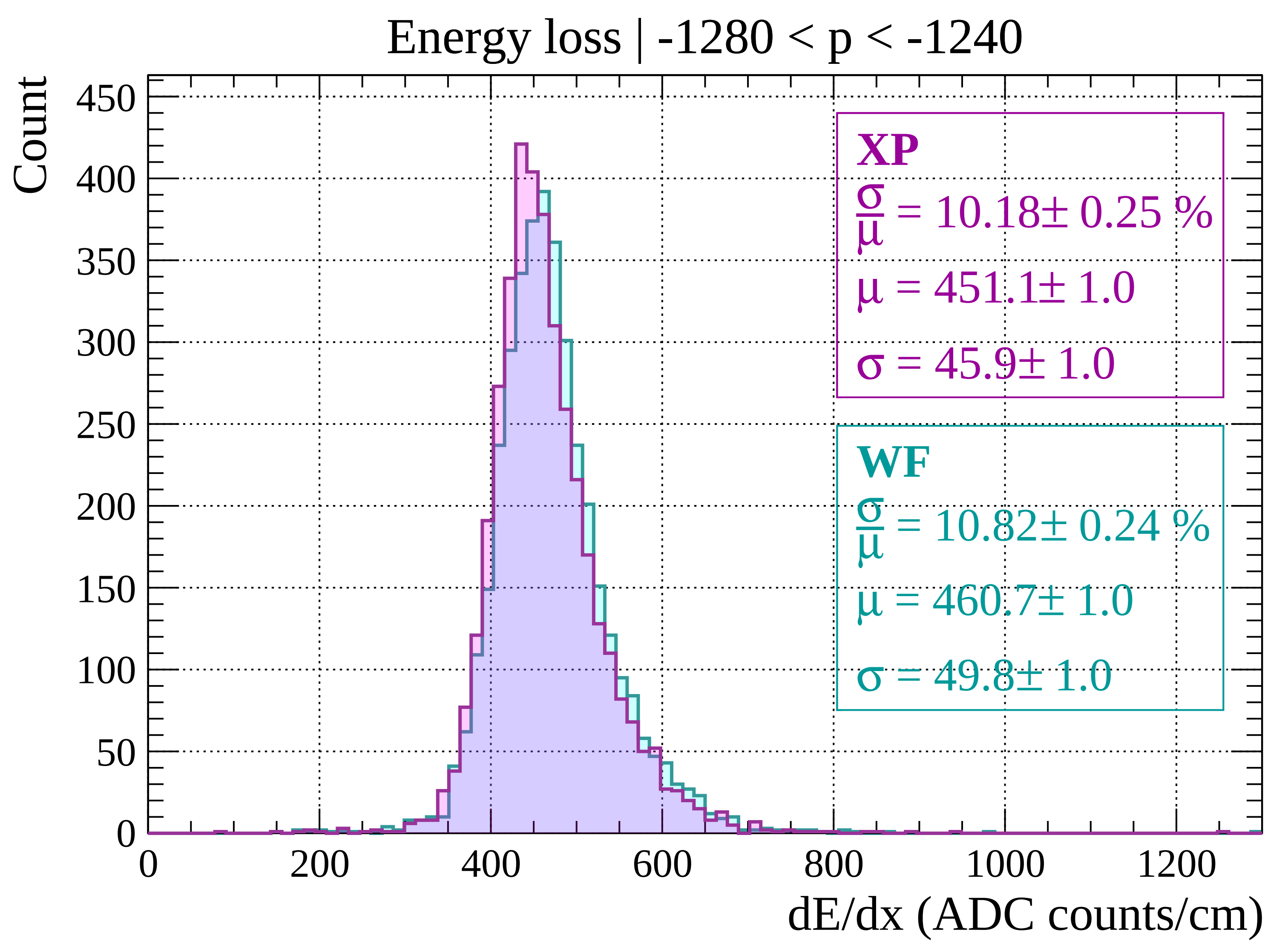


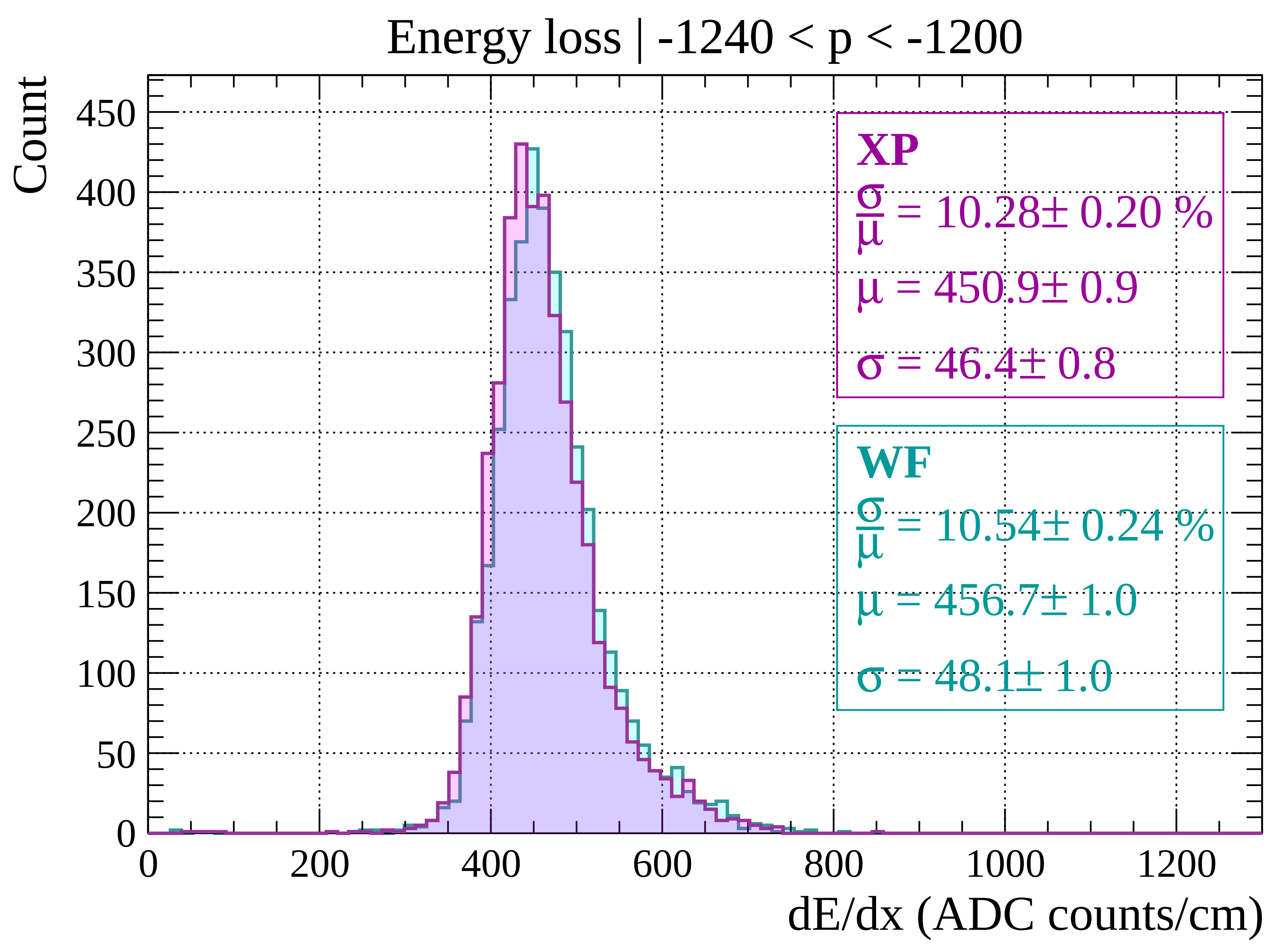
Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 10.75± 0.23 %
μ = 459.4± 1.0
σ = 49.4± 1.0
WF
σ/μ = 11.02± 0.23 %
μ = 467.7± 1.0
σ = 51.5± 1.0

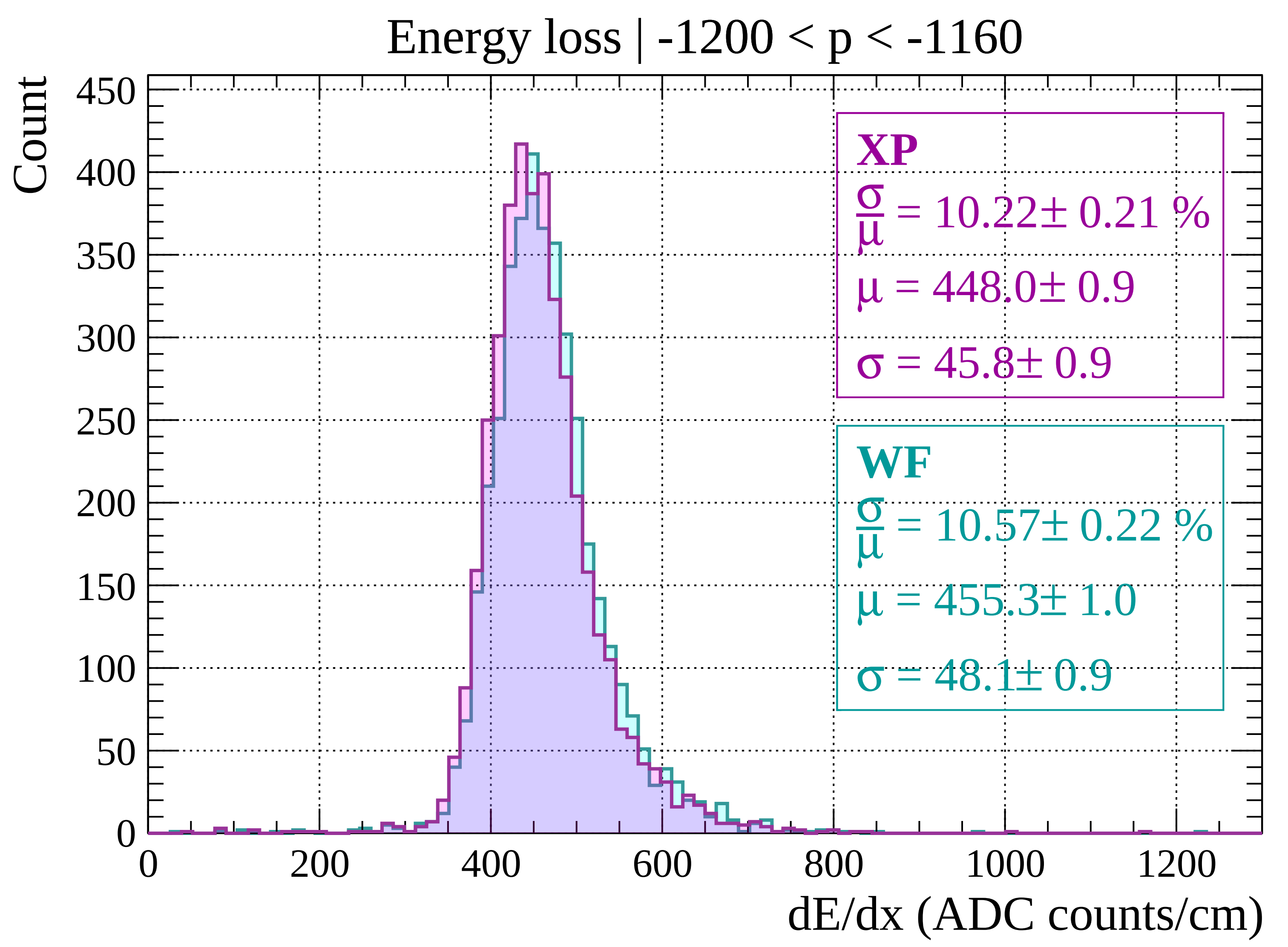


Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 10.58± 0.25 %
μ = 458.6± 1.0
σ = 48.5± 1.0
WF
σ/μ = 10.52± 0.23 %
μ = 467.3± 1.0
σ = 49.2± 1.0

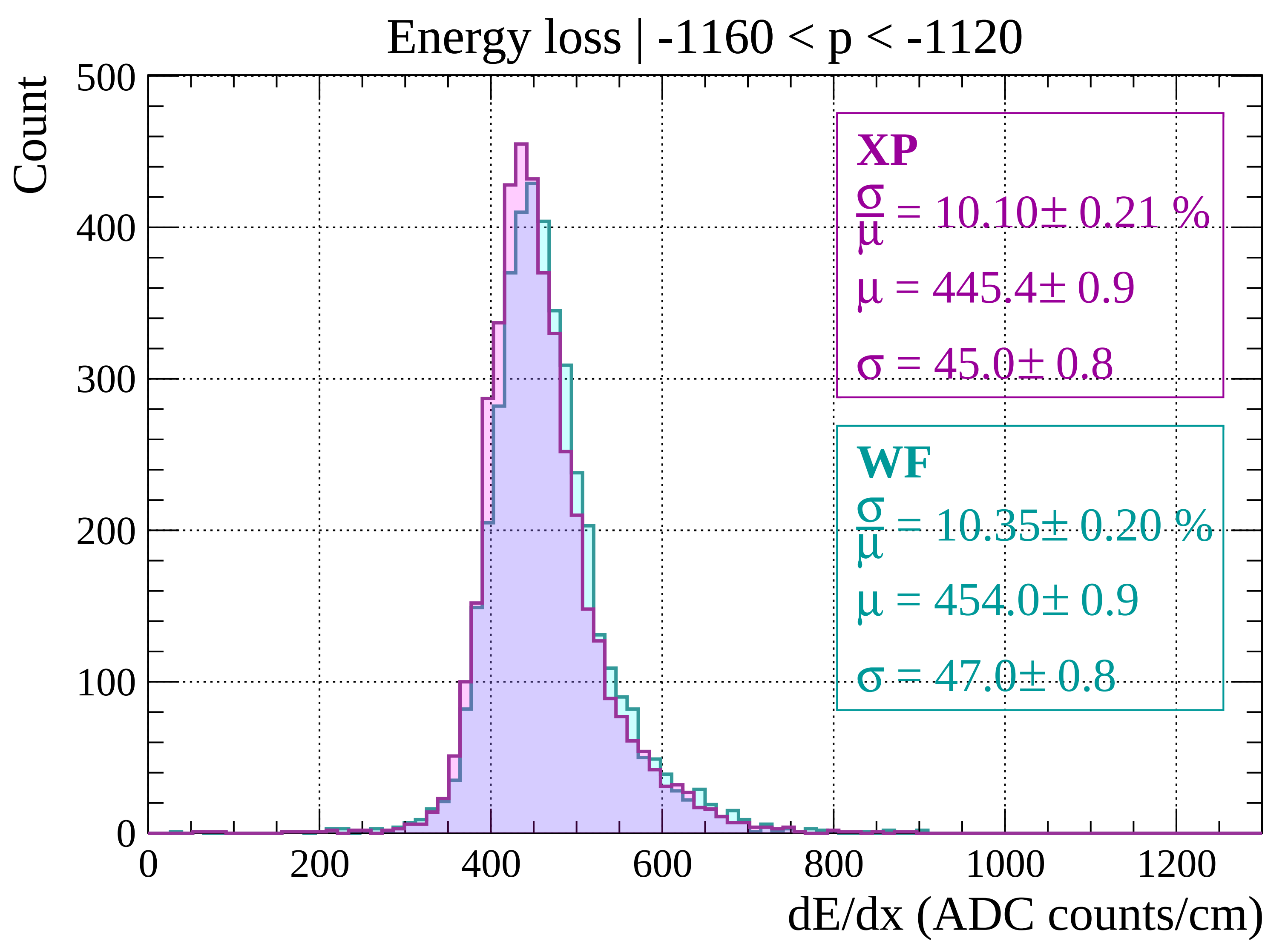

Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.12± 0.20 %
μ = 454.5± 0.9
σ = 46.0± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.53± 0.23 %
μ = 462.4± 1.0
σ = 48.7± 0.9


Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP
σ/μ = 10.18± 0.25 %
μ = 451.1± 1.0
σ = 45.9± 1.0
WF
σ/μ = 10.82± 0.24 %
μ = 460.7± 1.0
σ = 49.8± 1.0

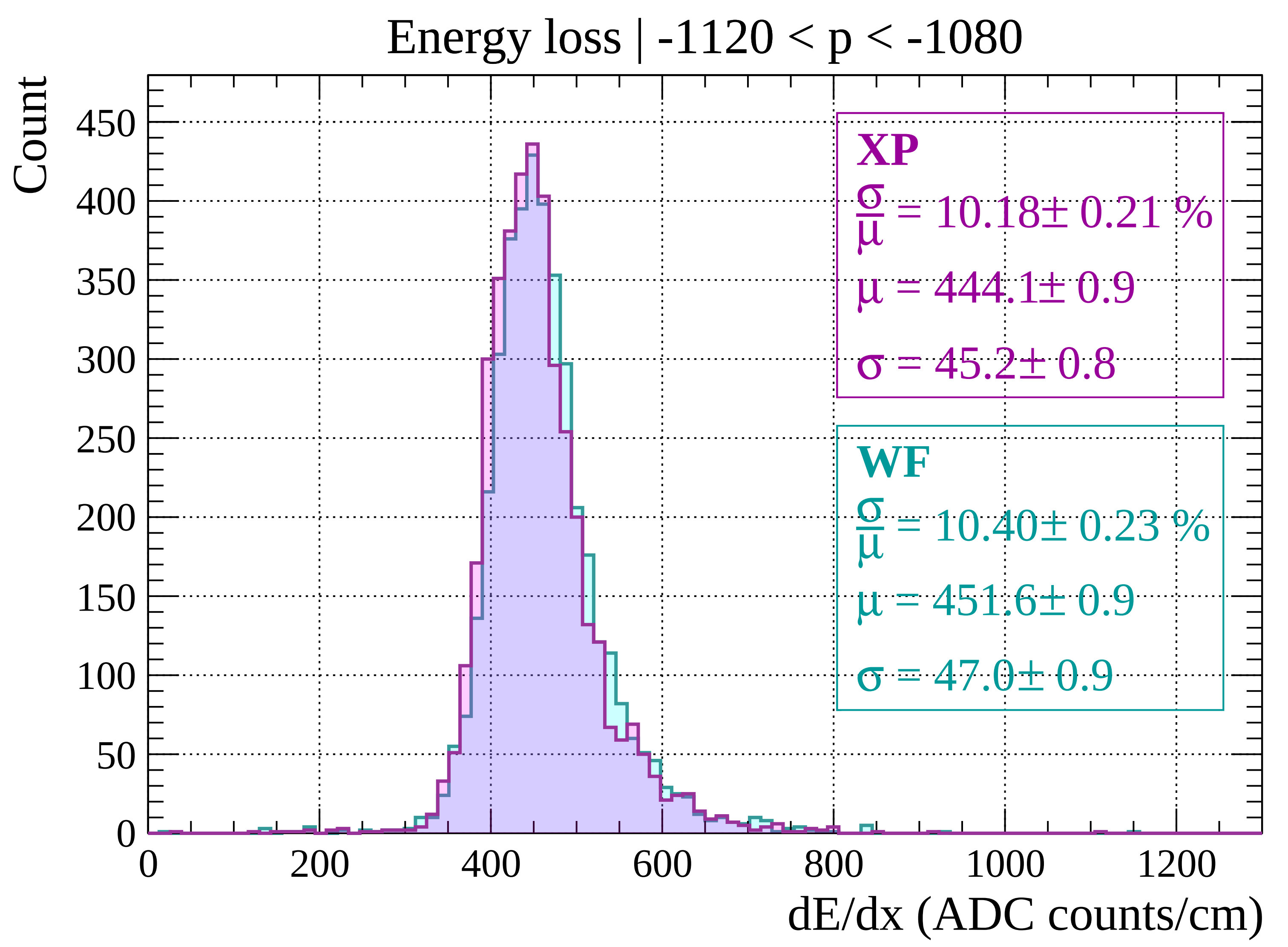

Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.28± 0.20 %
μ = 450.9± 0.9
σ = 46.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.54± 0.24 %
μ = 456.7± 1.0
σ = 48.1± 1.0



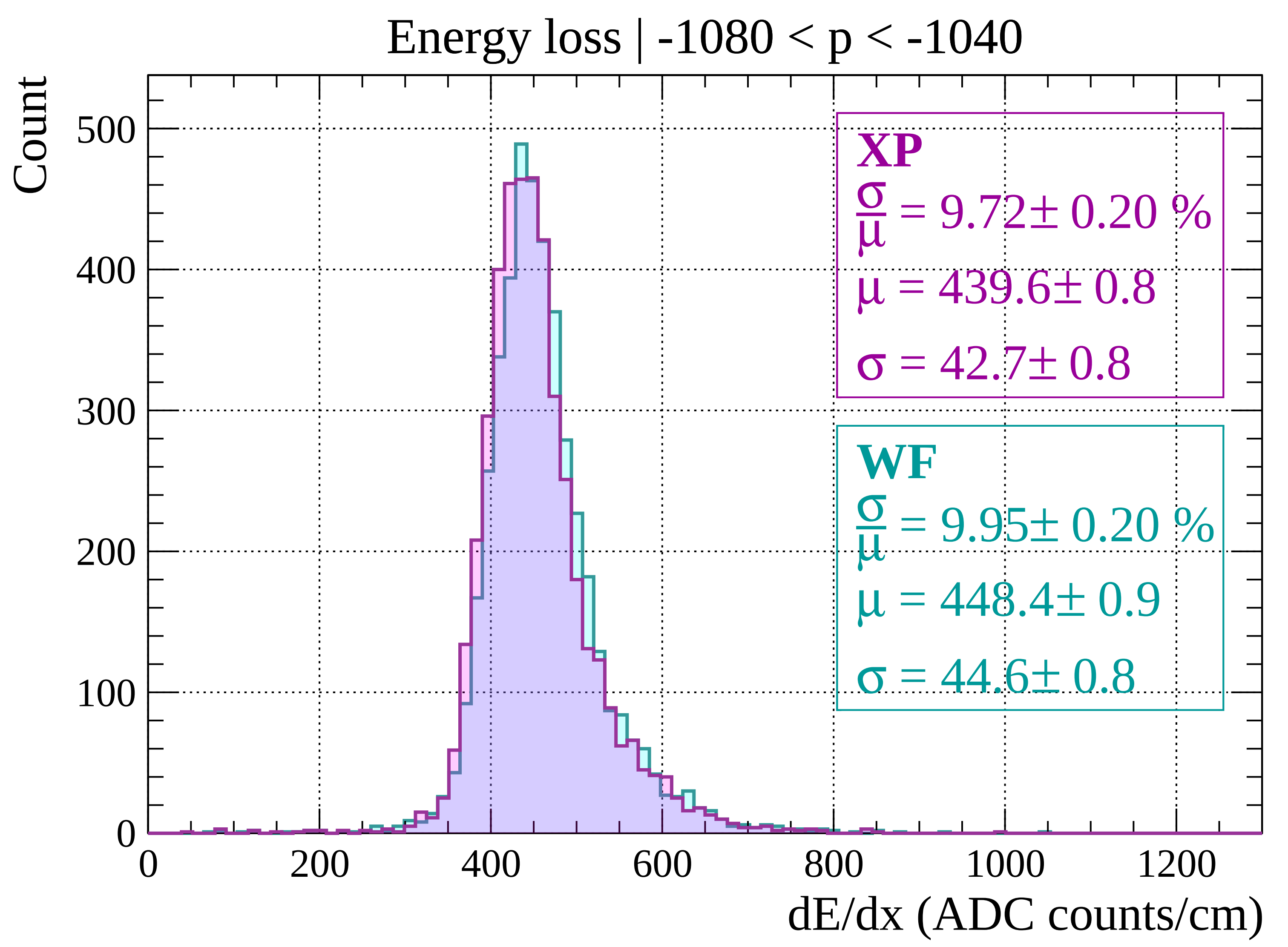
Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 10.22 ± 0.21 %
μ = 448.0 ± 0.9
σ = 45.8 ± 0.9
WF
σ/μ = 10.57 ± 0.22 %
μ = 455.3 ± 1.0
σ = 48.1 ± 0.9

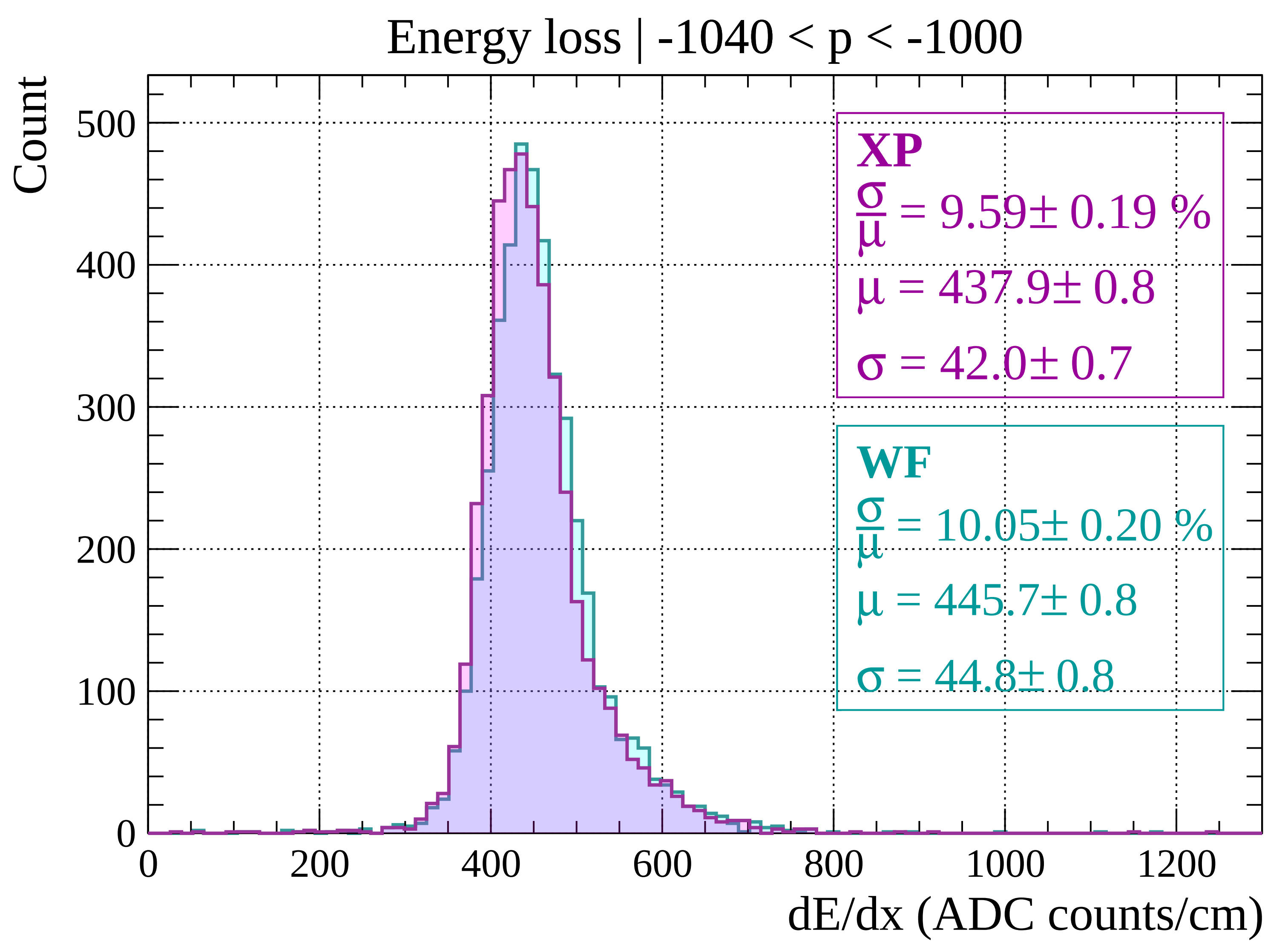


Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 10.10± 0.21 %
μ = 445.4± 0.9
σ = 45.0± 0.8
WF
σ/μ = 10.35± 0.20 %
μ = 454.0± 0.9
σ = 47.0± 0.8

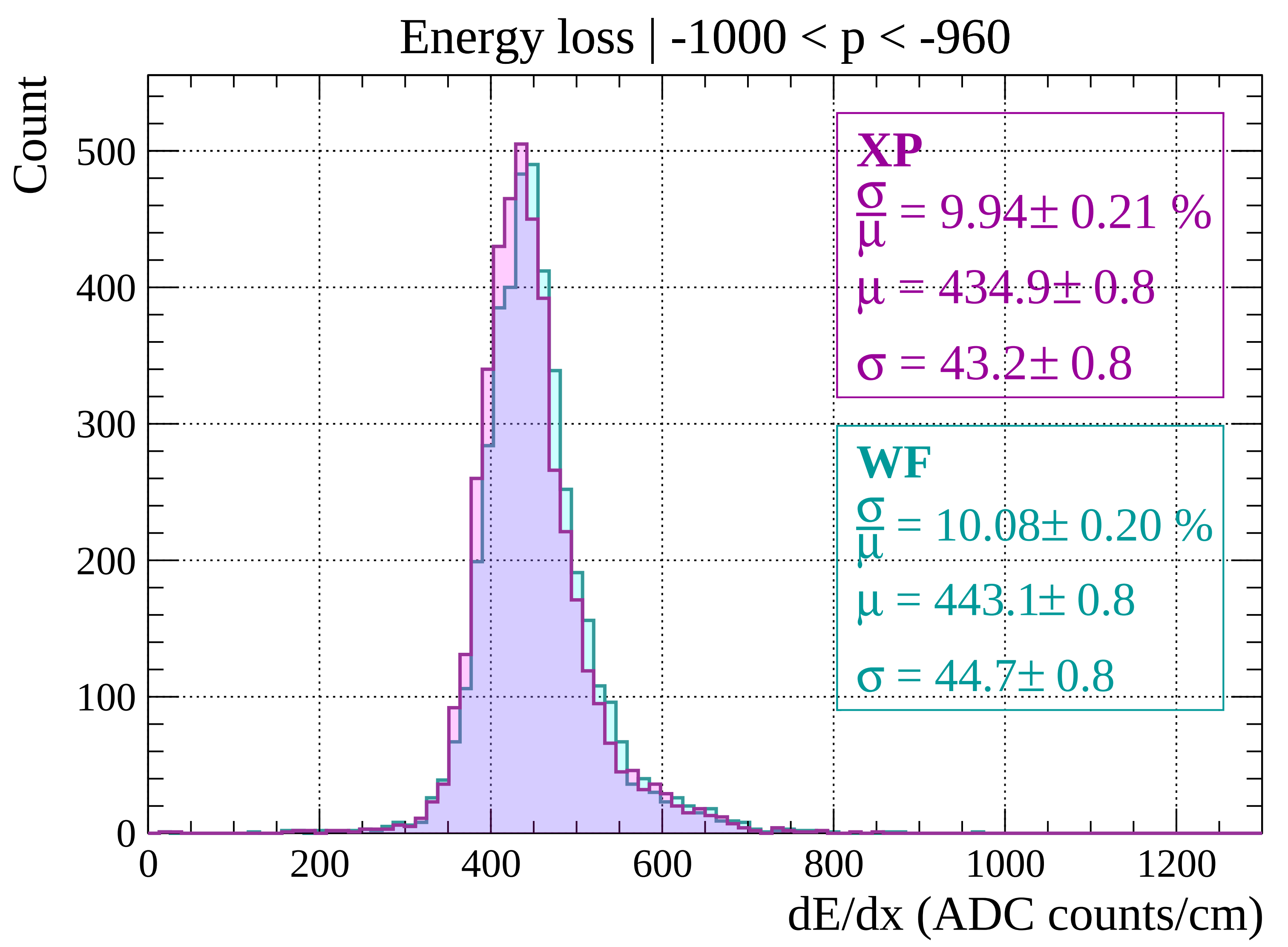


Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.18± 0.21 %
μ = 444.1± 0.9
σ = 45.2± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.40± 0.23 %
μ = 451.6± 0.9
σ = 47.0± 0.9

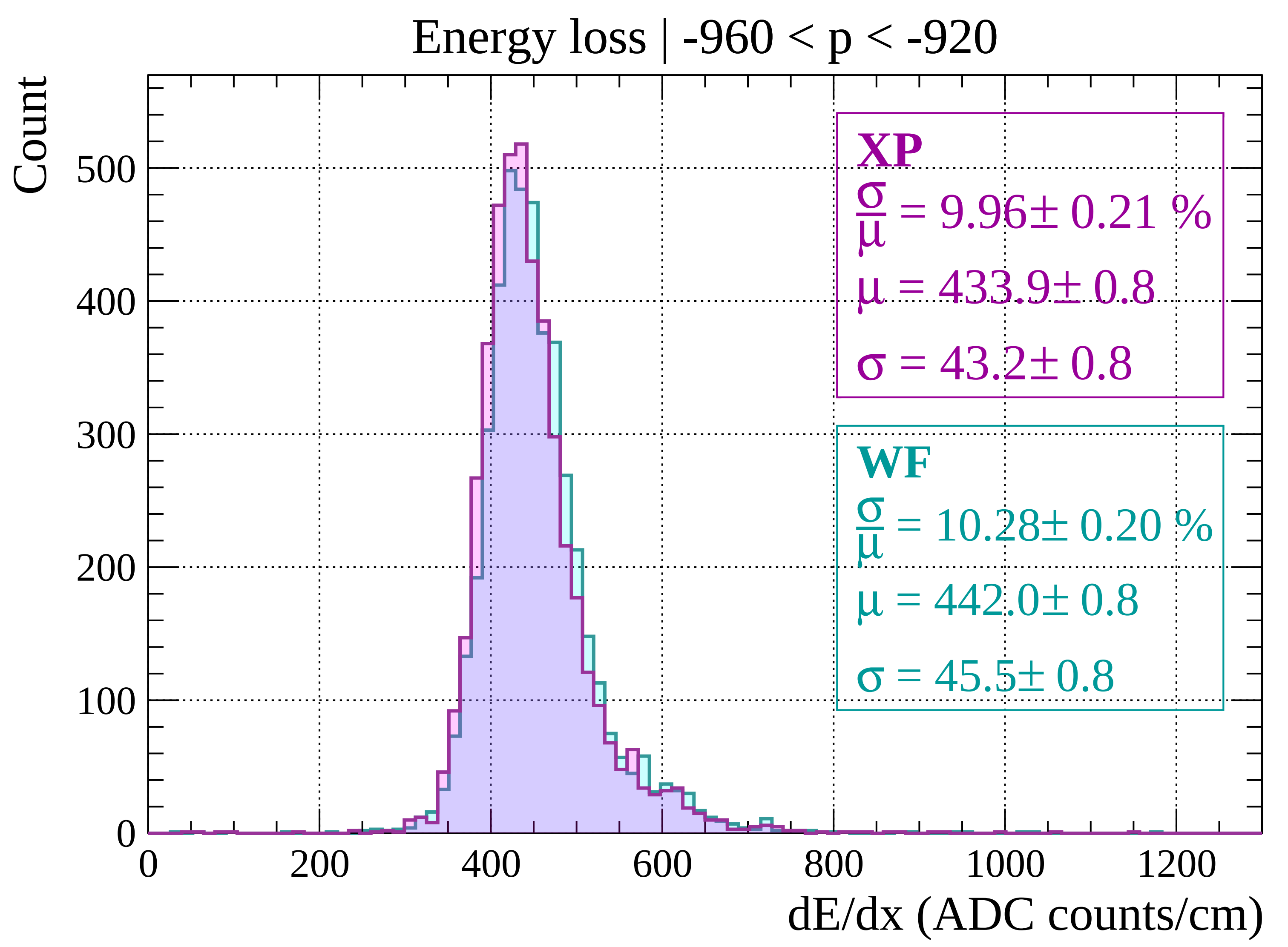

Energy loss | -1080 < p < -1040
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.72± 0.20 %
μ = 439.6± 0.8
σ = 42.7± 0.8
WF
σ/μ = 9.95± 0.20 %
μ = 448.4± 0.9
σ = 44.6± 0.8

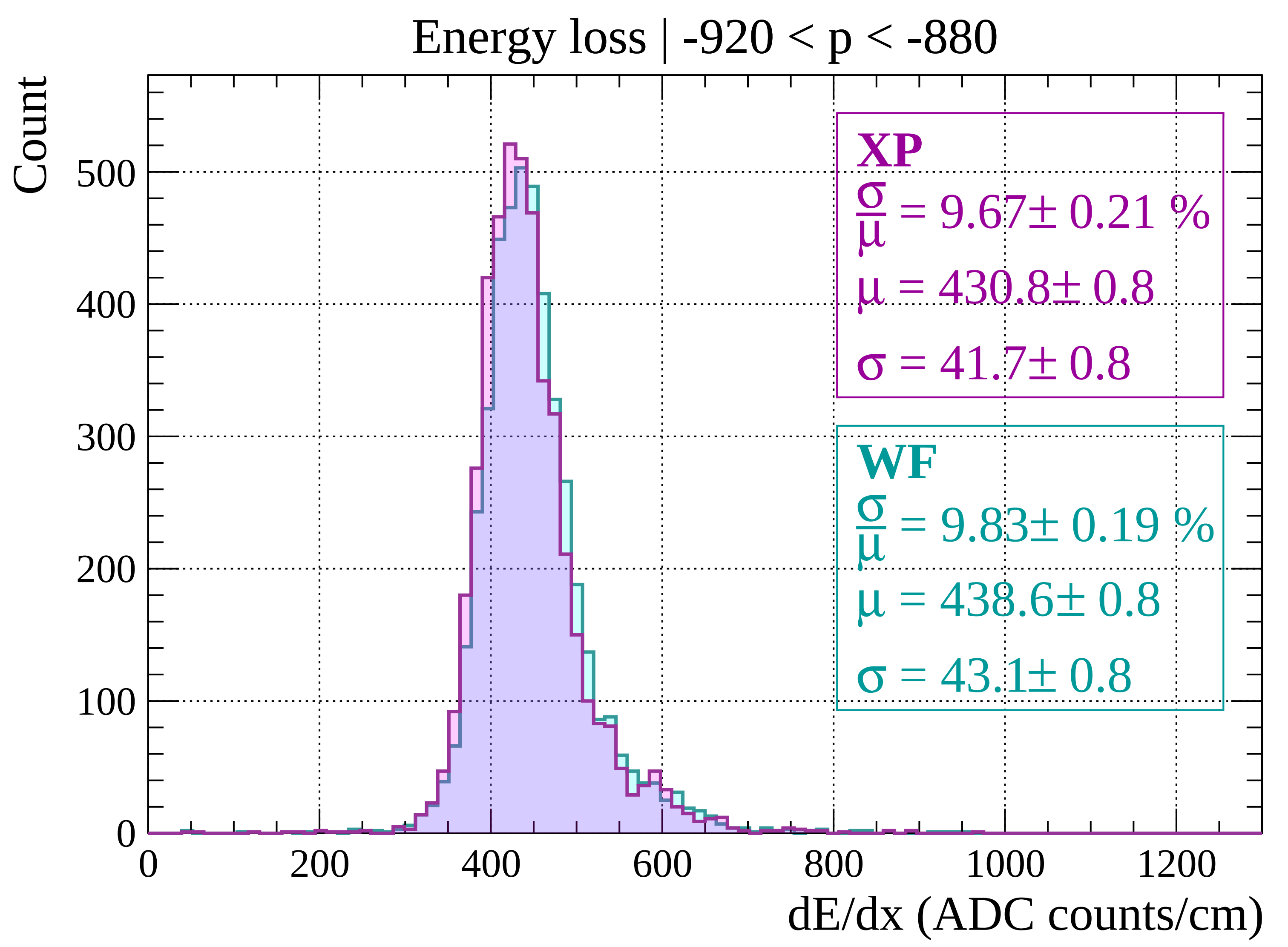


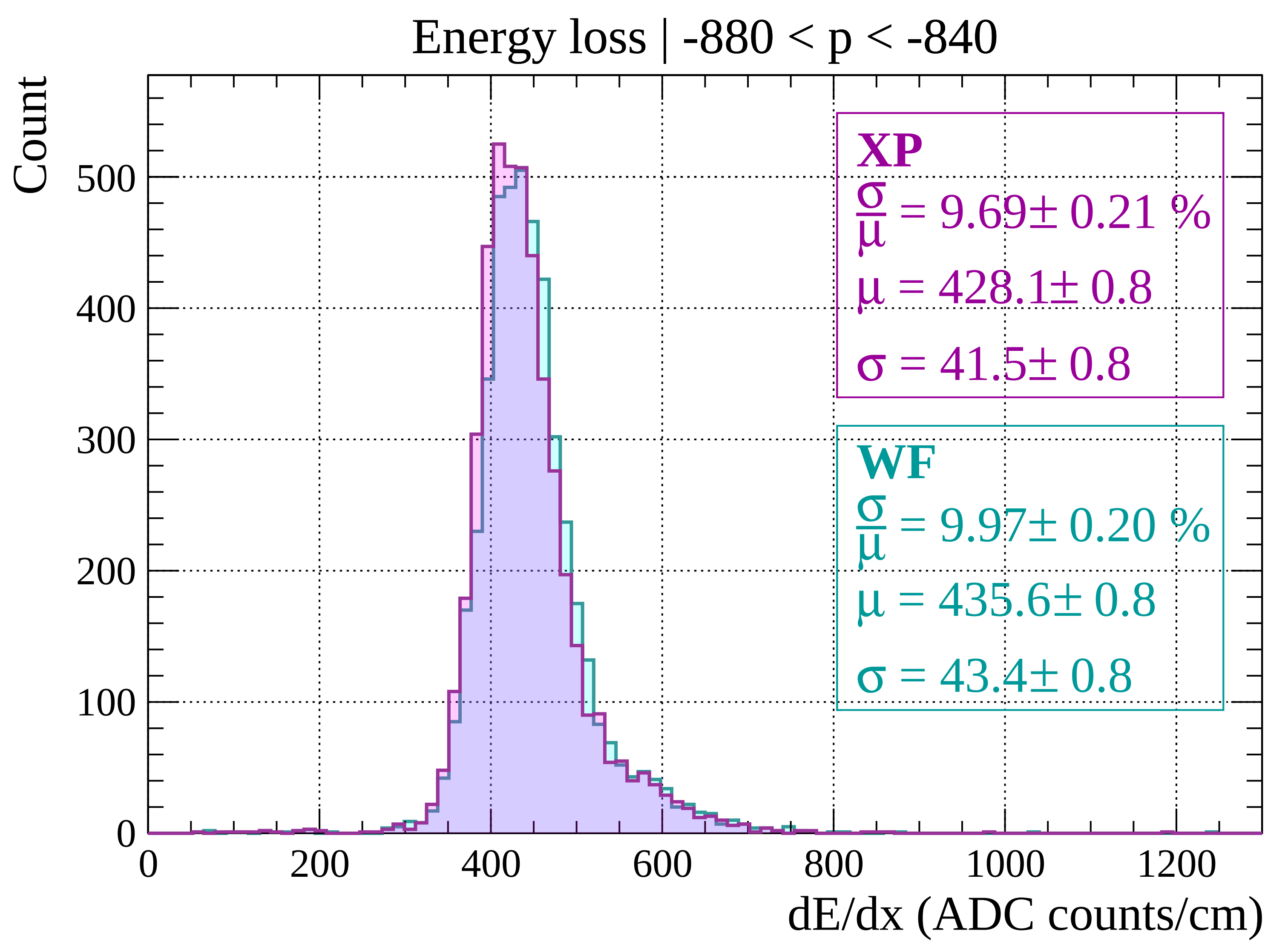
Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.59± 0.19 %
μ = 437.9± 0.8
σ = 42.0± 0.7
WF
σ/μ = 10.05± 0.20 %
μ = 445.7± 0.8
σ = 44.8± 0.8

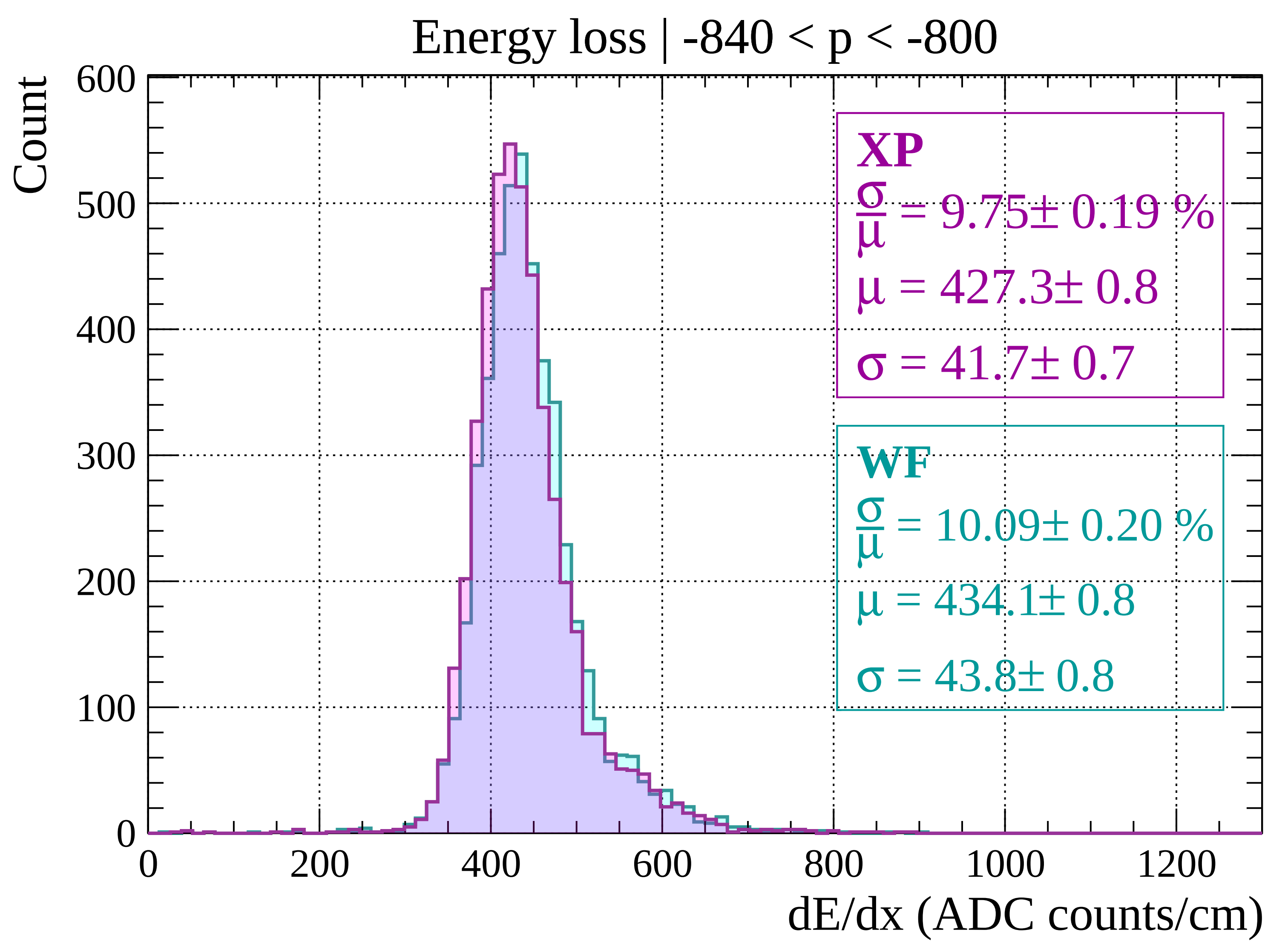


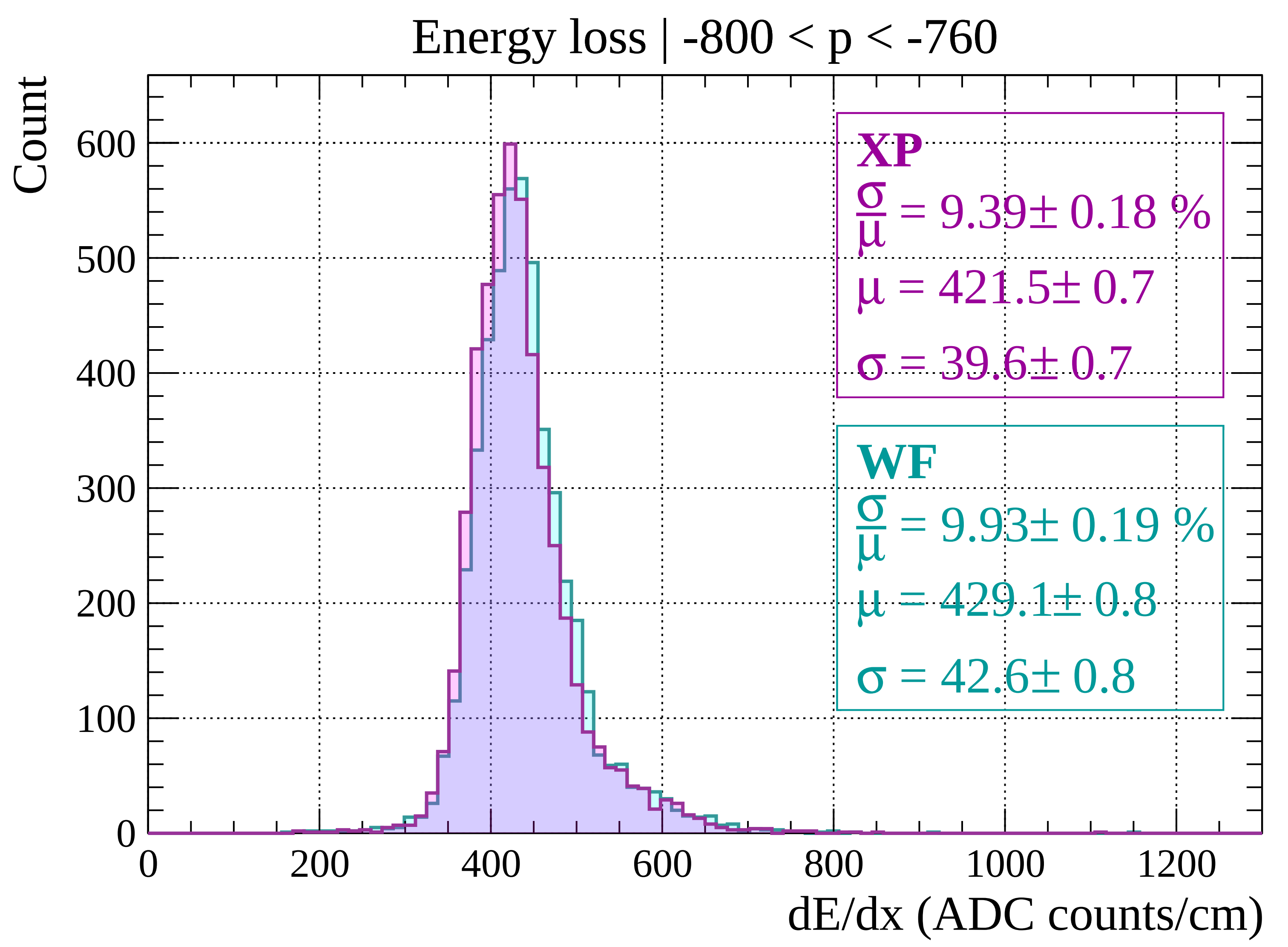
Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.94± 0.21 %
μ = 434.9± 0.8
σ = 43.2± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.08± 0.20 %
μ = 443.1± 0.8
σ = 44.7± 0.8


Energy loss | -960 < p < -920
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.96± 0.21 %
μ = 433.9± 0.8
σ = 43.2± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.28± 0.20 %
μ = 442.0± 0.8
σ = 45.5± 0.8

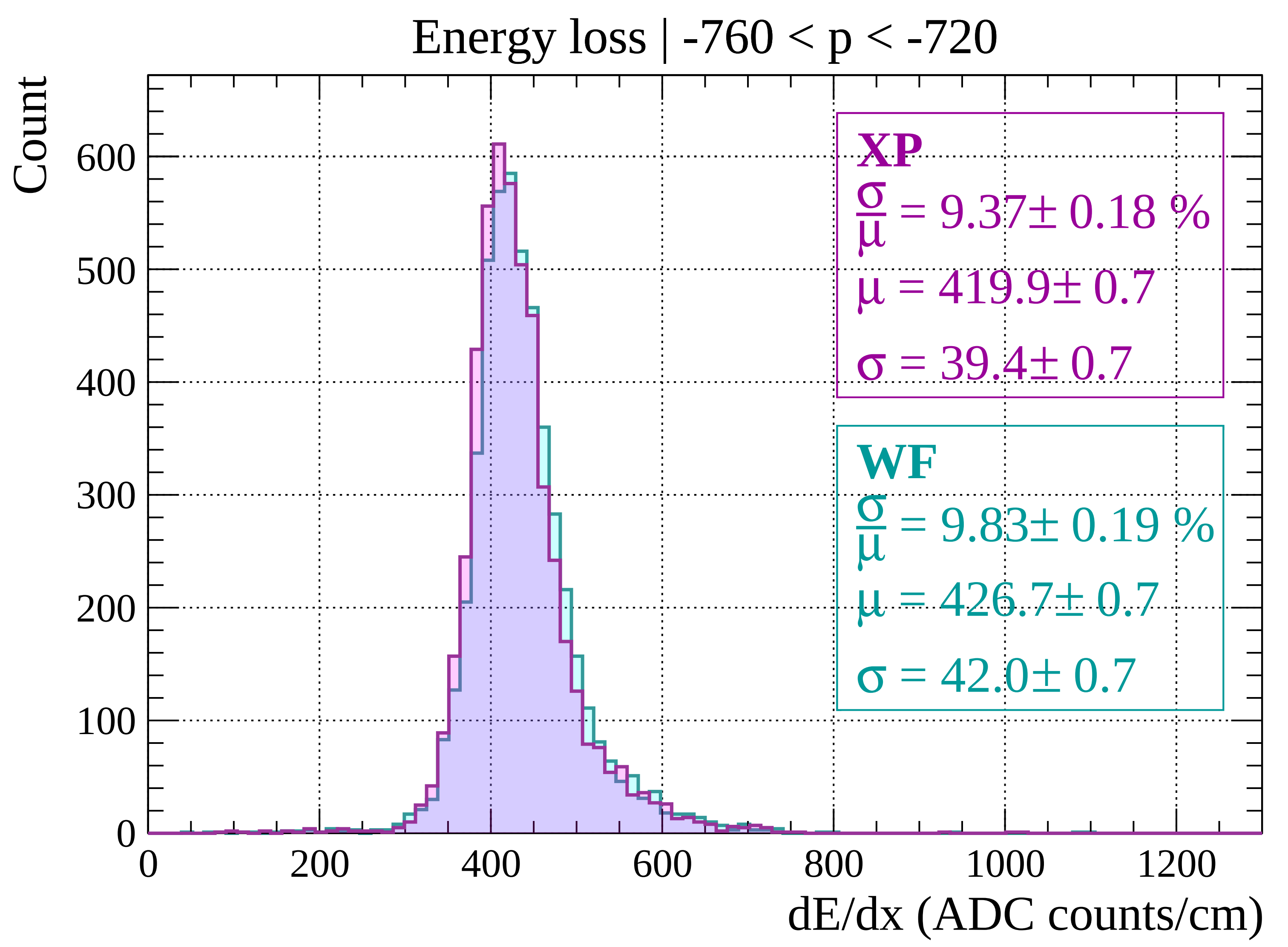

Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.67± 0.21 %
μ = 430.8± 0.8
σ = 41.7± 0.8
WF
σ/μ = 9.83± 0.19 %
μ = 438.6± 0.8
σ = 43.1± 0.8
0
100
200
300
400
500
0
200
400
600
800
1000
1200

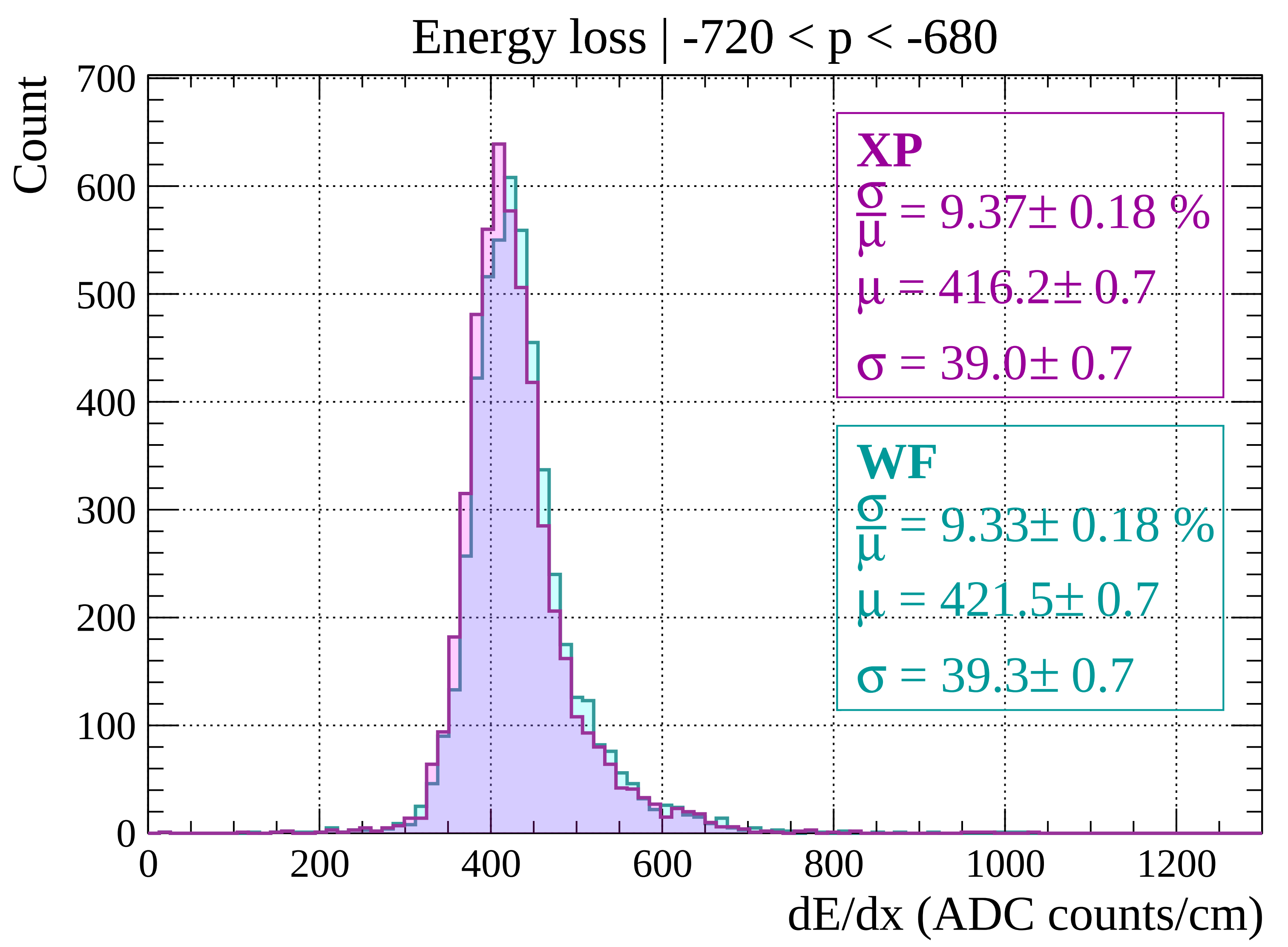

Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.69 \pm 0.21$ %
$\mu = 428.1 \pm 0.8$
$\sigma = 41.5 \pm 0.8$
WF
$\frac{\sigma}{\mu} = 9.97 \pm 0.20$ %
$\mu = 435.6 \pm 0.8$
$\sigma = 43.4 \pm 0.8$


Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.75± 0.19 %
μ = 427.3± 0.8
σ = 41.7± 0.7
WF
σ/μ = 10.09± 0.20 %
μ = 434.1± 0.8
σ = 43.8± 0.8

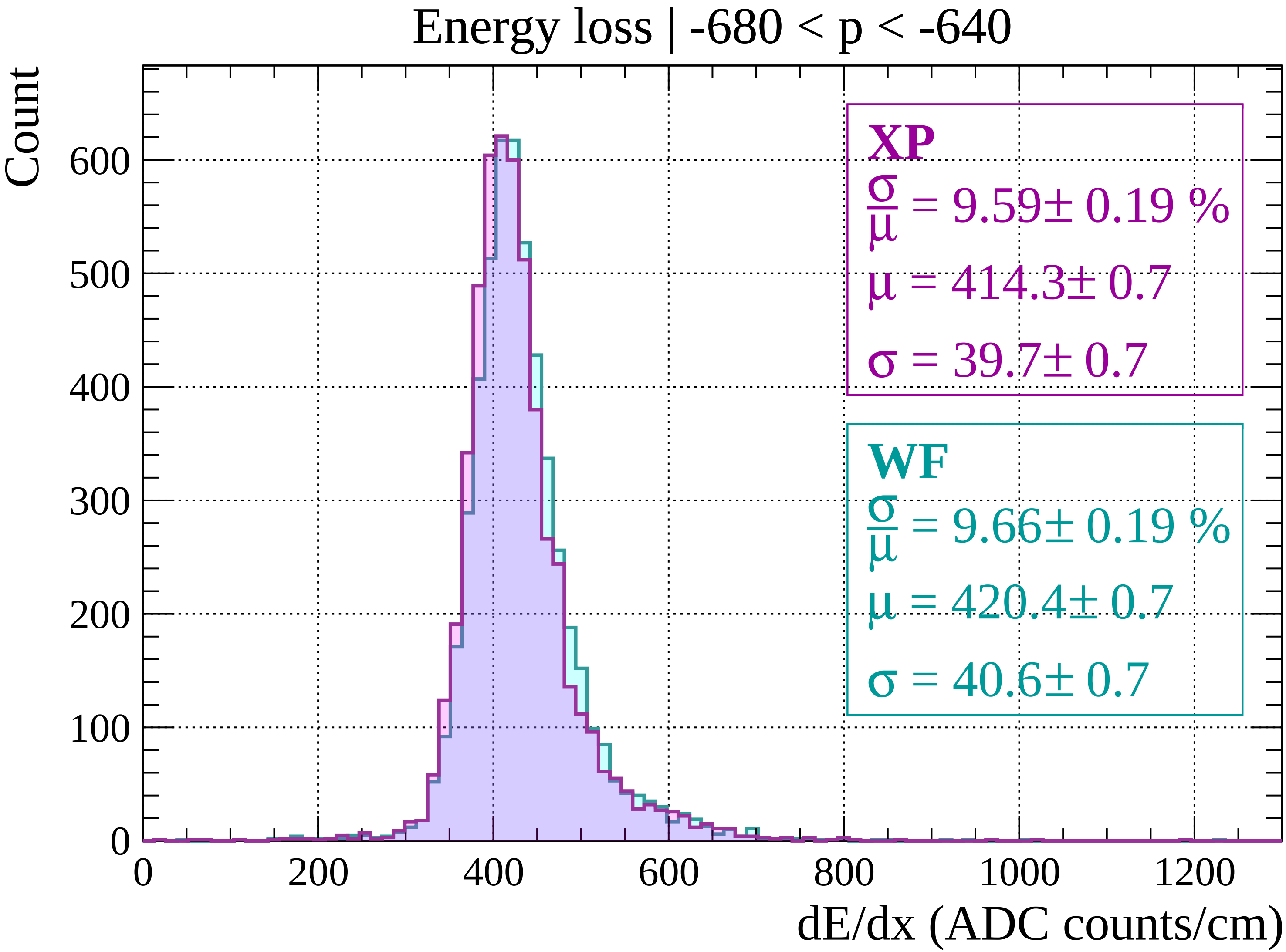


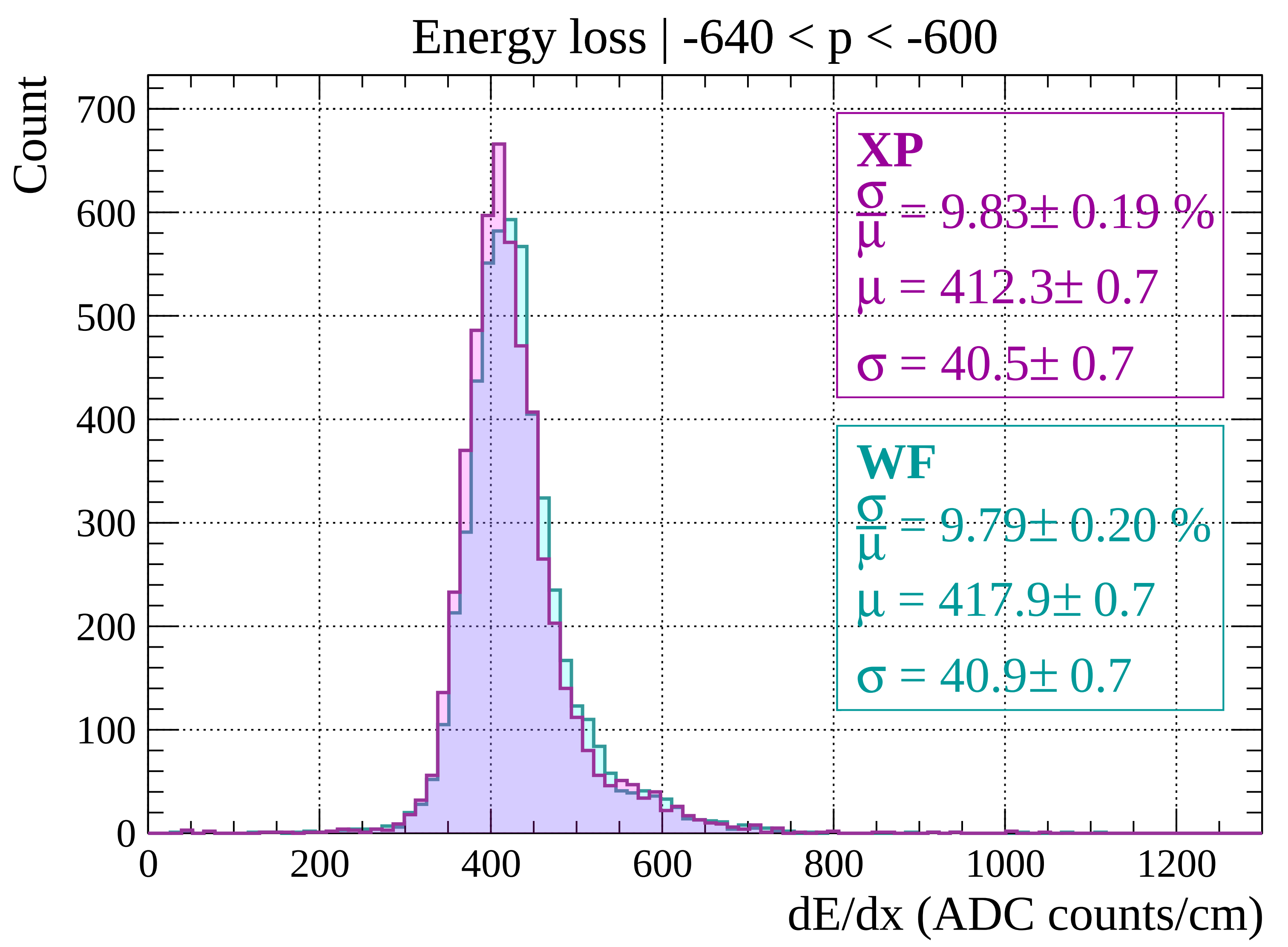
Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.39± 0.18 %
μ = 421.5± 0.7
σ = 39.6± 0.7
WF
σ/μ = 9.93± 0.19 %
μ = 429.1± 0.8
σ = 42.6± 0.8

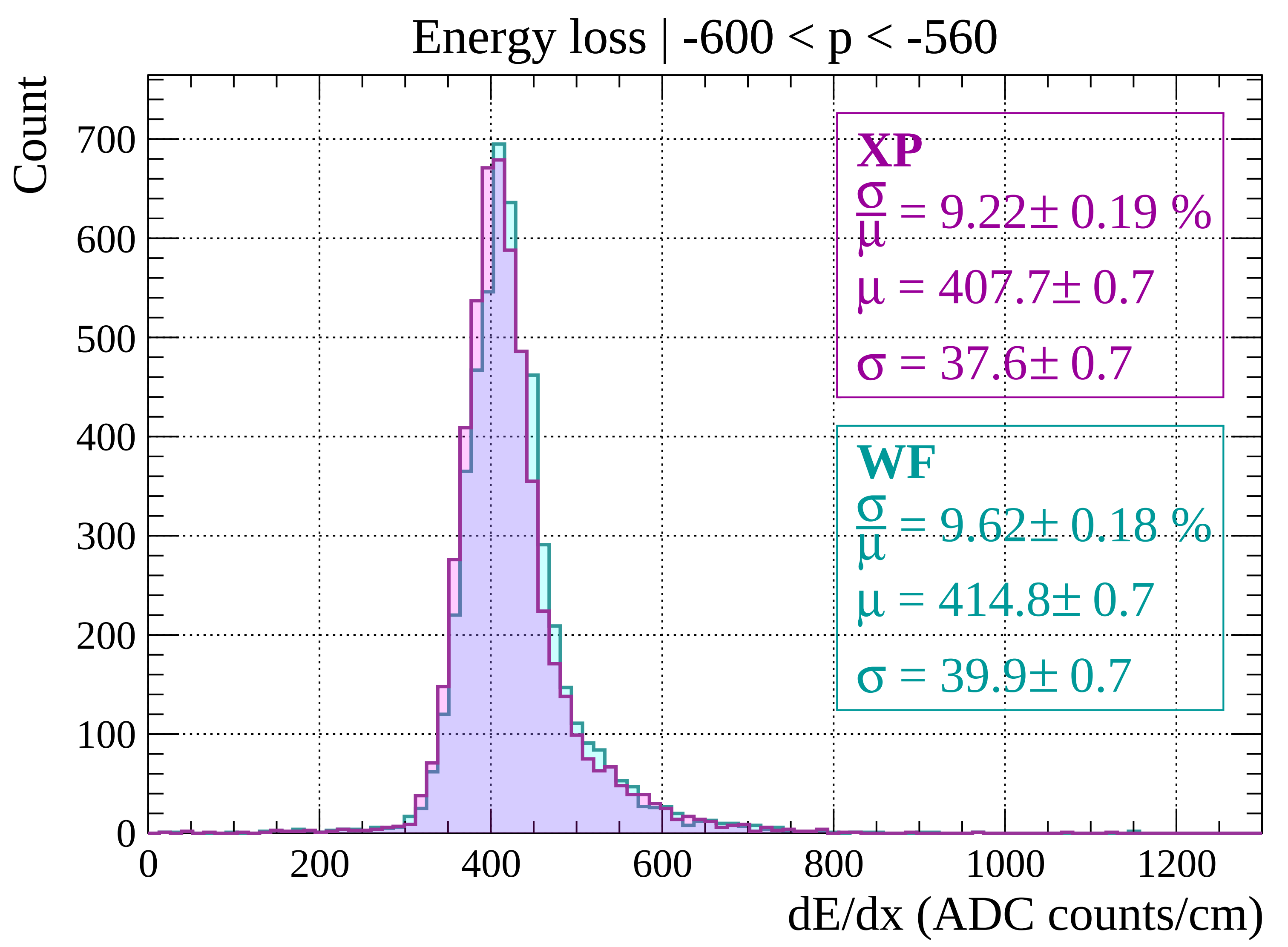

Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.37± 0.18 %
μ = 419.9± 0.7
σ = 39.4± 0.7
WF
σ/μ = 9.83± 0.19 %
μ = 426.7± 0.7
σ = 42.0± 0.7



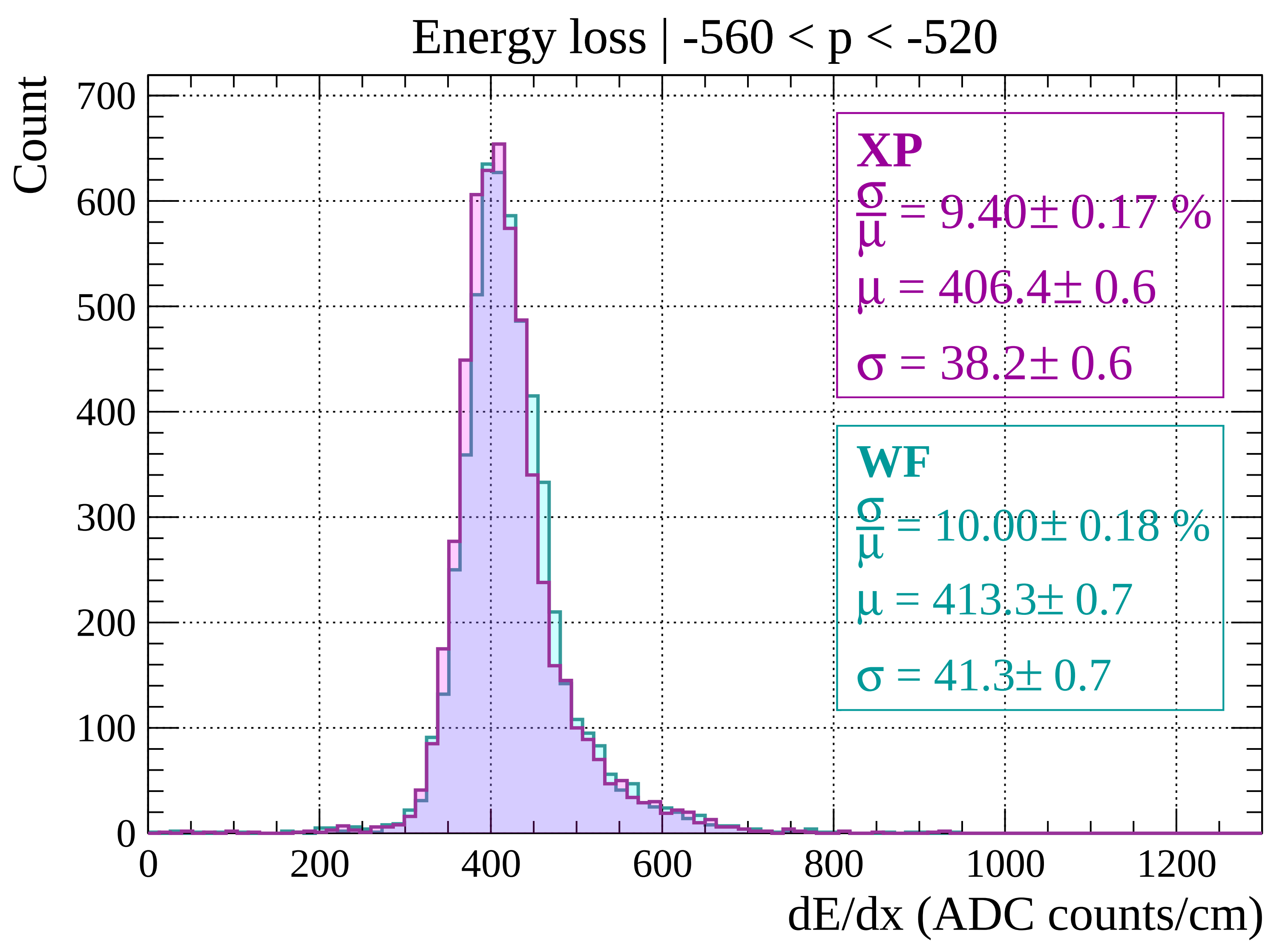
Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.37 \pm 0.18$ %
$\mu = 416.2 \pm 0.7$
$\sigma = 39.0 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 9.33 \pm 0.18$ %
$\mu = 421.5 \pm 0.7$
$\sigma = 39.3 \pm 0.7$


Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.59± 0.19 %
μ = 414.3± 0.7
σ = 39.7± 0.7
WF
σ/μ = 9.66± 0.19 %
μ = 420.4± 0.7
σ = 40.6± 0.7
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

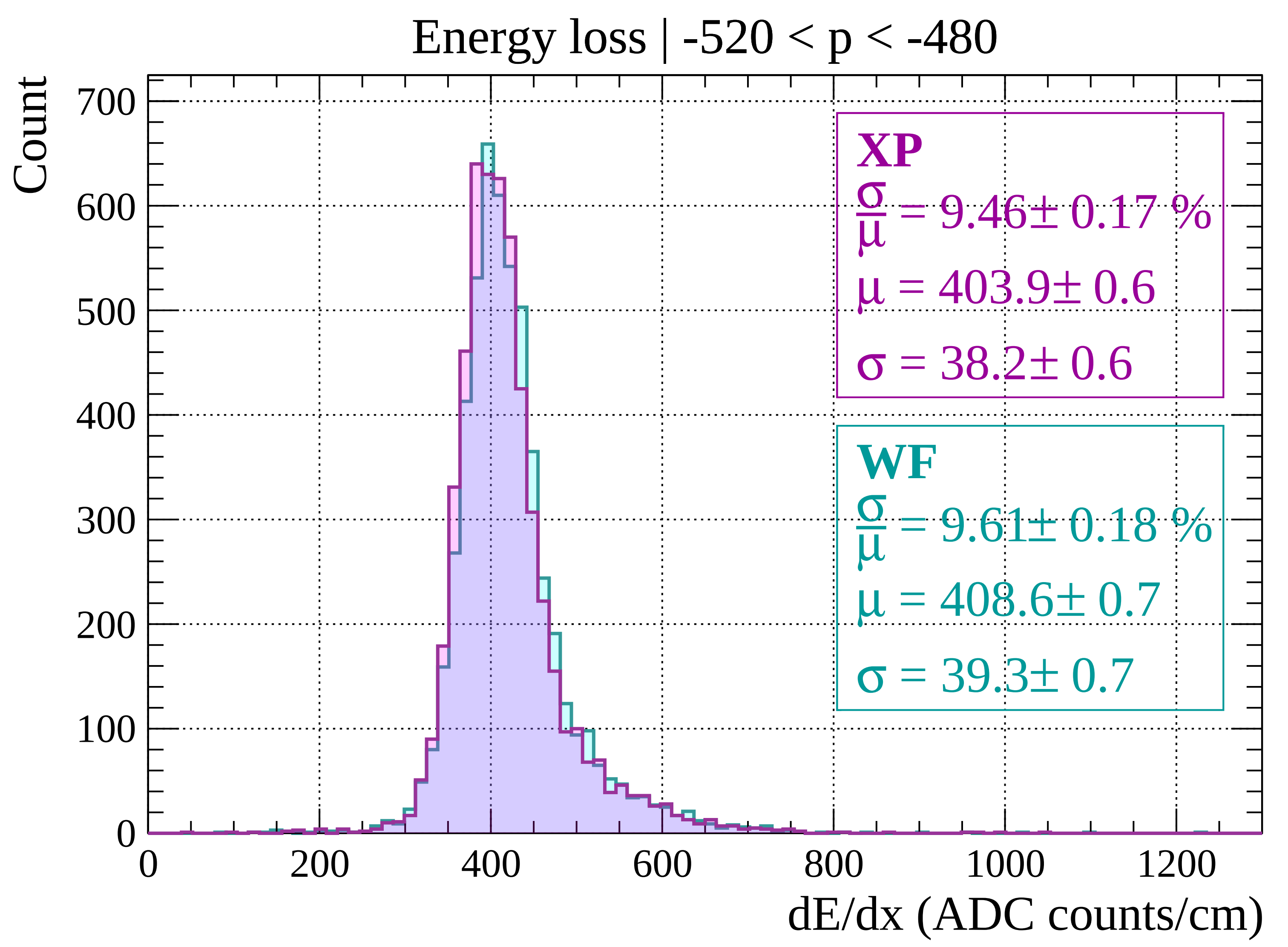

Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.83± 0.19 %
μ = 412.3± 0.7
σ = 40.5± 0.7
WF
σ/μ = 9.79± 0.20 %
μ = 417.9± 0.7
σ = 40.9± 0.7



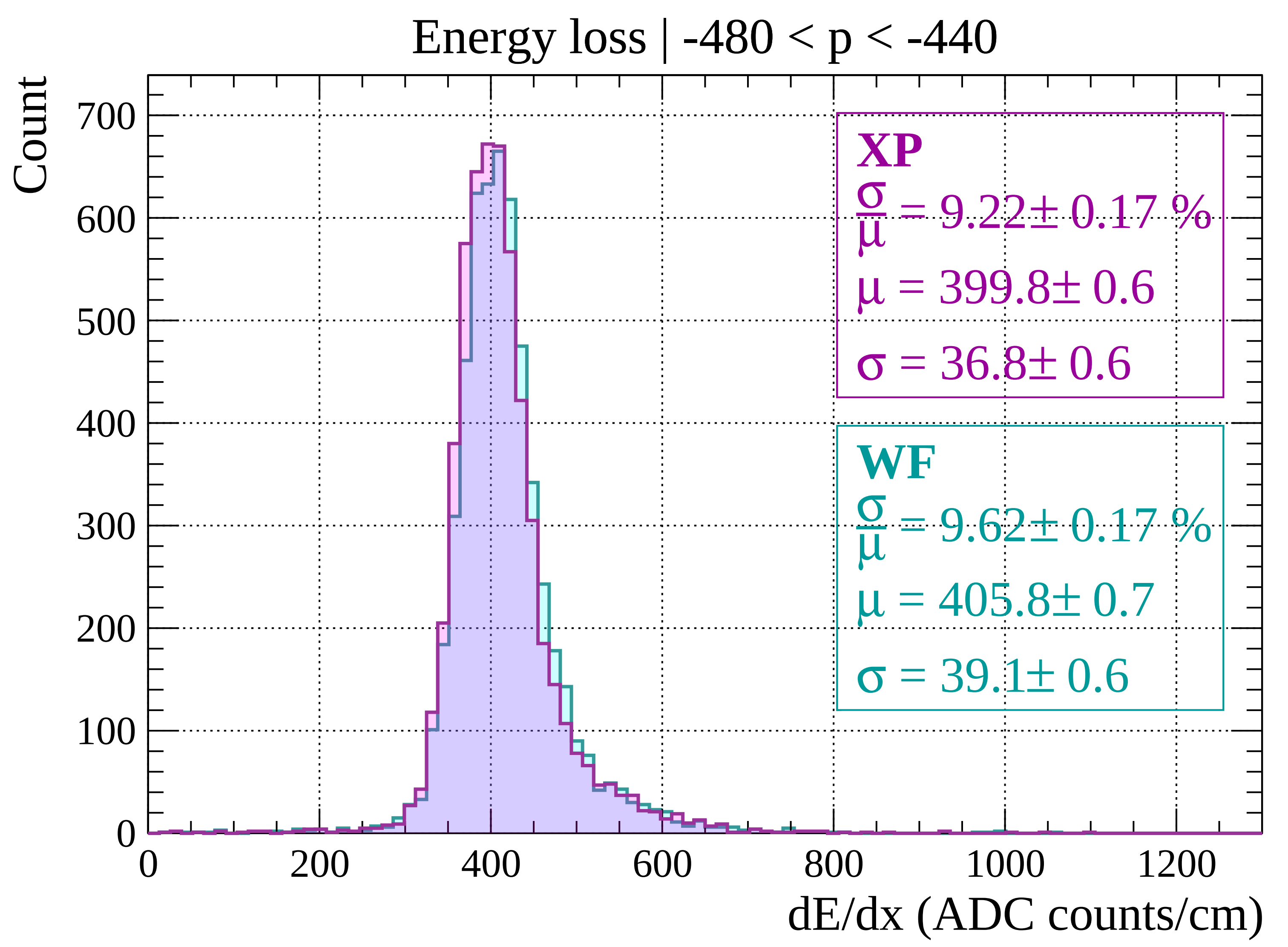
Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.22± 0.19 %
μ = 407.7± 0.7
σ = 37.6± 0.7
WF
σ/μ = 9.62± 0.18 %
μ = 414.8± 0.7
σ = 39.9± 0.7



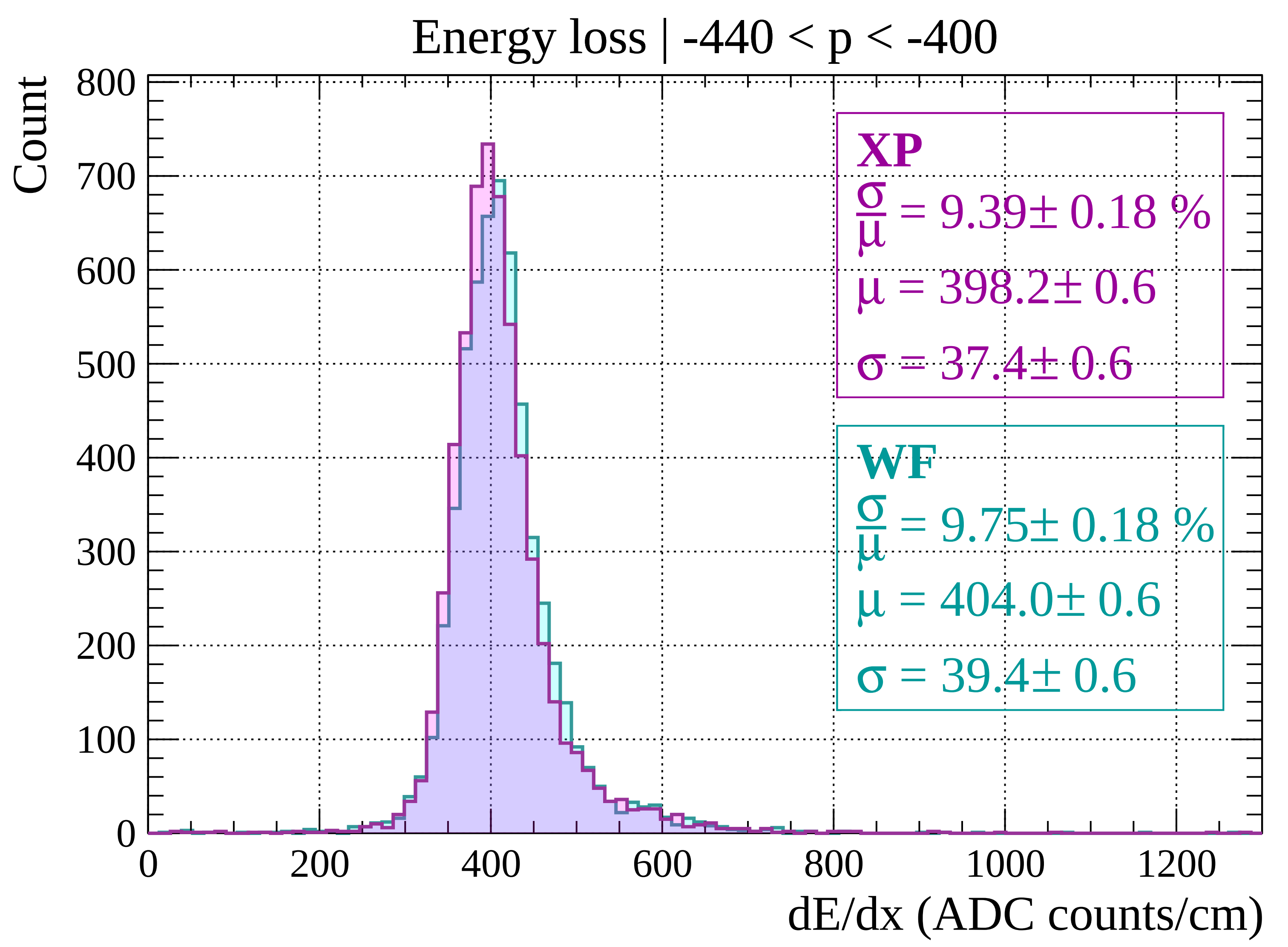
Energy loss | -560 < p < -520
Count
700
600
500
400
300
200
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.40± 0.17 %
μ = 406.4± 0.6
σ = 38.2± 0.6
WF
$\frac{\sigma}{\mu}$ = 10.00± 0.18 %
μ = 413.3± 0.7
σ = 41.3± 0.7



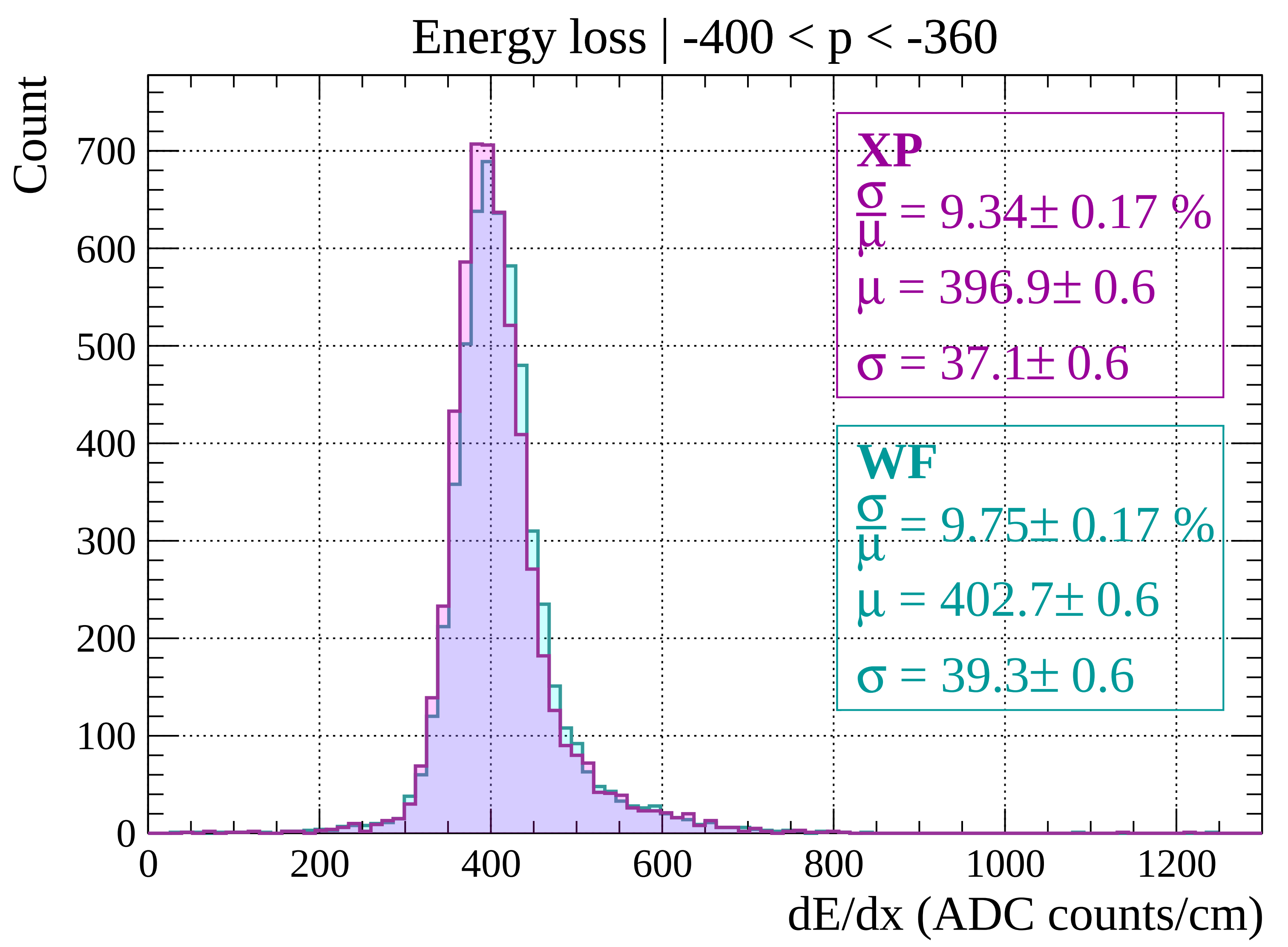
Energy loss | -520 < p < -480
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.46± 0.17 %
μ = 403.9± 0.6
σ = 38.2± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.61± 0.18 %
μ = 408.6± 0.7
σ = 39.3± 0.7



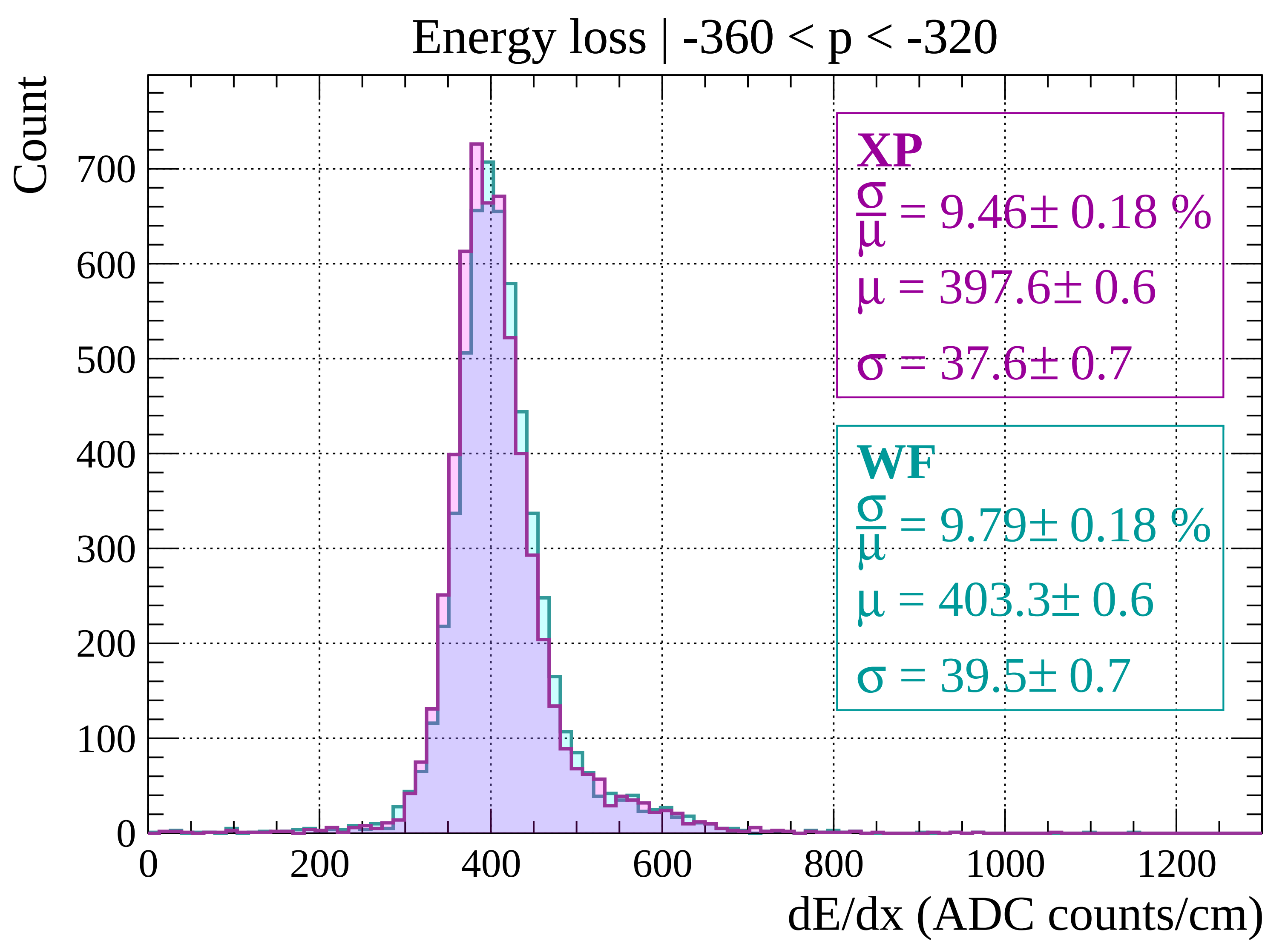
Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.22± 0.17 %
μ = 399.8± 0.6
σ = 36.8± 0.6
WF
σ/μ = 9.62± 0.17 %
μ = 405.8± 0.7
σ = 39.1± 0.6
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200



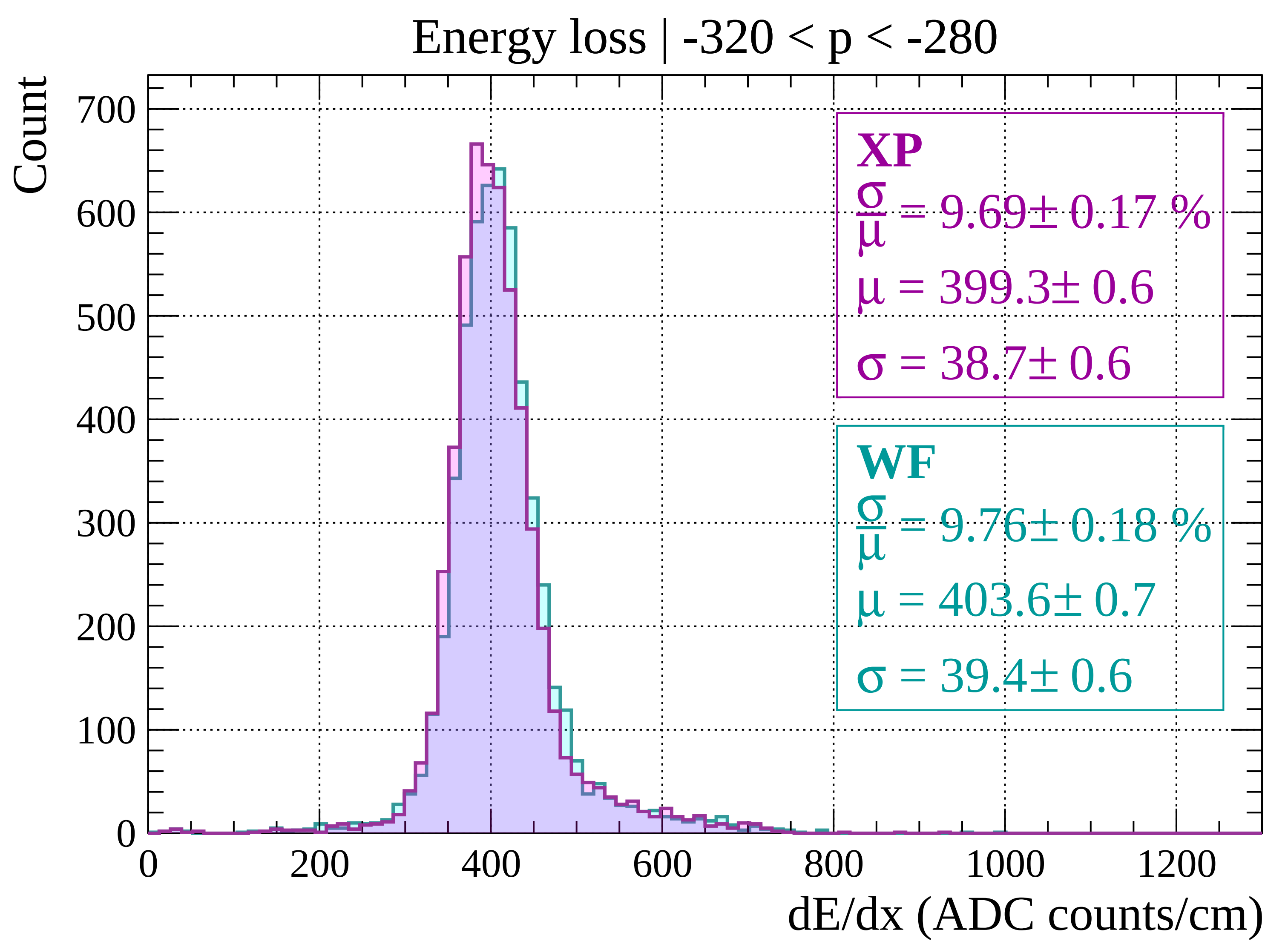
Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.39± 0.18 %
μ = 398.2± 0.6
σ = 37.4± 0.6
WF
σ/μ = 9.75± 0.18 %
μ = 404.0± 0.6
σ = 39.4± 0.6

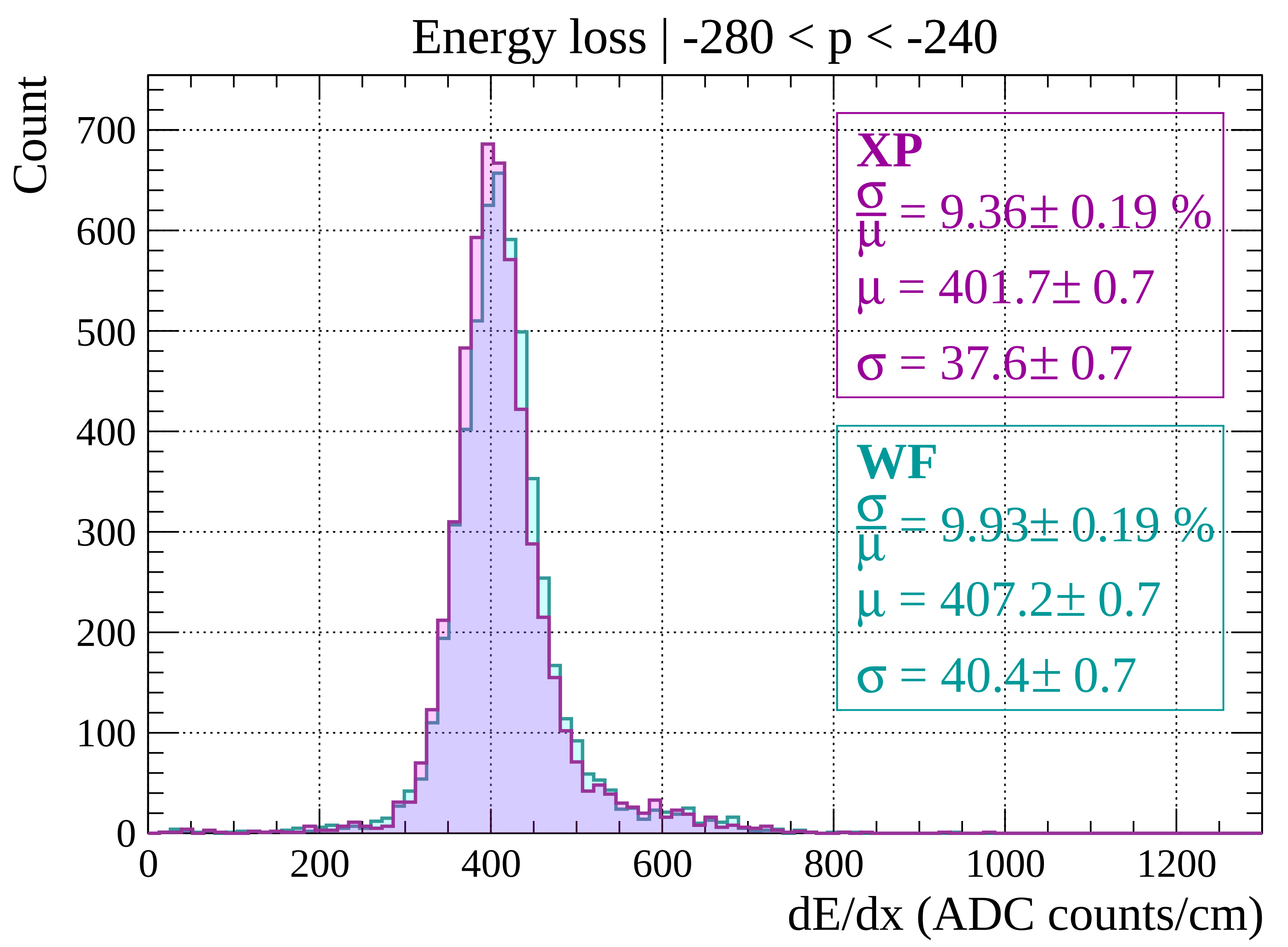


Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.34± 0.17 %
μ = 396.9± 0.6
σ = 37.1± 0.6
WF
σ/μ = 9.75± 0.17 %
μ = 402.7± 0.6
σ = 39.3± 0.6

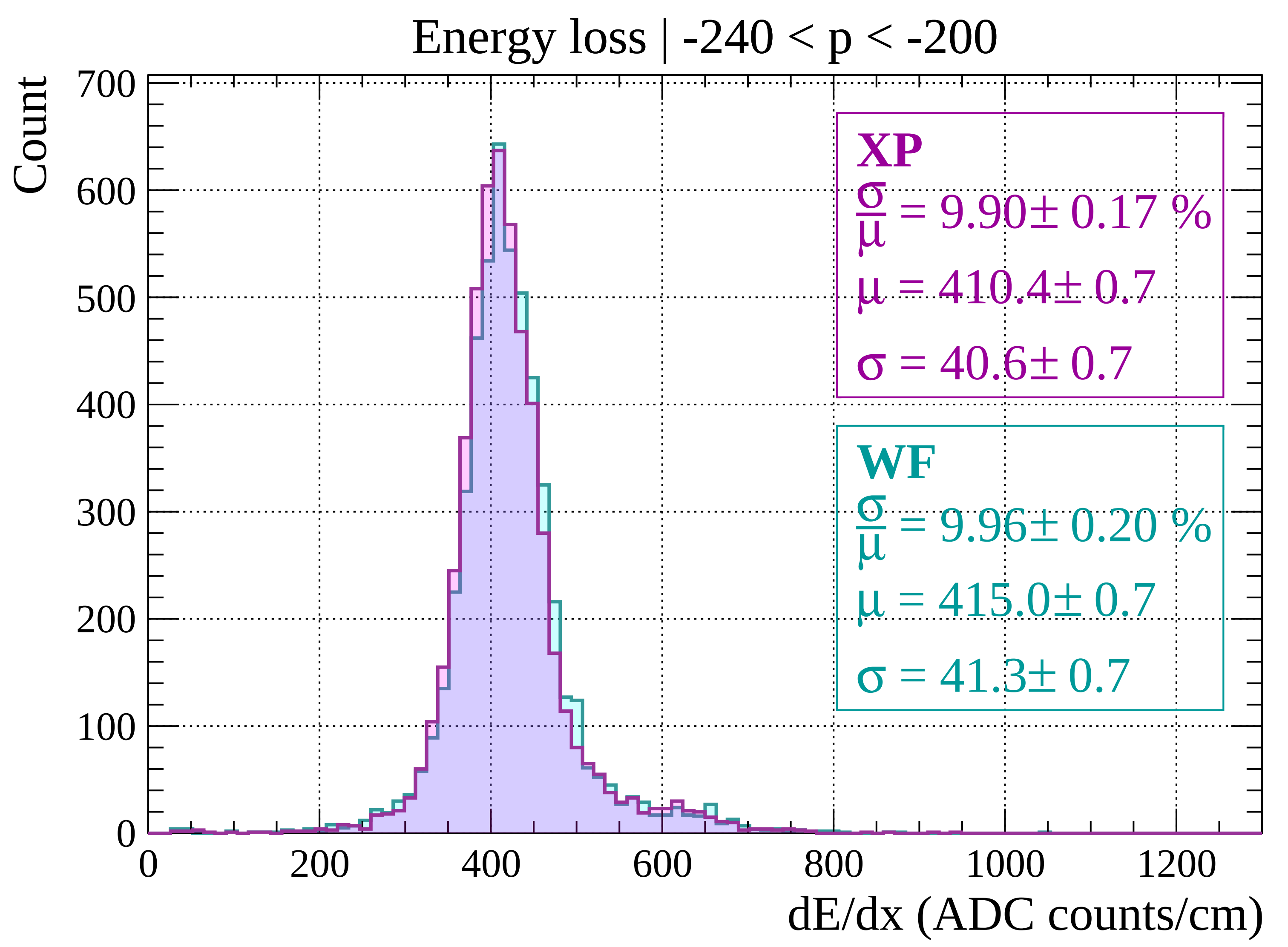


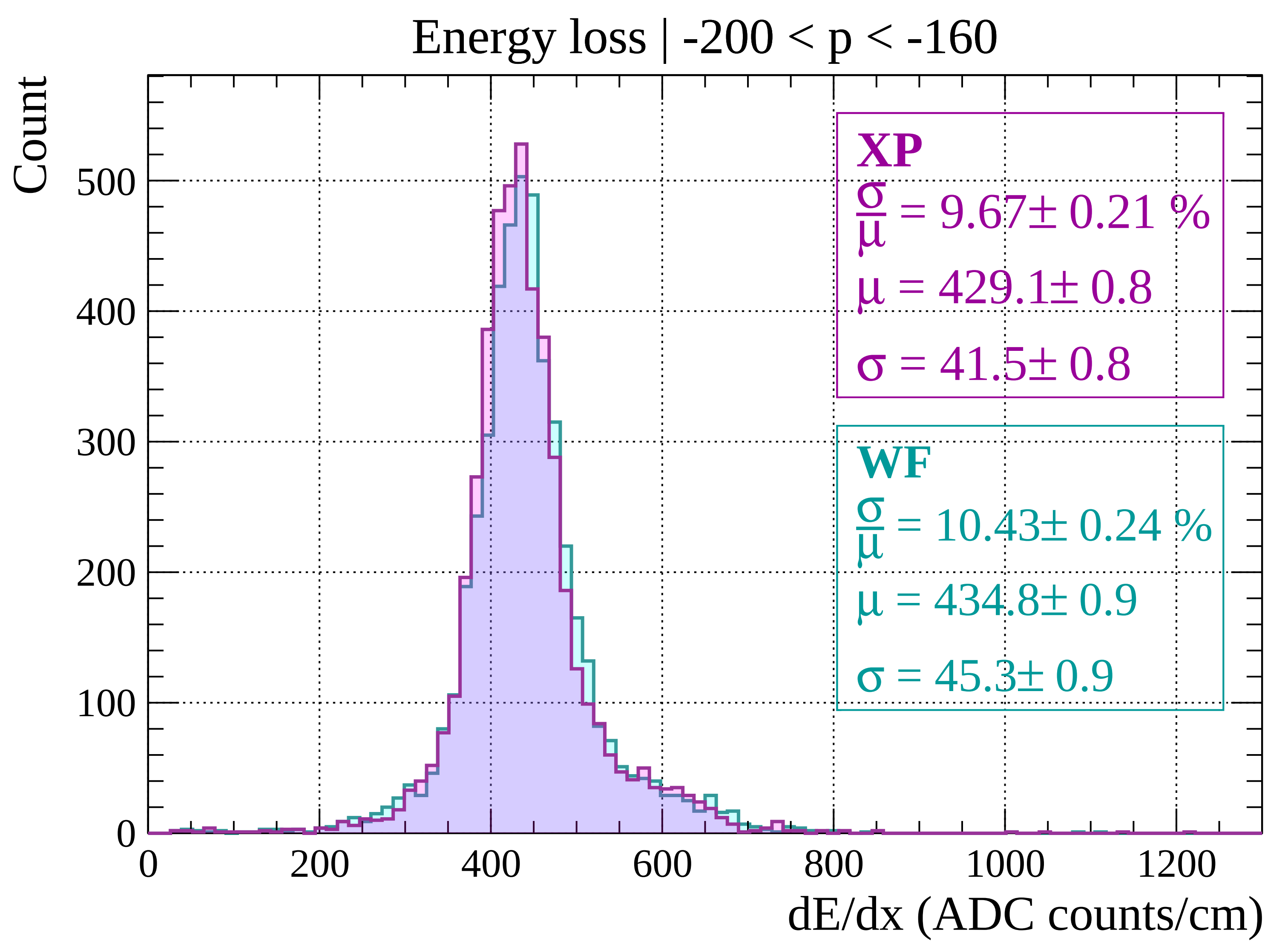
Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.46± 0.18 %
μ = 397.6± 0.6
σ = 37.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.79± 0.18 %
μ = 403.3± 0.6
σ = 39.5± 0.7


Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.69± 0.17 %
μ = 399.3± 0.6
σ = 38.7± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.76± 0.18 %
μ = 403.6± 0.7
σ = 39.4± 0.6

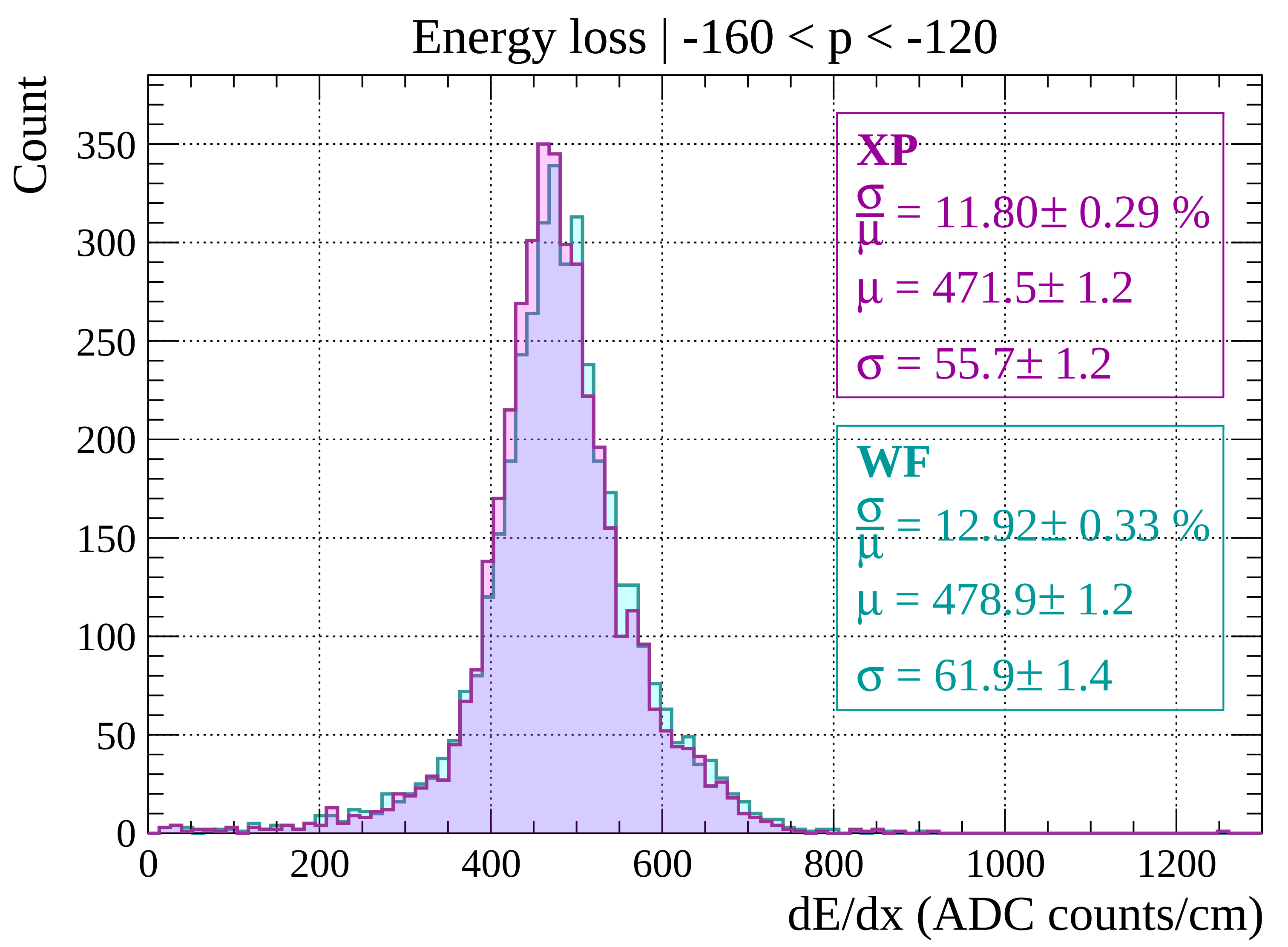


Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.36± 0.19 %
μ = 401.7± 0.7
σ = 37.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.93± 0.19 %
μ = 407.2± 0.7
σ = 40.4± 0.7

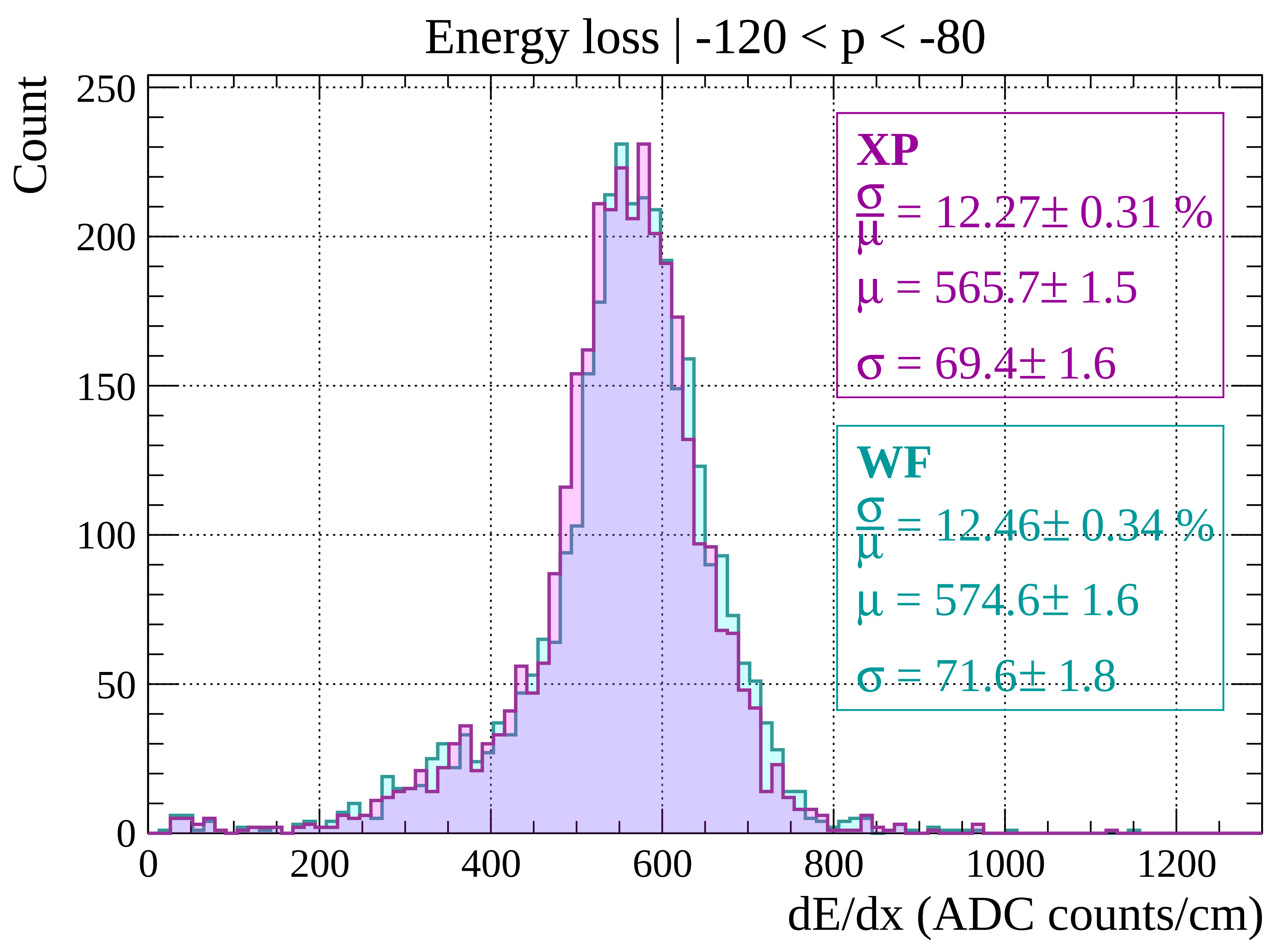

Energy loss | -240 < p < -200
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.90 \pm 0.17$ %
$\mu = 410.4 \pm 0.7$
$\sigma = 40.6 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 9.96 \pm 0.20$ %
$\mu = 415.0 \pm 0.7$
$\sigma = 41.3 \pm 0.7$

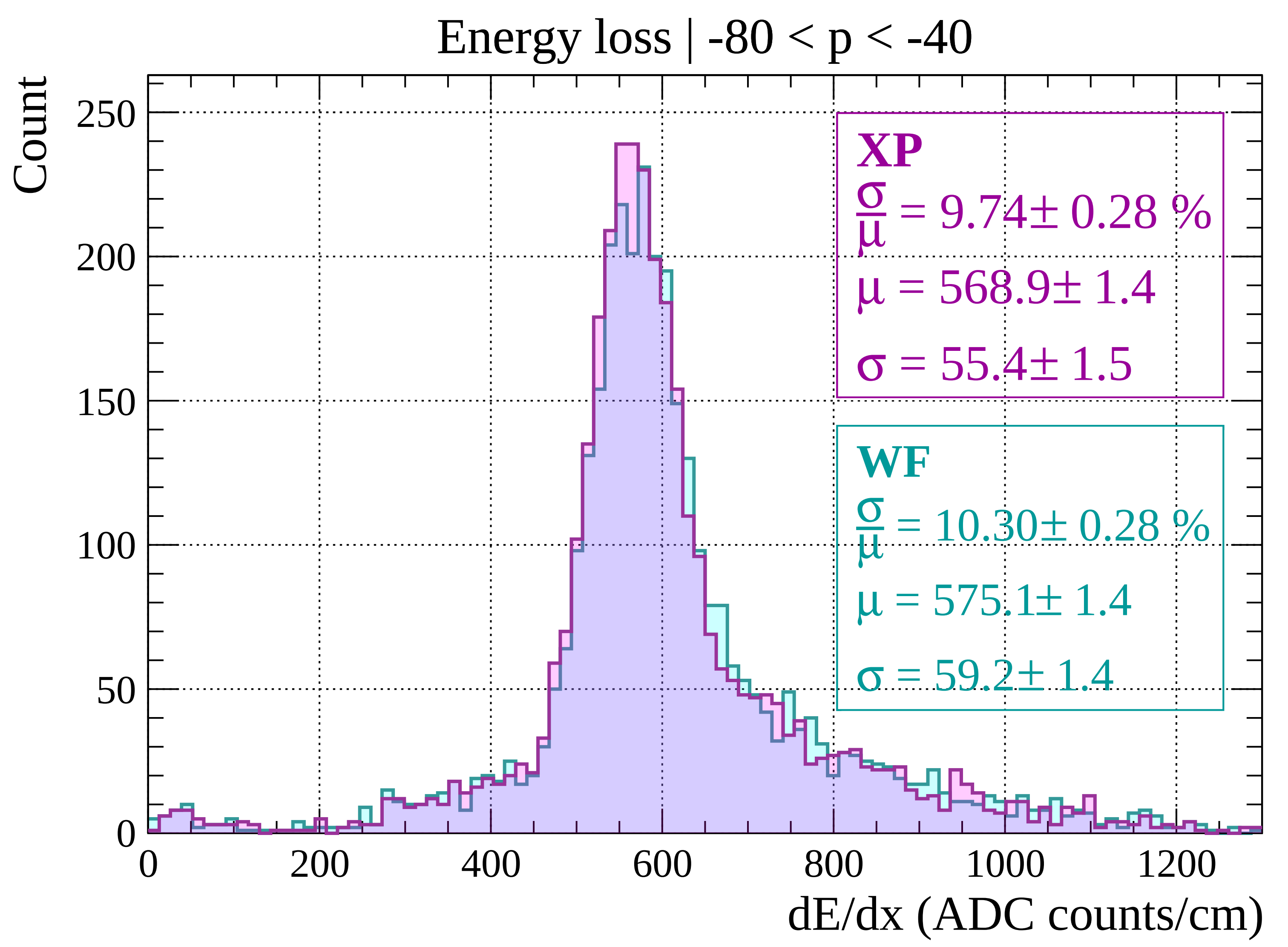


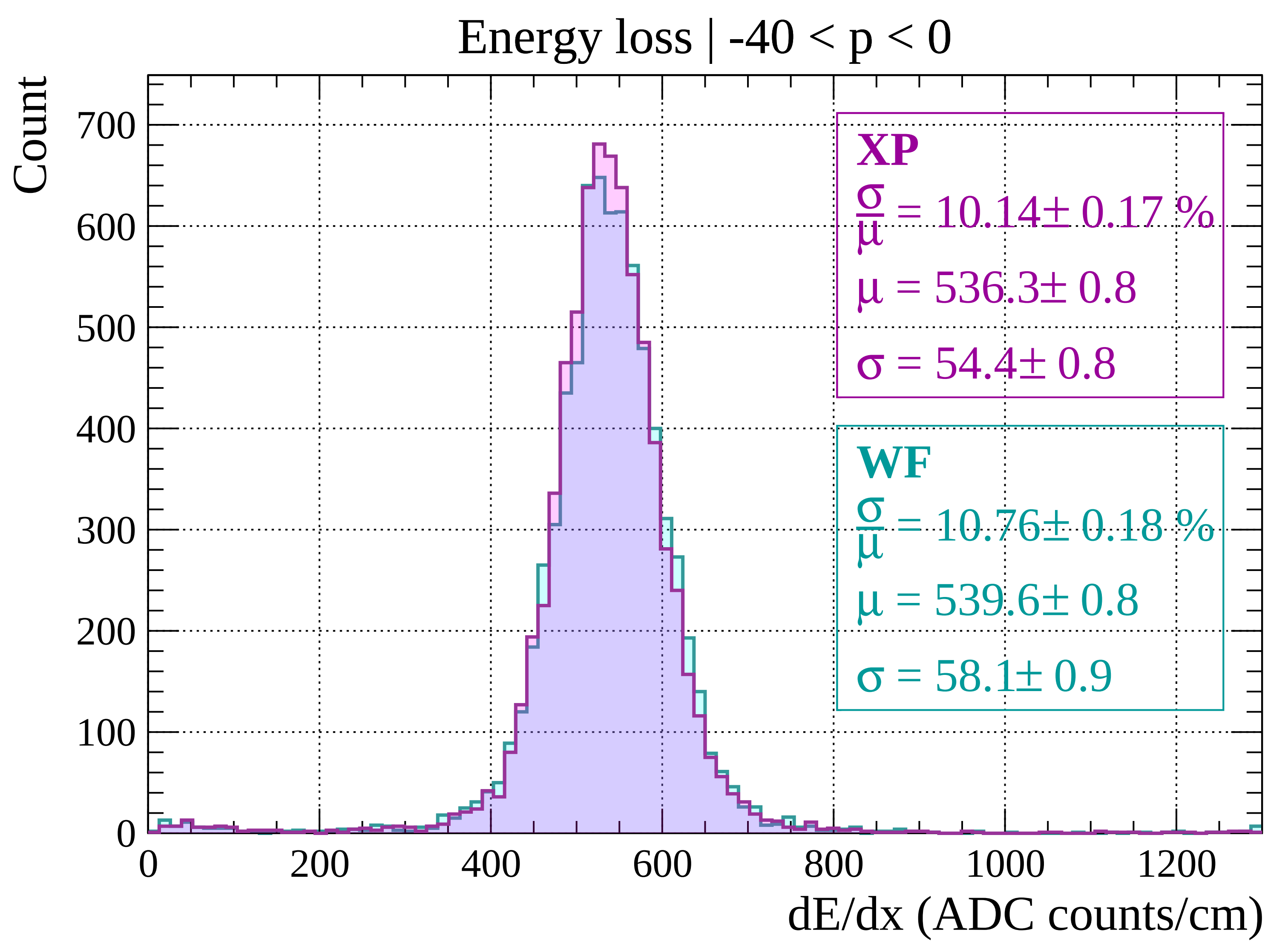
Energy loss | -200 < p < -160
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.67± 0.21 %
μ = 429.1± 0.8
σ = 41.5± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.43± 0.24 %
μ = 434.8± 0.9
σ = 45.3± 0.9



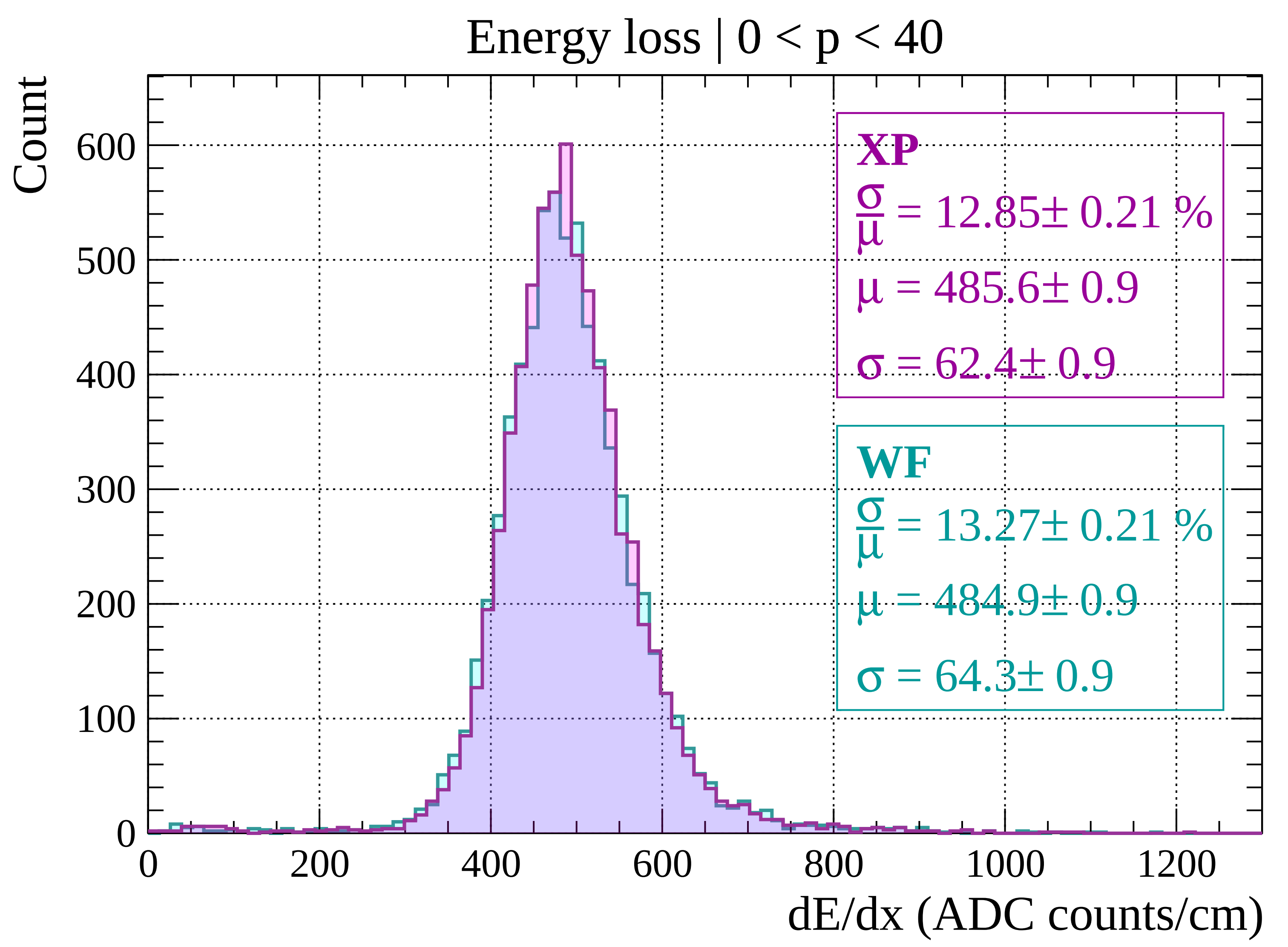
Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
σ/μ = 11.80± 0.29 %
μ = 471.5± 1.2
σ = 55.7± 1.2
WF
σ/μ = 12.92± 0.33 %
μ = 478.9± 1.2
σ = 61.9± 1.4


Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.27± 0.31 %
μ = 565.7± 1.5
σ = 69.4± 1.6
WF
σ/μ = 12.46± 0.34 %
μ = 574.6± 1.6
σ = 71.6± 1.8
0
50
100
150
200
250
0
200
400
600
800
1000
1200

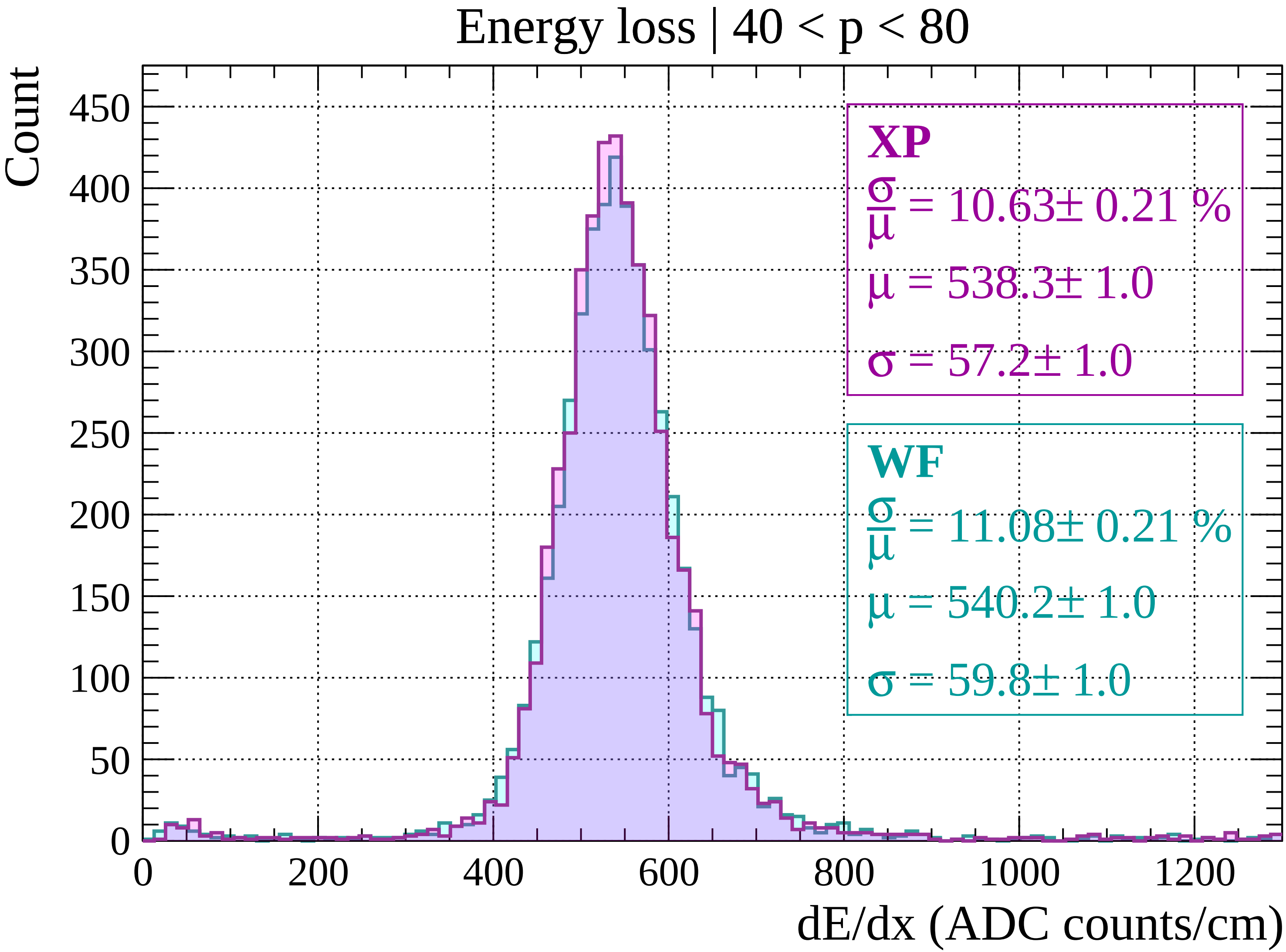

Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP
σ/μ = 9.74± 0.28 %
μ = 568.9± 1.4
σ = 55.4± 1.5
WF
σ/μ = 10.30± 0.28 %
μ = 575.1± 1.4
σ = 59.2± 1.4



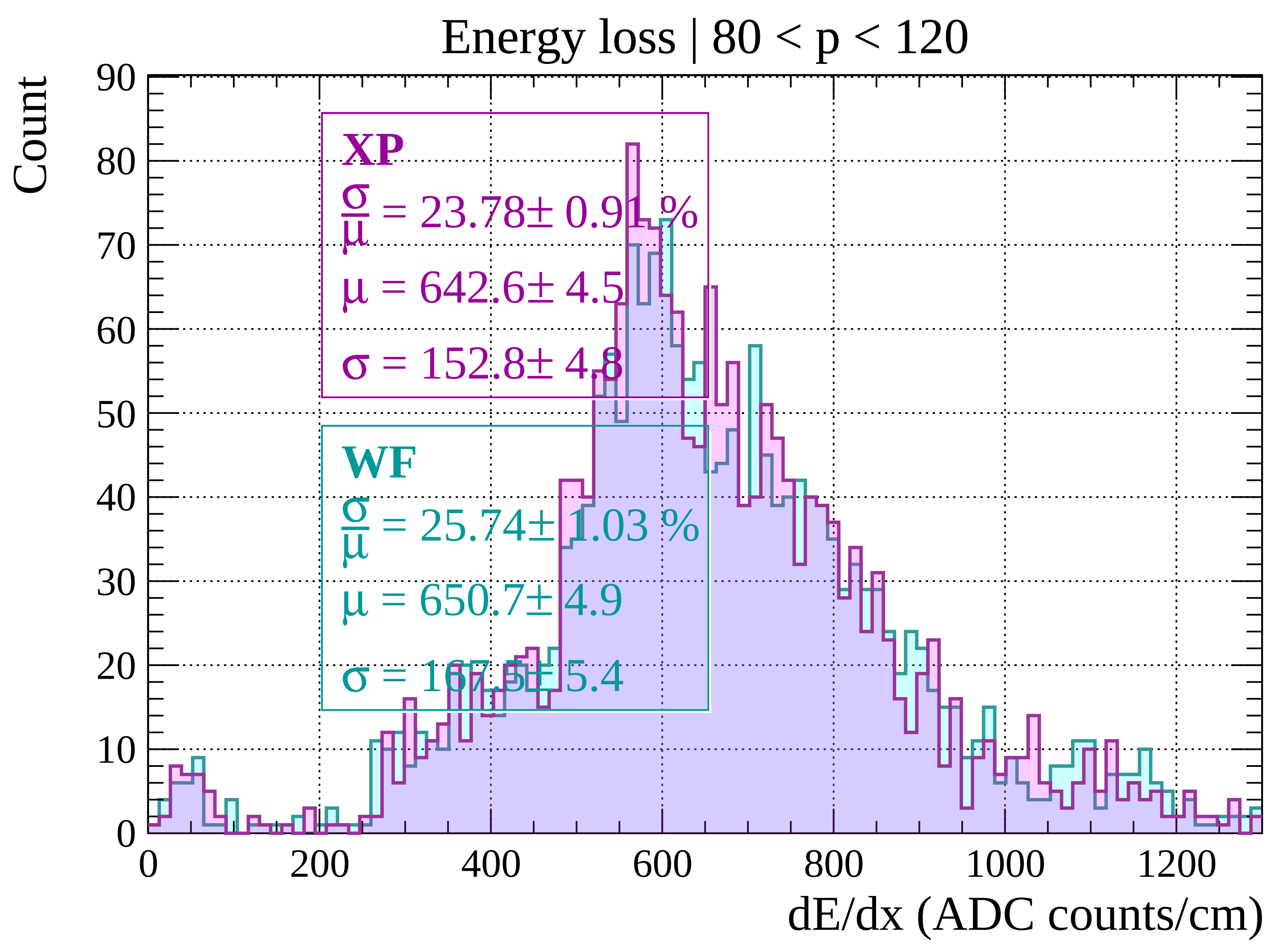
Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.14± 0.17 %
μ = 536.3± 0.8
σ = 54.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.76± 0.18 %
μ = 539.6± 0.8
σ = 58.1± 0.9



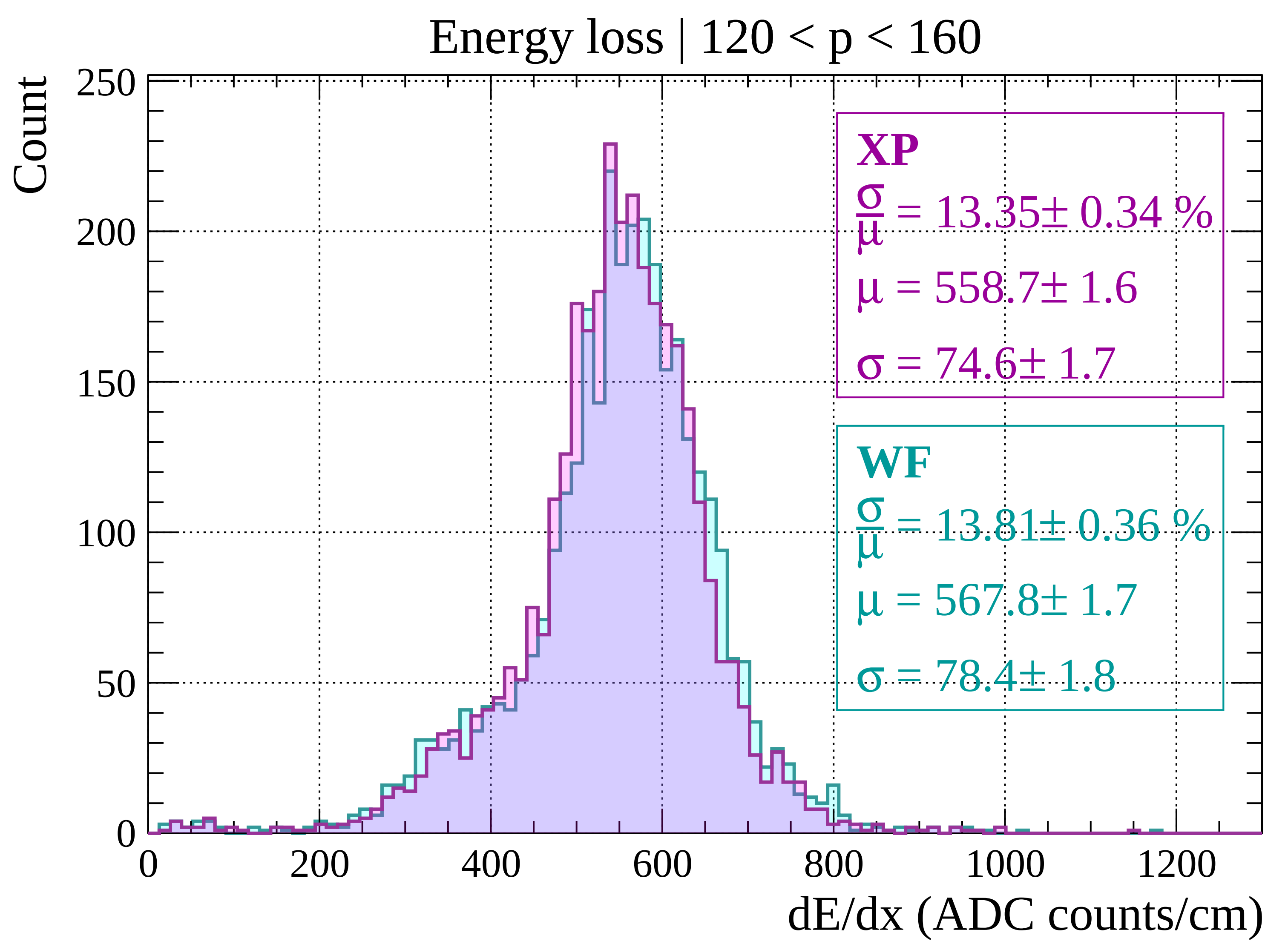
Energy loss | 0 < p < 40
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 12.85± 0.21 %
μ = 485.6± 0.9
σ = 62.4± 0.9
WF
$\frac{\sigma}{\mu}$ = 13.27± 0.21 %
μ = 484.9± 0.9
σ = 64.3± 0.9



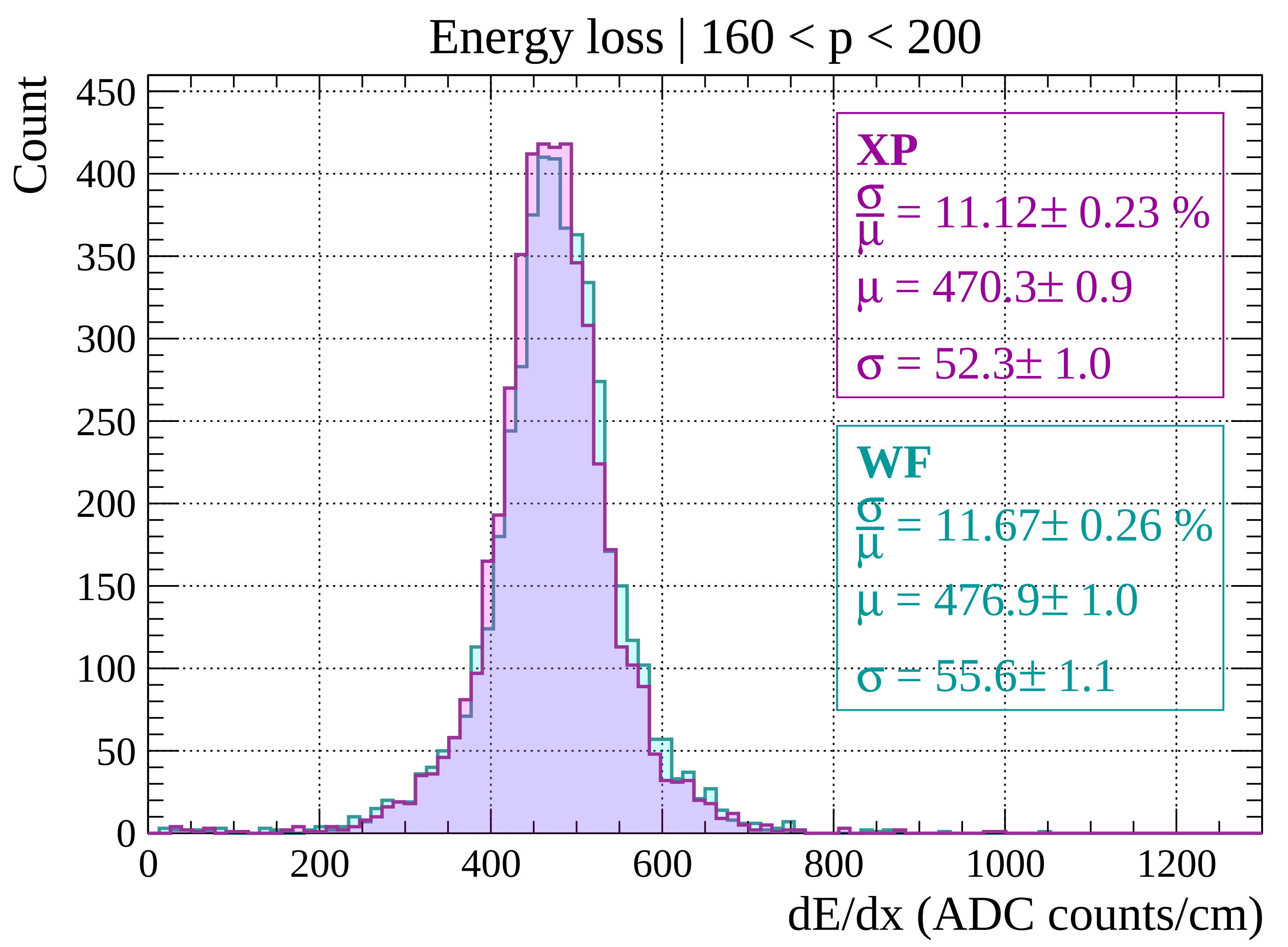
Energy loss | 40 < p < 80
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.63± 0.21 %
μ = 538.3± 1.0
σ = 57.2± 1.0
WF
$\frac{\sigma}{\mu}$ = 11.08± 0.21 %
μ = 540.2± 1.0
σ = 59.8± 1.0

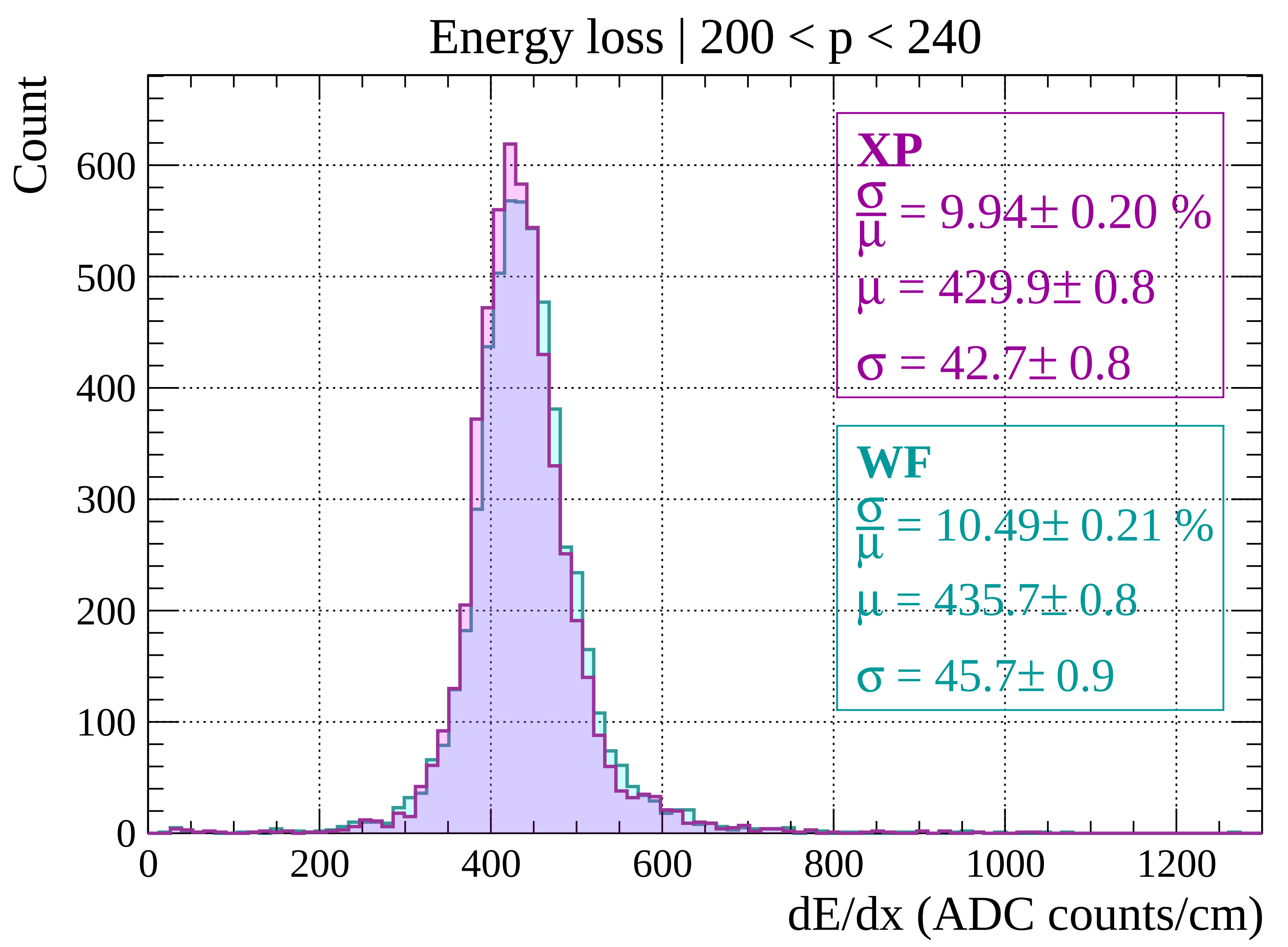


Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 23.78± 0.91 %
μ = 642.6± 4.5
σ = 152.8± 4.8
WF
σ/μ = 25.74± 1.03 %
μ = 650.7± 4.9
σ = 167.5± 5.4
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200

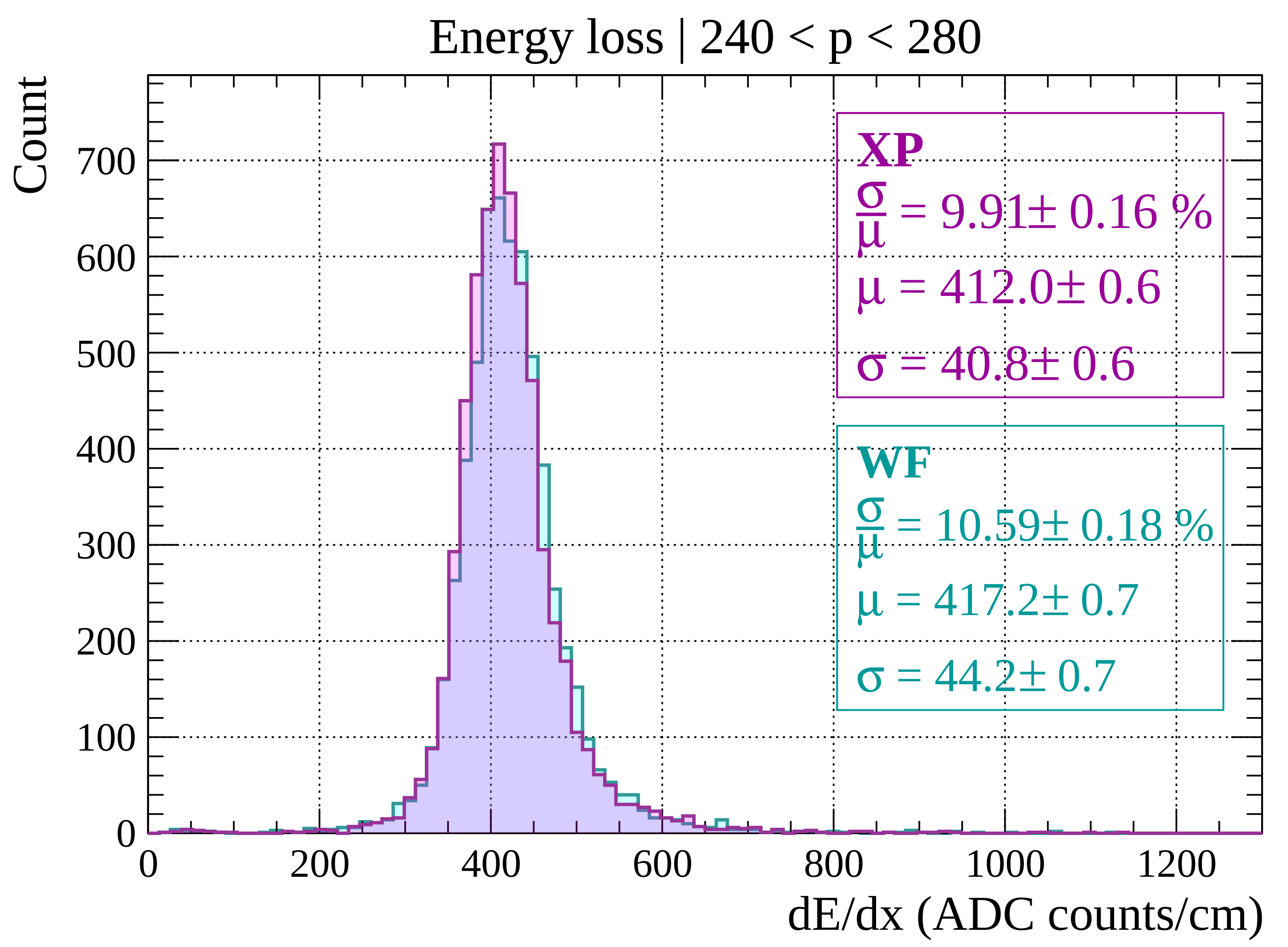


Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.35± 0.34 %
μ = 558.7± 1.6
σ = 74.6± 1.7
WF
$\frac{\sigma}{\mu}$ = 13.81± 0.36 %
μ = 567.8± 1.7
σ = 78.4± 1.8

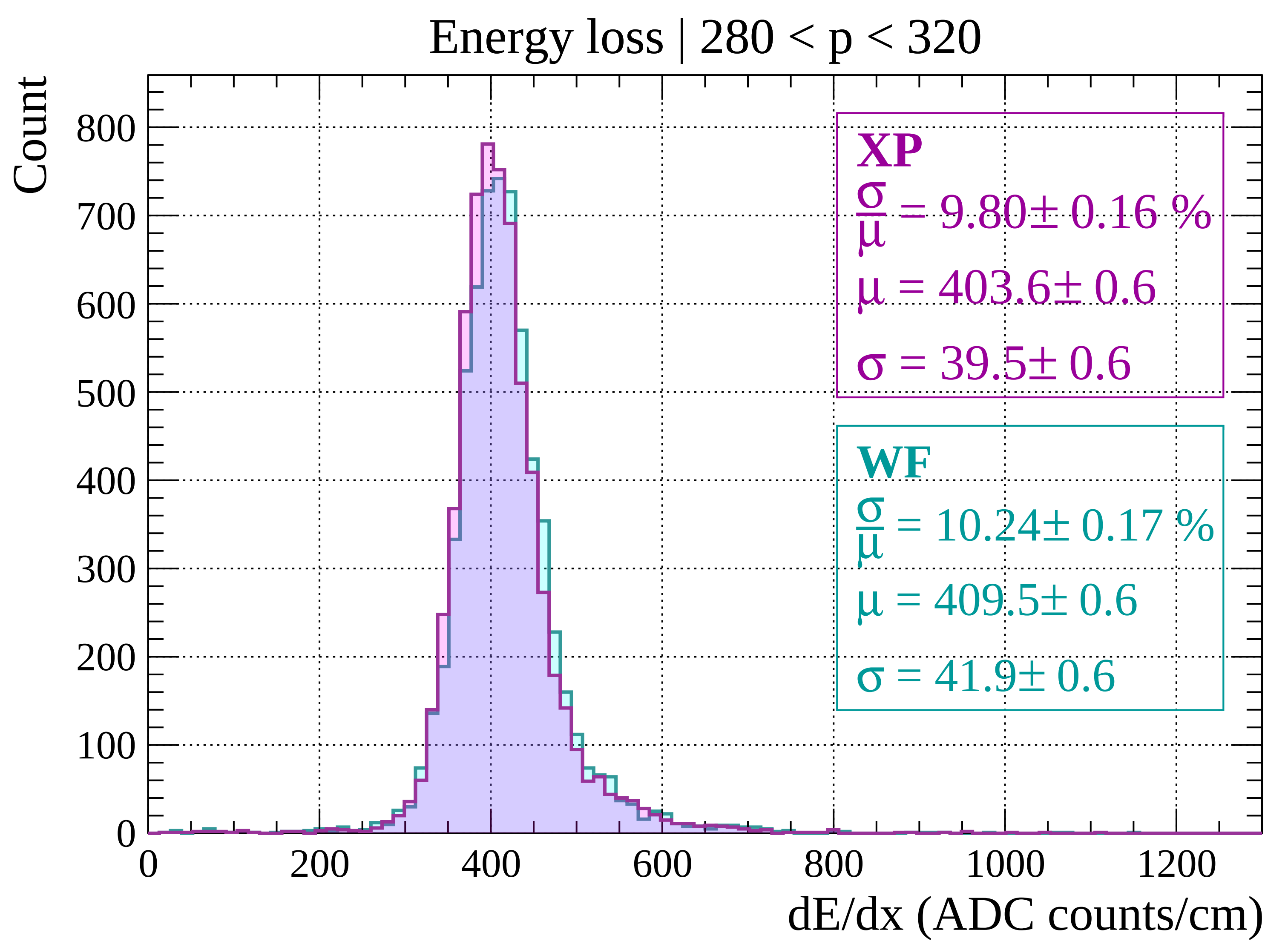

Energy loss | 160 < p < 200
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.12± 0.23 %
μ = 470.3± 0.9
σ = 52.3± 1.0
WF
σ/μ = 11.67± 0.26 %
μ = 476.9± 1.0
σ = 55.6± 1.1



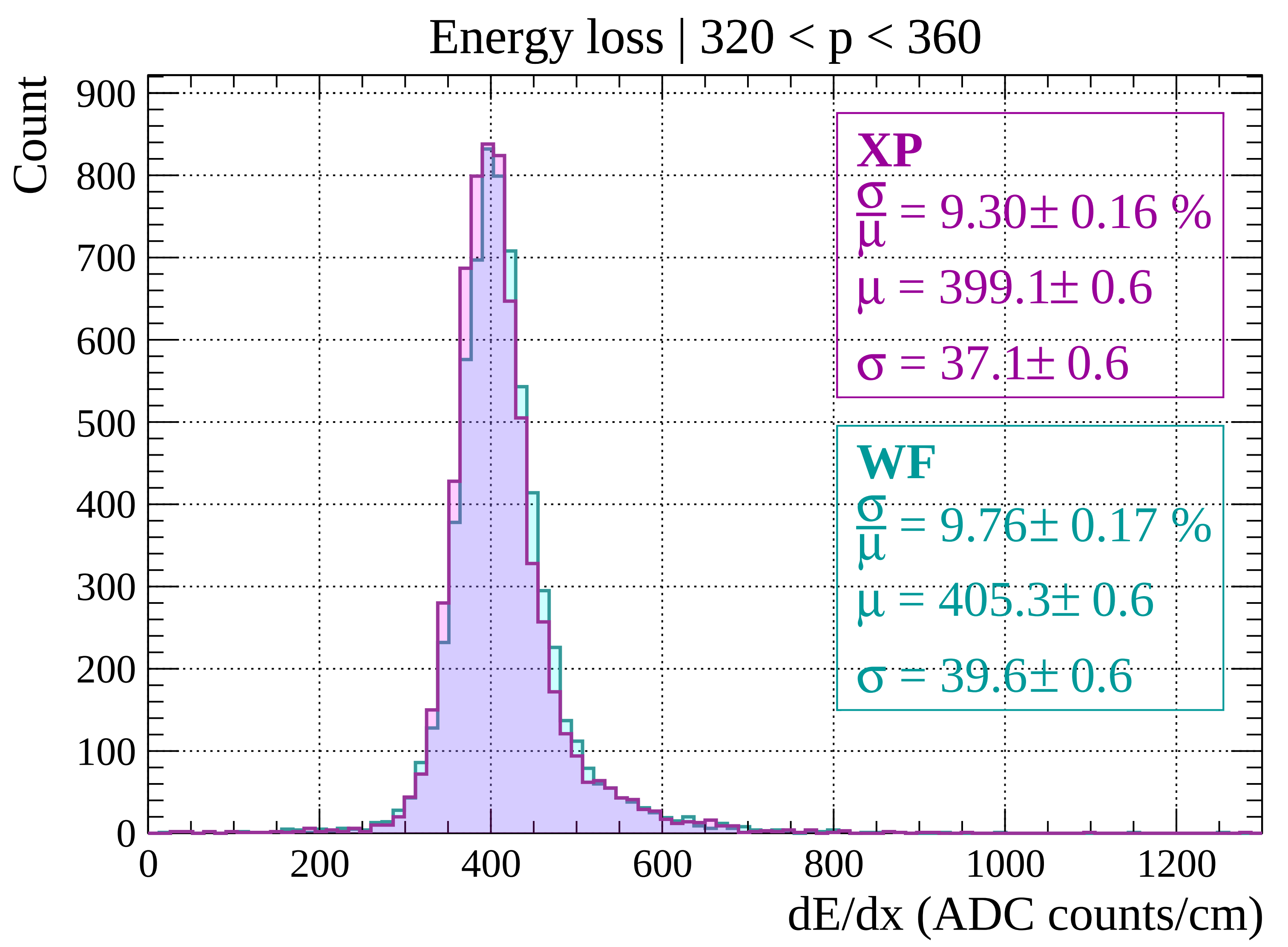
Energy loss | 200 < p < 240
Count
XP
σ/μ = 9.94± 0.20 %
μ = 429.9± 0.8
σ = 42.7± 0.8
WF
σ/μ = 10.49± 0.21 %
μ = 435.7± 0.8
σ = 45.7± 0.9
dE/dx (ADC counts/cm)



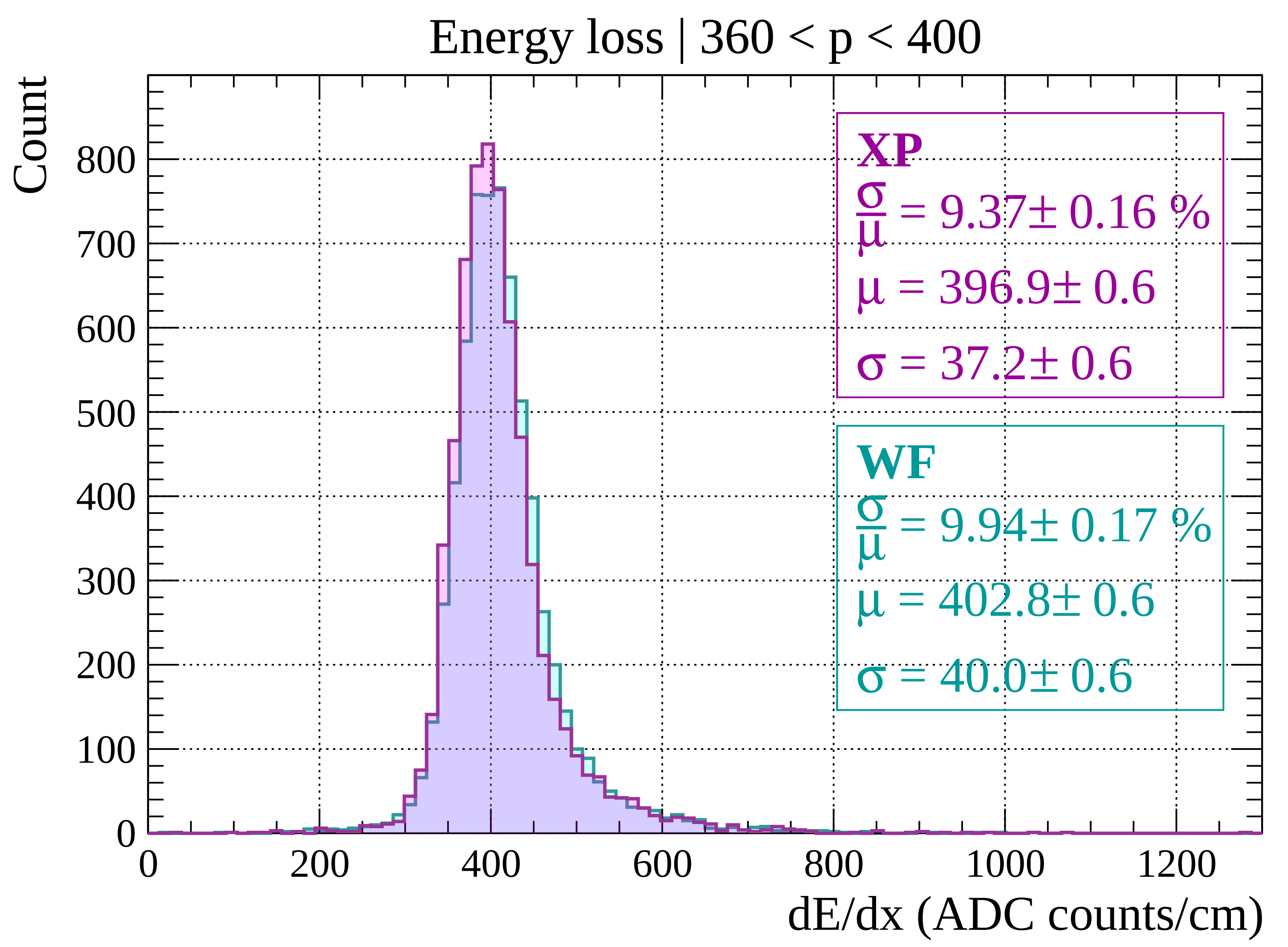
Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.91± 0.16 %
μ = 412.0± 0.6
σ = 40.8± 0.6
WF
$\frac{\sigma}{\mu}$ = 10.59± 0.18 %
μ = 417.2± 0.7
σ = 44.2± 0.7



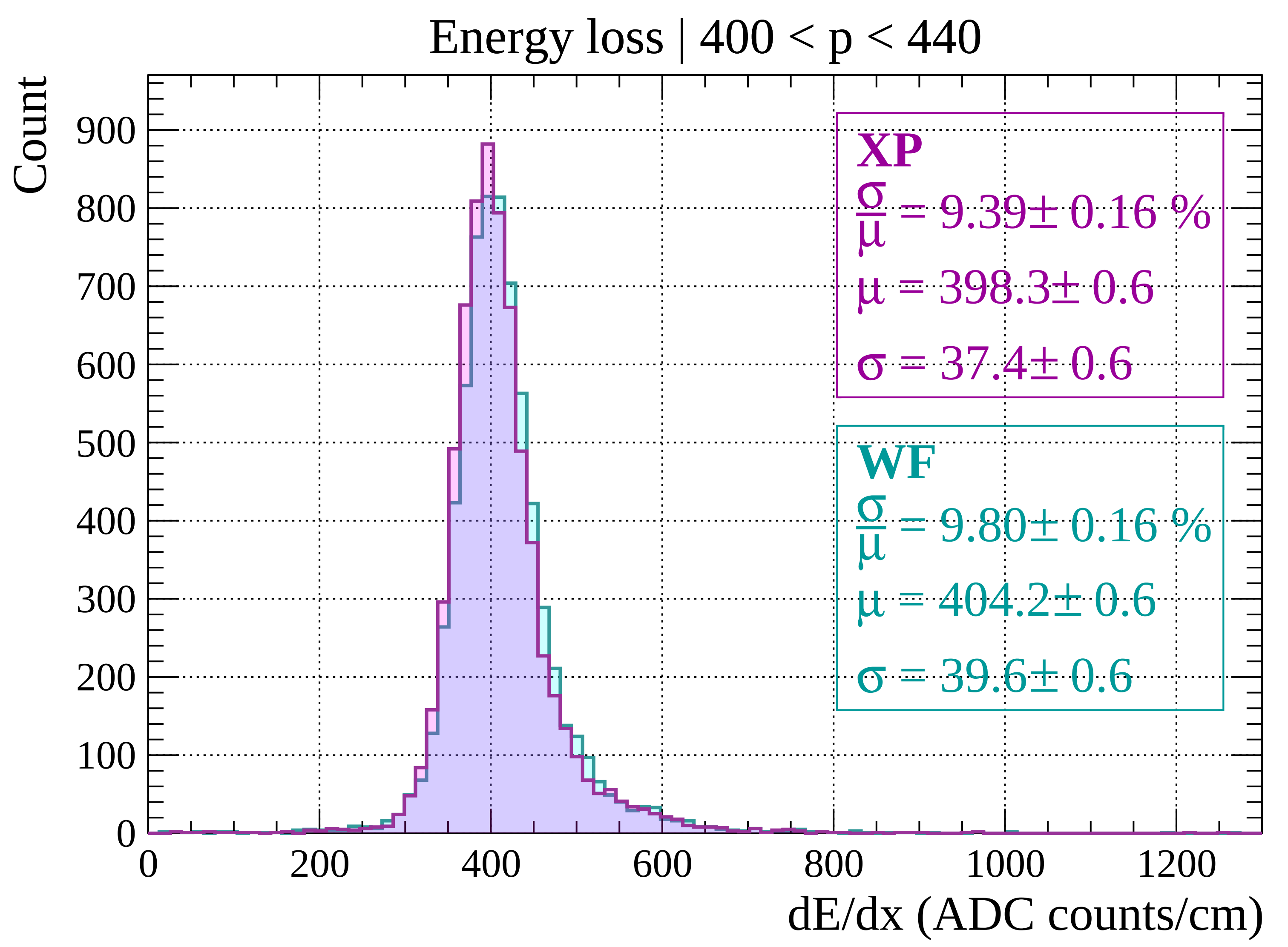
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 9.80± 0.16 %
μ = 403.6± 0.6
σ = 39.5± 0.6
WF
σ/μ = 10.24± 0.17 %
μ = 409.5± 0.6
σ = 41.9± 0.6



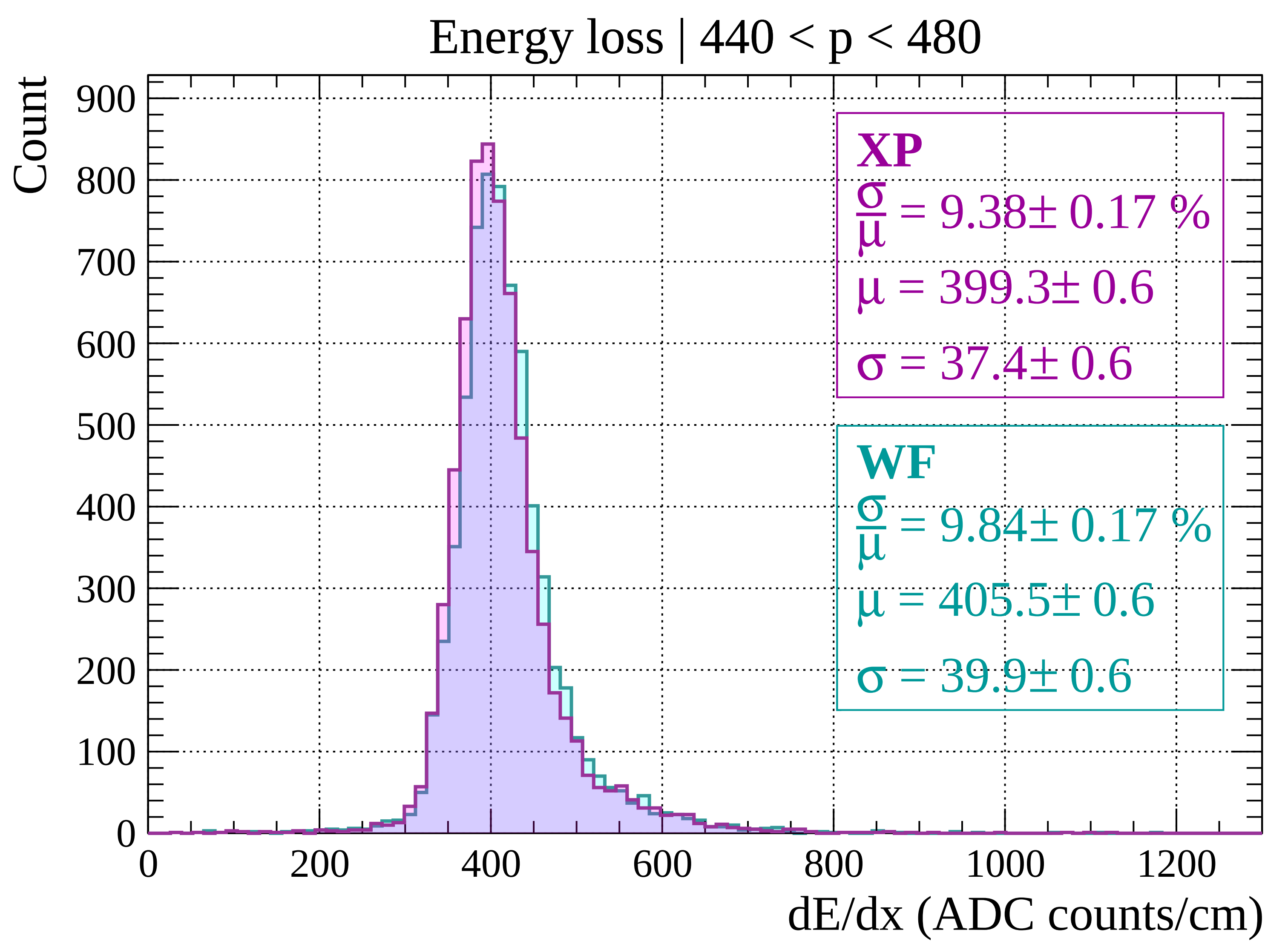
Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.30± 0.16 %
μ = 399.1± 0.6
σ = 37.1± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.76± 0.17 %
μ = 405.3± 0.6
σ = 39.6± 0.6



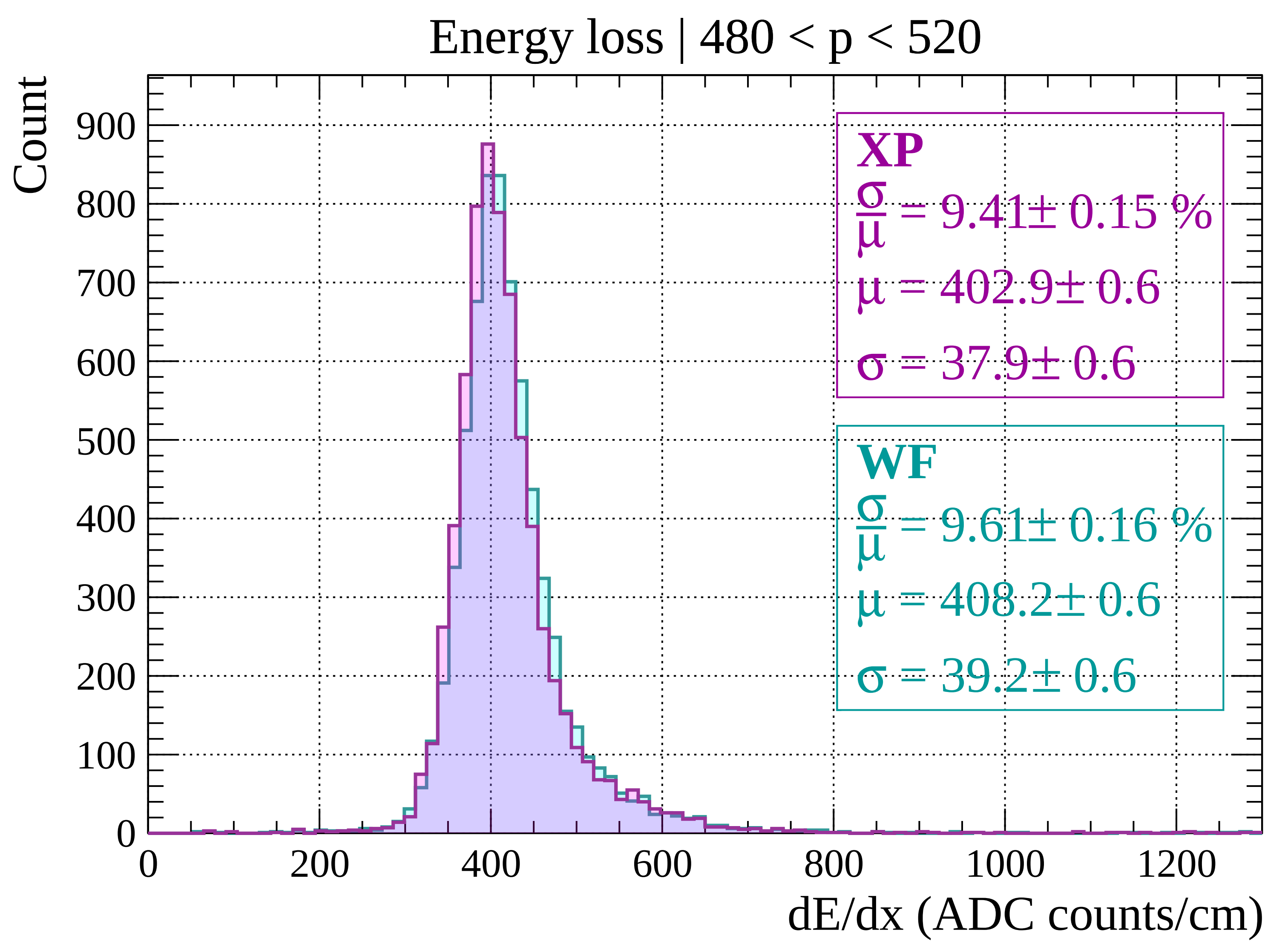
Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 9.37± 0.16 %
μ = 396.9± 0.6
σ = 37.2± 0.6
WF
σ/μ = 9.94± 0.17 %
μ = 402.8± 0.6
σ = 40.0± 0.6



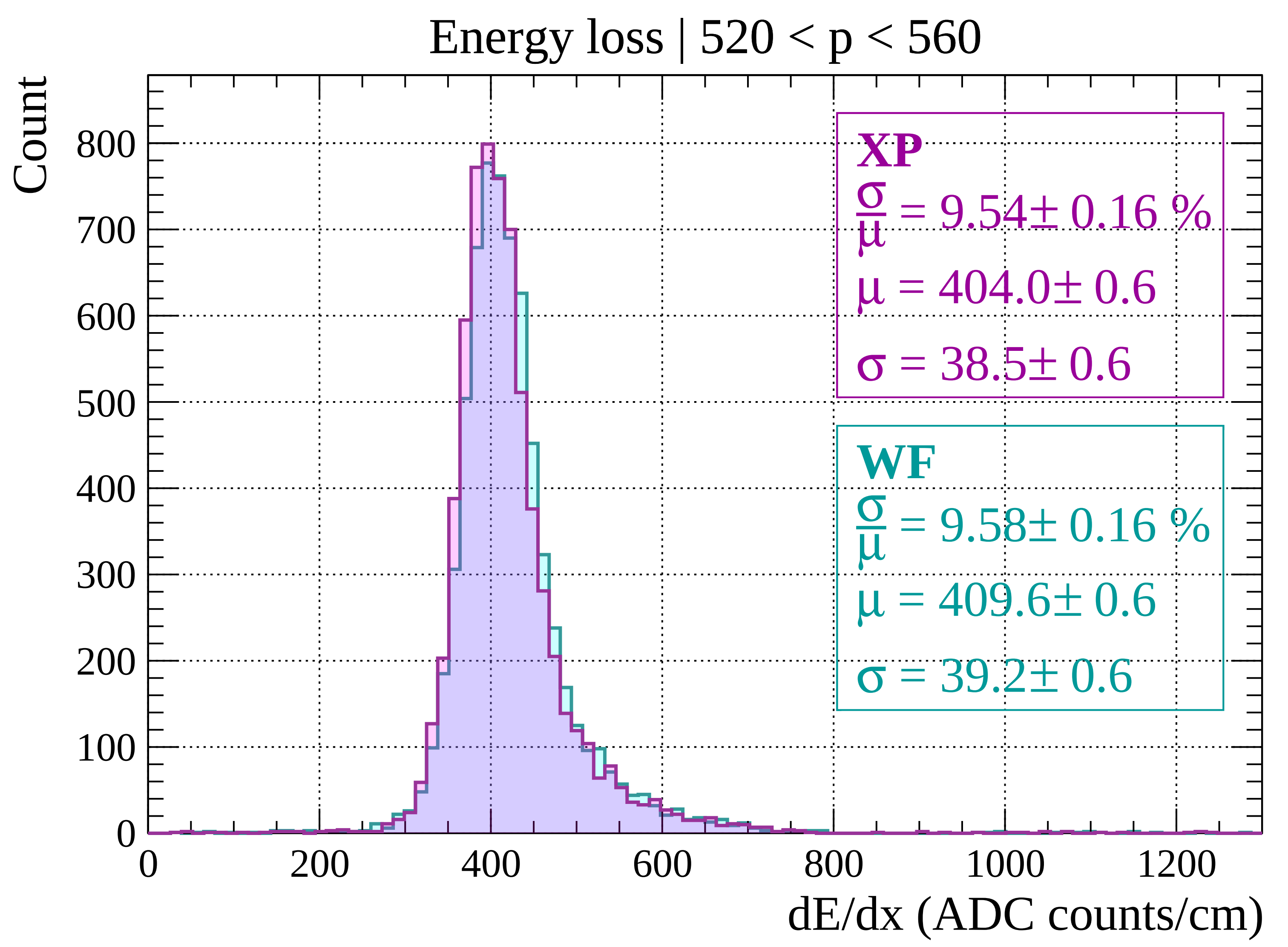
Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 9.39± 0.16 %
μ = 398.3± 0.6
σ = 37.4± 0.6
WF
σ/μ = 9.80± 0.16 %
μ = 404.2± 0.6
σ = 39.6± 0.6



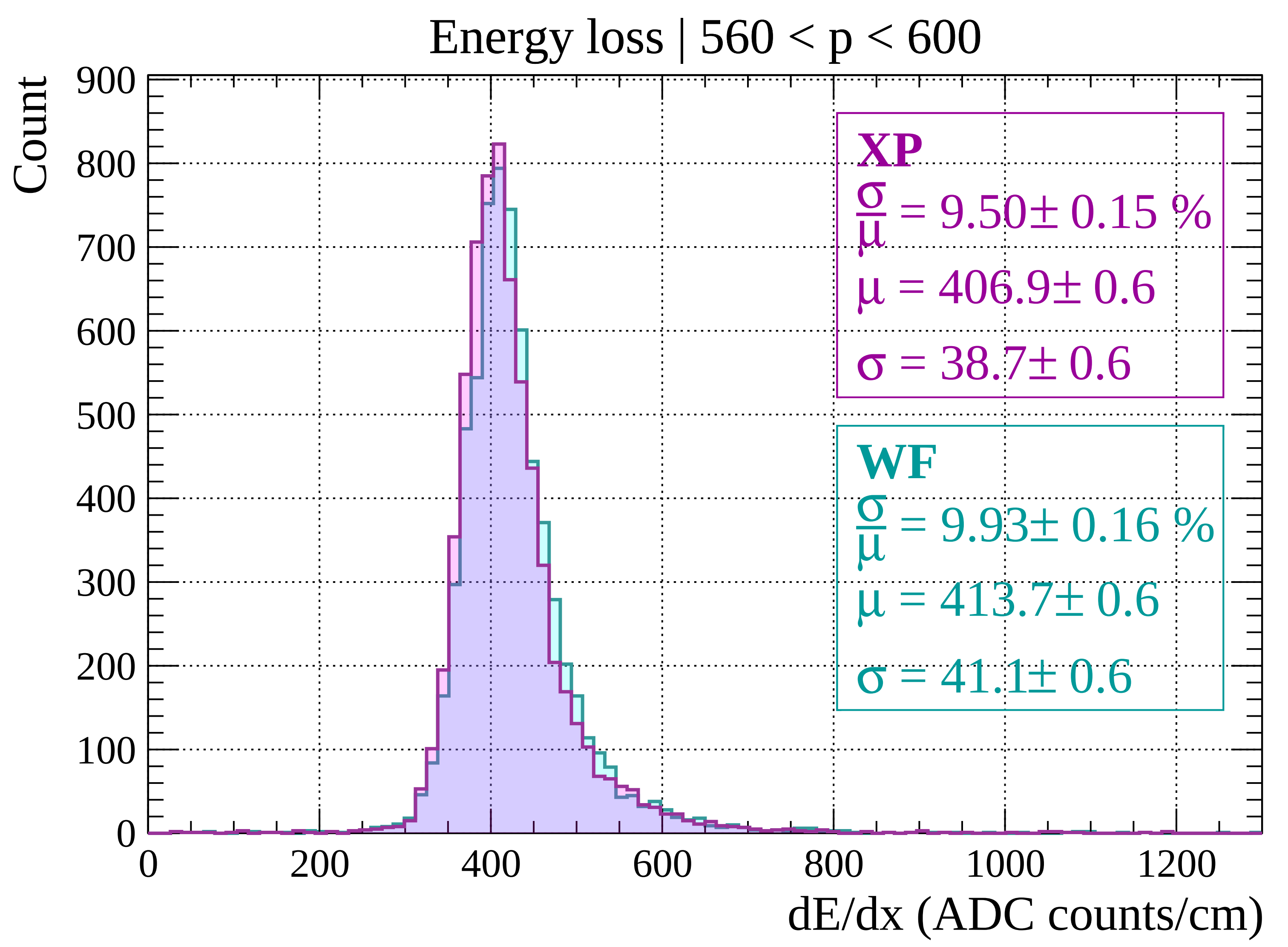
Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 9.38± 0.17 %
μ = 399.3± 0.6
σ = 37.4± 0.6
WF
σ/μ = 9.84± 0.17 %
μ = 405.5± 0.6
σ = 39.9± 0.6



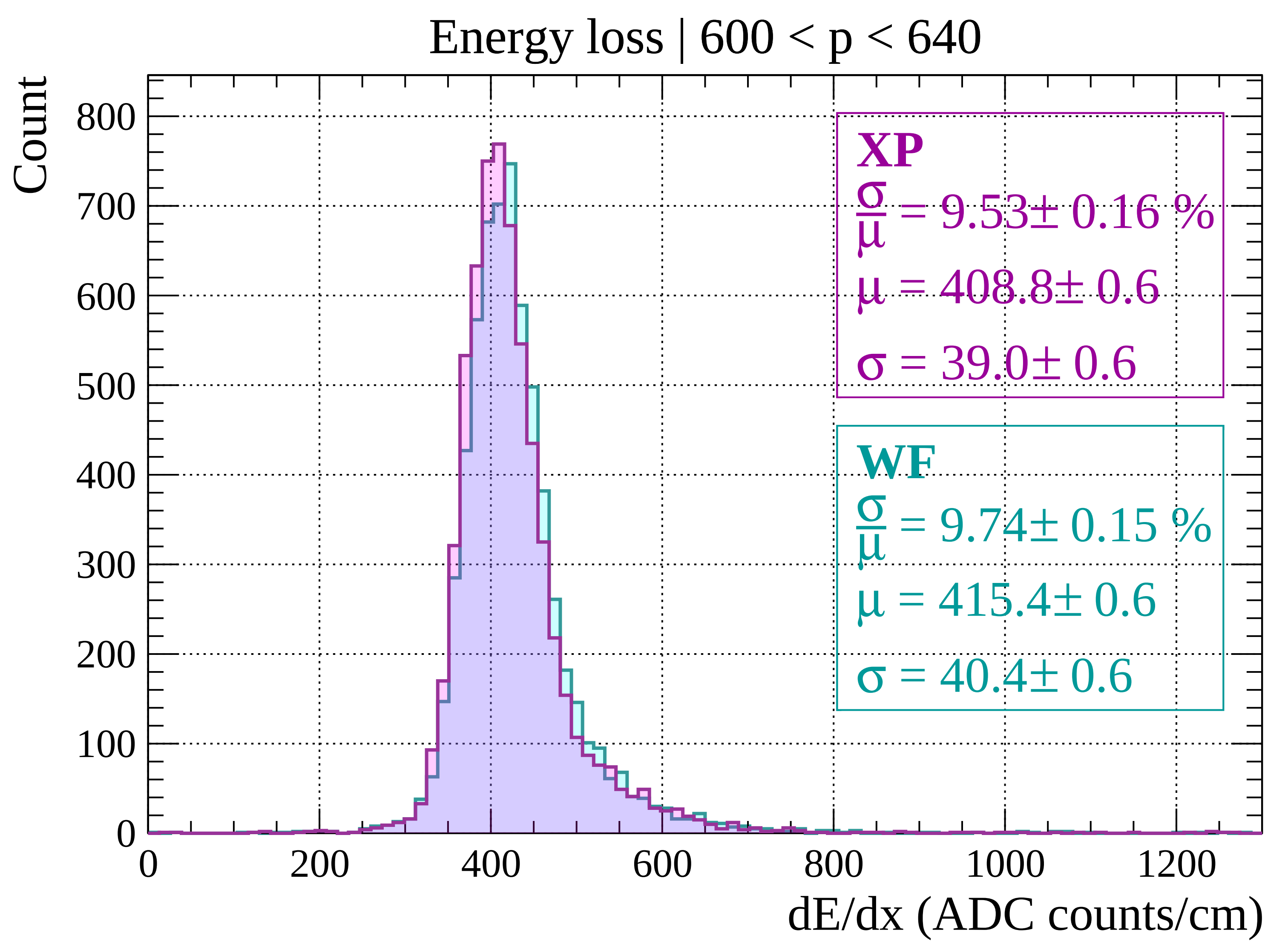
Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.41 \pm 0.15$ %
$\mu = 402.9 \pm 0.6$
$\sigma = 37.9 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.61 \pm 0.16$ %
$\mu = 408.2 \pm 0.6$
$\sigma = 39.2 \pm 0.6$



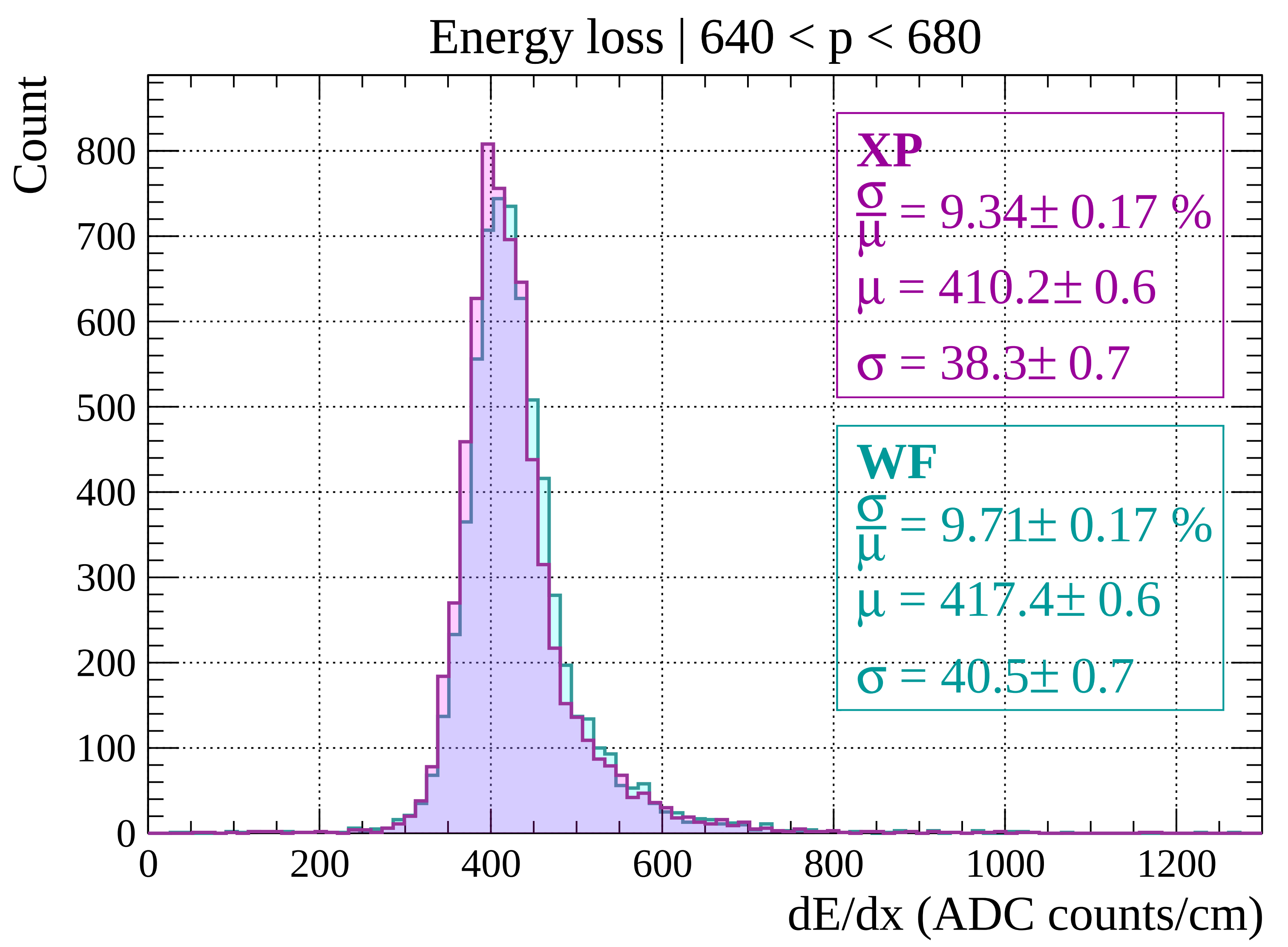
Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 9.54± 0.16 %
μ = 404.0± 0.6
σ = 38.5± 0.6
WF
σ/μ = 9.58± 0.16 %
μ = 409.6± 0.6
σ = 39.2± 0.6

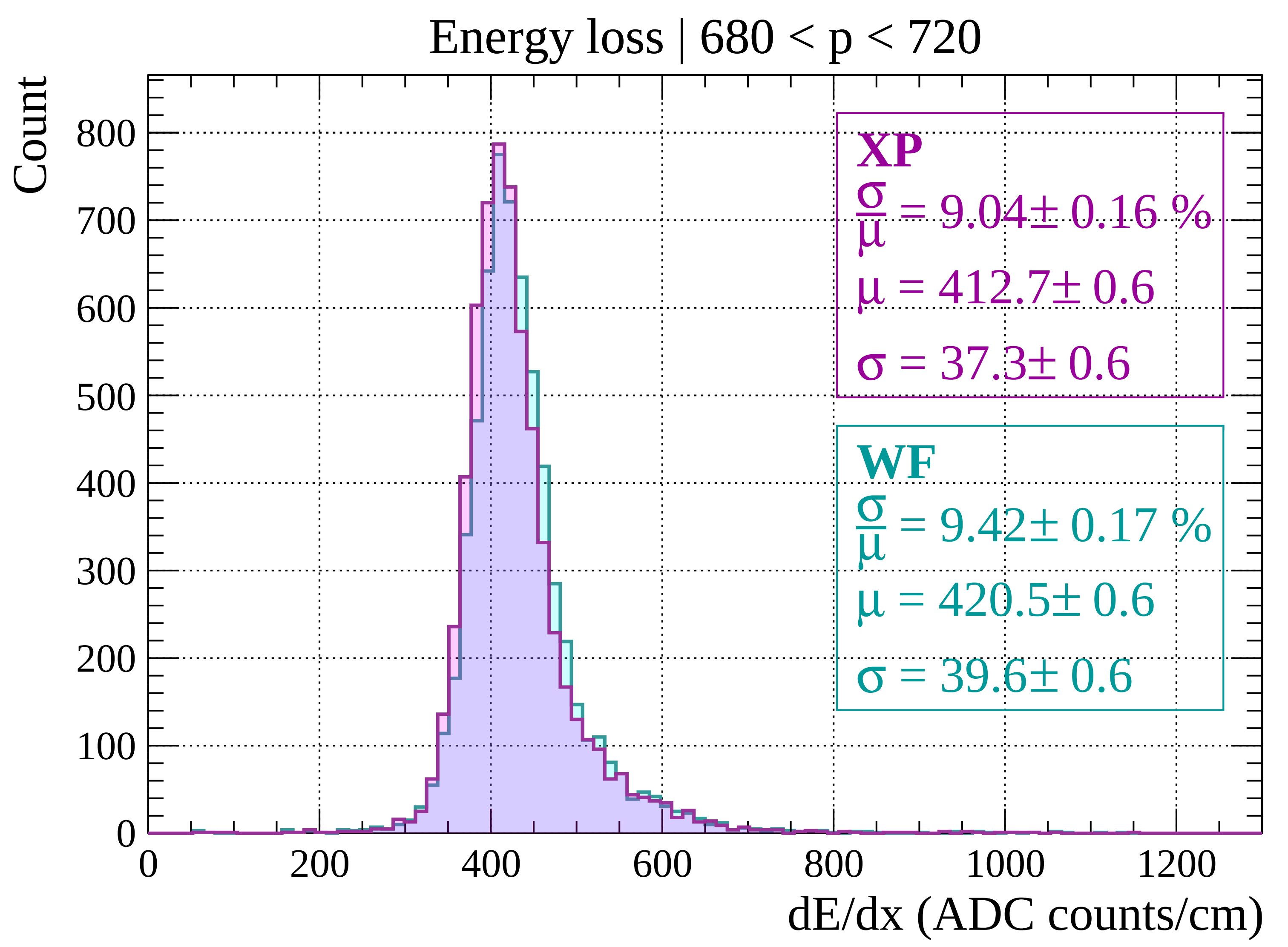


Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 9.50± 0.15 %
μ = 406.9± 0.6
σ = 38.7± 0.6
WF
σ/μ = 9.93± 0.16 %
μ = 413.7± 0.6
σ = 41.1± 0.6

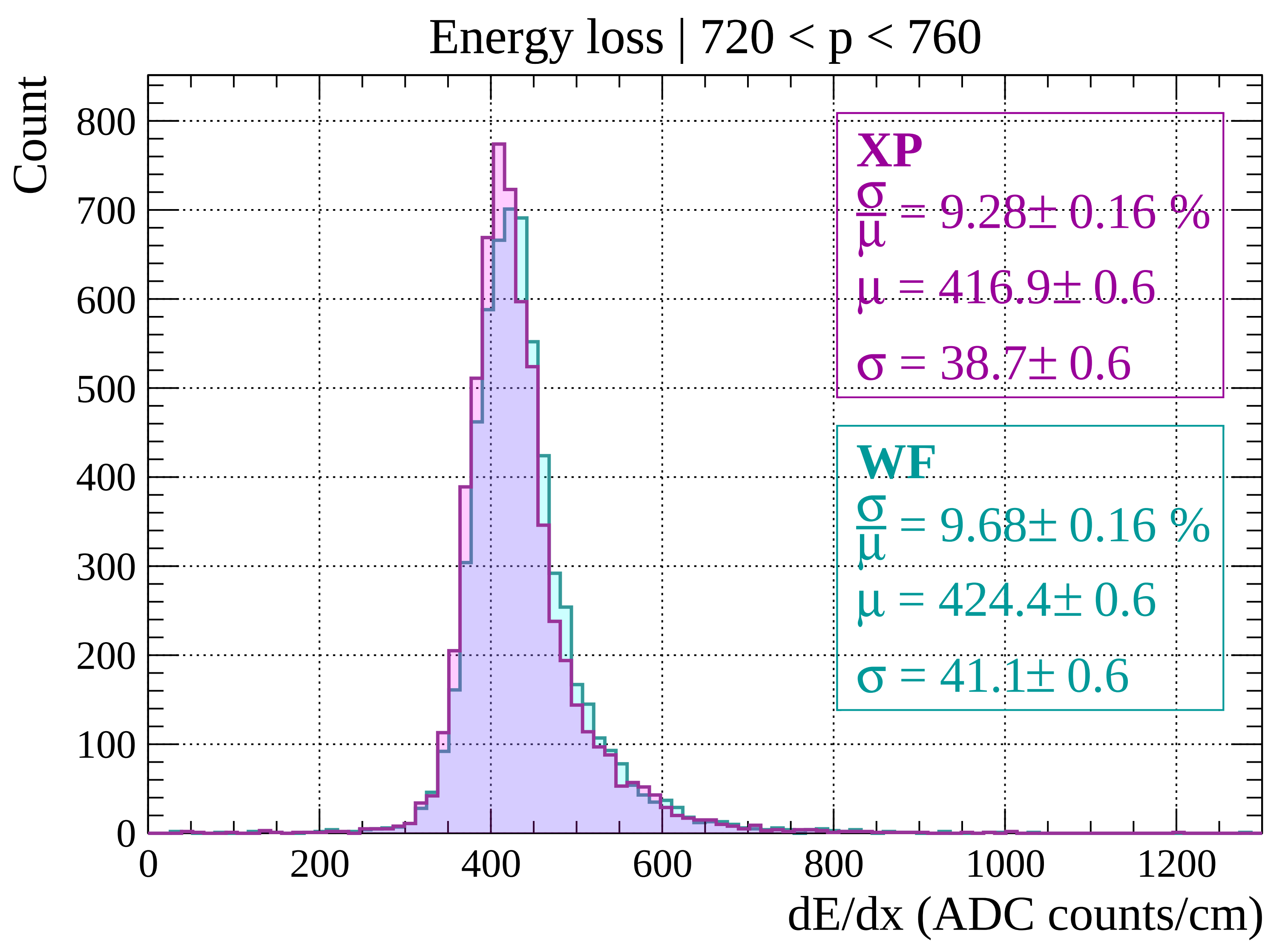


Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.53± 0.16 %
μ = 408.8± 0.6
σ = 39.0± 0.6
WF
σ/μ = 9.74± 0.15 %
μ = 415.4± 0.6
σ = 40.4± 0.6
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

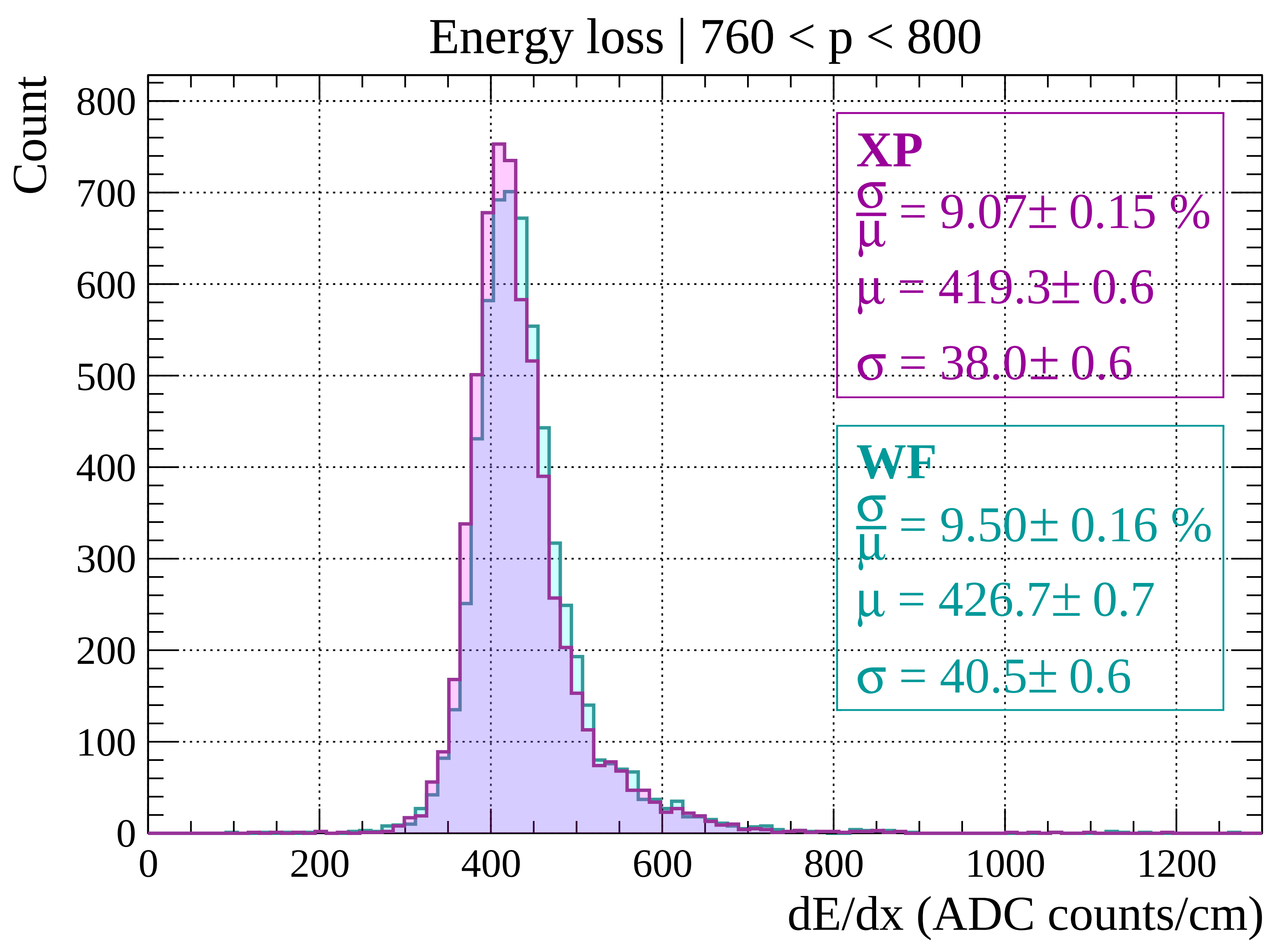

Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 9.34± 0.17 %
μ = 410.2± 0.6
σ = 38.3± 0.7
WF
σ/μ = 9.71± 0.17 %
μ = 417.4± 0.6
σ = 40.5± 0.7



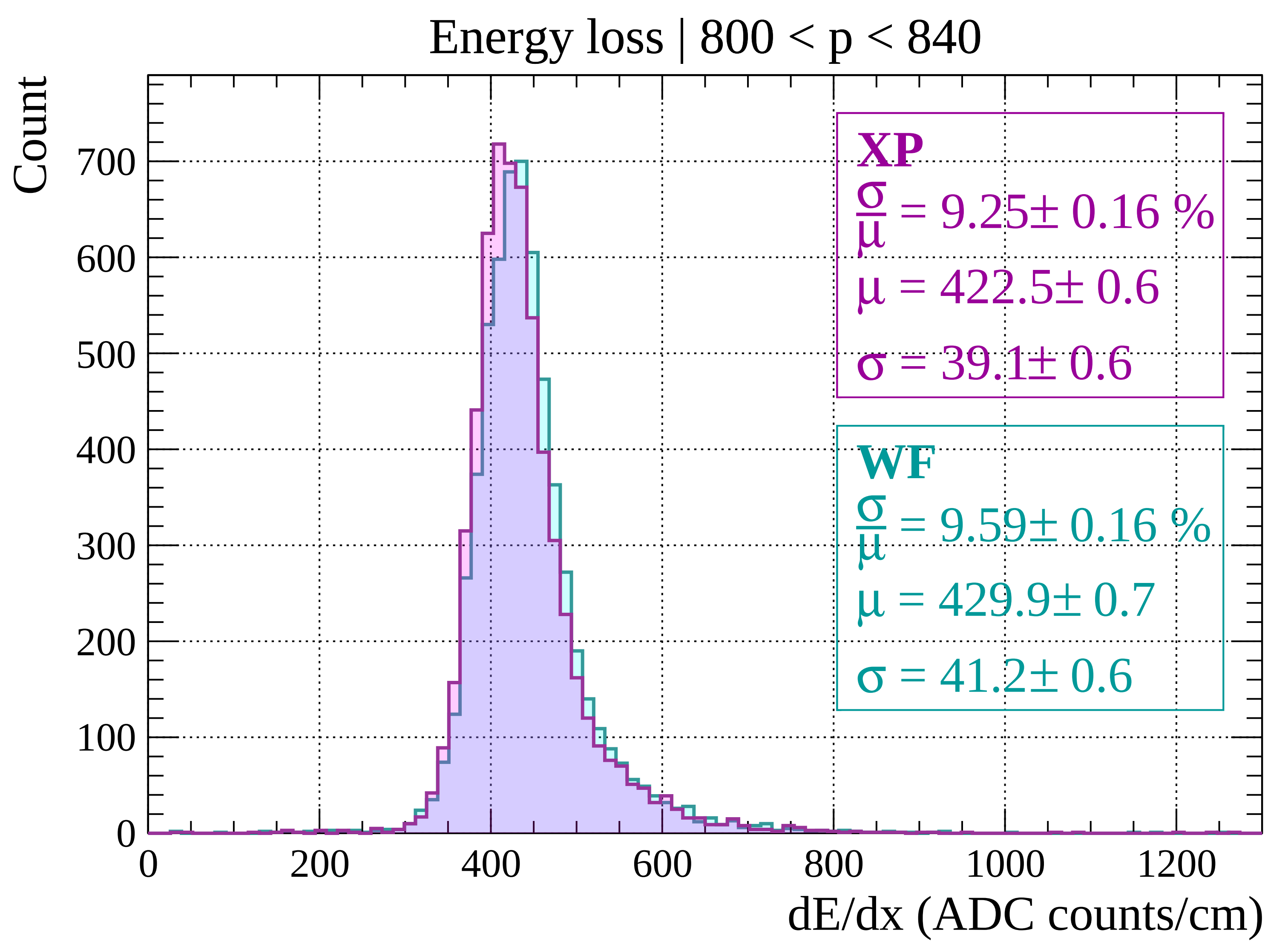
Energy loss | 680 < p < 720
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.04± 0.16 %
μ = 412.7± 0.6
σ = 37.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.42± 0.17 %
μ = 420.5± 0.6
σ = 39.6± 0.6

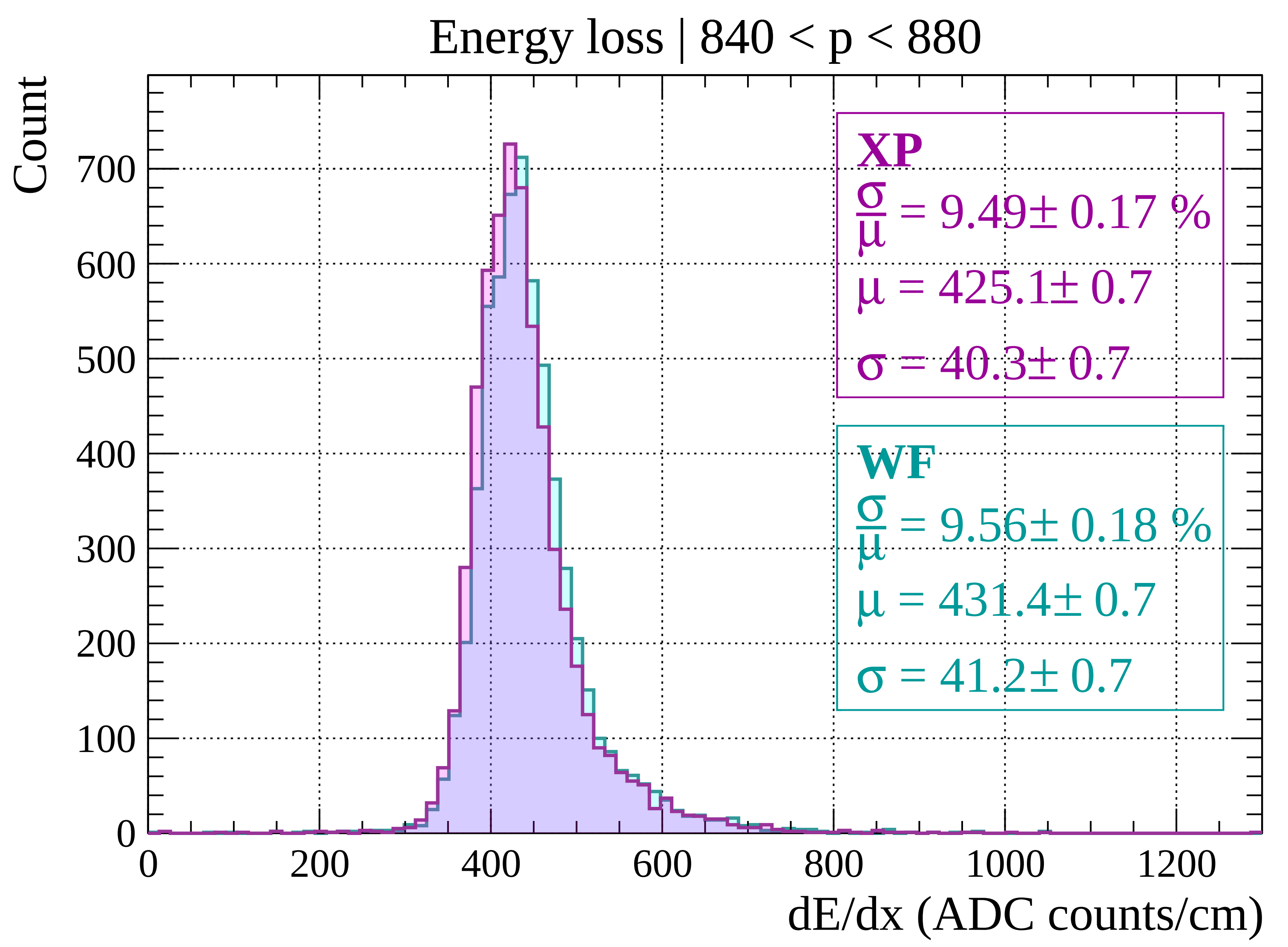


Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.28 \pm 0.16$ %
$\mu = 416.9 \pm 0.6$
$\sigma = 38.7 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.68 \pm 0.16$ %
$\mu = 424.4 \pm 0.6$
$\sigma = 41.1 \pm 0.6$

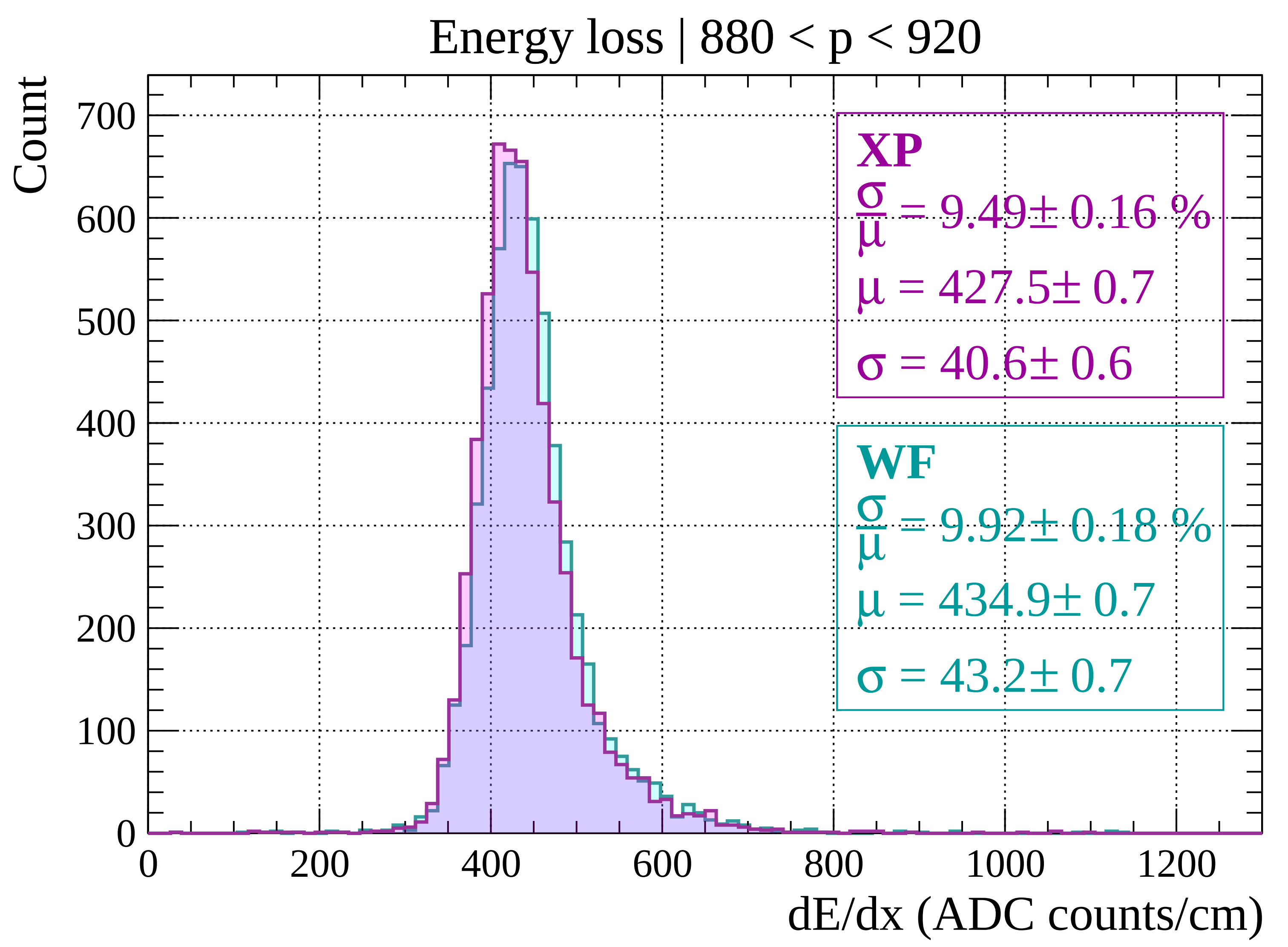


Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.07± 0.15 %
μ = 419.3± 0.6
σ = 38.0± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.50± 0.16 %
μ = 426.7± 0.7
σ = 40.5± 0.6
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

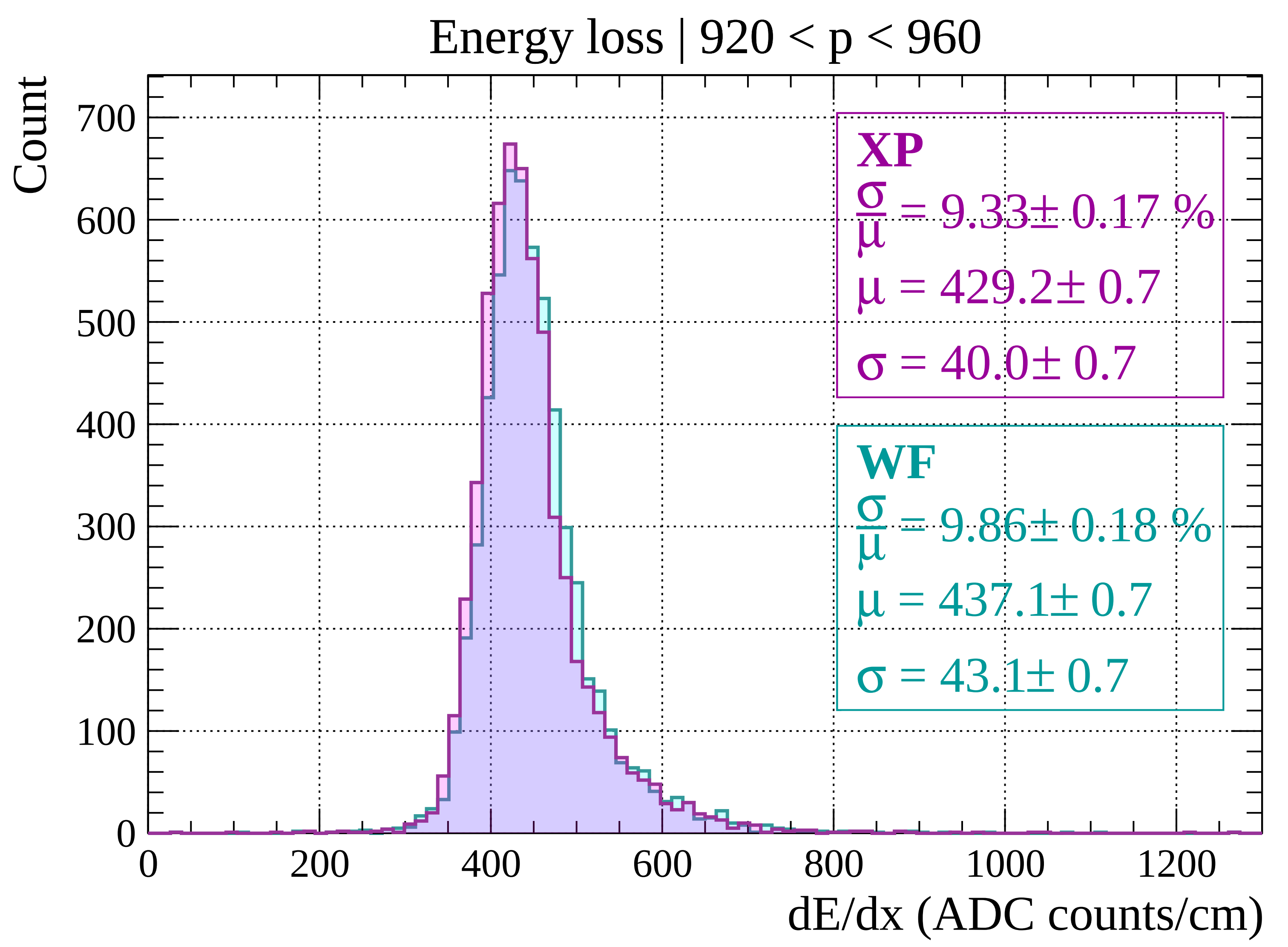

Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.25 \pm 0.16$ %
$\mu = 422.5 \pm 0.6$
$\sigma = 39.1 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.59 \pm 0.16$ %
$\mu = 429.9 \pm 0.7$
$\sigma = 41.2 \pm 0.6$



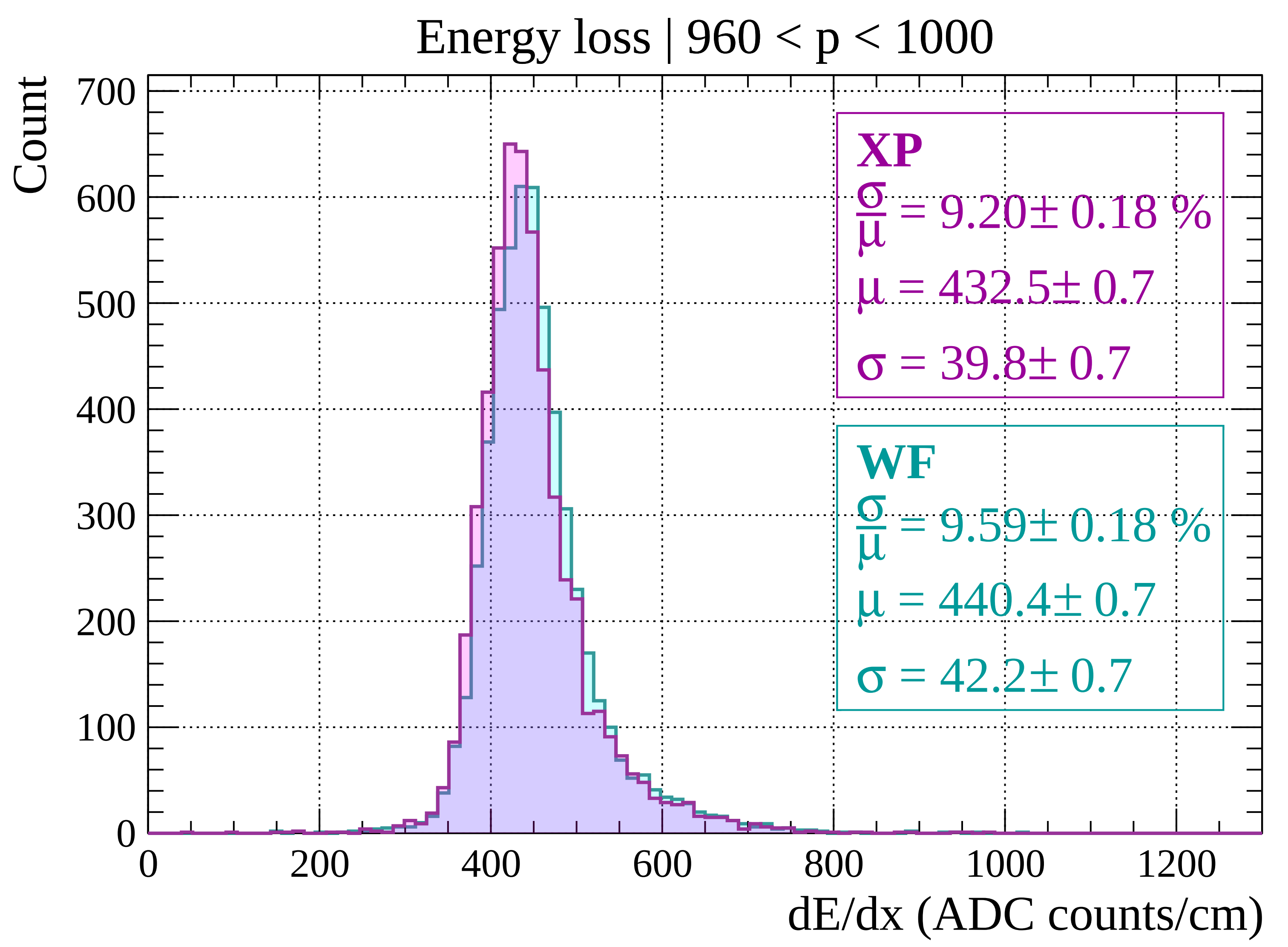
Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.49± 0.17 %
μ = 425.1± 0.7
σ = 40.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.56± 0.18 %
μ = 431.4± 0.7
σ = 41.2± 0.7

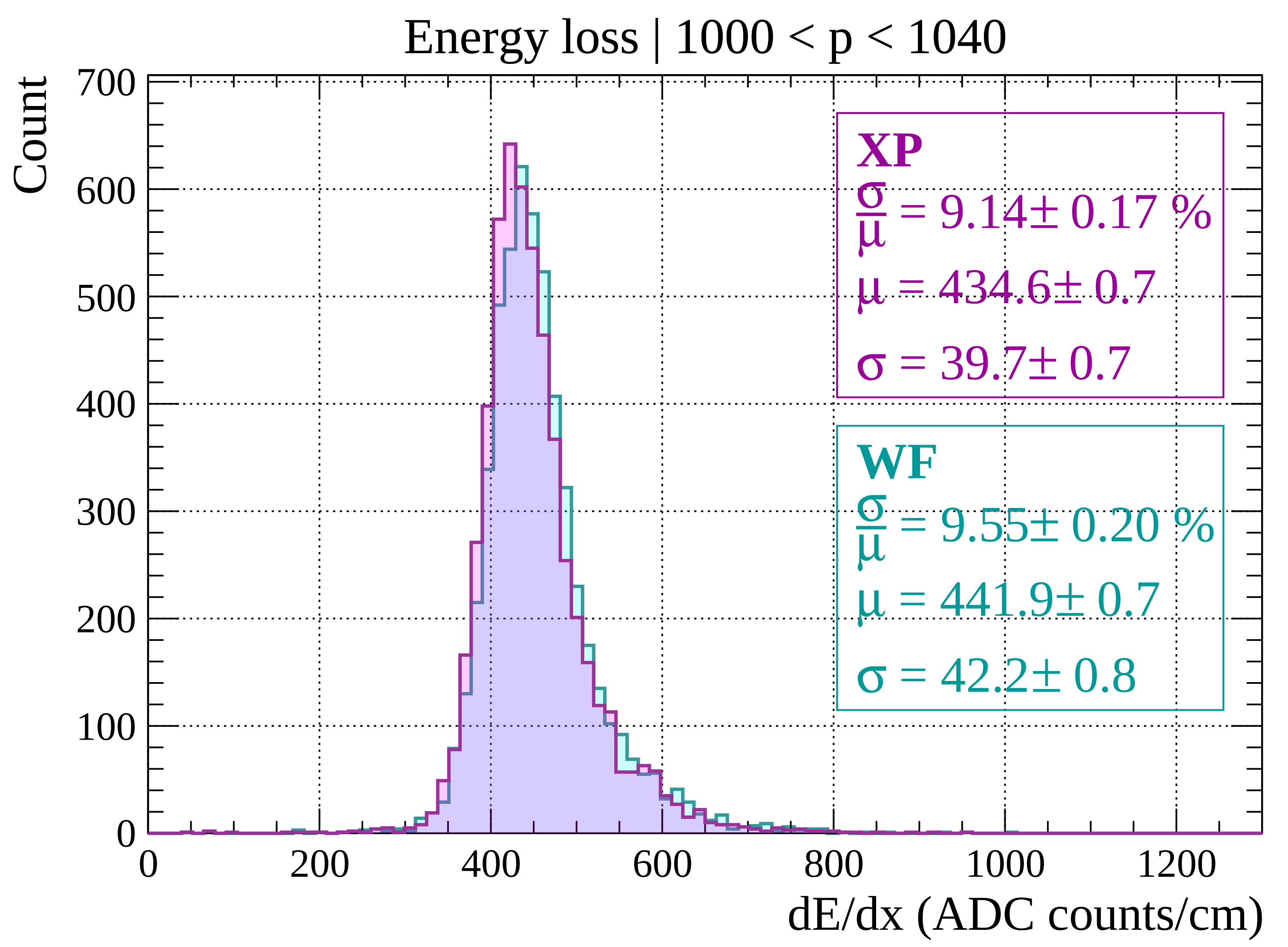


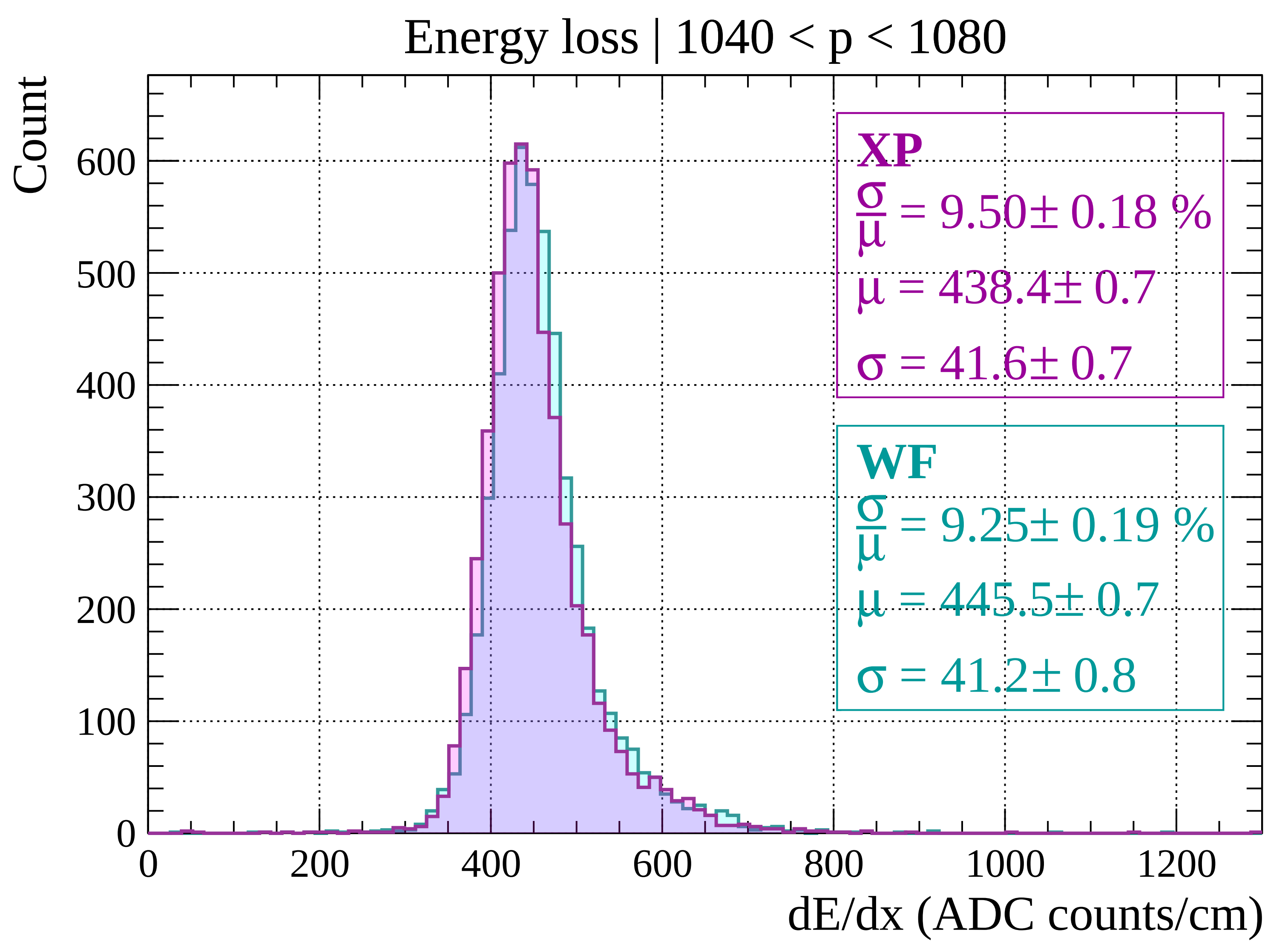
Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.49 \pm 0.16$ %
$\mu = 427.5 \pm 0.7$
$\sigma = 40.6 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.92 \pm 0.18$ %
$\mu = 434.9 \pm 0.7$
$\sigma = 43.2 \pm 0.7$



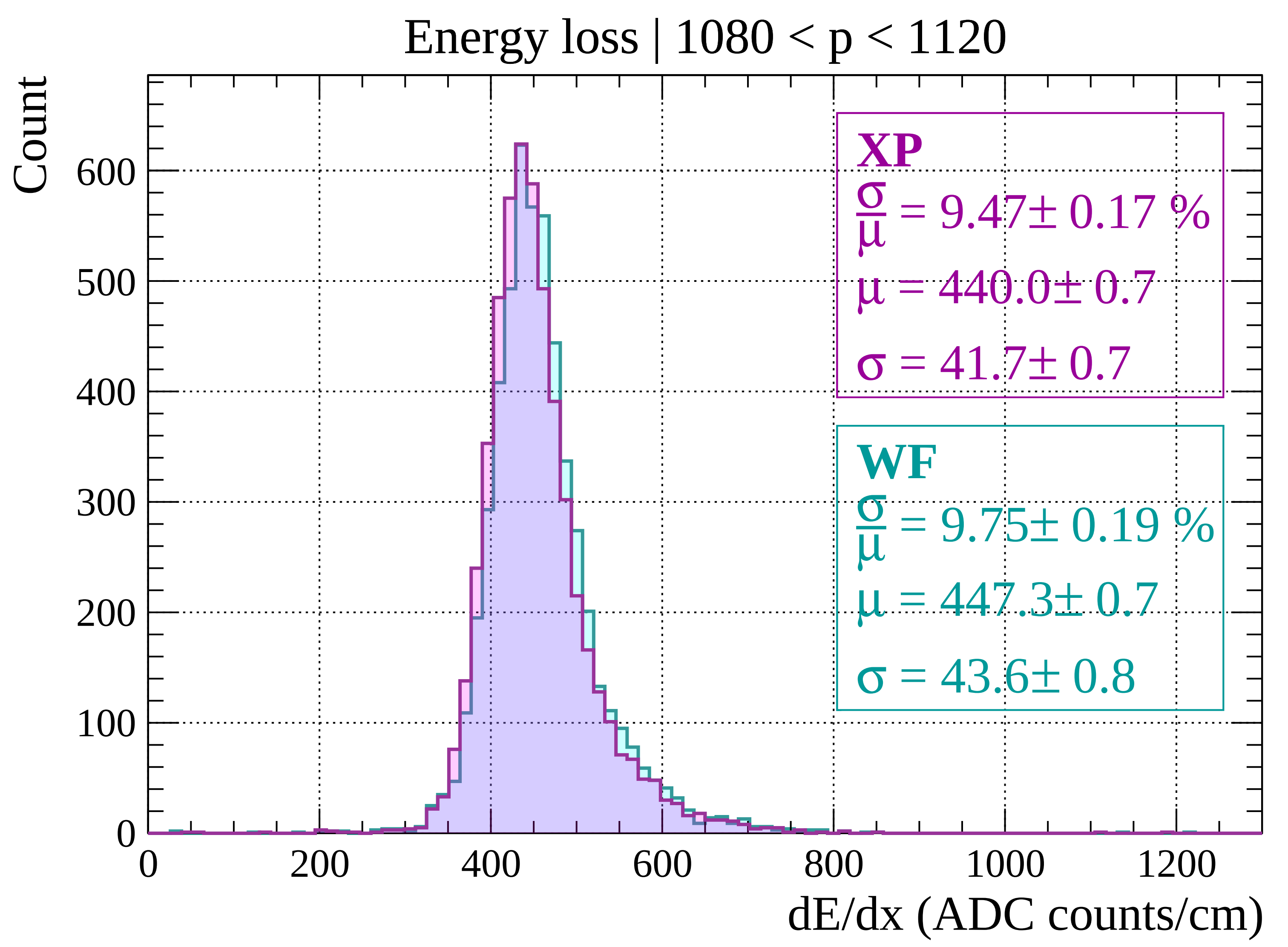
Energy loss | 920 < p < 960
Count
XP
$\frac{\sigma}{\mu}$ = 9.33± 0.17 %
μ = 429.2± 0.7
σ = 40.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.86± 0.18 %
μ = 437.1± 0.7
σ = 43.1± 0.7
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

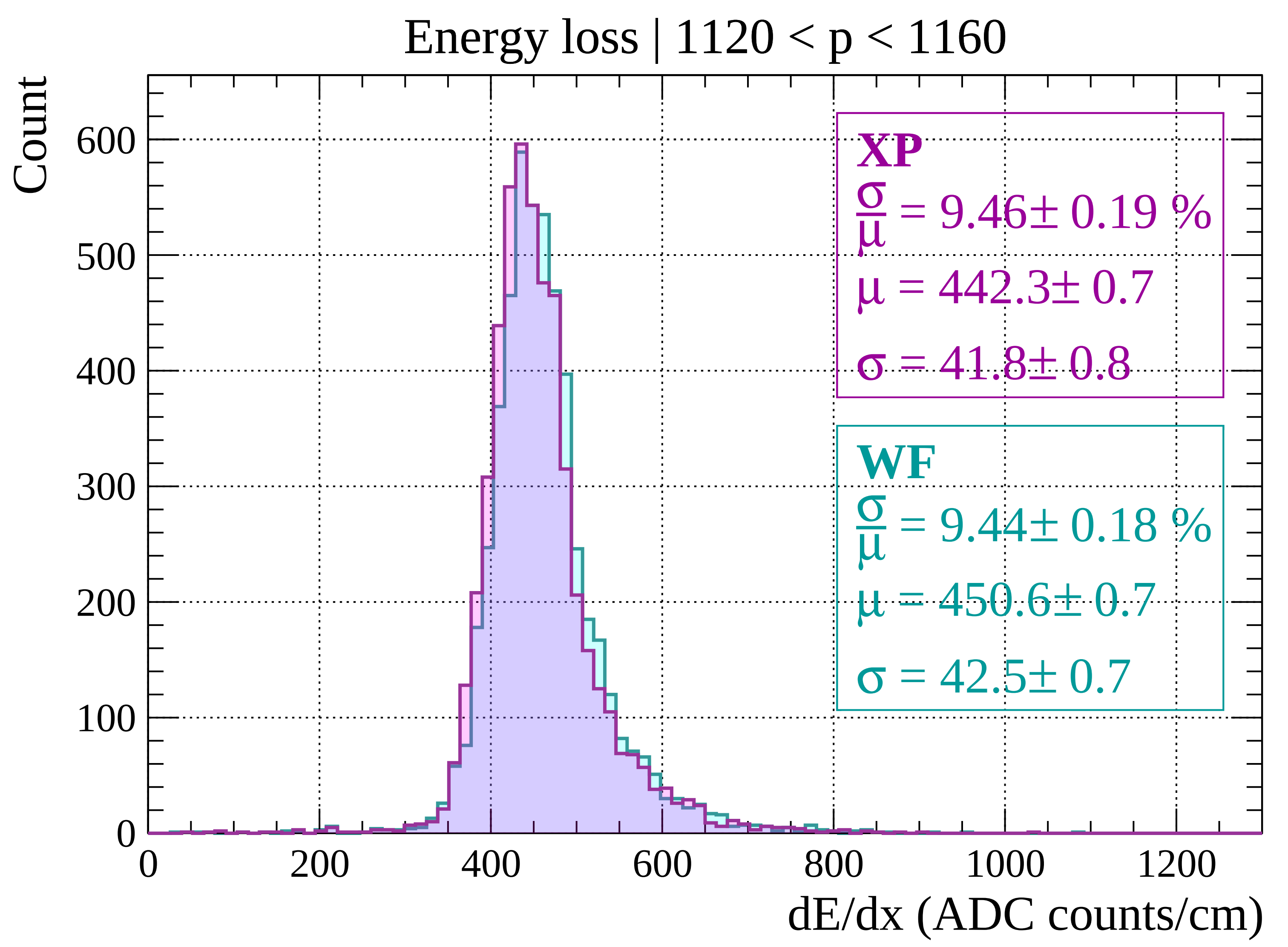

Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.20± 0.18 %
μ = 432.5± 0.7
σ = 39.8± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.59± 0.18 %
μ = 440.4± 0.7
σ = 42.2± 0.7
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200


Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.14± 0.17 %
μ = 434.6± 0.7
σ = 39.7± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.55± 0.20 %
μ = 441.9± 0.7
σ = 42.2± 0.8

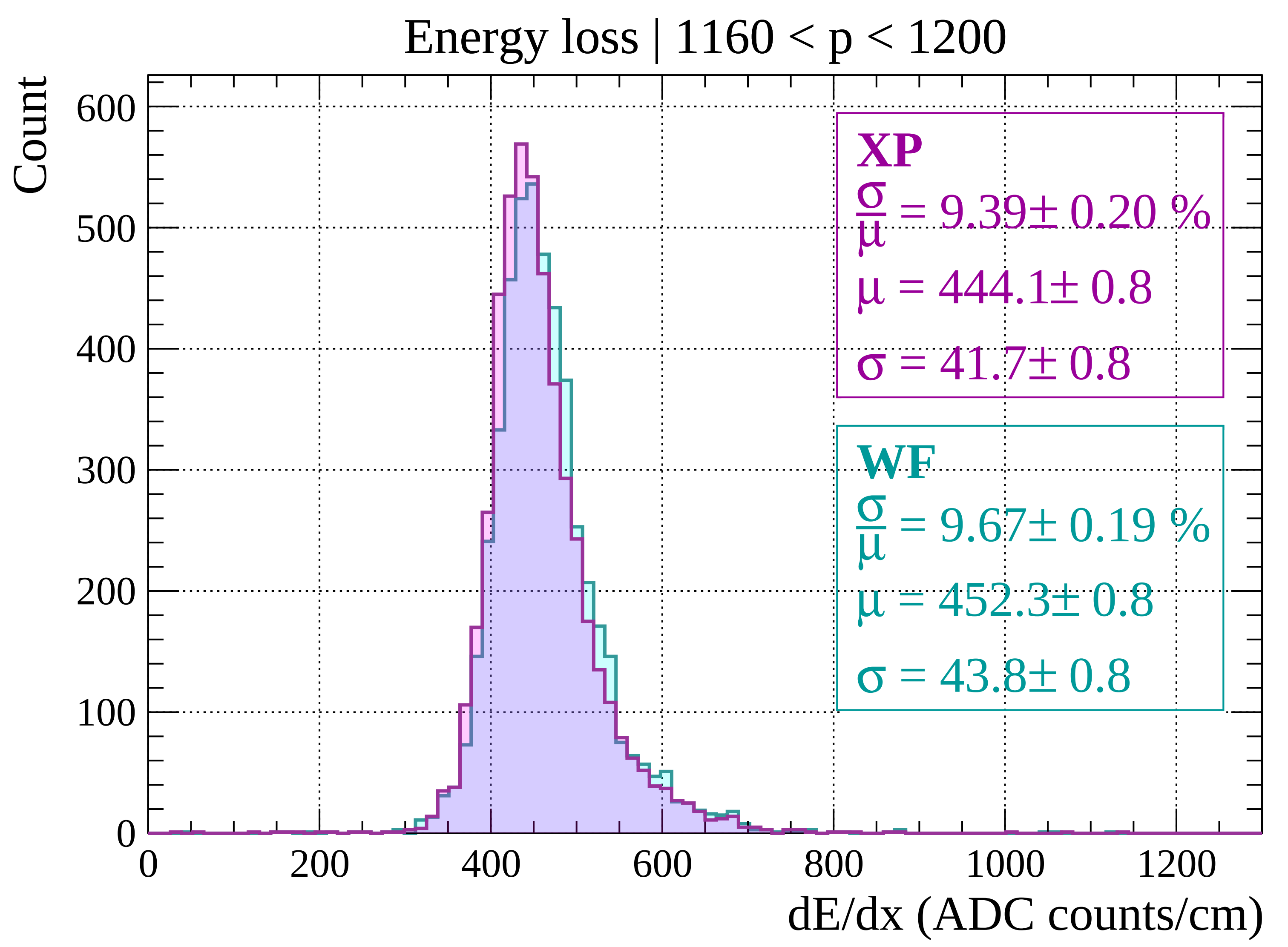


Energy loss | 1040 < p < 1080
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.50± 0.18 %
μ = 438.4± 0.7
σ = 41.6± 0.7
WF
σ/μ = 9.25± 0.19 %
μ = 445.5± 0.7
σ = 41.2± 0.8

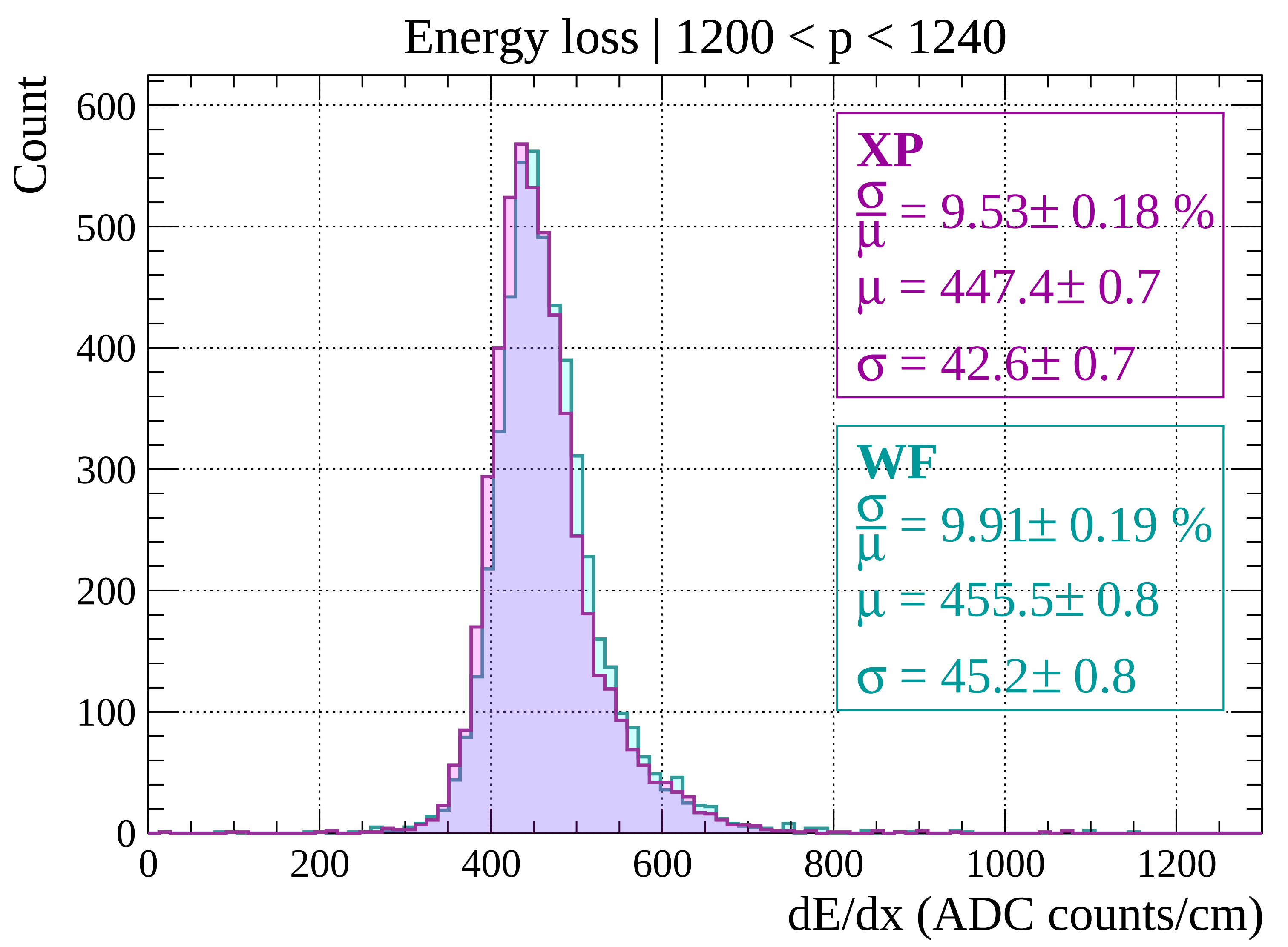

Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.47± 0.17 %
μ = 440.0± 0.7
σ = 41.7± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.75± 0.19 %
μ = 447.3± 0.7
σ = 43.6± 0.8



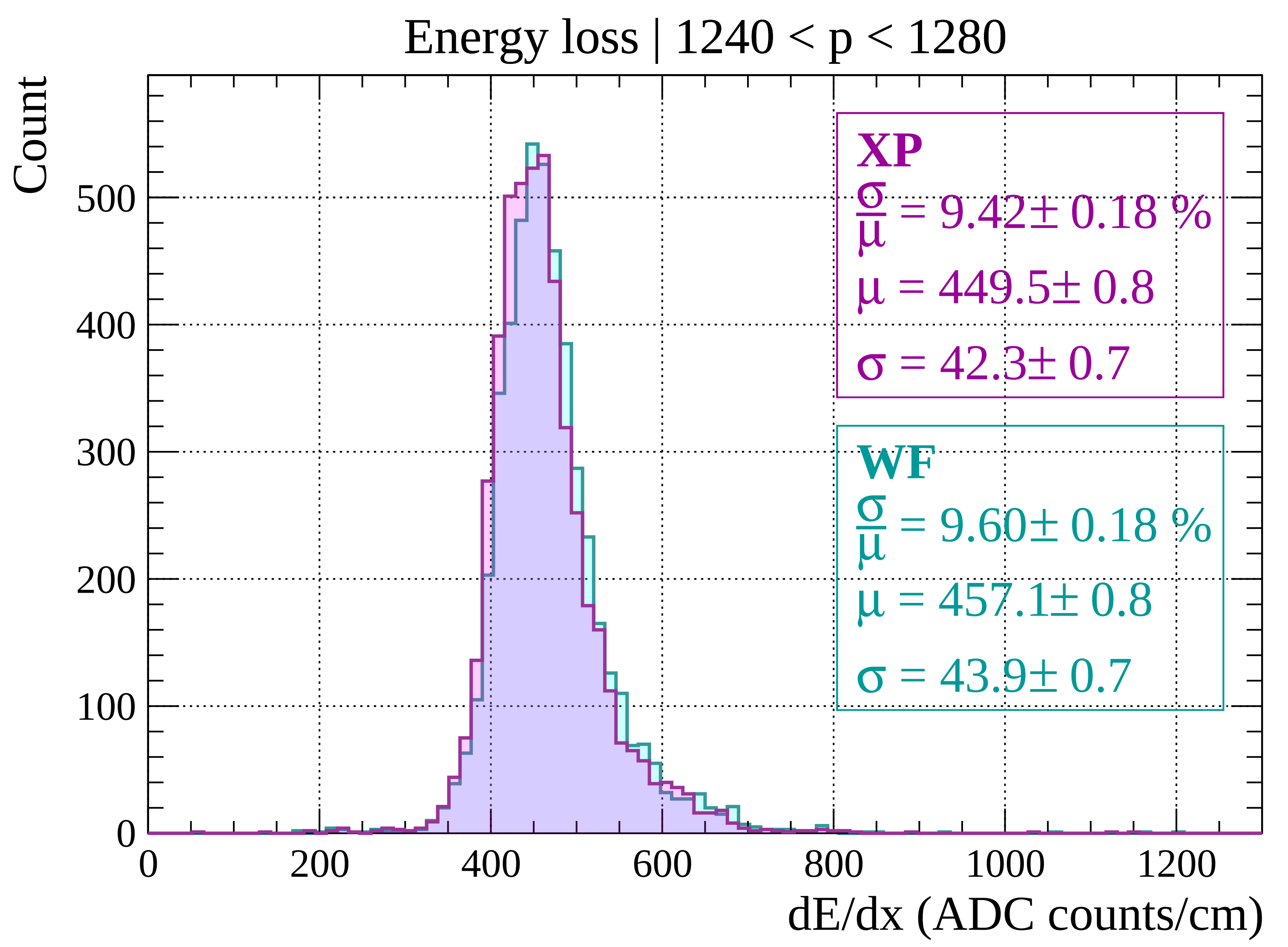
Energy loss | 1120 < p < 1160
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.46± 0.19 %
μ = 442.3± 0.7
σ = 41.8± 0.8
WF
σ/μ = 9.44± 0.18 %
μ = 450.6± 0.7
σ = 42.5± 0.7



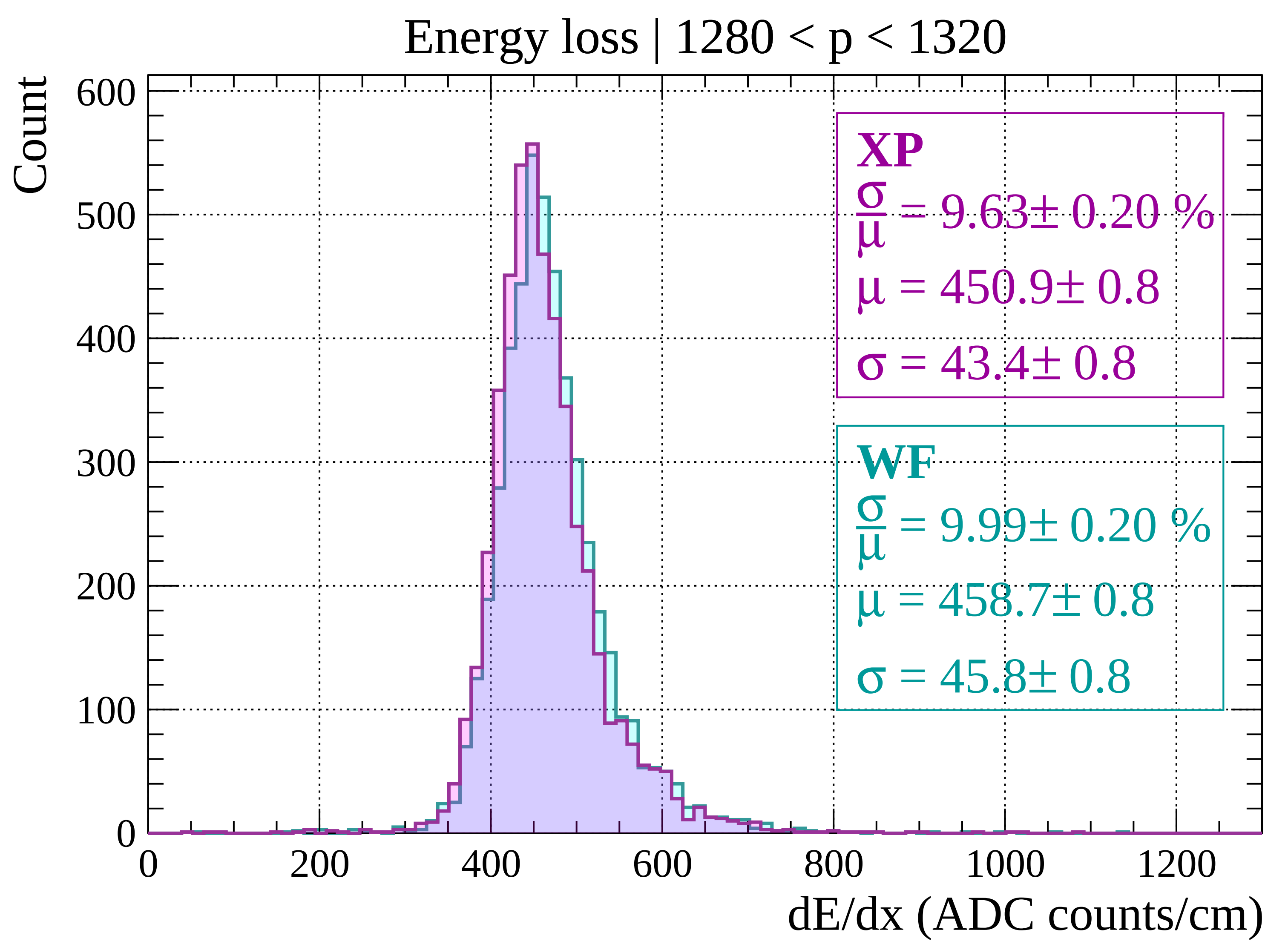
Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.39± 0.20 %
μ = 444.1± 0.8
σ = 41.7± 0.8
WF
σ/μ = 9.67± 0.19 %
μ = 452.3± 0.8
σ = 43.8± 0.8

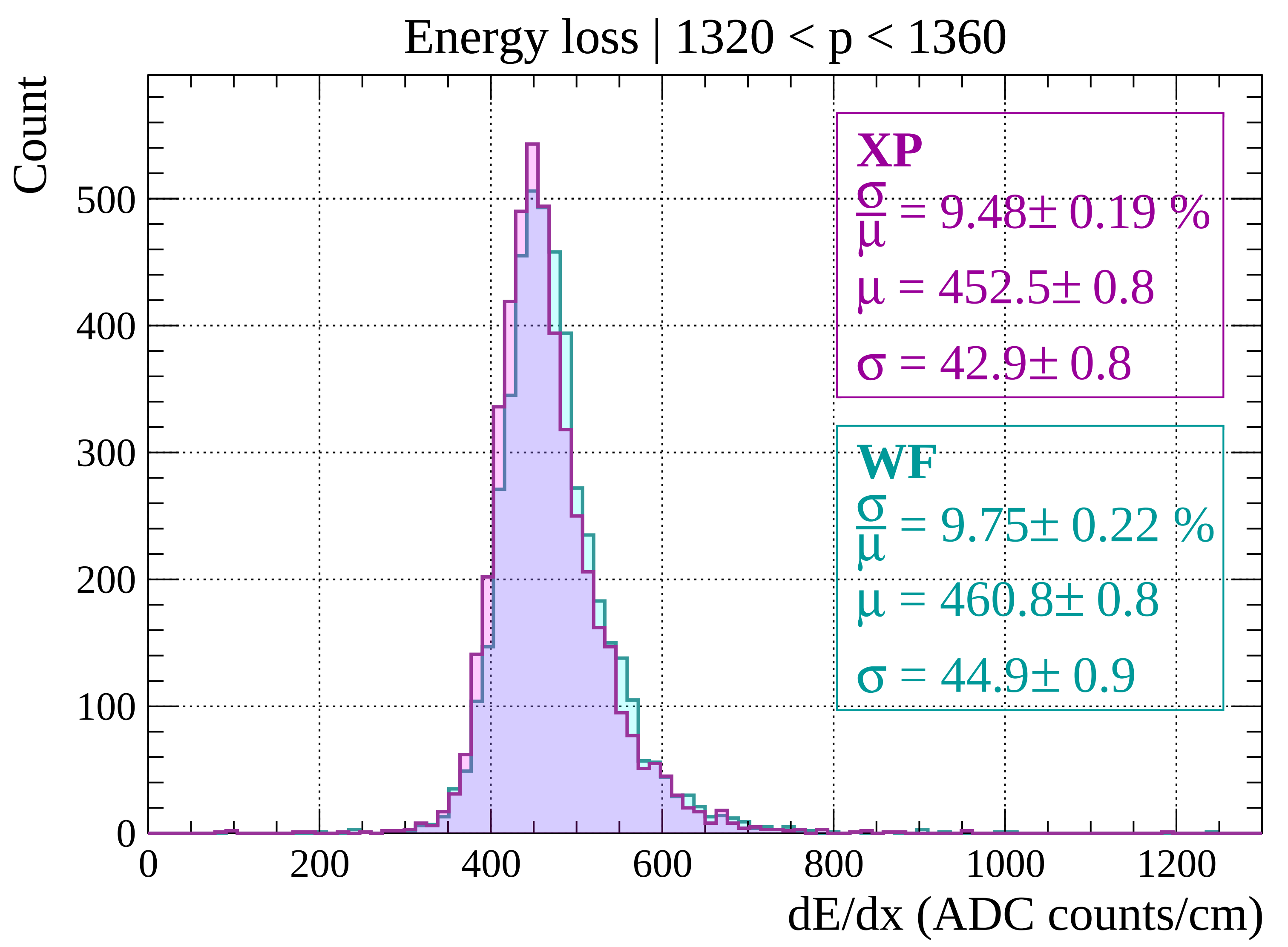


Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.53± 0.18 %
μ = 447.4± 0.7
σ = 42.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.91± 0.19 %
μ = 455.5± 0.8
σ = 45.2± 0.8

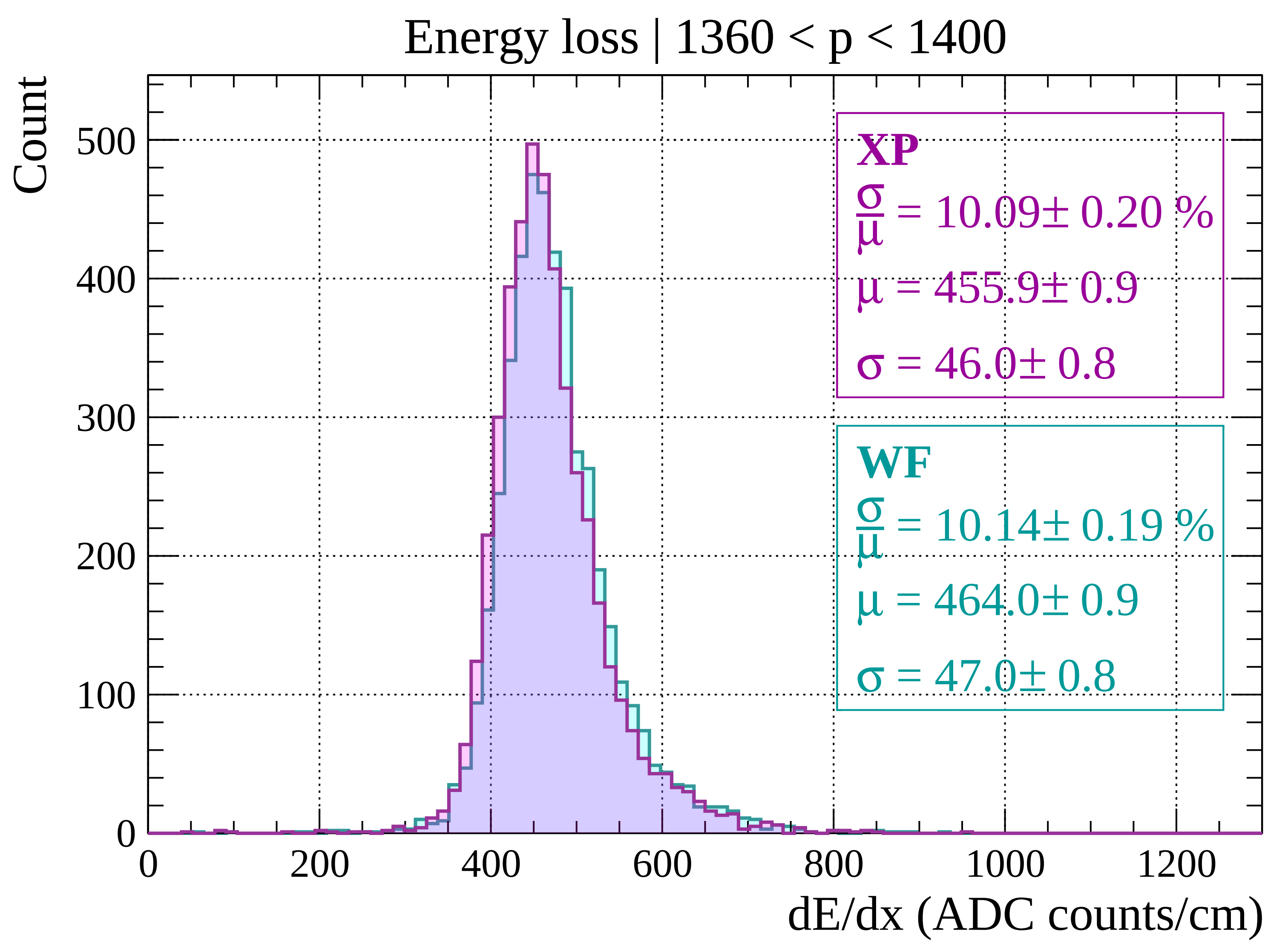


Energy loss | 1240 < p < 1280
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.42± 0.18 %
μ = 449.5± 0.8
σ = 42.3± 0.7
WF
σ/μ = 9.60± 0.18 %
μ = 457.1± 0.8
σ = 43.9± 0.7

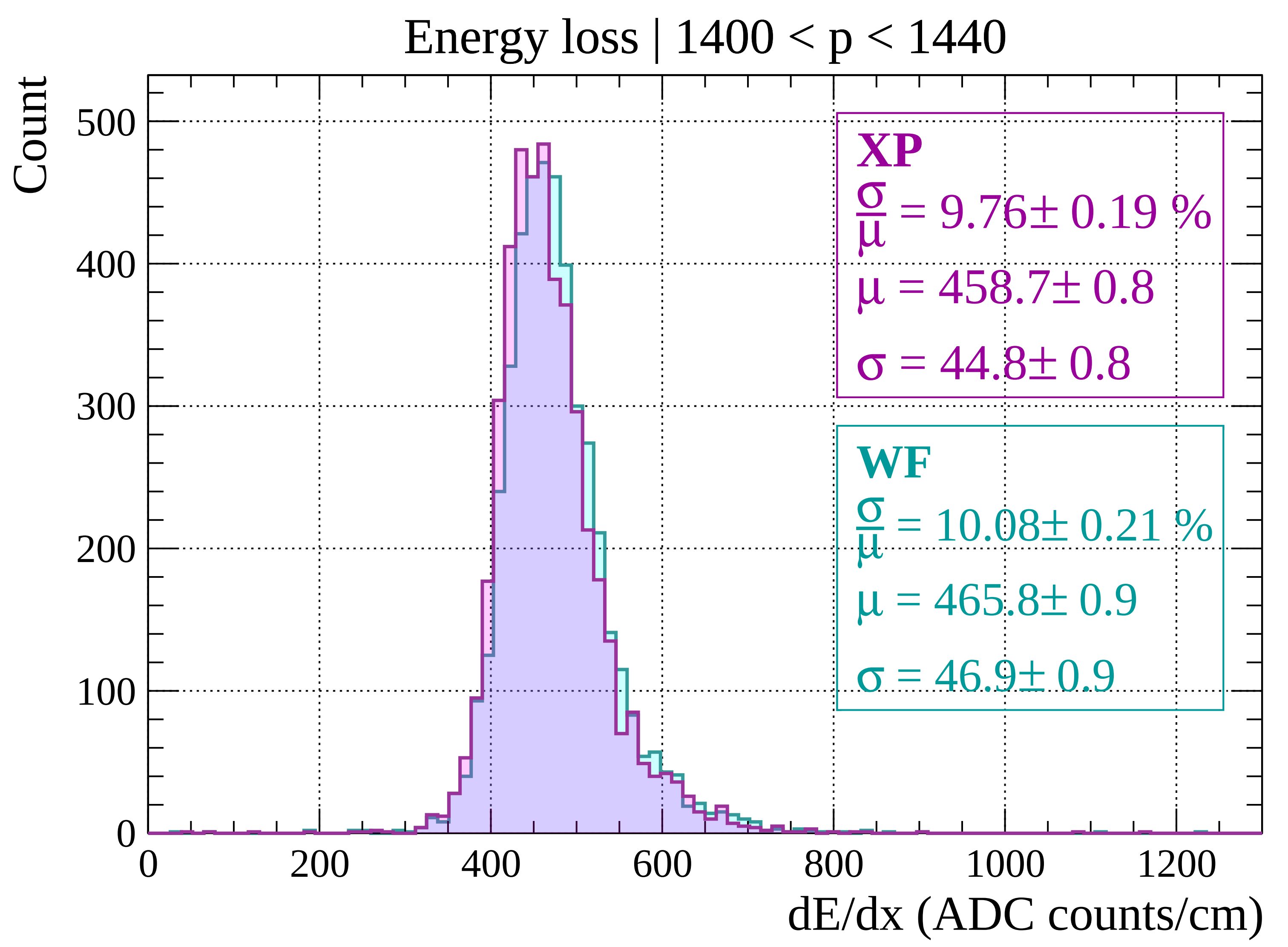

Energy loss | 1280 < p < 1320
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.63± 0.20 %
μ = 450.9± 0.8
σ = 43.4± 0.8
WF
σ/μ = 9.99± 0.20 %
μ = 458.7± 0.8
σ = 45.8± 0.8



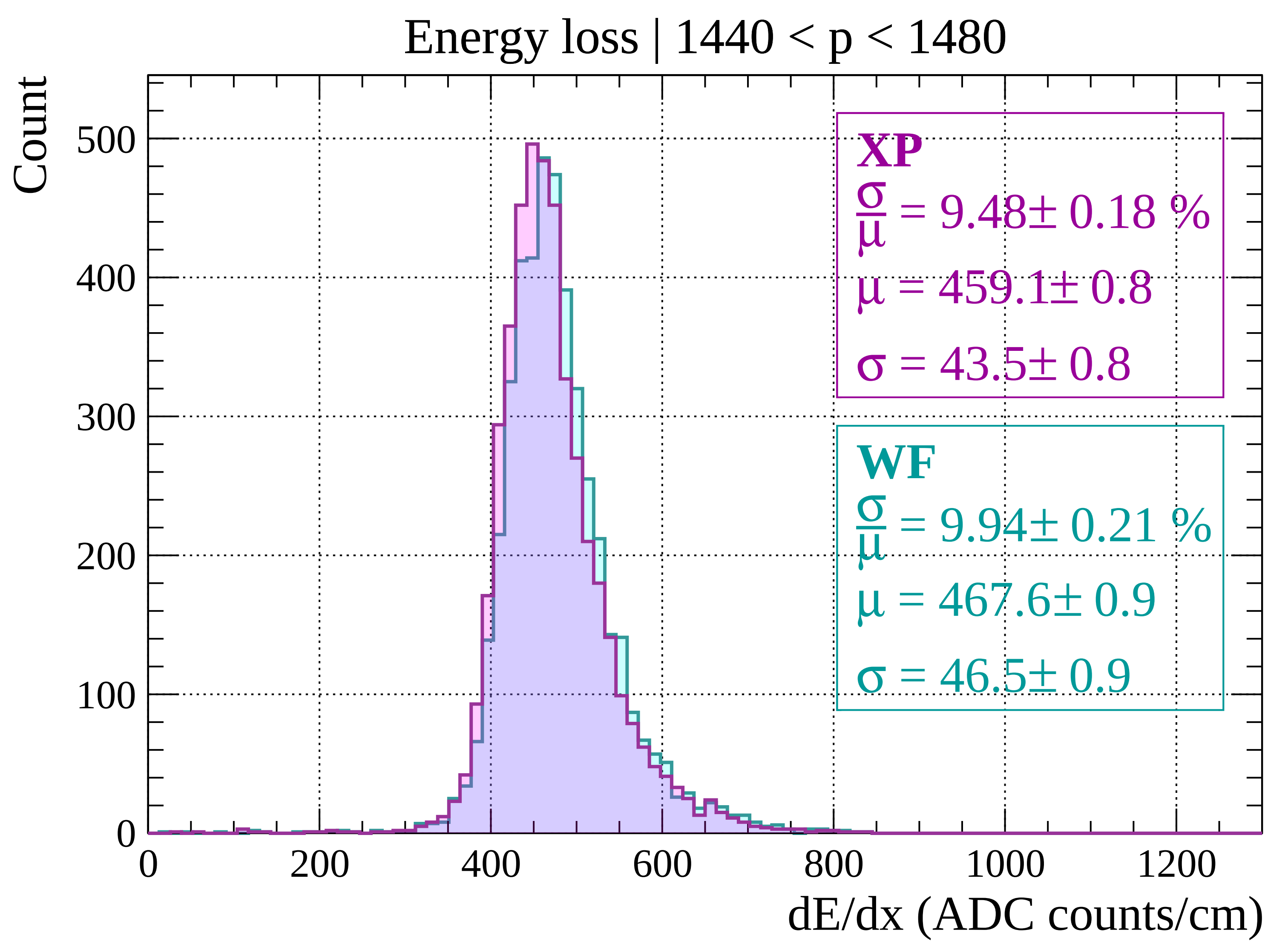
Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.48± 0.19 %
μ = 452.5± 0.8
σ = 42.9± 0.8
WF
σ/μ = 9.75± 0.22 %
μ = 460.8± 0.8
σ = 44.9± 0.9



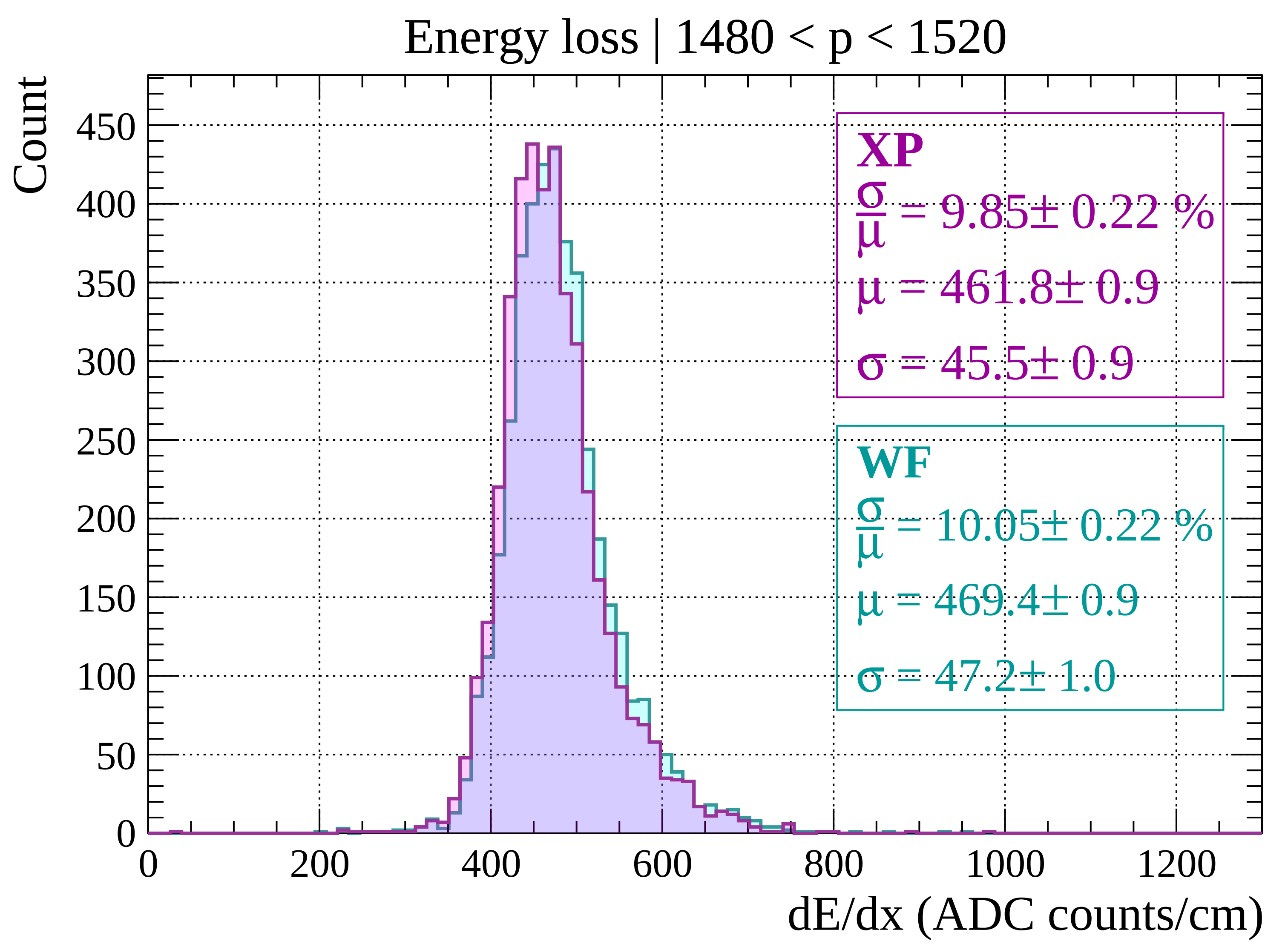
Energy loss | 1360 < p < 1400
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.09± 0.20 %
μ = 455.9± 0.9
σ = 46.0± 0.8
WF
σ/μ = 10.14± 0.19 %
μ = 464.0± 0.9
σ = 47.0± 0.8



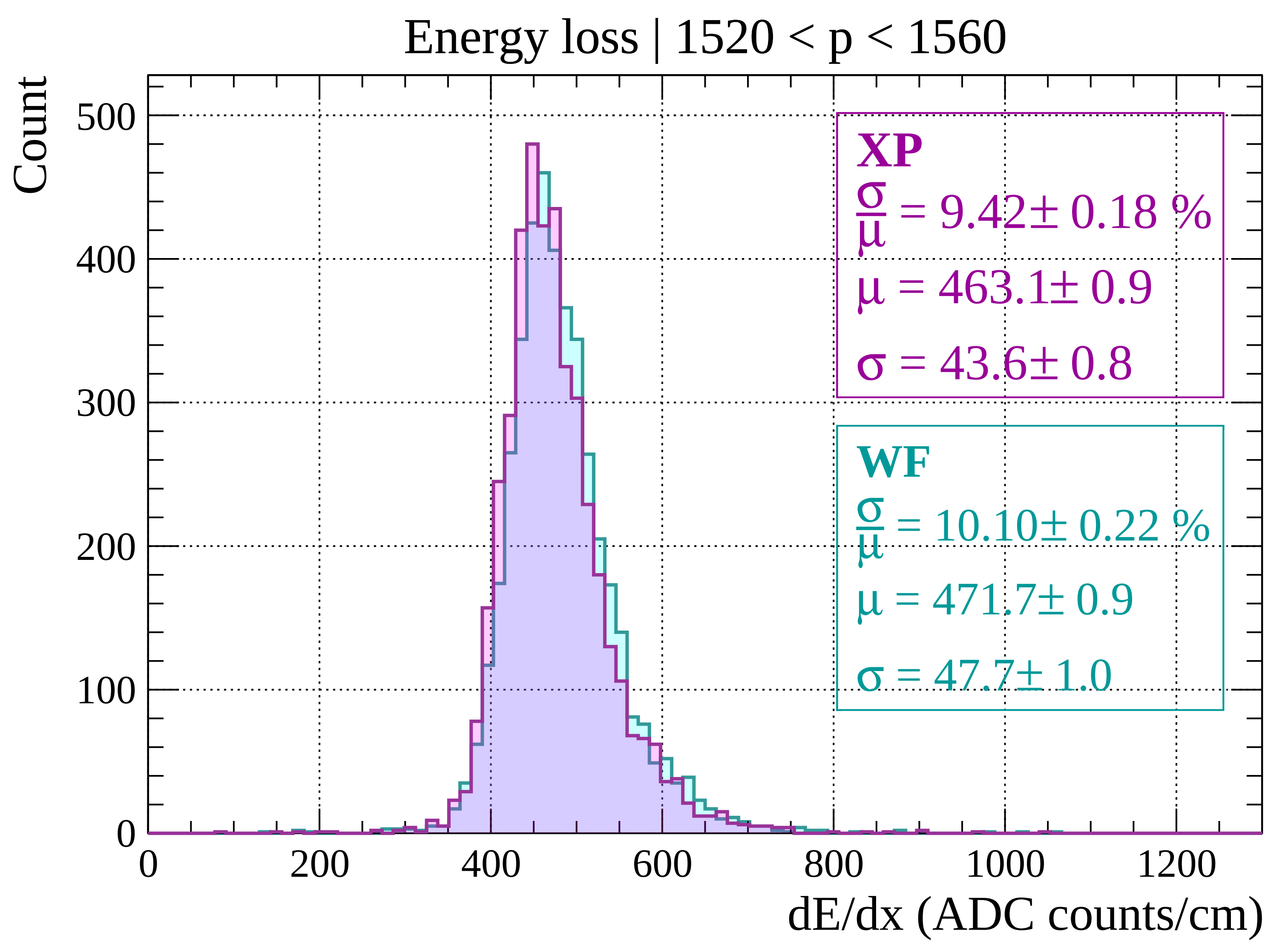
Energy loss | 1400 < p < 1440
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.76± 0.19 %
μ = 458.7± 0.8
σ = 44.8± 0.8
WF
σ/μ = 10.08± 0.21 %
μ = 465.8± 0.9
σ = 46.9± 0.9



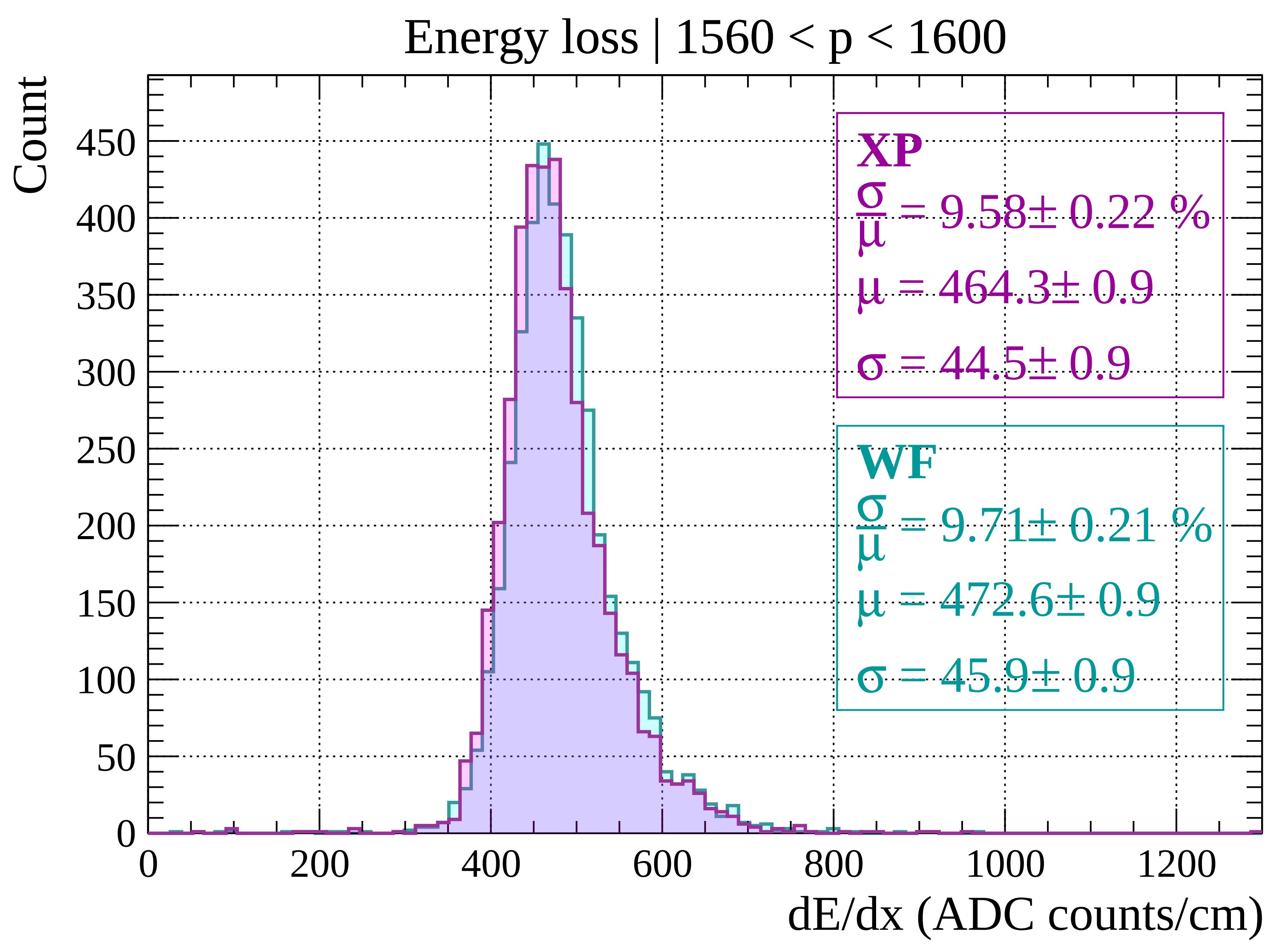
Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.48± 0.18 %
μ = 459.1± 0.8
σ = 43.5± 0.8
WF
σ/μ = 9.94± 0.21 %
μ = 467.6± 0.9
σ = 46.5± 0.9



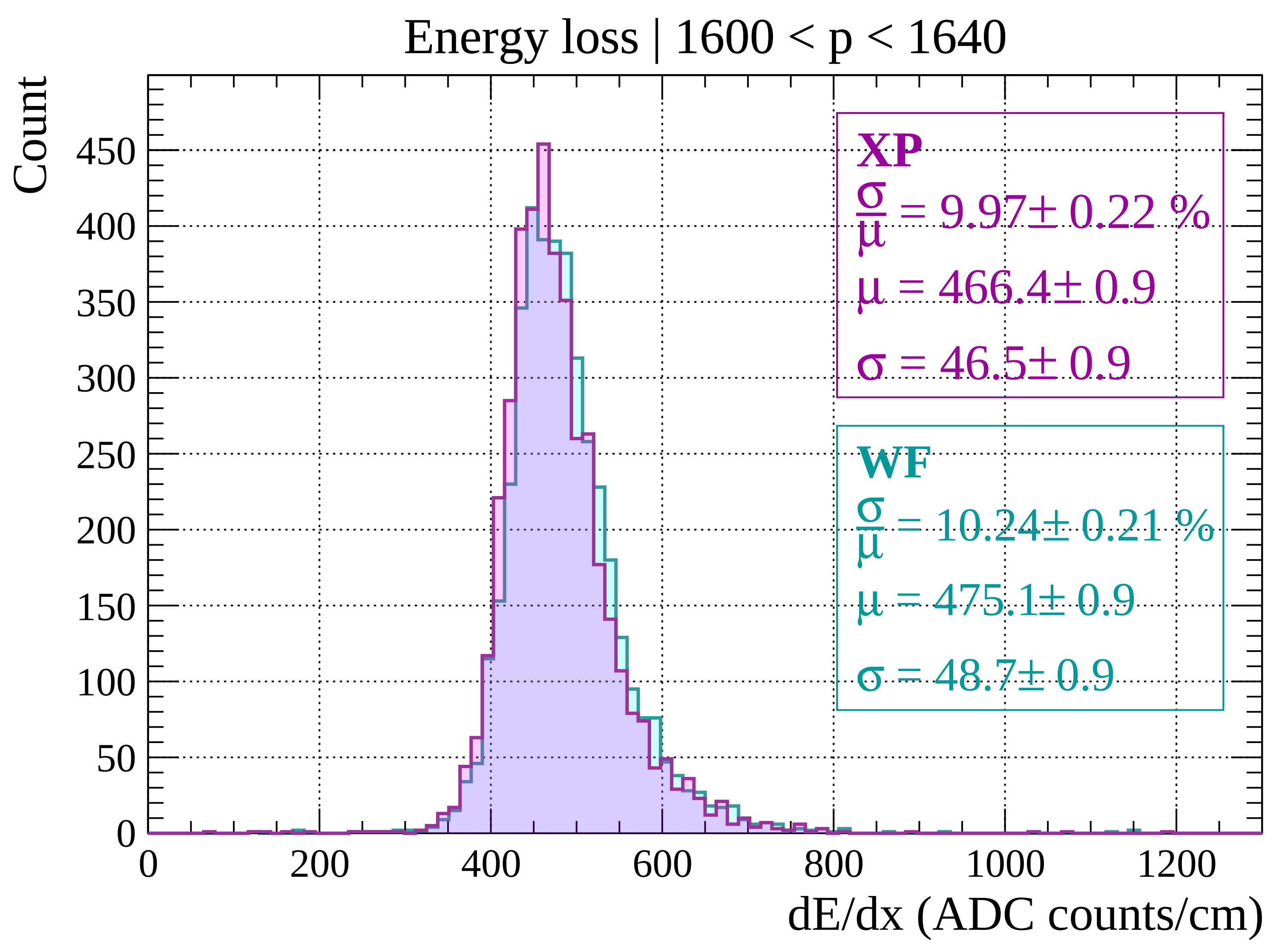
Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
450
XP
σ/μ = 9.85± 0.22 %
μ = 461.8± 0.9
σ = 45.5± 0.9
WF
σ/μ = 10.05± 0.22 %
μ = 469.4± 0.9
σ = 47.2± 1.0



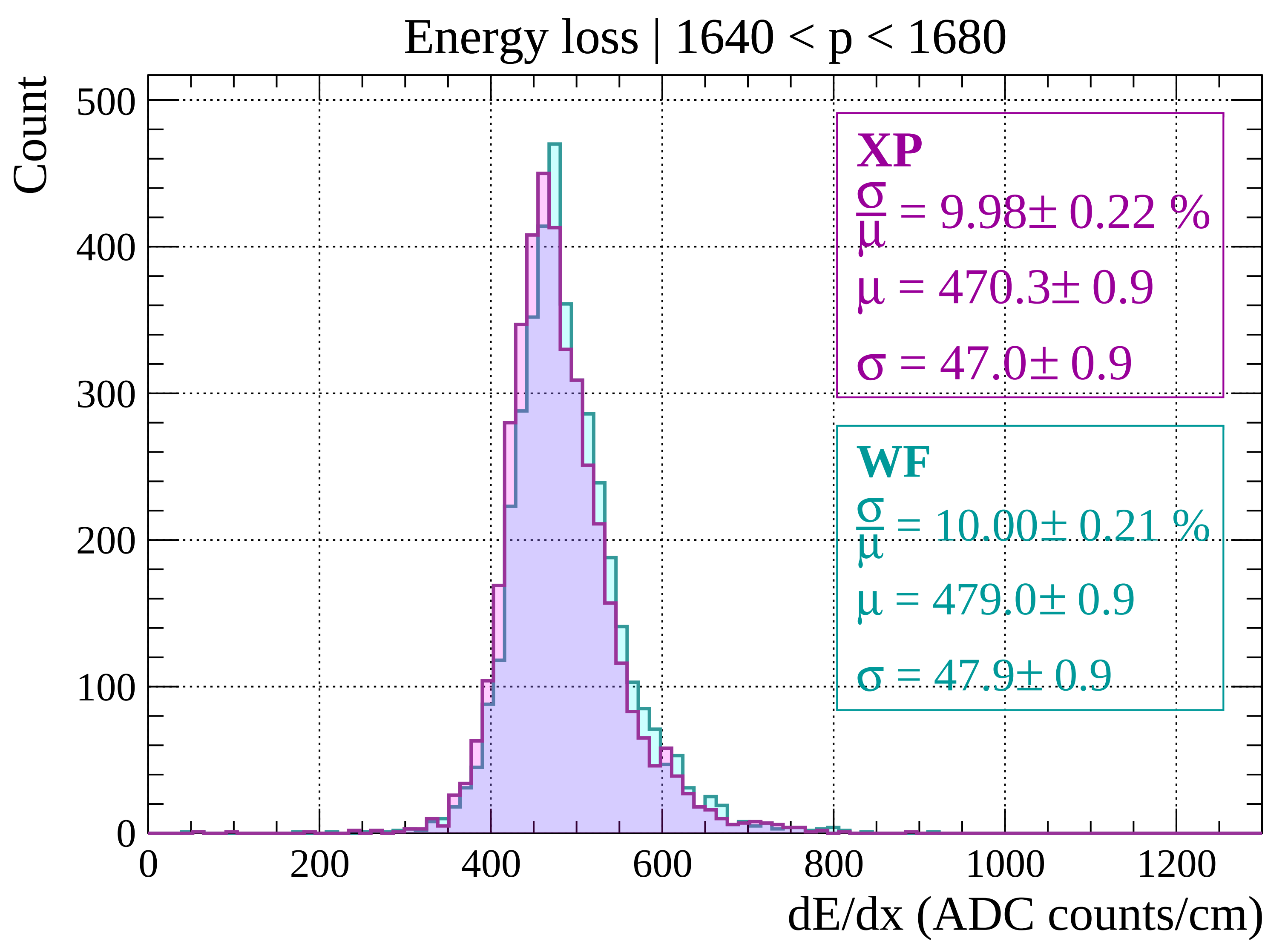
Energy loss | 1520 < p < 1560
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.42± 0.18 %
μ = 463.1± 0.9
σ = 43.6± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.10± 0.22 %
μ = 471.7± 0.9
σ = 47.7± 1.0



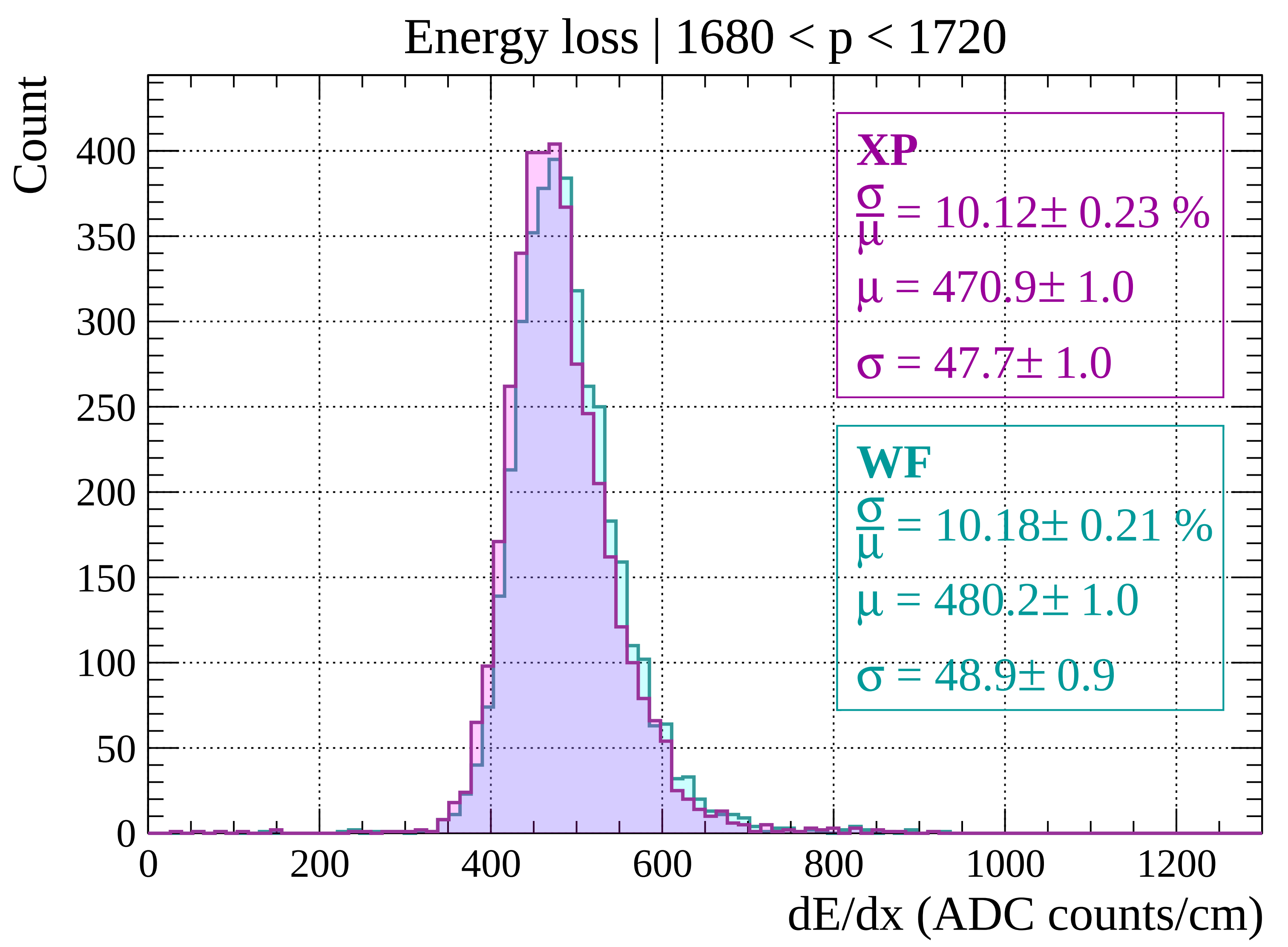
Energy loss | 1560 < p < 1600
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.58 \pm 0.22$ %
$\mu = 464.3 \pm 0.9$
$\sigma = 44.5 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 9.71 \pm 0.21$ %
$\mu = 472.6 \pm 0.9$
$\sigma = 45.9 \pm 0.9$

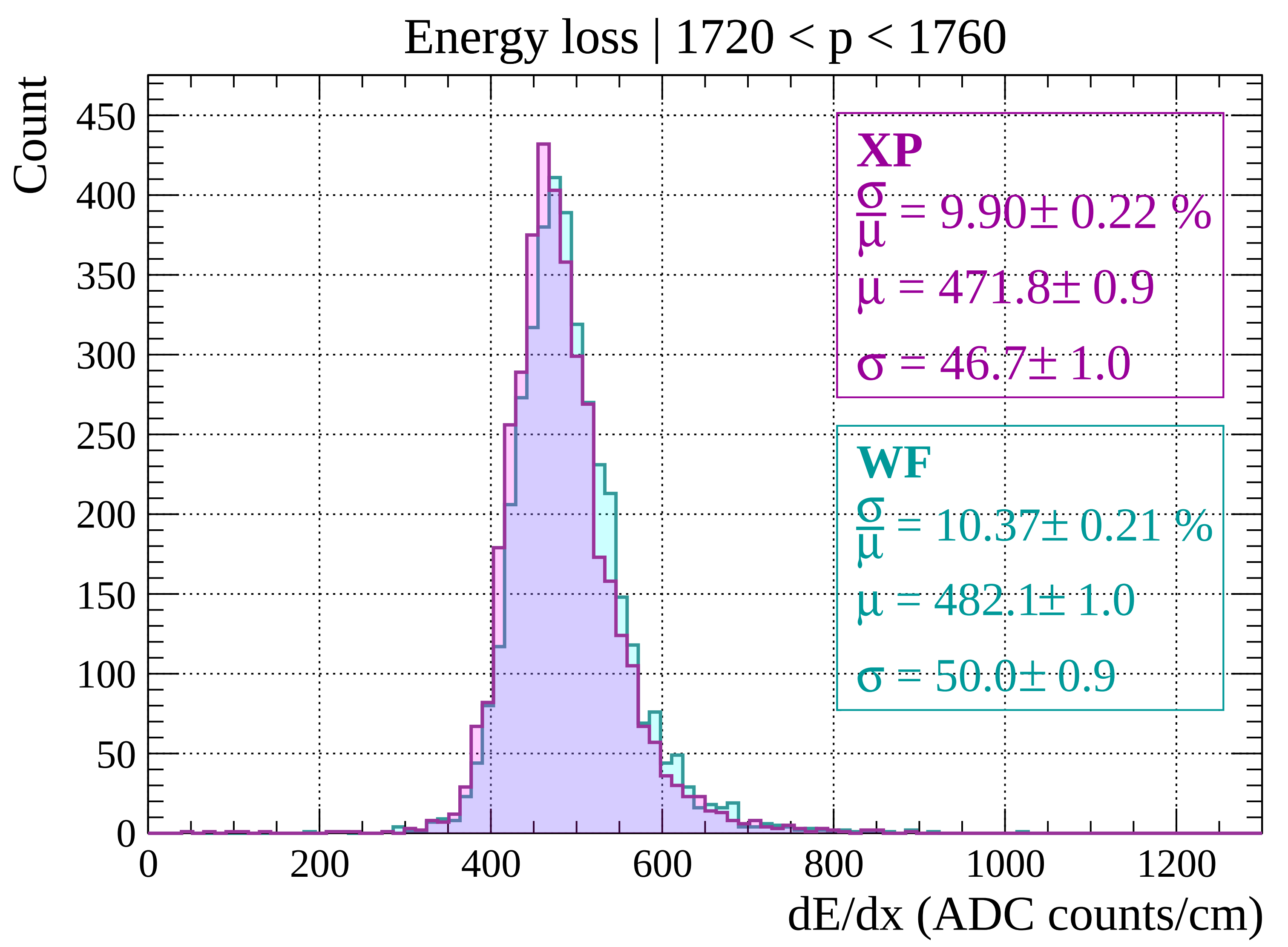


Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
450
XP
$\frac{\sigma}{\mu} = 9.97 \pm 0.22$ %
$\mu = 466.4 \pm 0.9$
$\sigma = 46.5 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 10.24 \pm 0.21$ %
$\mu = 475.1 \pm 0.9$
$\sigma = 48.7 \pm 0.9$

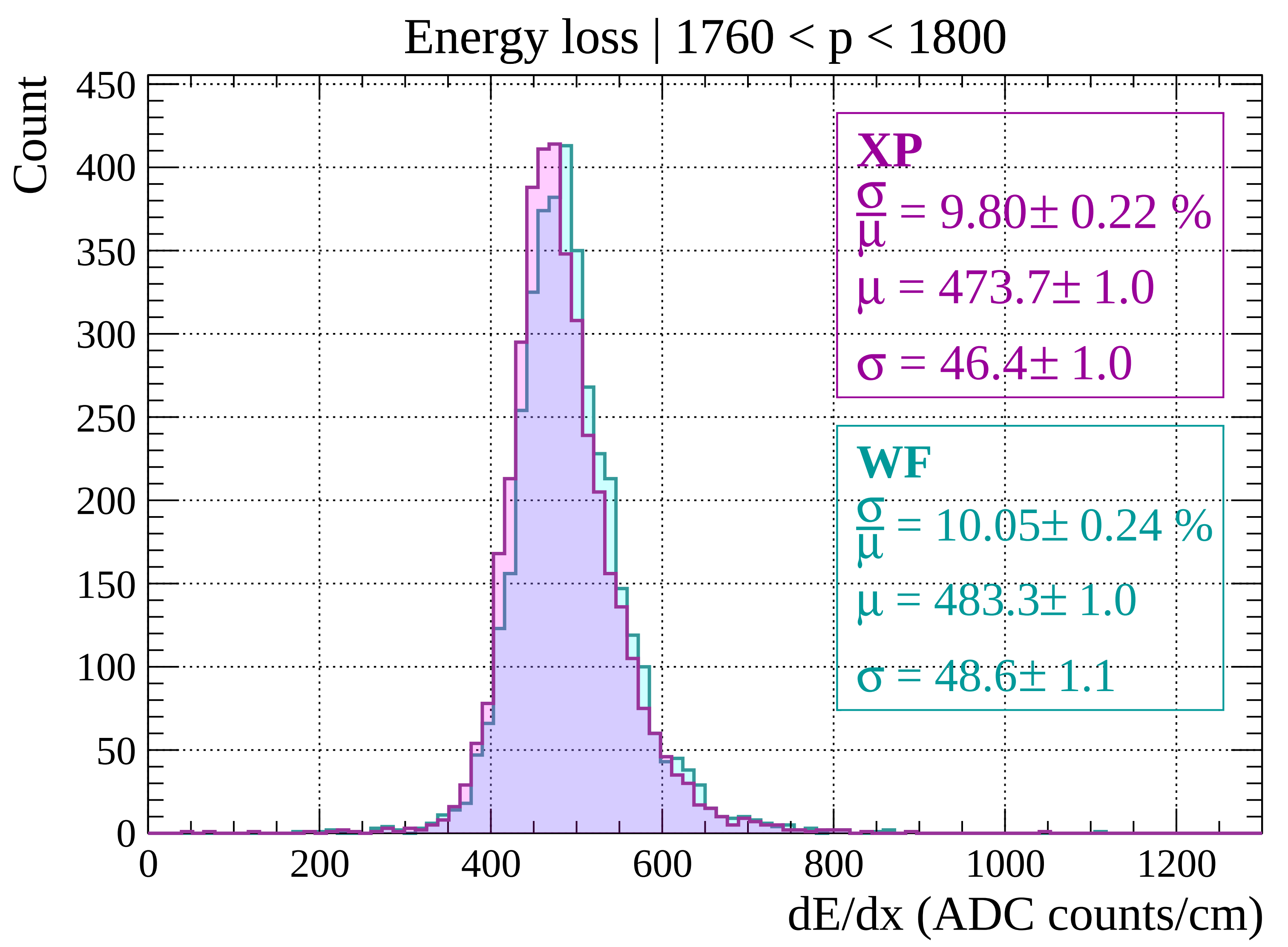


Energy loss | 1640 < p < 1680
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.98± 0.22 %
μ = 470.3± 0.9
σ = 47.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.00± 0.21 %
μ = 479.0± 0.9
σ = 47.9± 0.9

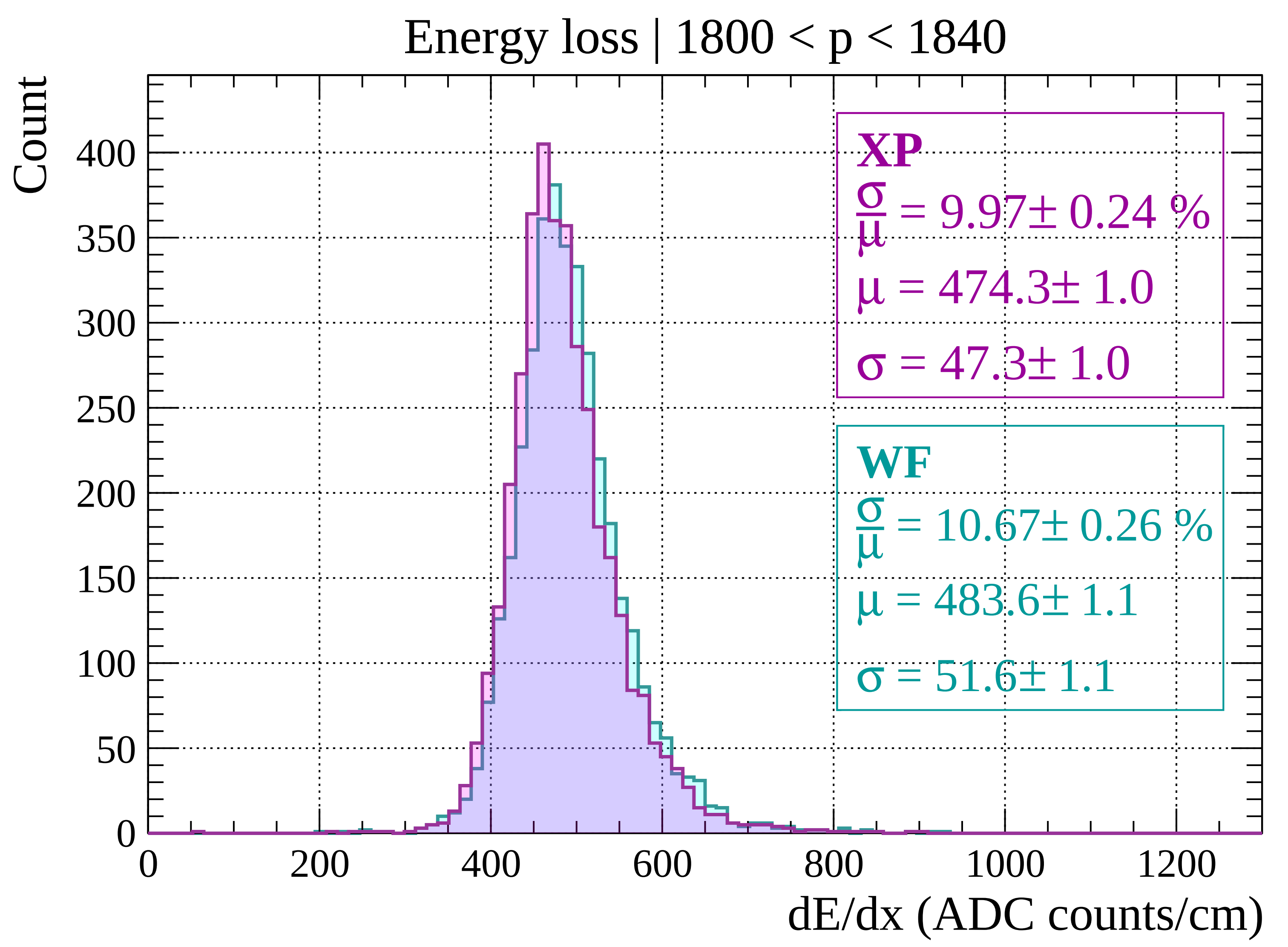

Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.12± 0.23 %
μ = 470.9± 1.0
σ = 47.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 10.18± 0.21 %
μ = 480.2± 1.0
σ = 48.9± 0.9



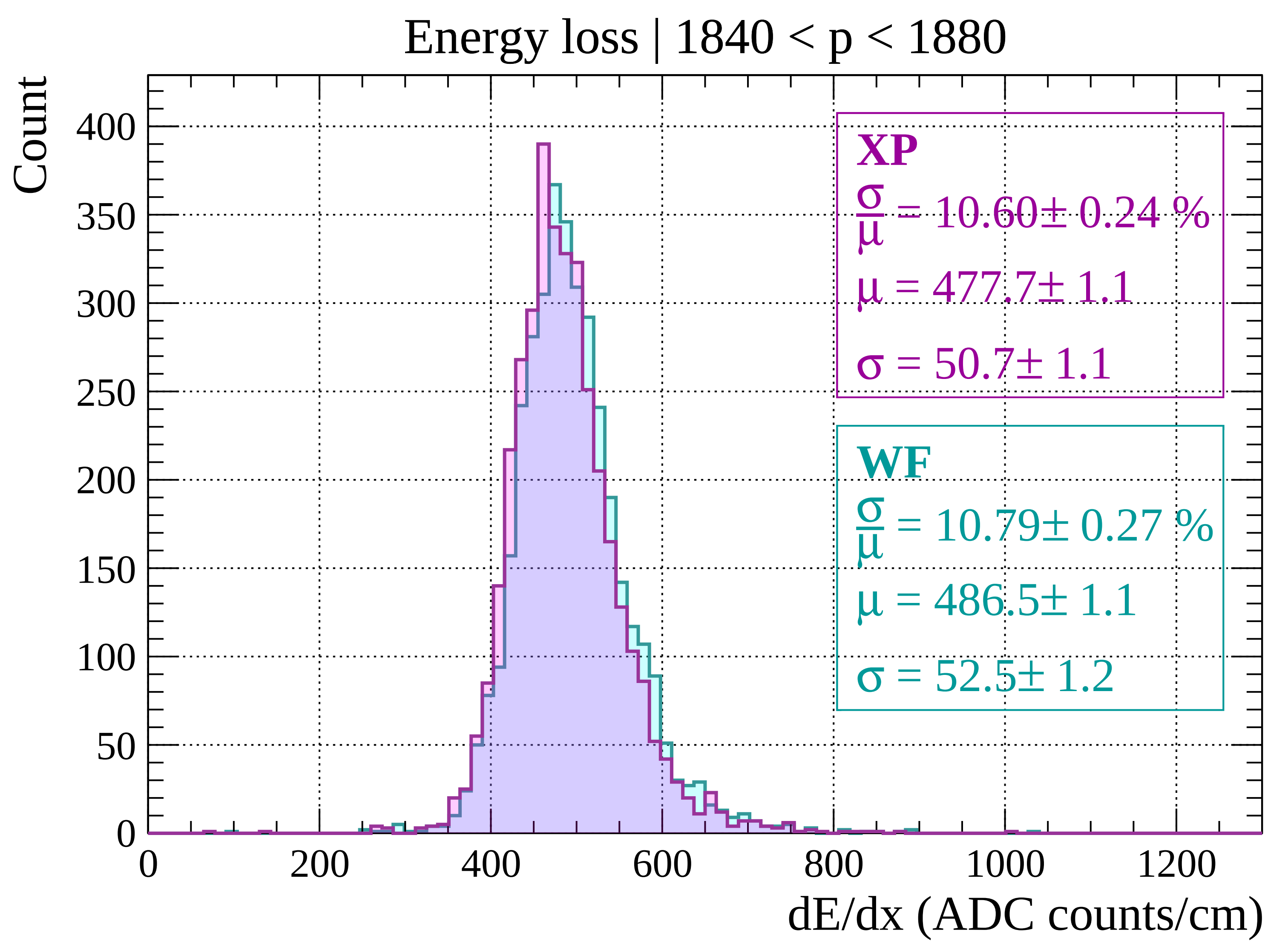
Energy loss | 1720 < p < 1760
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.90± 0.22 %
μ = 471.8± 0.9
σ = 46.7± 1.0
WF
σ/μ = 10.37± 0.21 %
μ = 482.1± 1.0
σ = 50.0± 0.9



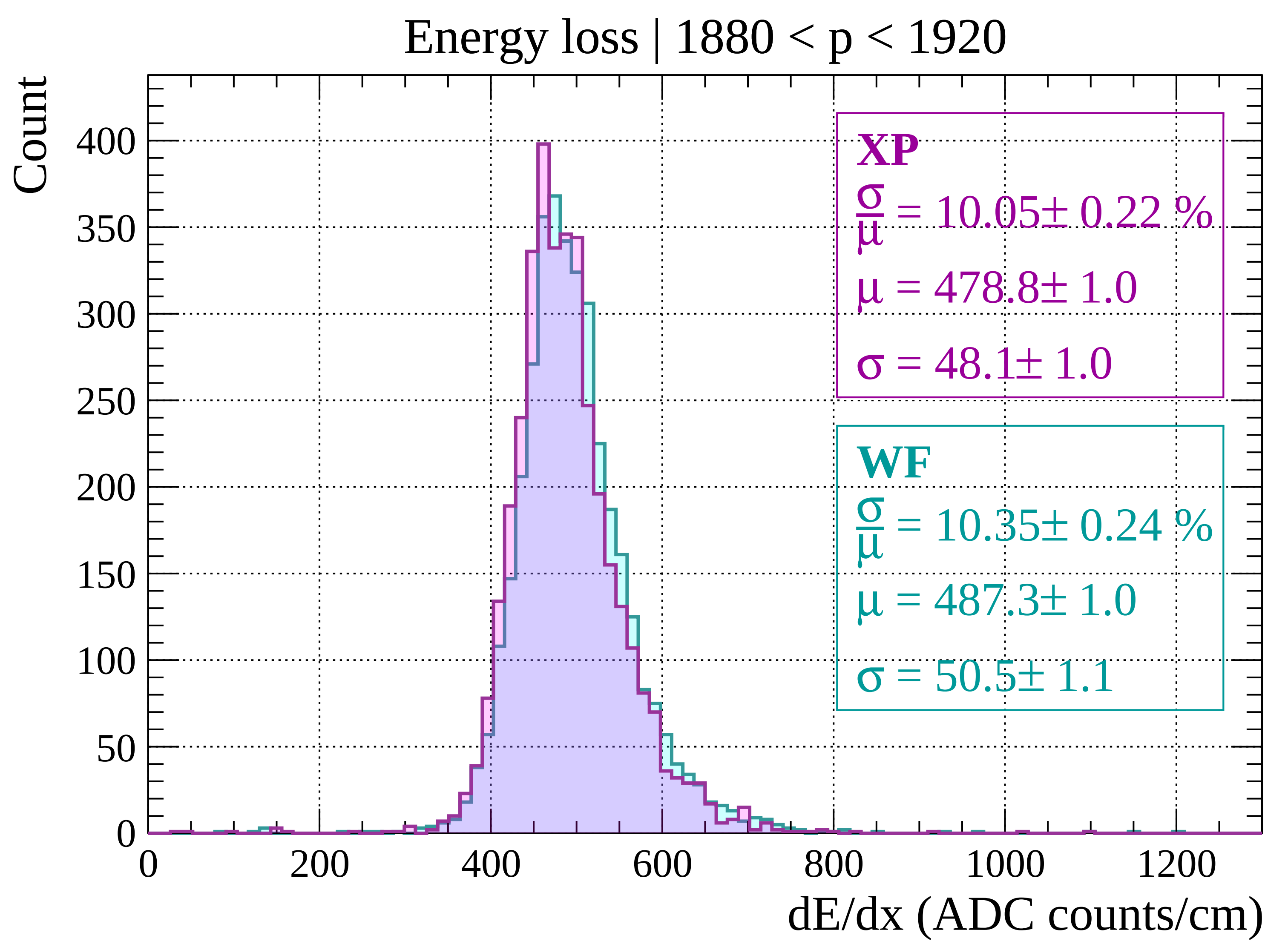
Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP
σ/μ = 9.80± 0.22 %
μ = 473.7± 1.0
σ = 46.4± 1.0
WF
σ/μ = 10.05± 0.24 %
μ = 483.3± 1.0
σ = 48.6± 1.1



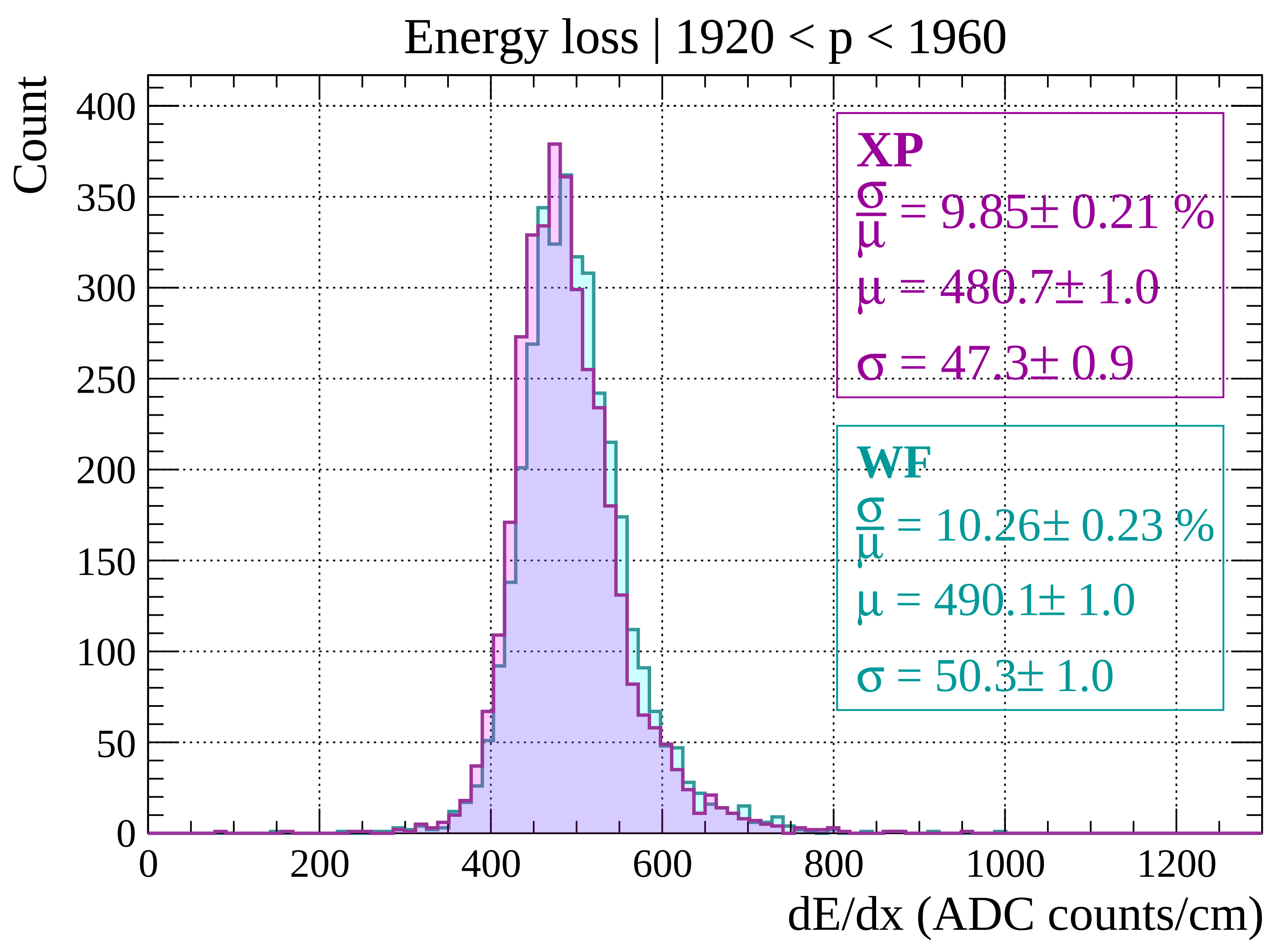
Energy loss | 1800 < p < 1840
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.97± 0.24 %
μ = 474.3± 1.0
σ = 47.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 10.67± 0.26 %
μ = 483.6± 1.1
σ = 51.6± 1.1



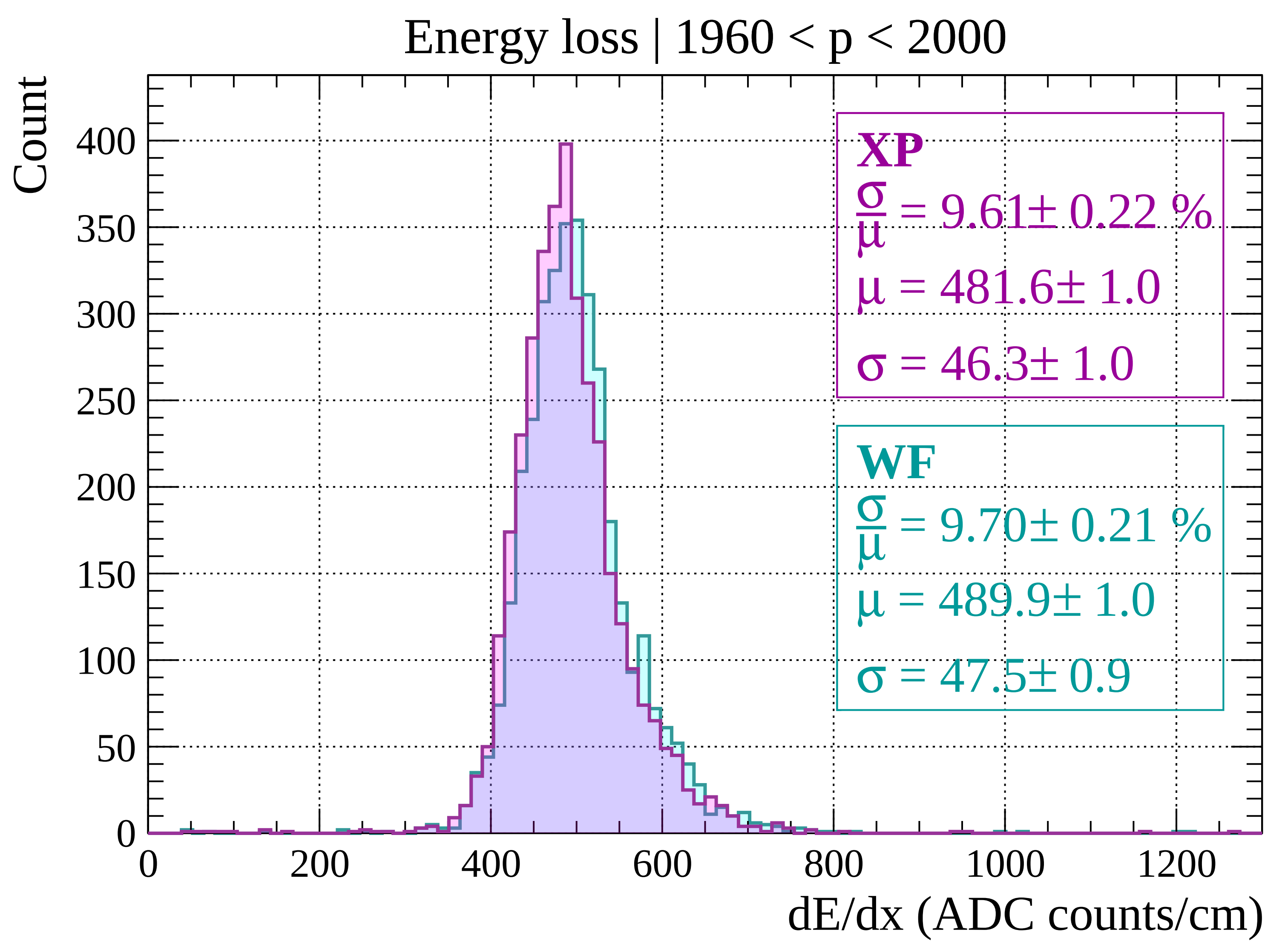
Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 10.60± 0.24 %
μ = 477.7± 1.1
σ = 50.7± 1.1
WF
σ/μ = 10.79± 0.27 %
μ = 486.5± 1.1
σ = 52.5± 1.2



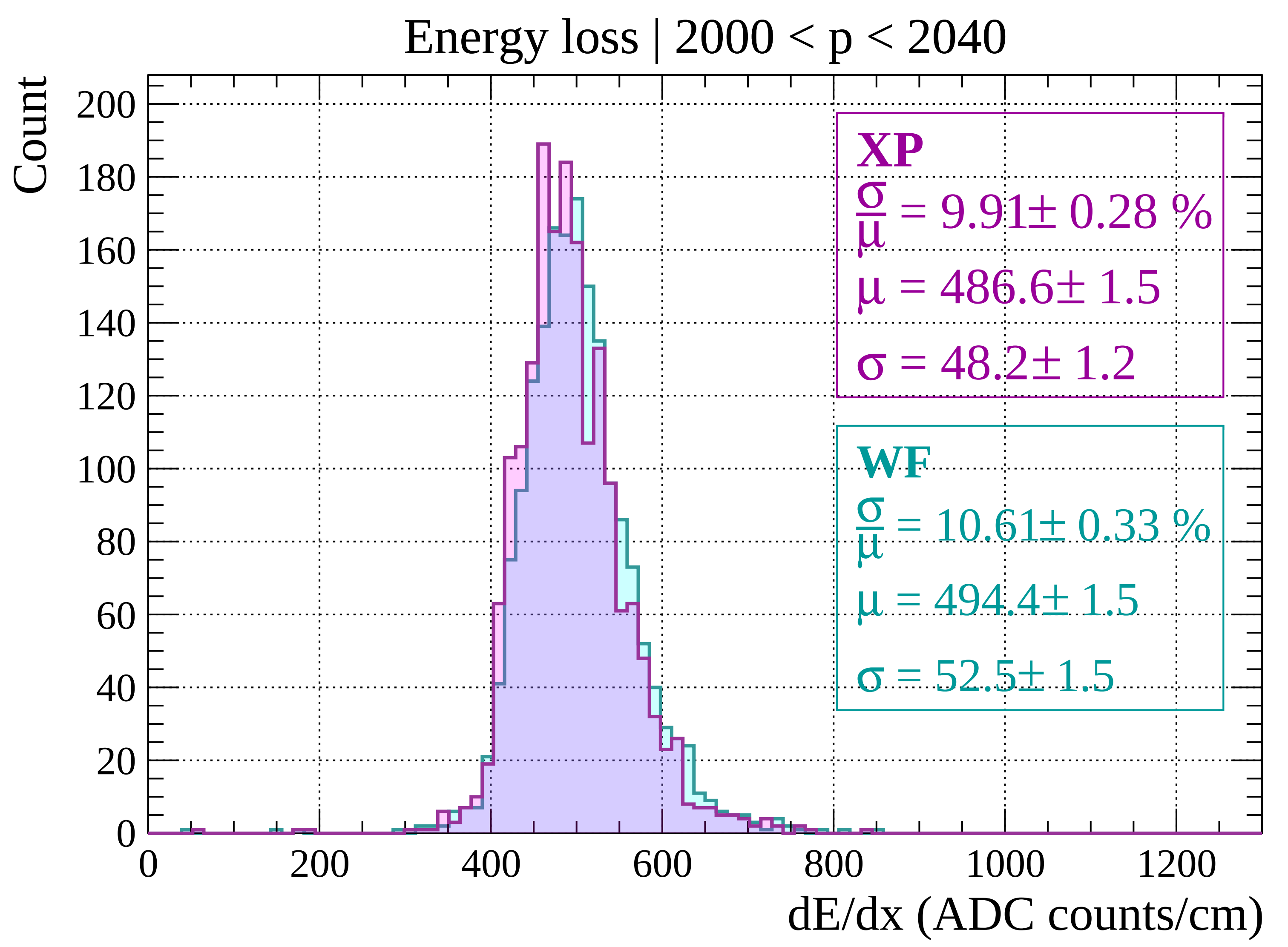
Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.05± 0.22 %
μ = 478.8± 1.0
σ = 48.1± 1.0
WF
$\frac{\sigma}{\mu}$ = 10.35± 0.24 %
μ = 487.3± 1.0
σ = 50.5± 1.1



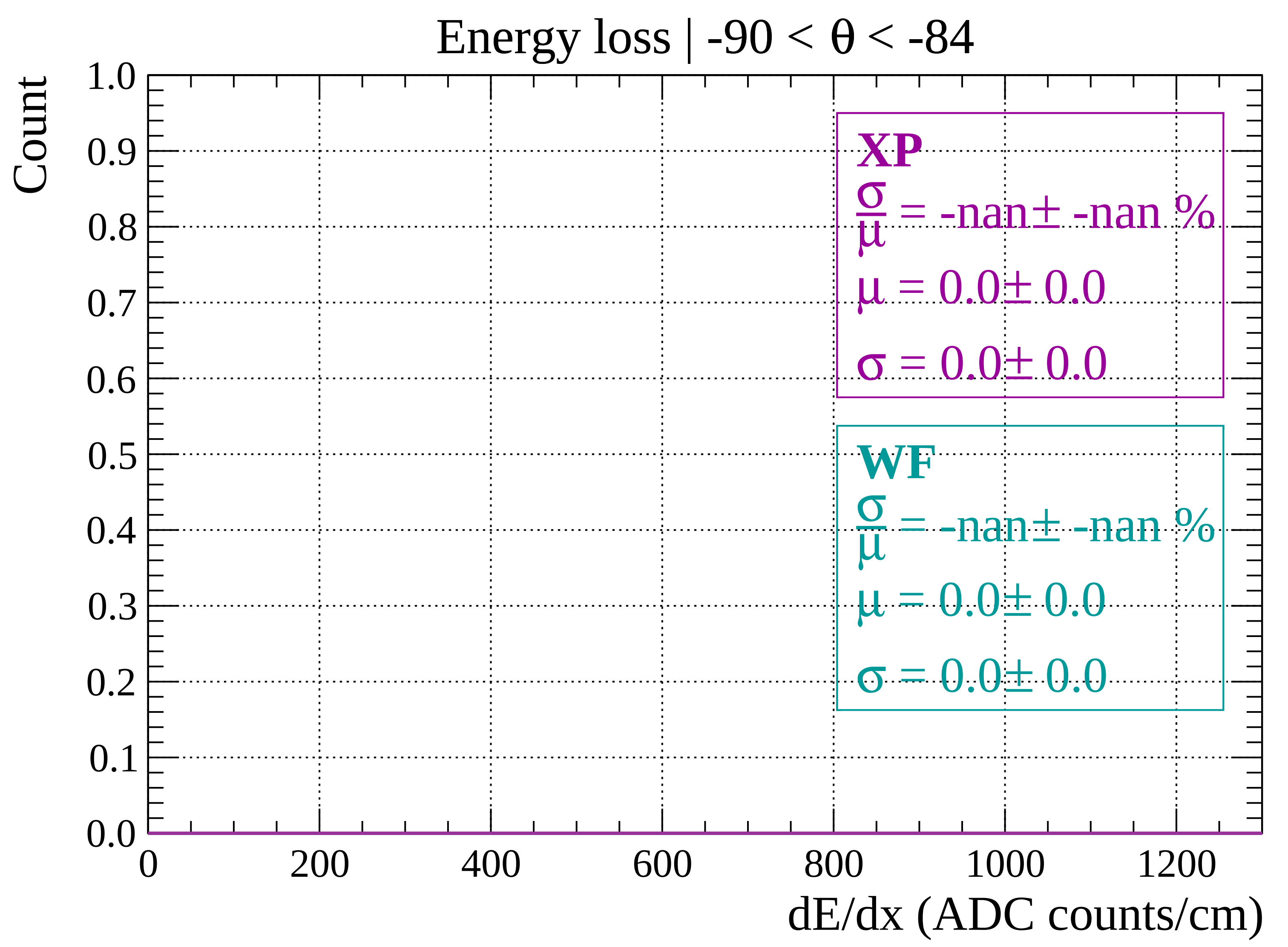
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 9.85± 0.21 %
μ = 480.7± 1.0
σ = 47.3± 0.9
WF
σ/μ = 10.26± 0.23 %
μ = 490.1± 1.0
σ = 50.3± 1.0

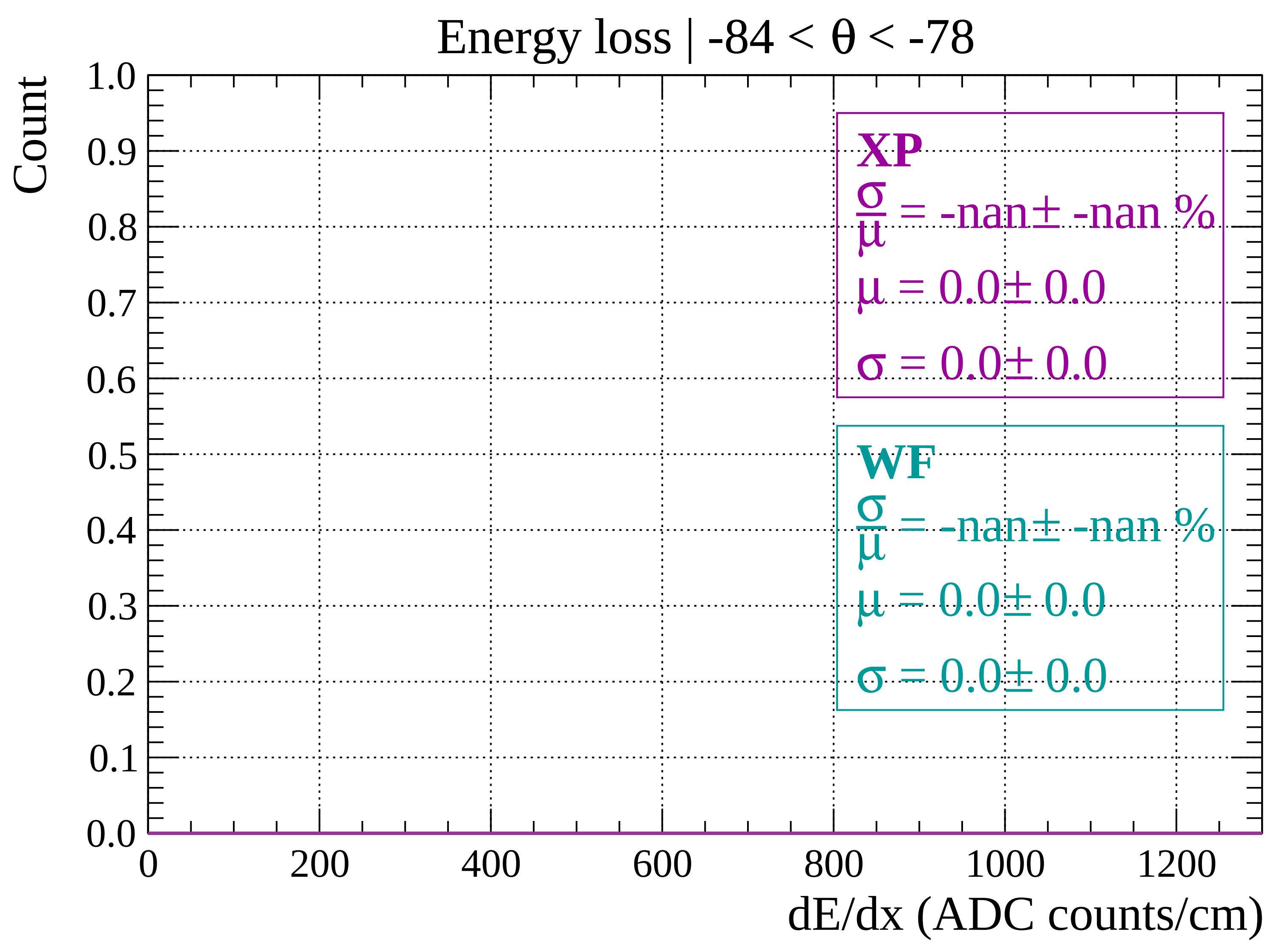


Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP
σ/μ = 9.61± 0.22 %
μ = 481.6± 1.0
σ = 46.3± 1.0
WF
σ/μ = 9.70± 0.21 %
μ = 489.9± 1.0
σ = 47.5± 0.9

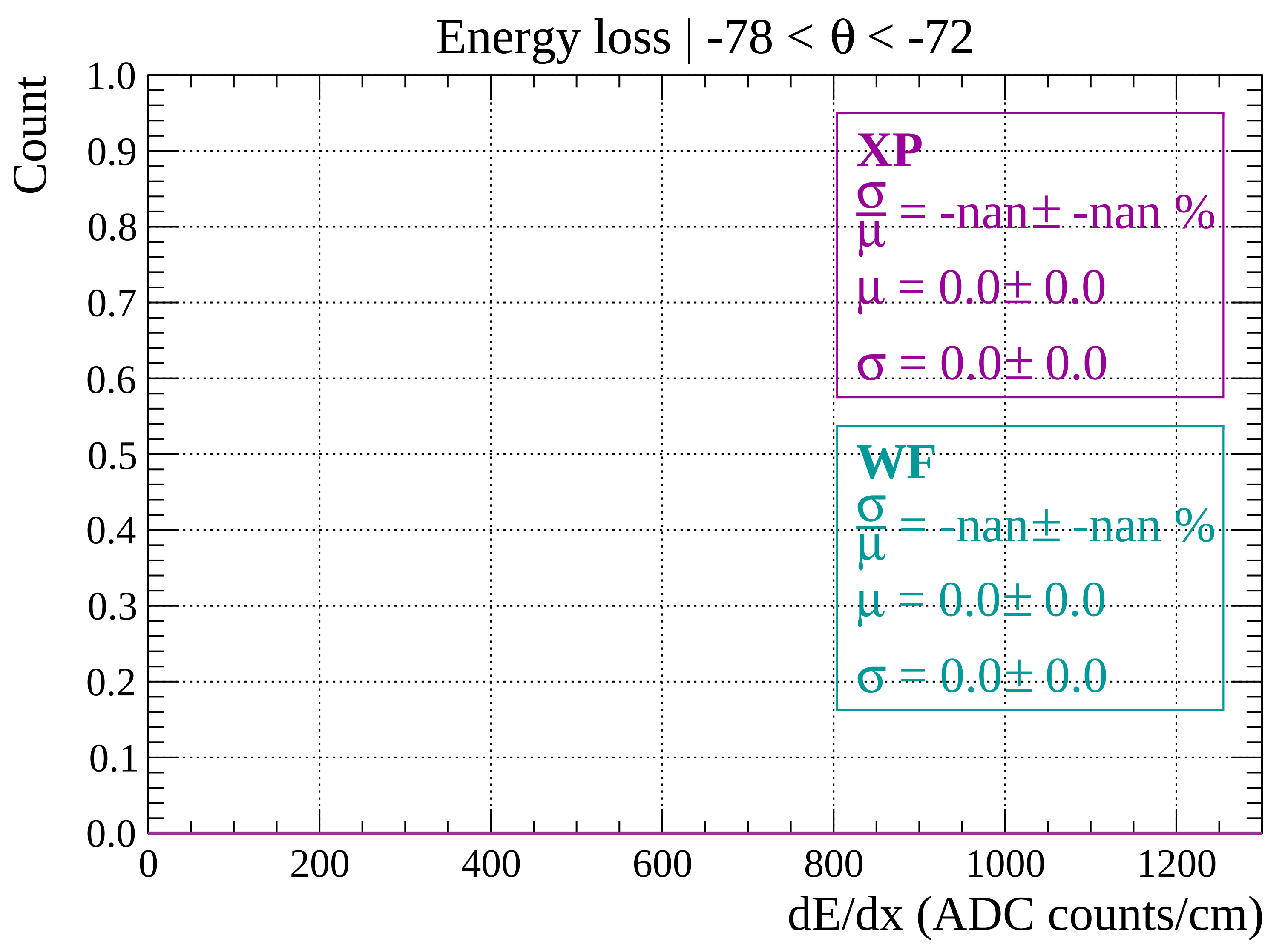


Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.91± 0.28 %
μ = 486.6± 1.5
σ = 48.2± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.61± 0.33 %
μ = 494.4± 1.5
σ = 52.5± 1.5

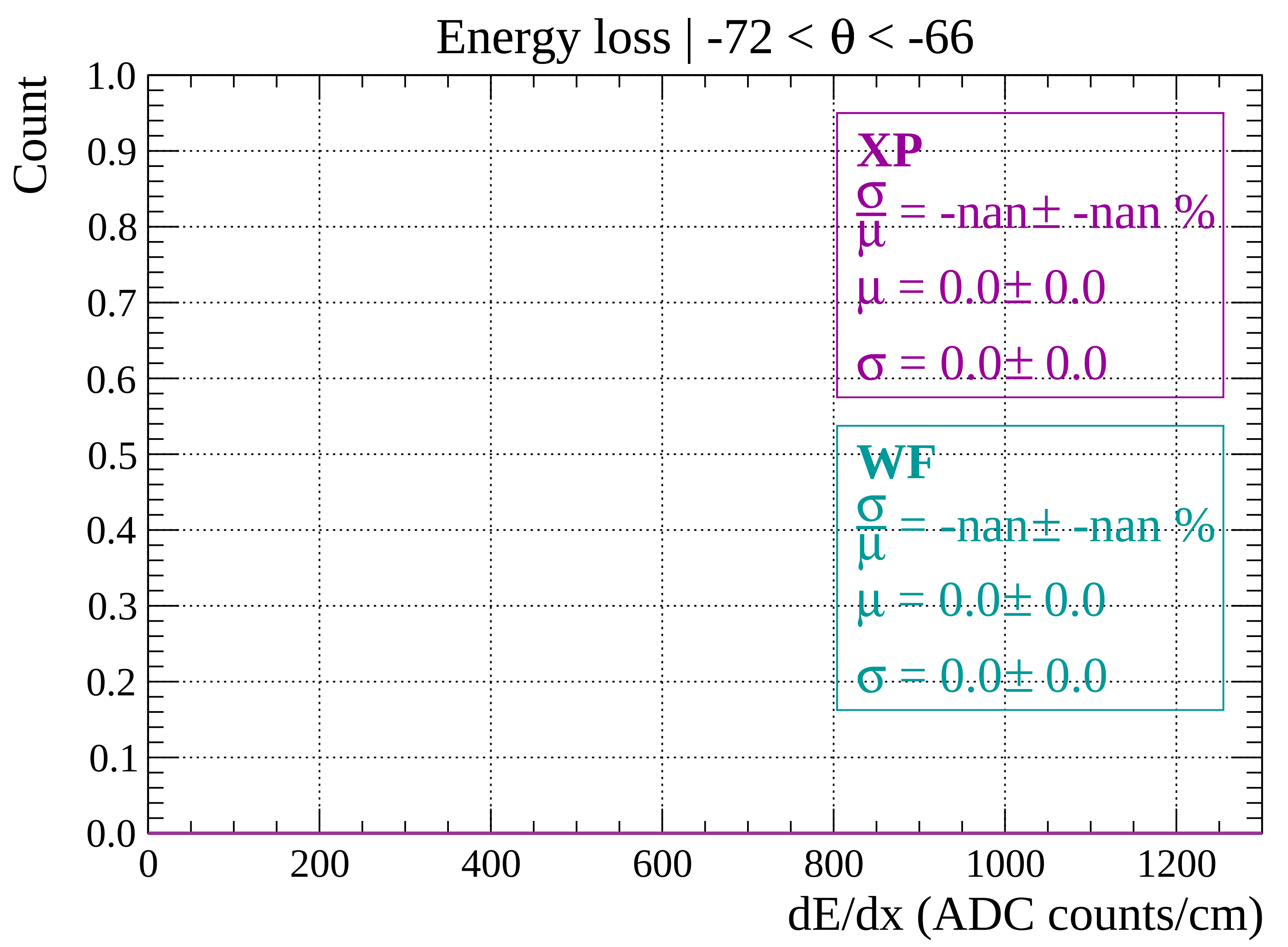

Energy loss | -90 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



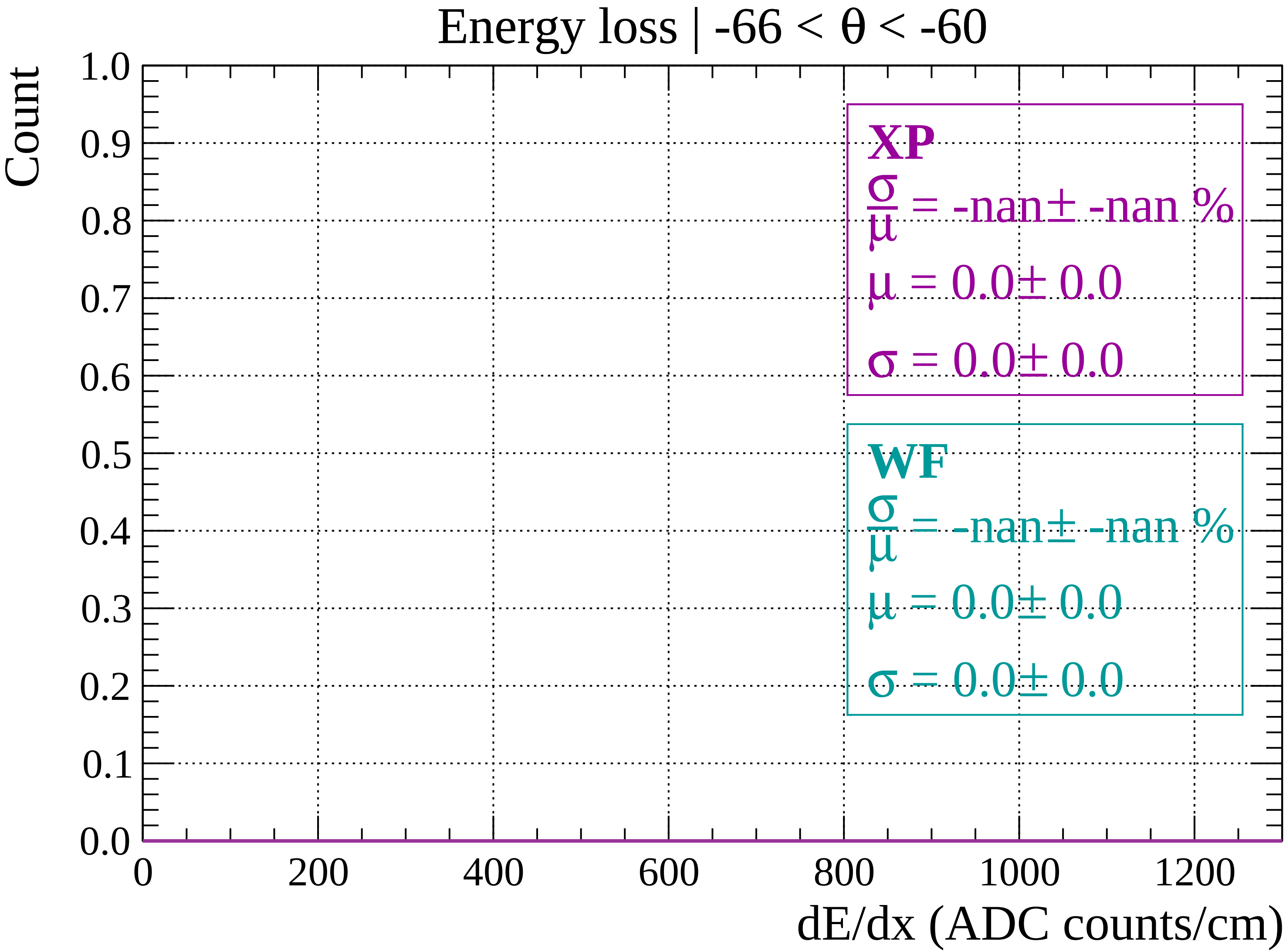
Energy loss | -84 < θ < -78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

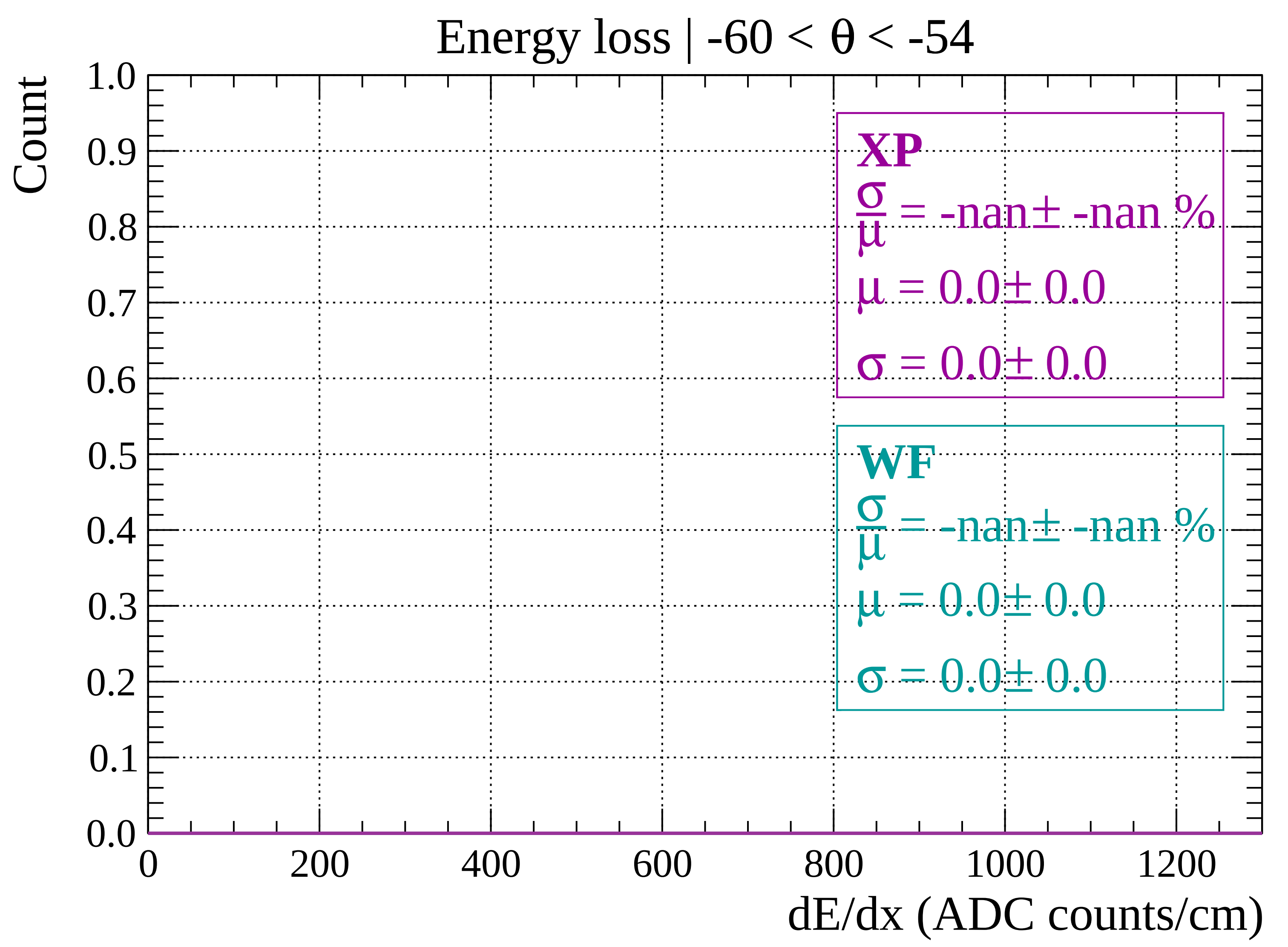


Energy loss | -78 < θ < -72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

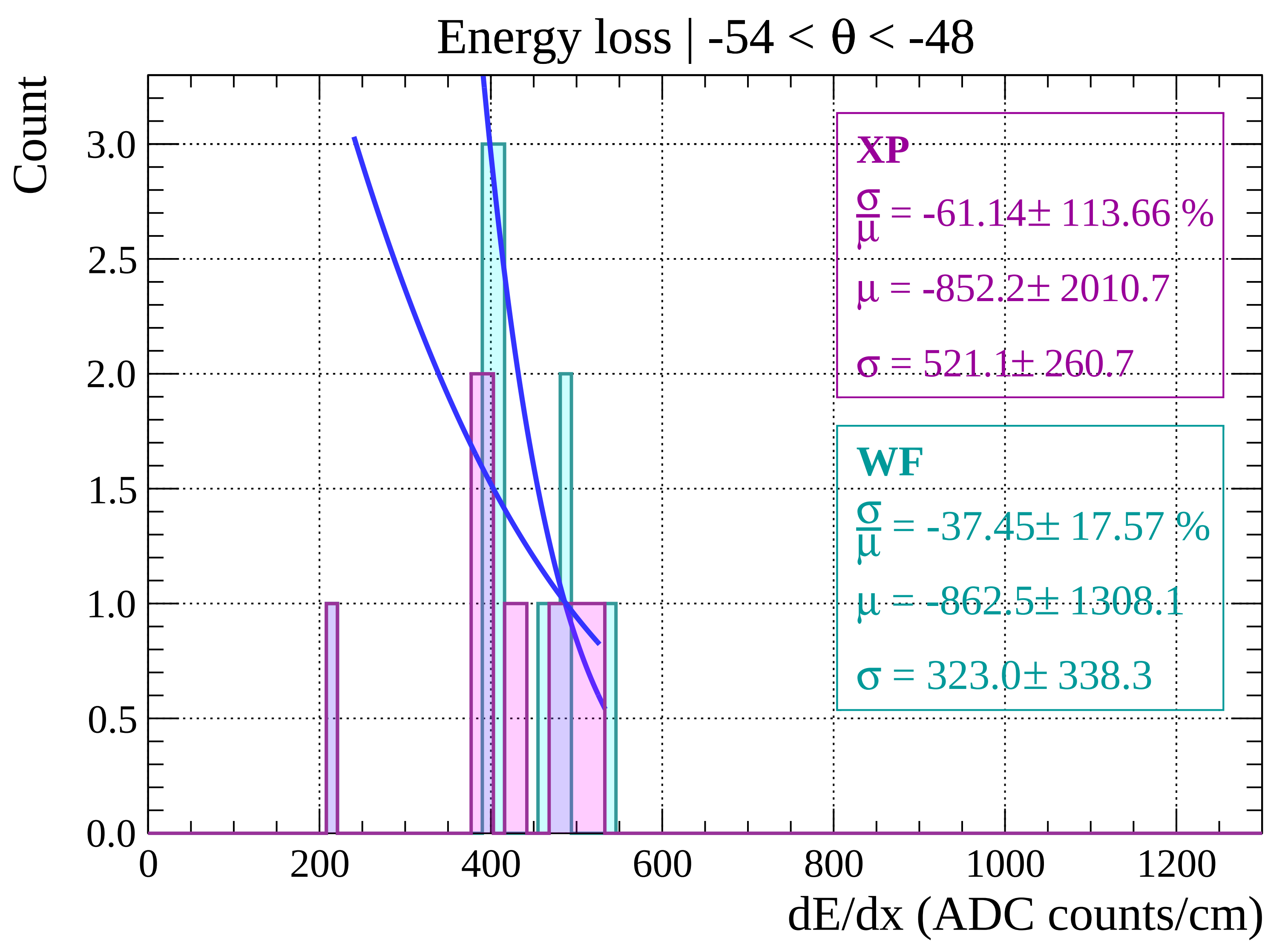


Energy loss | -72 < θ < -66
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

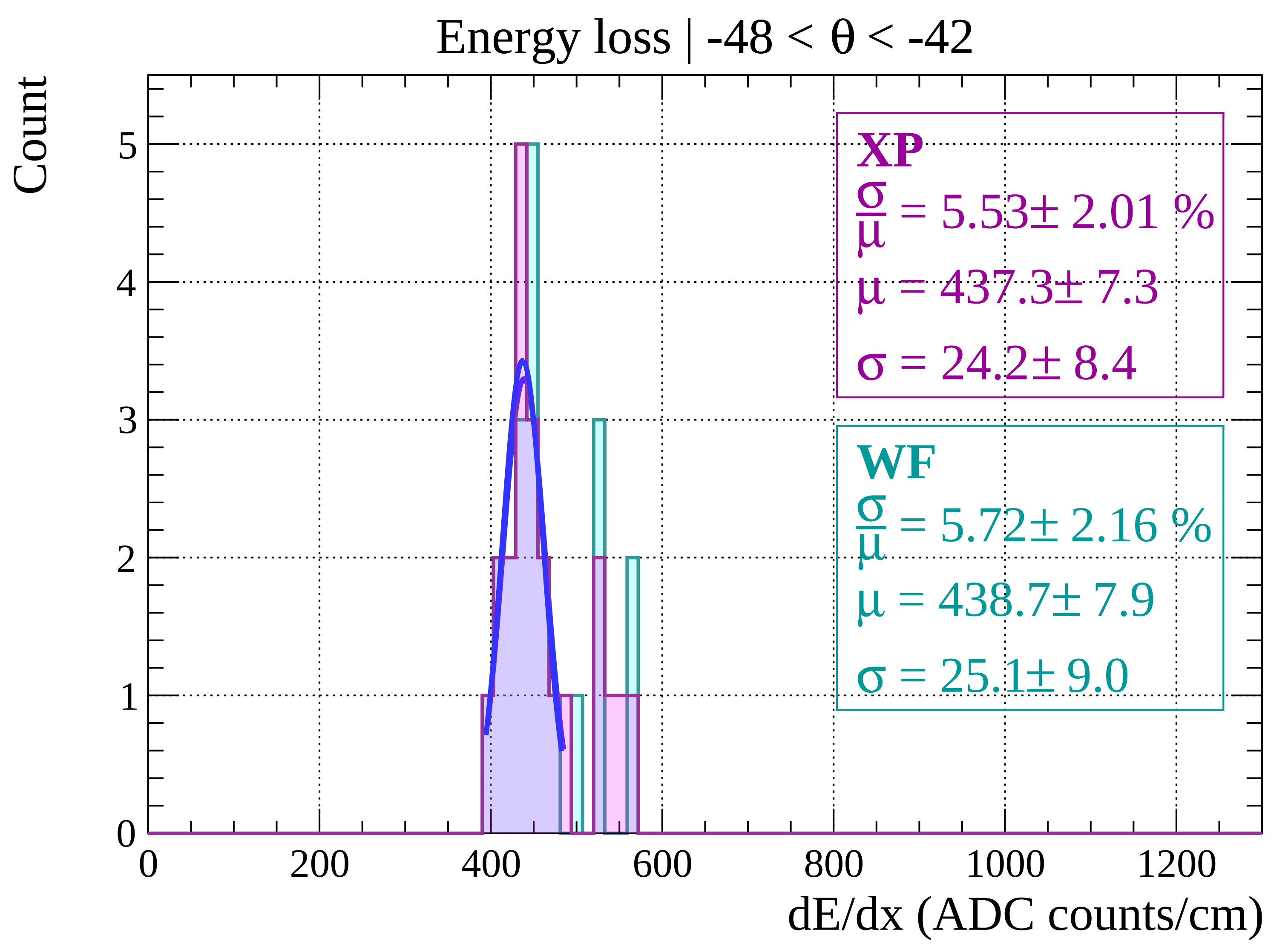

Energy loss | -66 < θ < -60
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



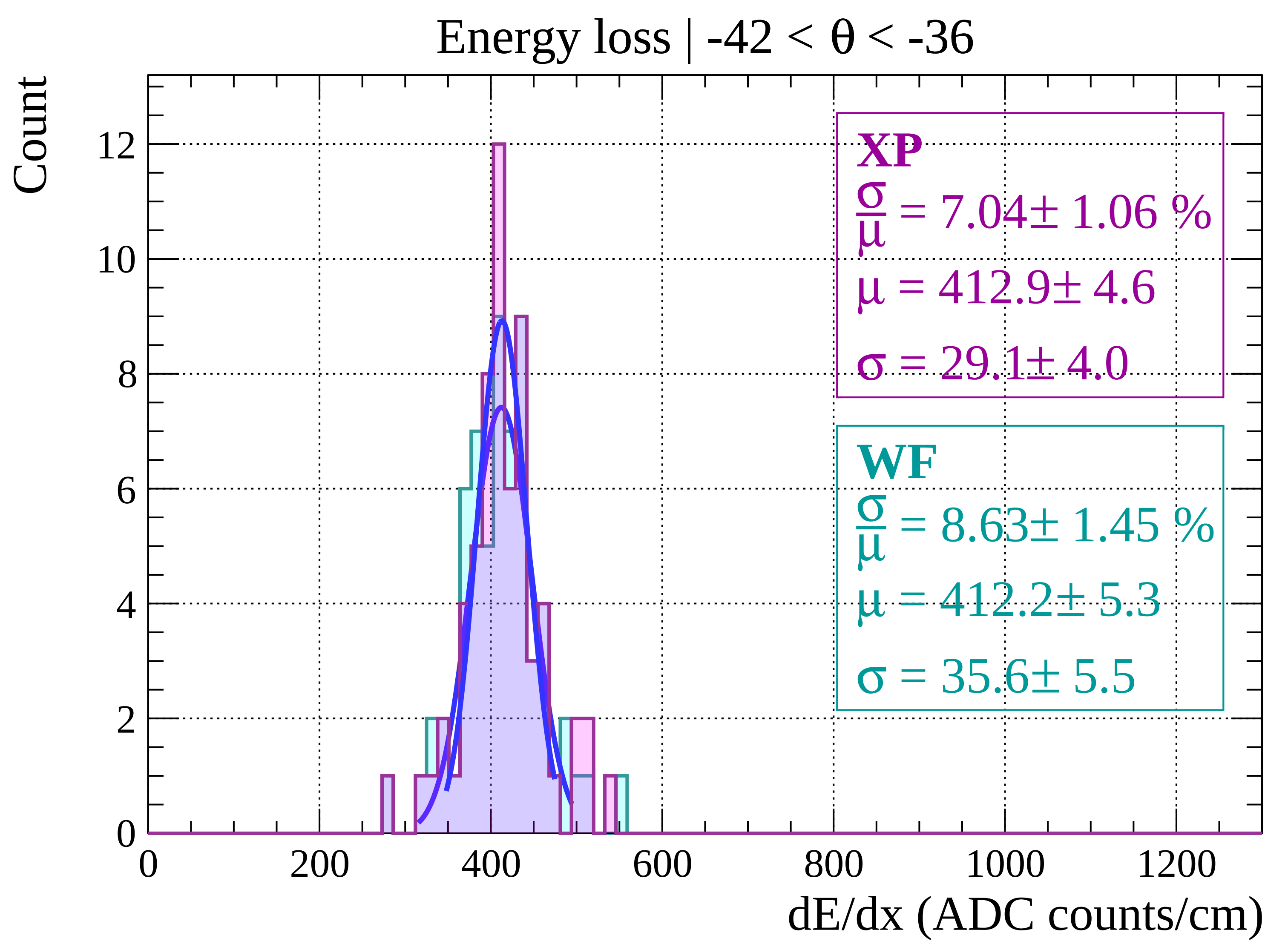
Energy loss | -60 < θ < -54
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



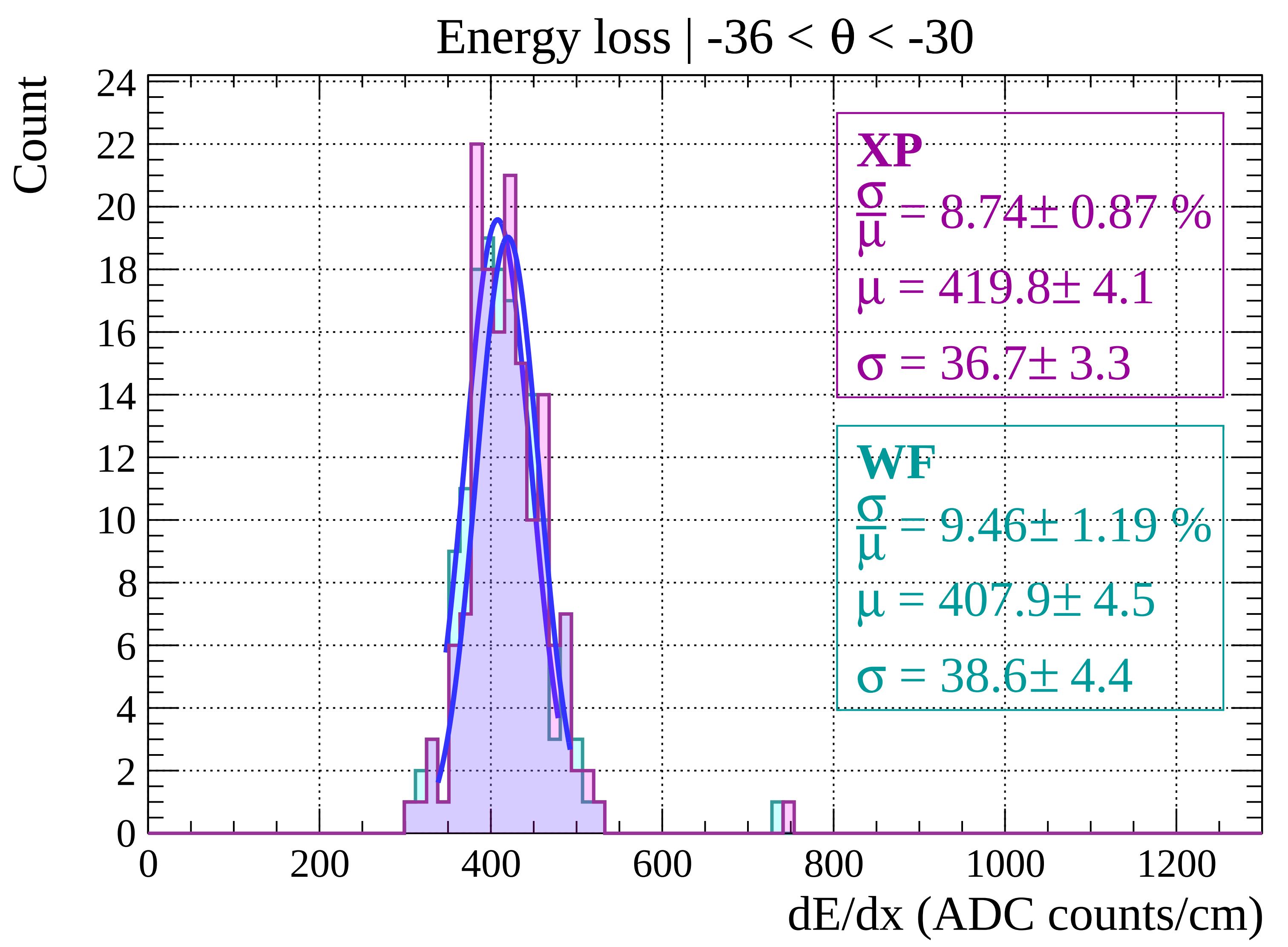
Energy loss | -54 < θ < -48
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -61.14± 113.66 %
μ = -852.2± 2010.7
σ = 521.1± 260.7
WF
σ/μ = -37.45± 17.57 %
μ = -862.5± 1308.1
σ = 323.0± 338.3

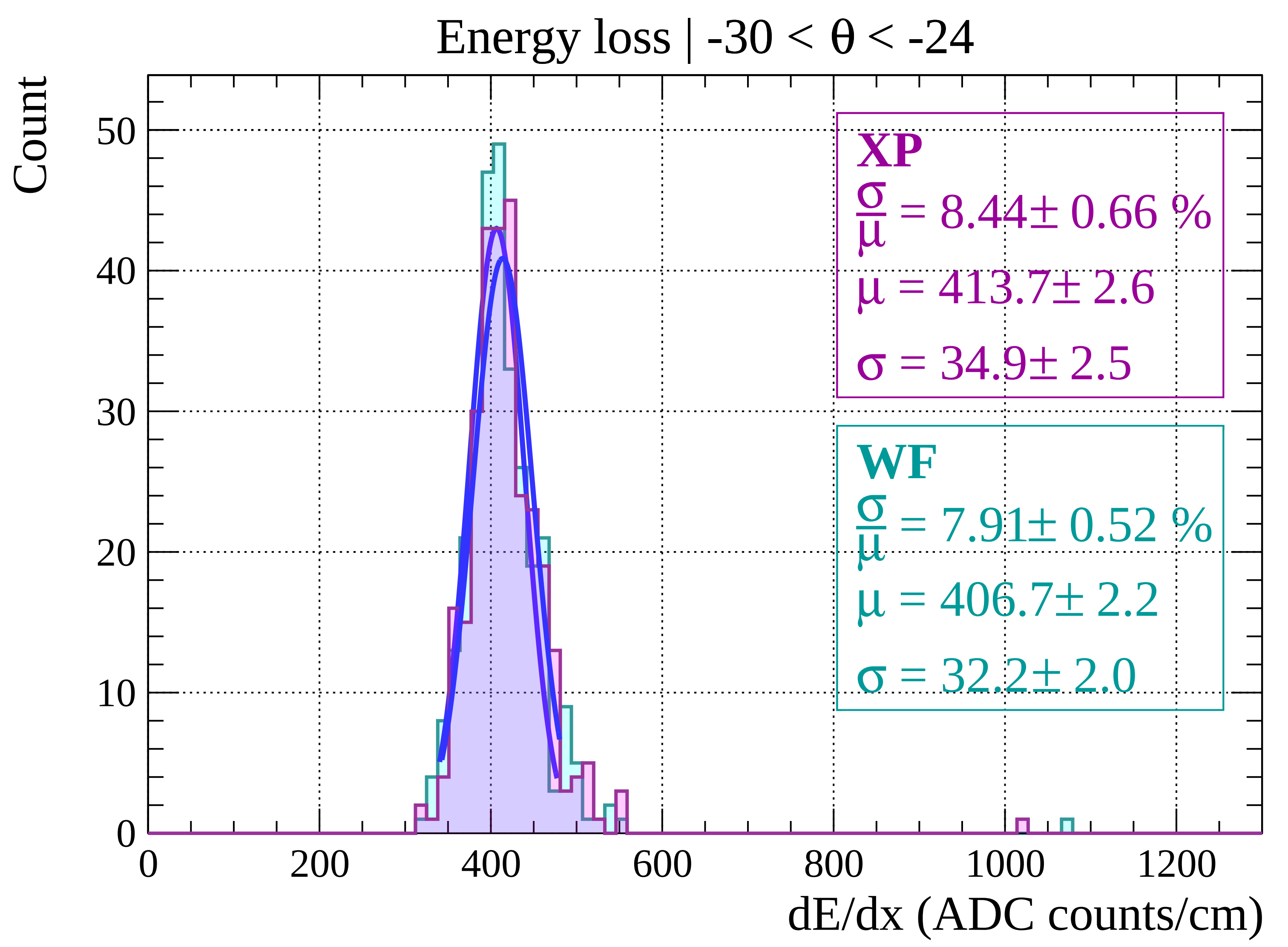


Energy loss | -48 < θ < -42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 5.53± 2.01 %
μ = 437.3± 7.3
σ = 24.2± 8.4
WF
σ/μ = 5.72± 2.16 %
μ = 438.7± 7.9
σ = 25.1± 9.0

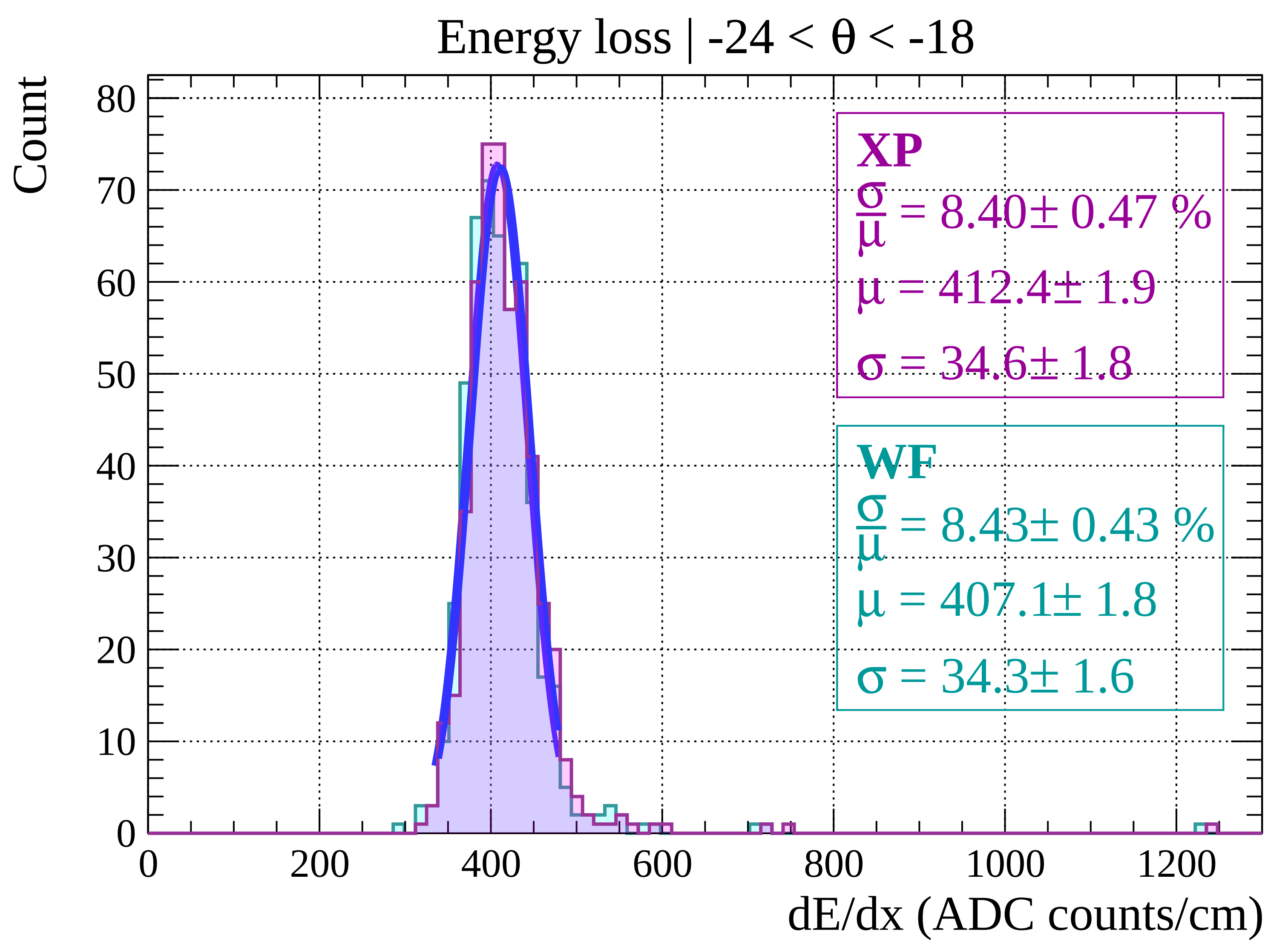


Energy loss | -42 < θ < -36
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 7.04± 1.06 %
μ = 412.9± 4.6
σ = 29.1± 4.0
WF
σ/μ = 8.63± 1.45 %
μ = 412.2± 5.3
σ = 35.6± 5.5

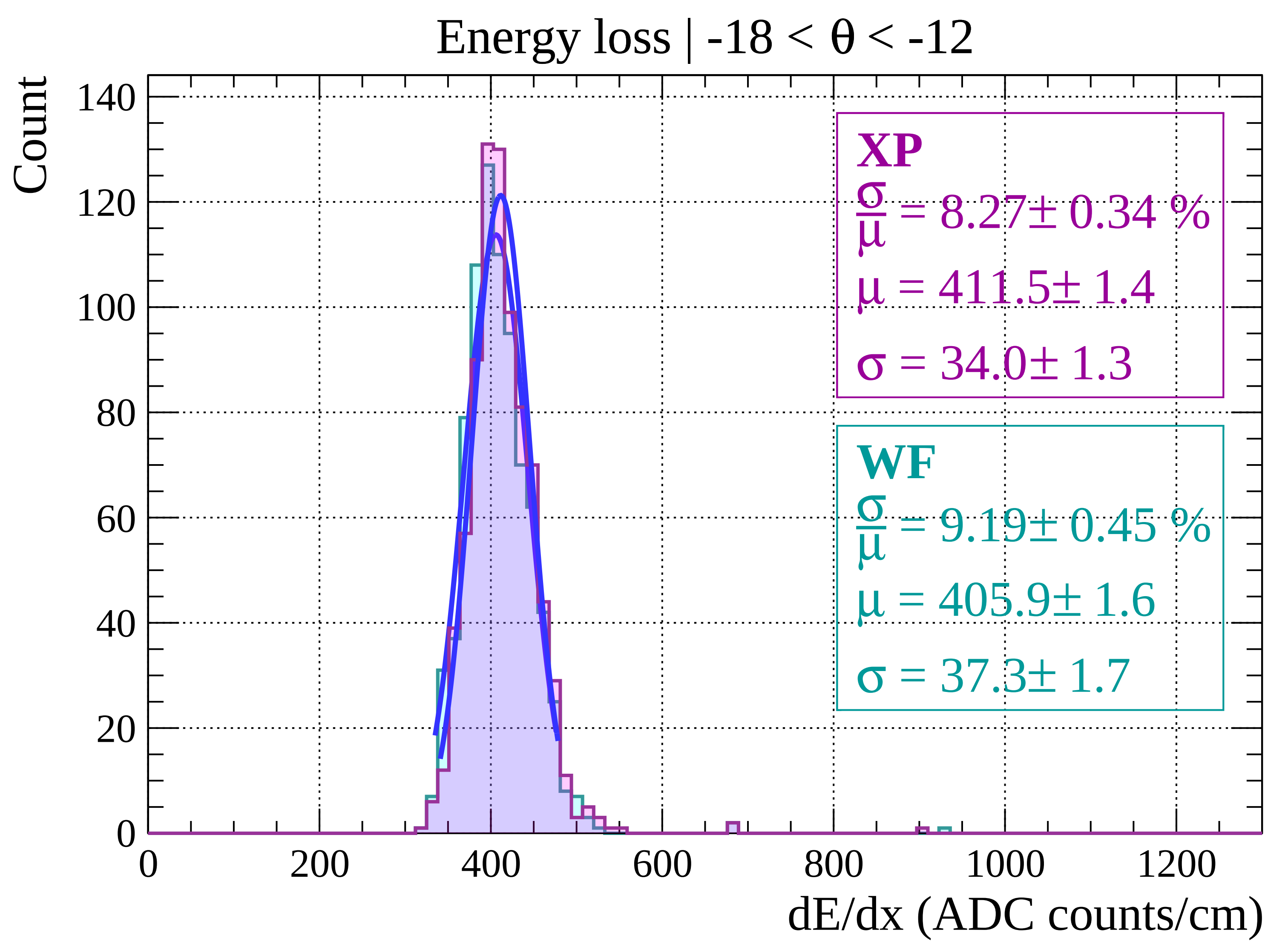

Energy loss | -36 < θ < -30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
22
24
XP
σ/μ = 8.74± 0.87 %
μ = 419.8± 4.1
σ = 36.7± 3.3
WF
σ/μ = 9.46± 1.19 %
μ = 407.9± 4.5
σ = 38.6± 4.4



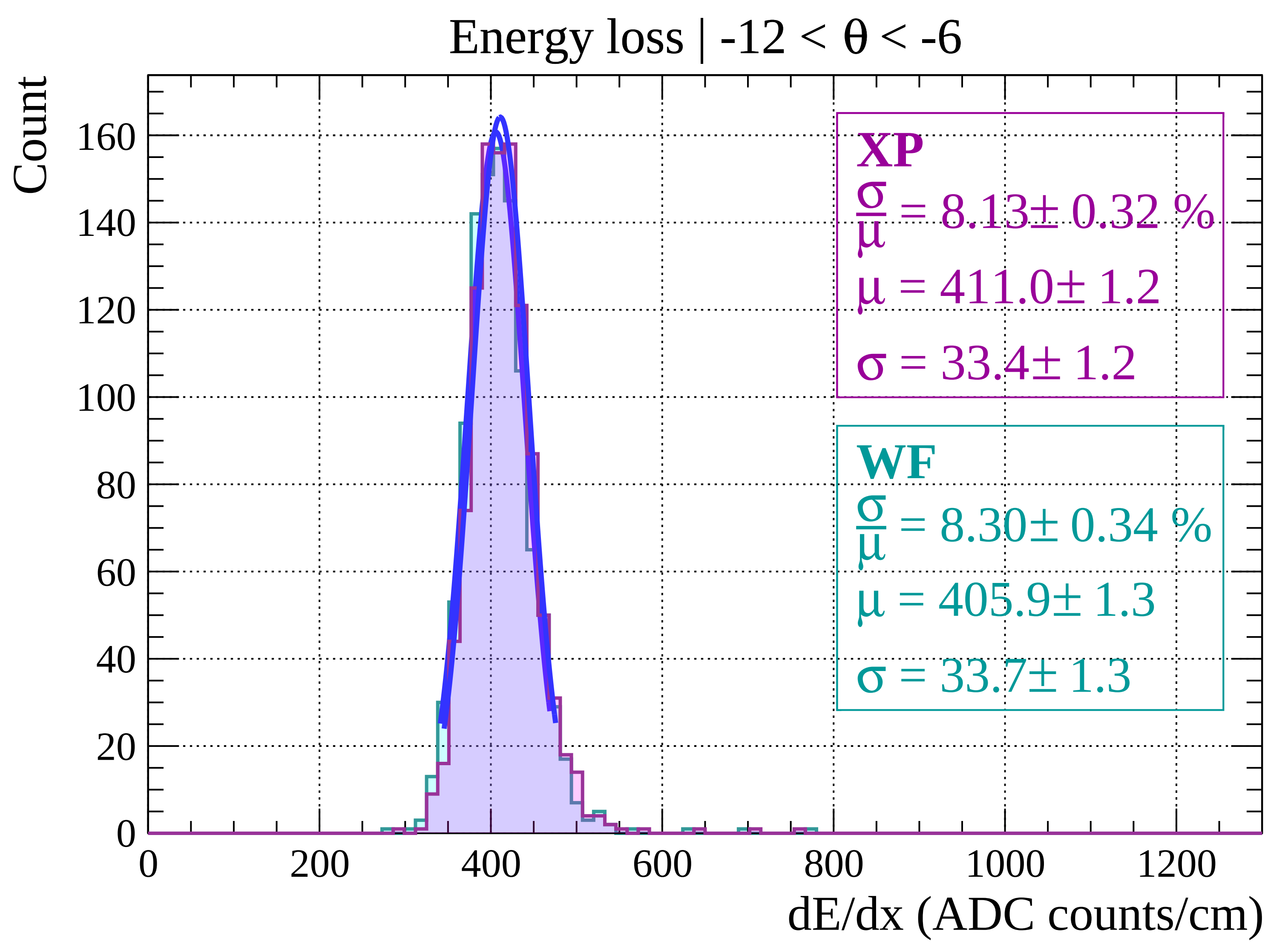
Energy loss | -30 < θ < -24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 8.44 ± 0.66 %
μ = 413.7 ± 2.6
σ = 34.9 ± 2.5
WF
σ/μ = 7.91 ± 0.52 %
μ = 406.7 ± 2.2
σ = 32.2 ± 2.0



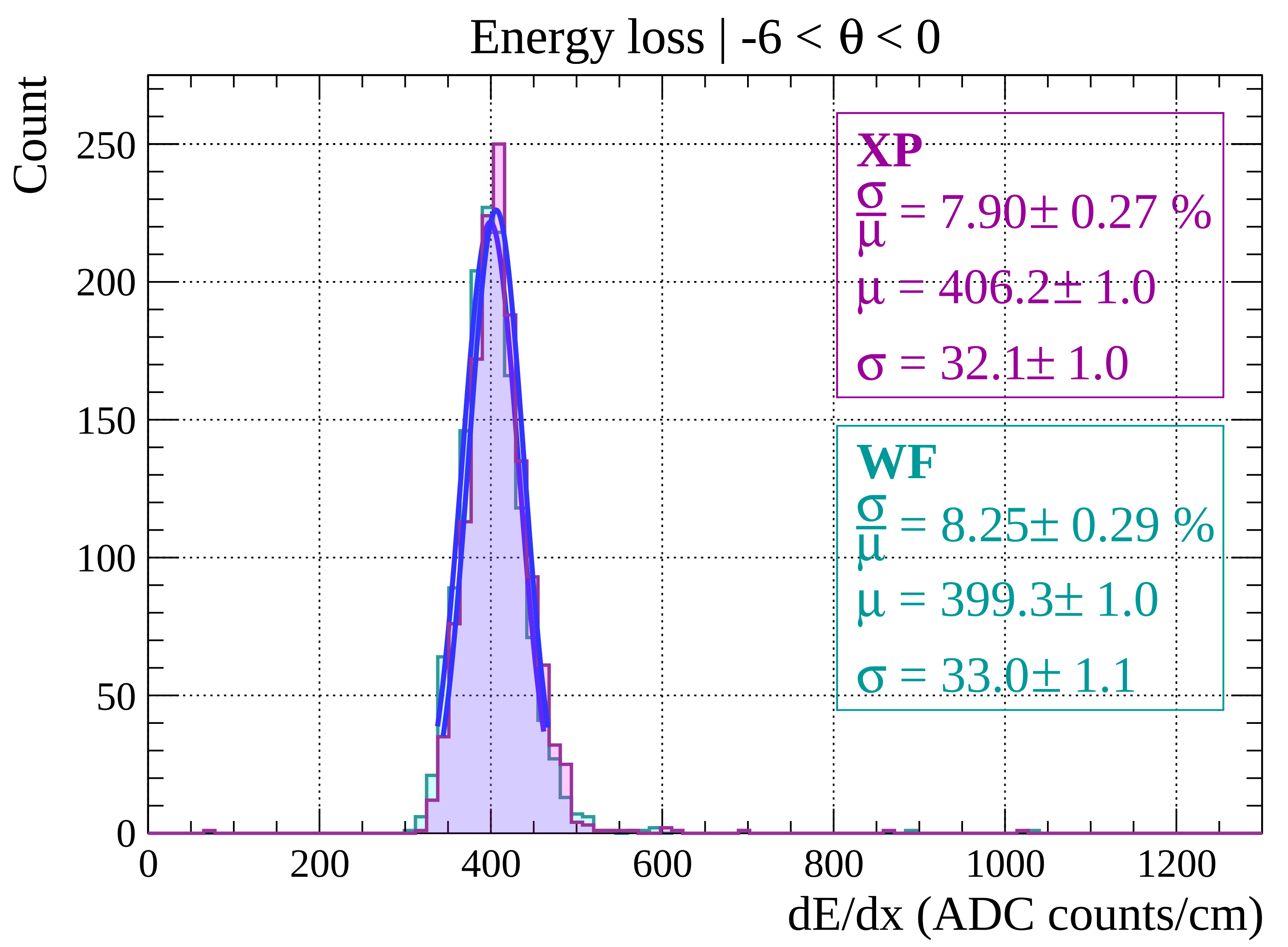
Energy loss | -24 < θ < -18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
XP
σ/μ = 8.40± 0.47 %
μ = 412.4± 1.9
σ = 34.6± 1.8
WF
σ/μ = 8.43± 0.43 %
μ = 407.1± 1.8
σ = 34.3± 1.6



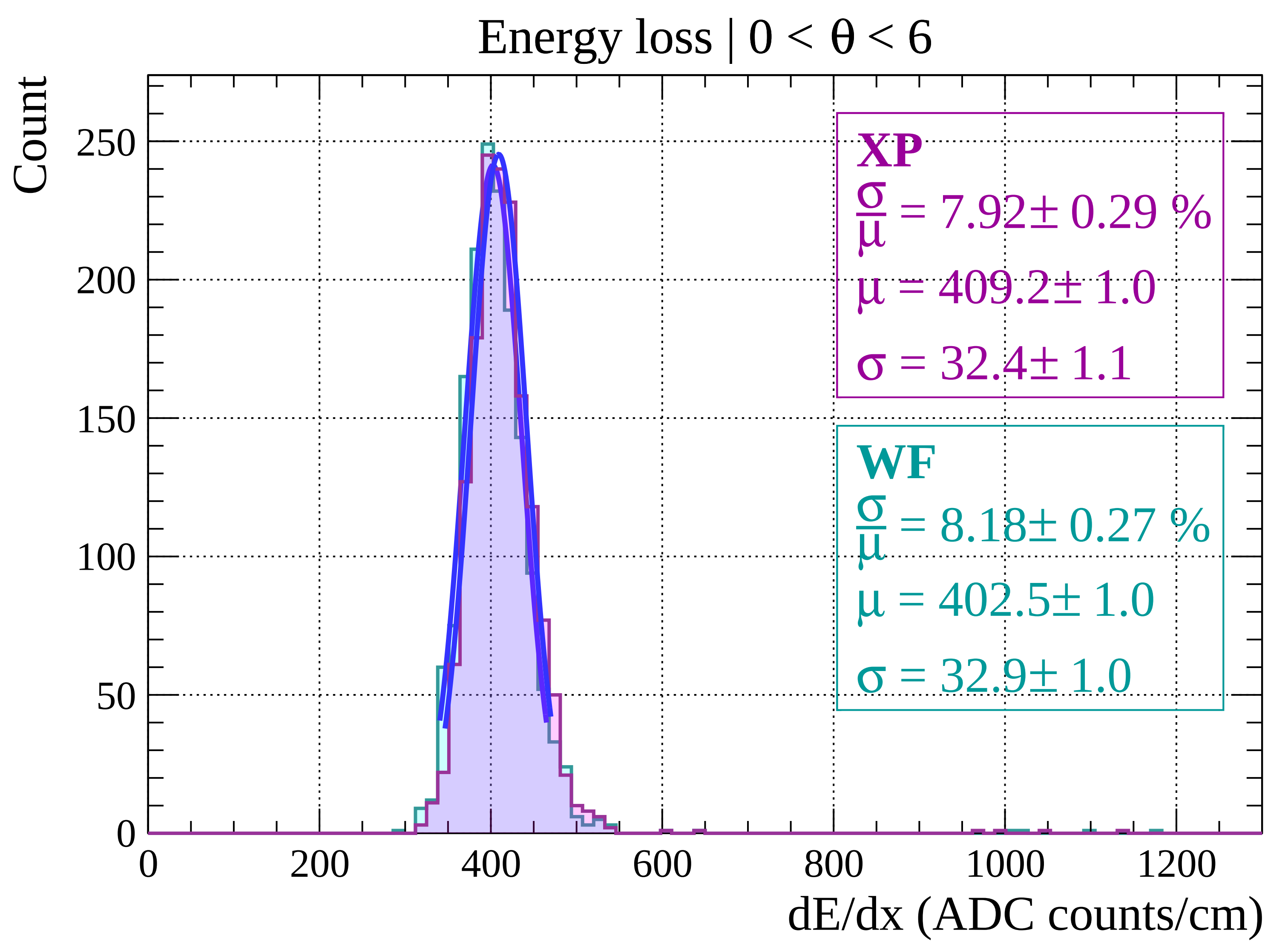
Energy loss | -18 < θ < -12
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 8.27± 0.34 %
μ = 411.5± 1.4
σ = 34.0± 1.3
WF
σ/μ = 9.19± 0.45 %
μ = 405.9± 1.6
σ = 37.3± 1.7



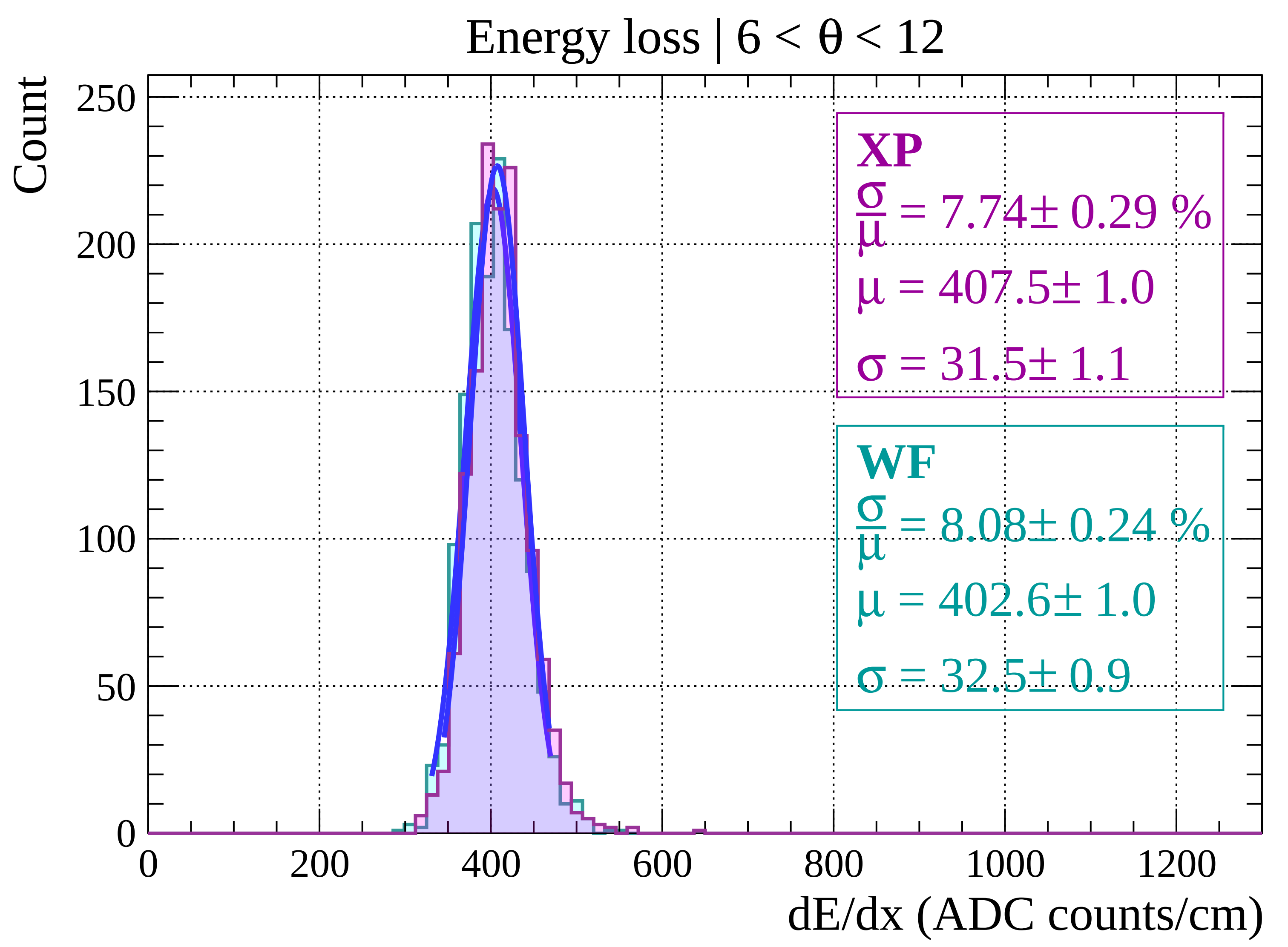
Energy loss | -12 < θ < -6
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.13± 0.32 %
μ = 411.0± 1.2
σ = 33.4± 1.2
WF
σ/μ = 8.30± 0.34 %
μ = 405.9± 1.3
σ = 33.7± 1.3
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160



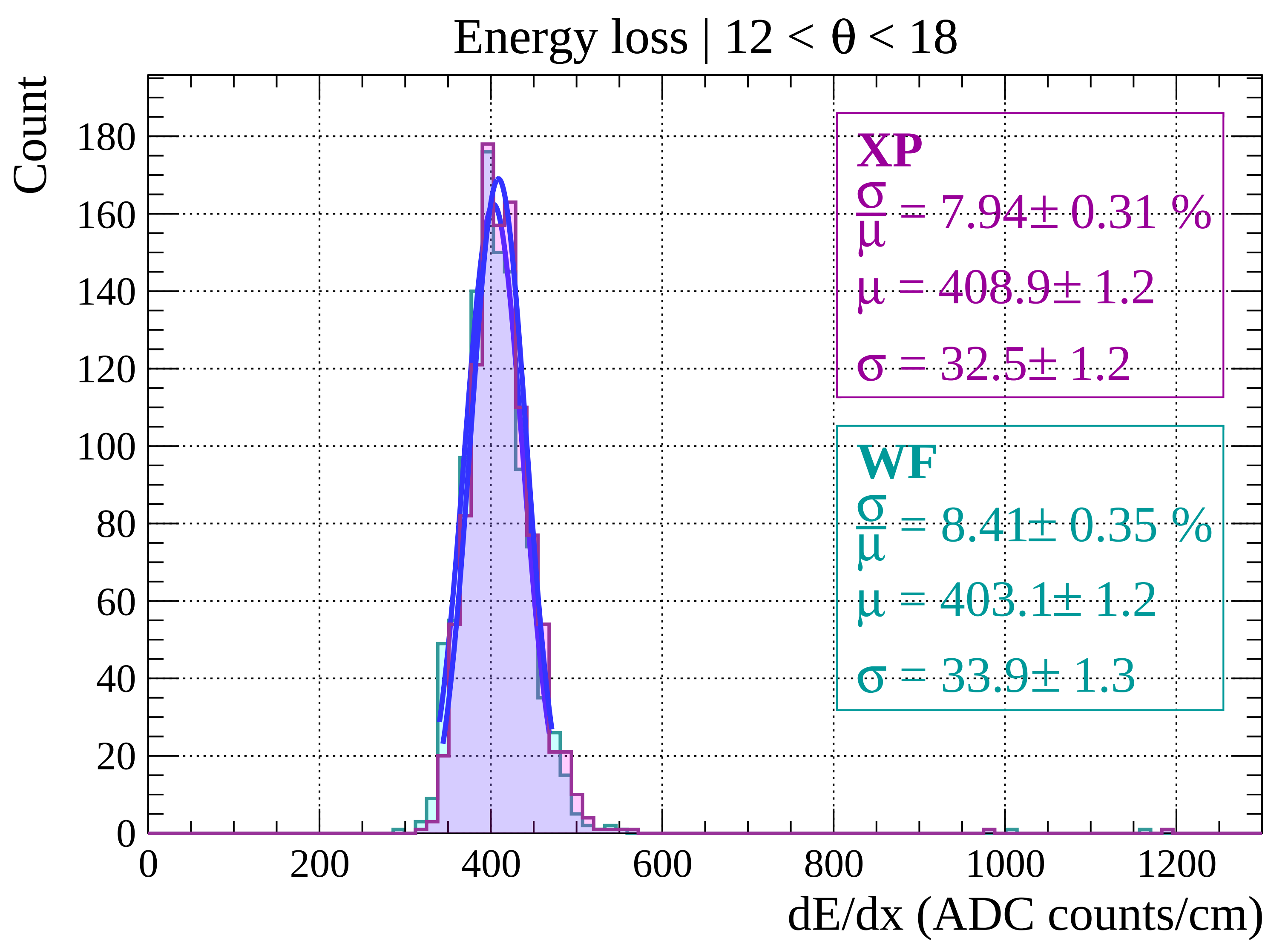
Energy loss | -6 < θ < 0
Count
XP
σ/μ = 7.90± 0.27 %
μ = 406.2± 1.0
σ = 32.1± 1.0
WF
σ/μ = 8.25± 0.29 %
μ = 399.3± 1.0
σ = 33.0± 1.1
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200

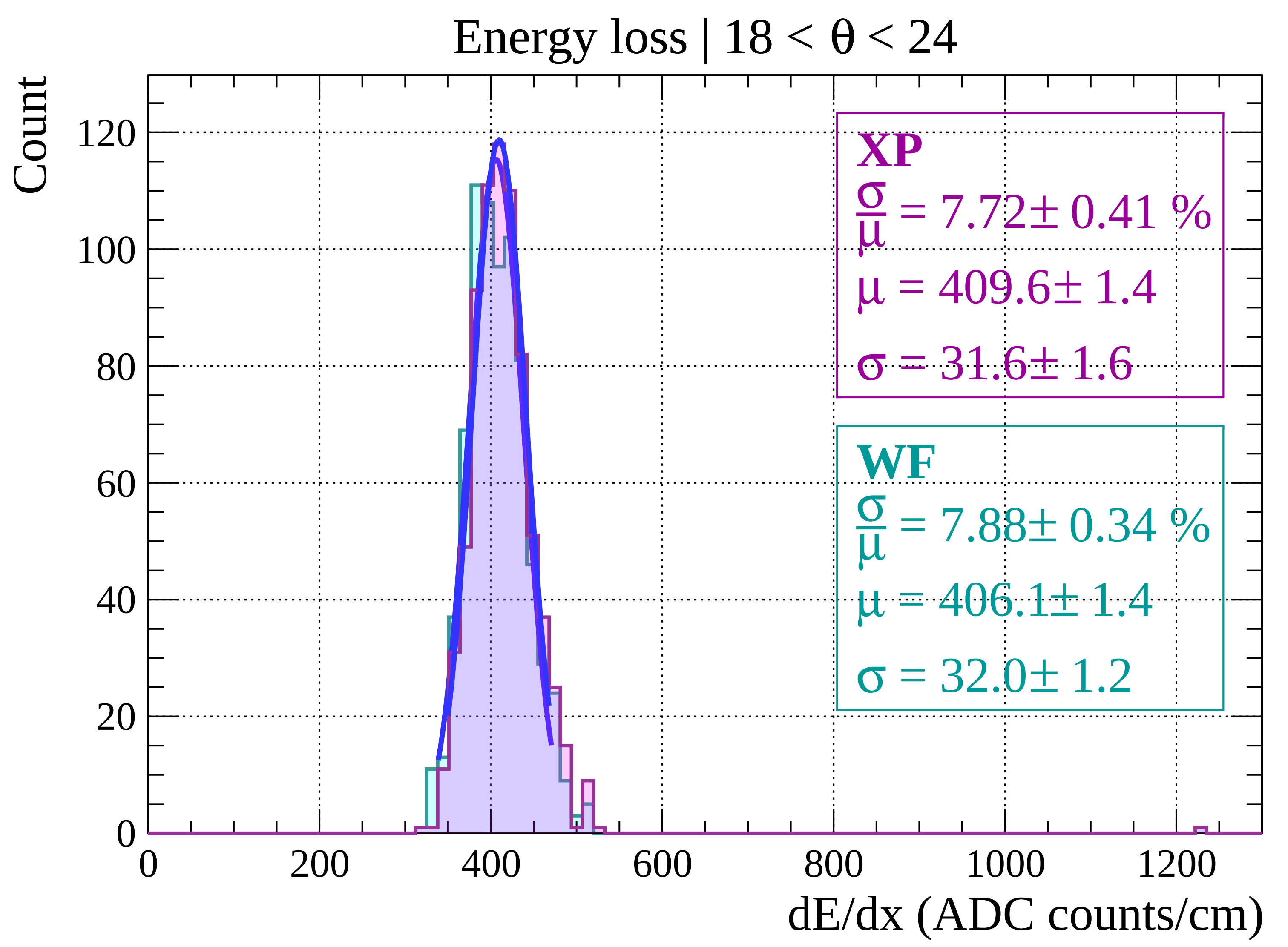


Energy loss | 0 < θ < 6
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 7.92± 0.29 %
μ = 409.2± 1.0
σ = 32.4± 1.1
WF
σ/μ = 8.18± 0.27 %
μ = 402.5± 1.0
σ = 32.9± 1.0

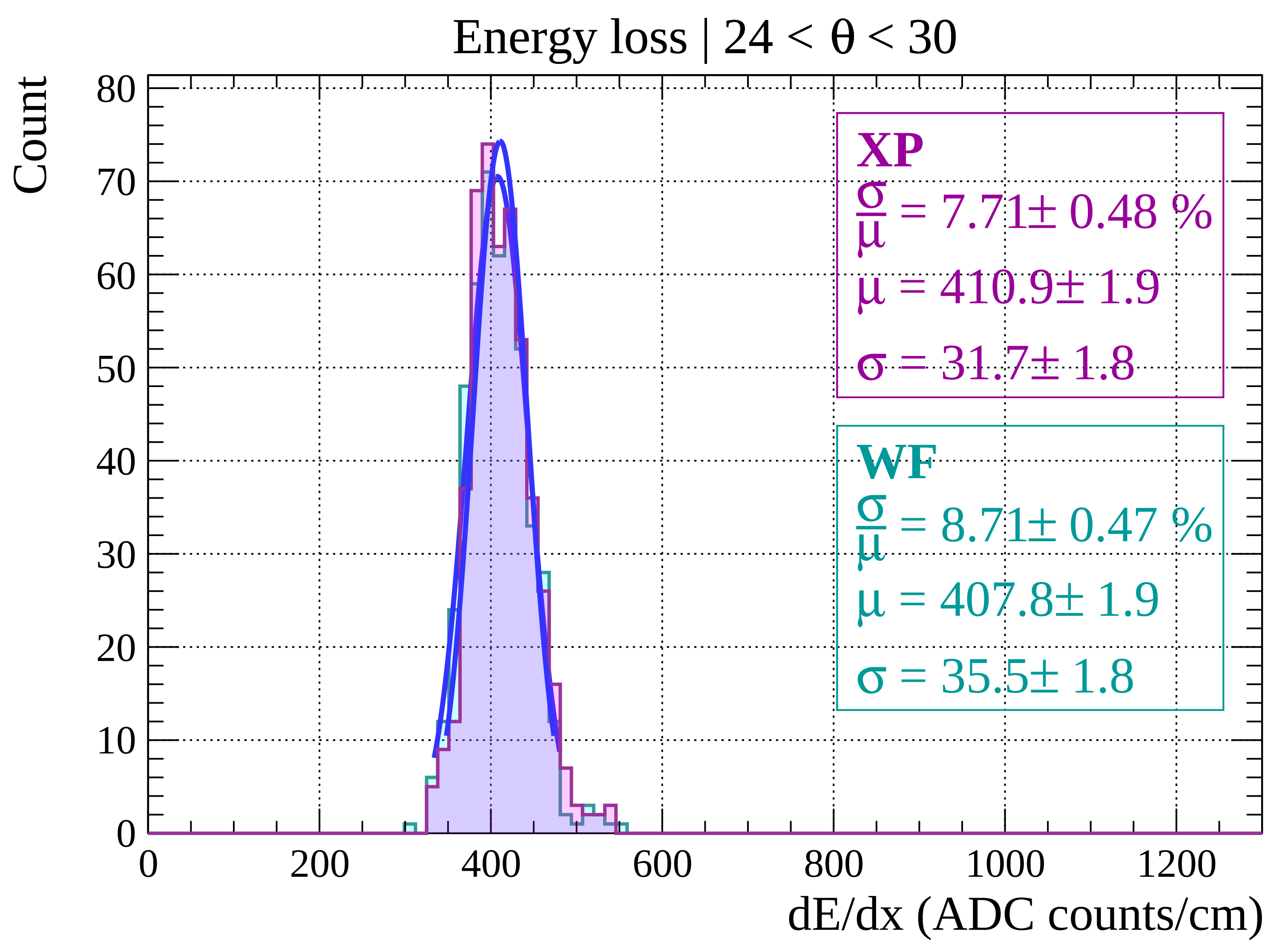


Energy loss | 6 < θ < 12
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.74± 0.29 %
μ = 407.5± 1.0
σ = 31.5± 1.1
WF
σ/μ = 8.08± 0.24 %
μ = 402.6± 1.0
σ = 32.5± 0.9

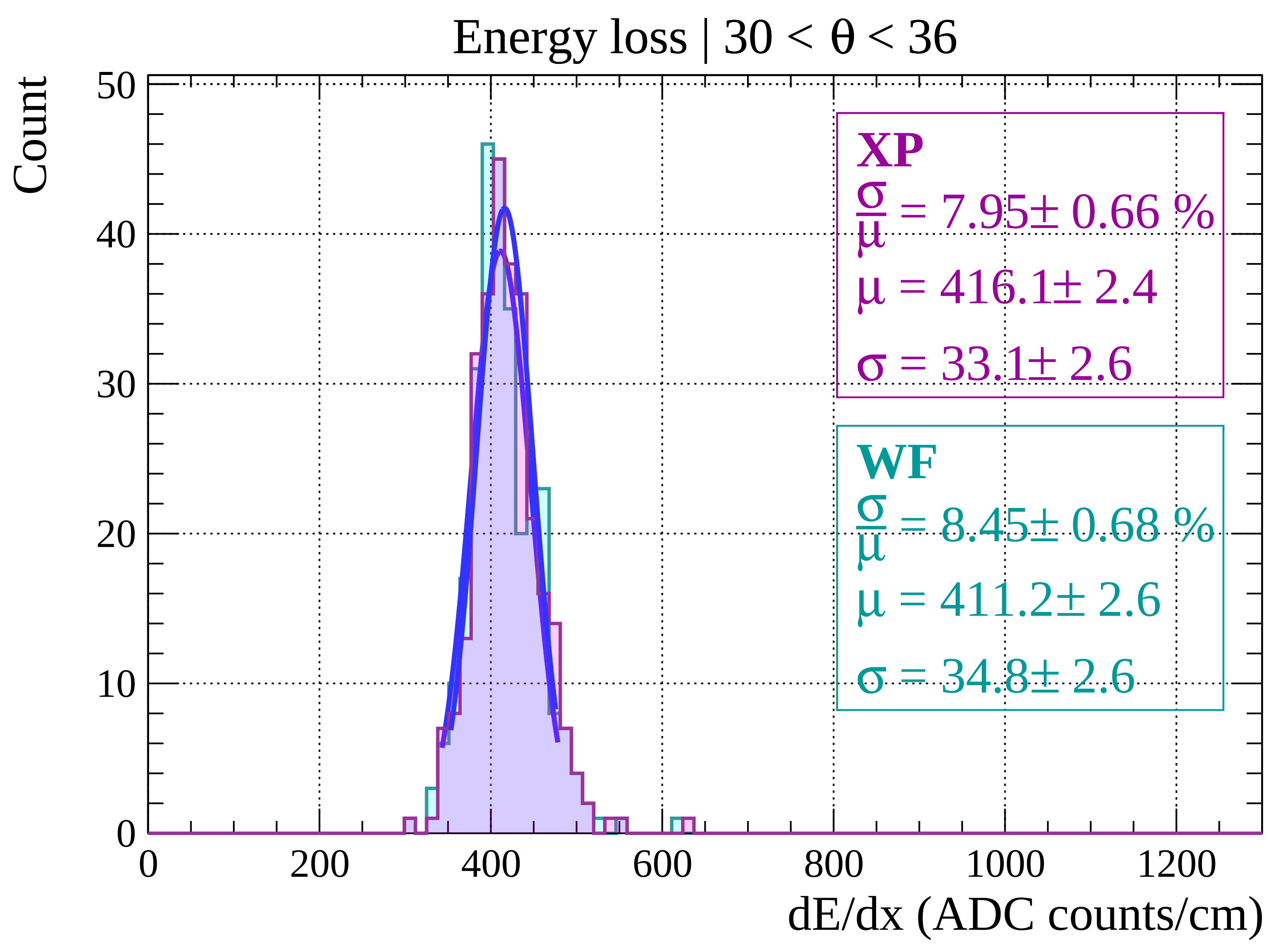

Energy loss | 12 < θ < 18
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 7.94± 0.31 %
μ = 408.9± 1.2
σ = 32.5± 1.2
WF
σ/μ = 8.41± 0.35 %
μ = 403.1± 1.2
σ = 33.9± 1.3



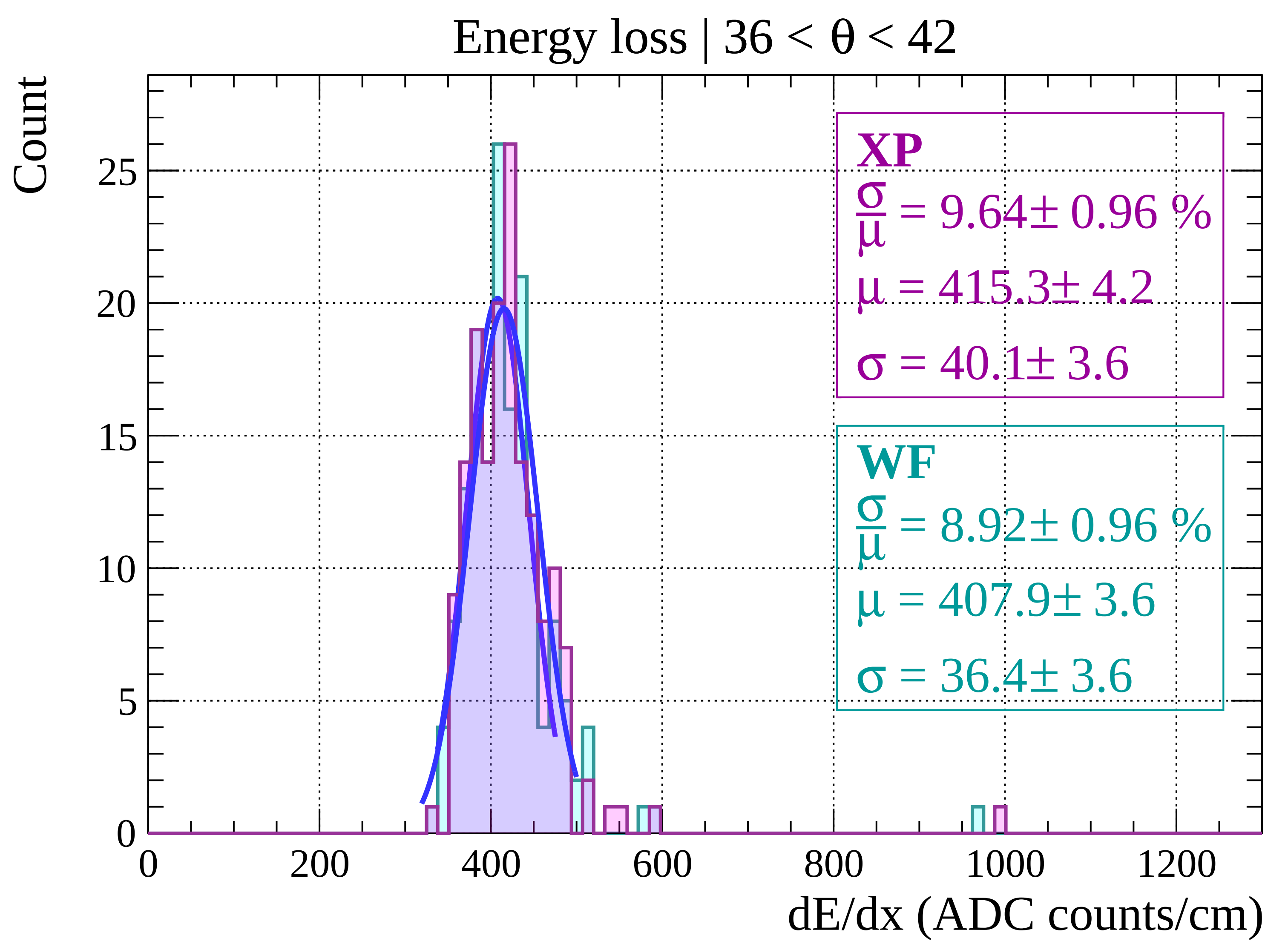
Energy loss | 18 < θ < 24
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.72± 0.41 %
μ = 409.6± 1.4
σ = 31.6± 1.6
WF
σ/μ = 7.88± 0.34 %
μ = 406.1± 1.4
σ = 32.0± 1.2



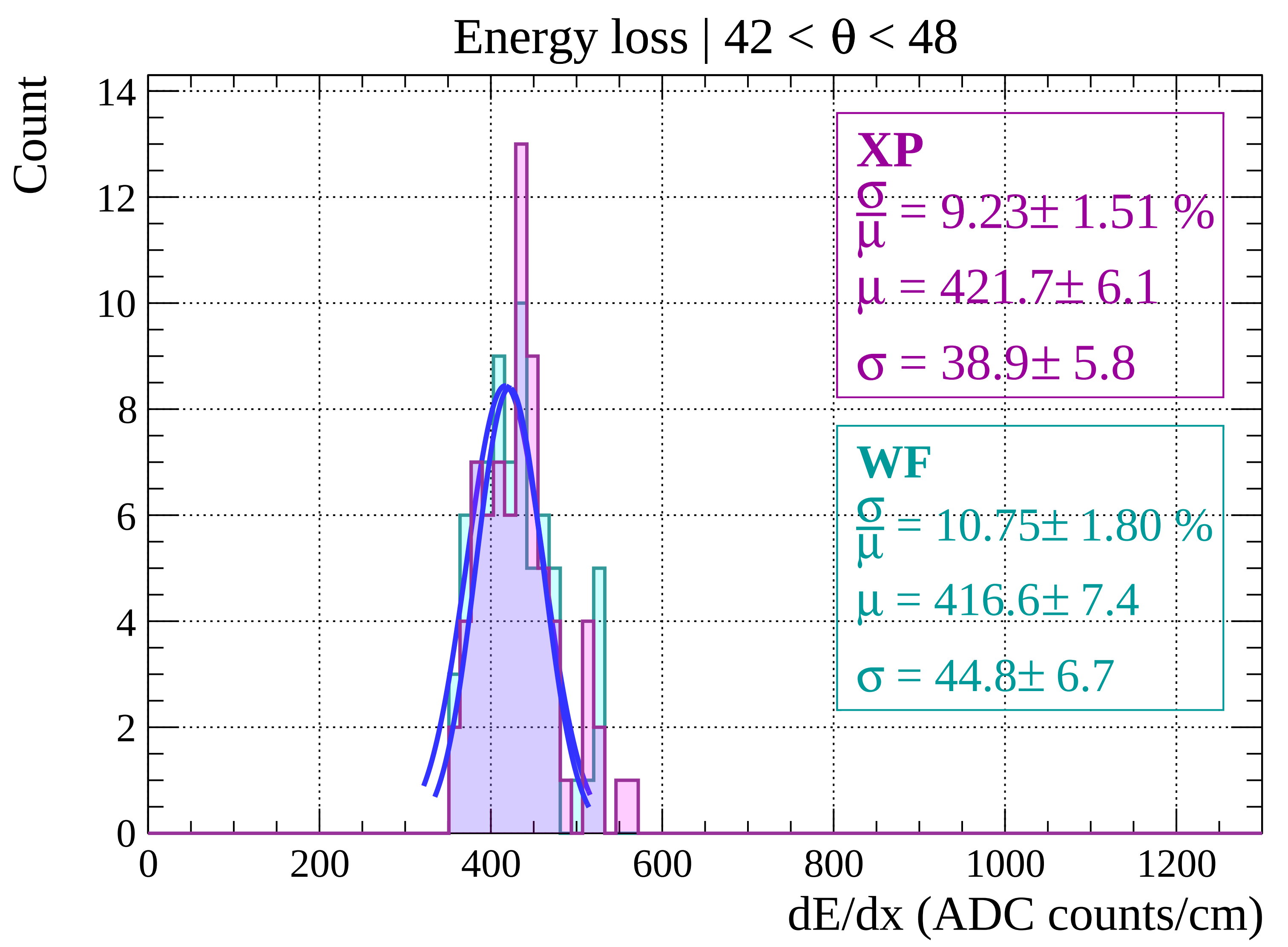
Energy loss | 24 < θ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.71± 0.48 %
μ = 410.9± 1.9
σ = 31.7± 1.8
WF
σ/μ = 8.71± 0.47 %
μ = 407.8± 1.9
σ = 35.5± 1.8
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200



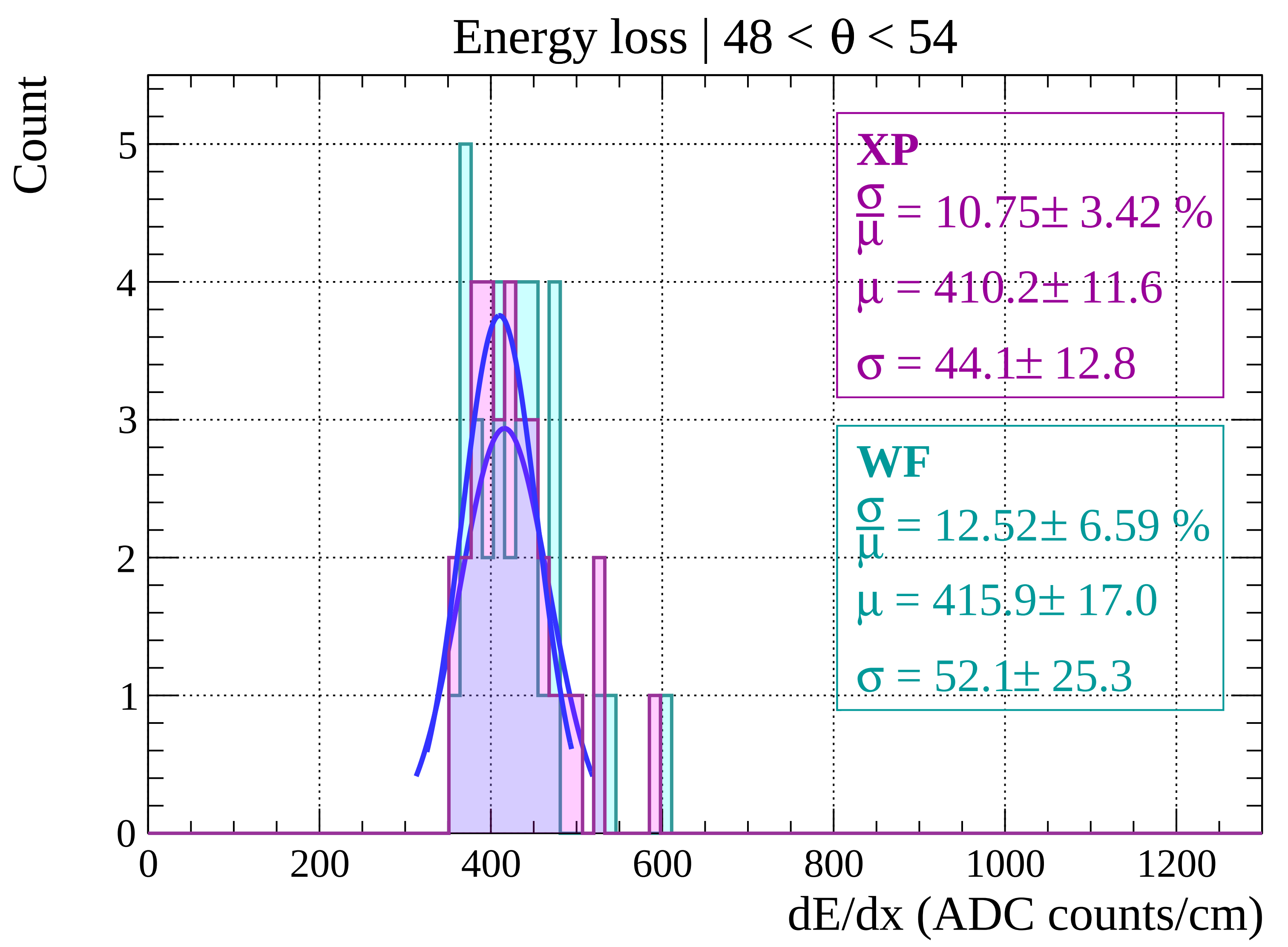
Energy loss | 30 < θ < 36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 7.95± 0.66 %
μ = 416.1± 2.4
σ = 33.1± 2.6
WF
σ/μ = 8.45± 0.68 %
μ = 411.2± 2.6
σ = 34.8± 2.6



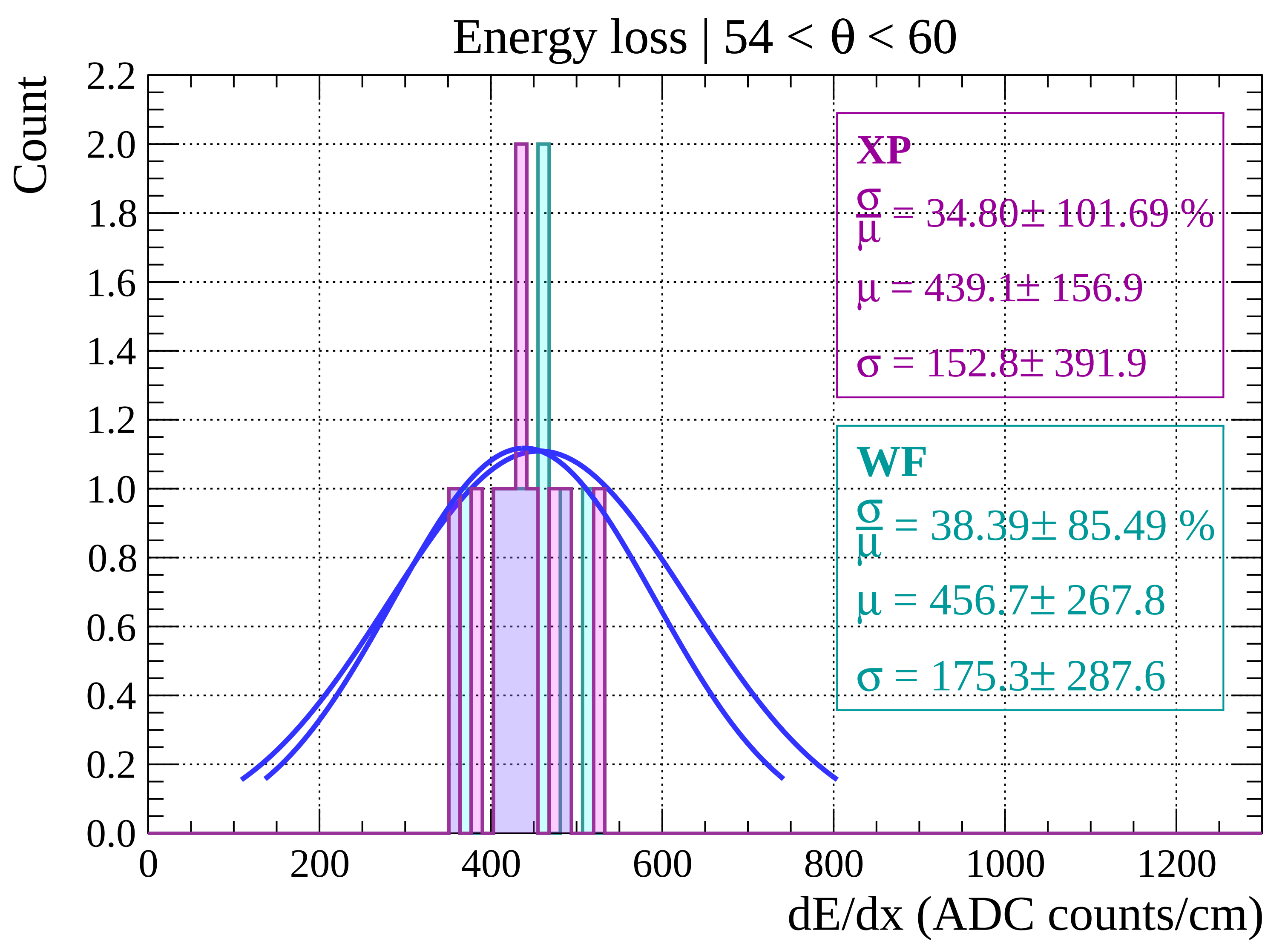
Energy loss | 36 < θ < 42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
XP
σ/μ = 9.64± 0.96 %
μ = 415.3± 4.2
σ = 40.1± 3.6
WF
σ/μ = 8.92± 0.96 %
μ = 407.9± 3.6
σ = 36.4± 3.6



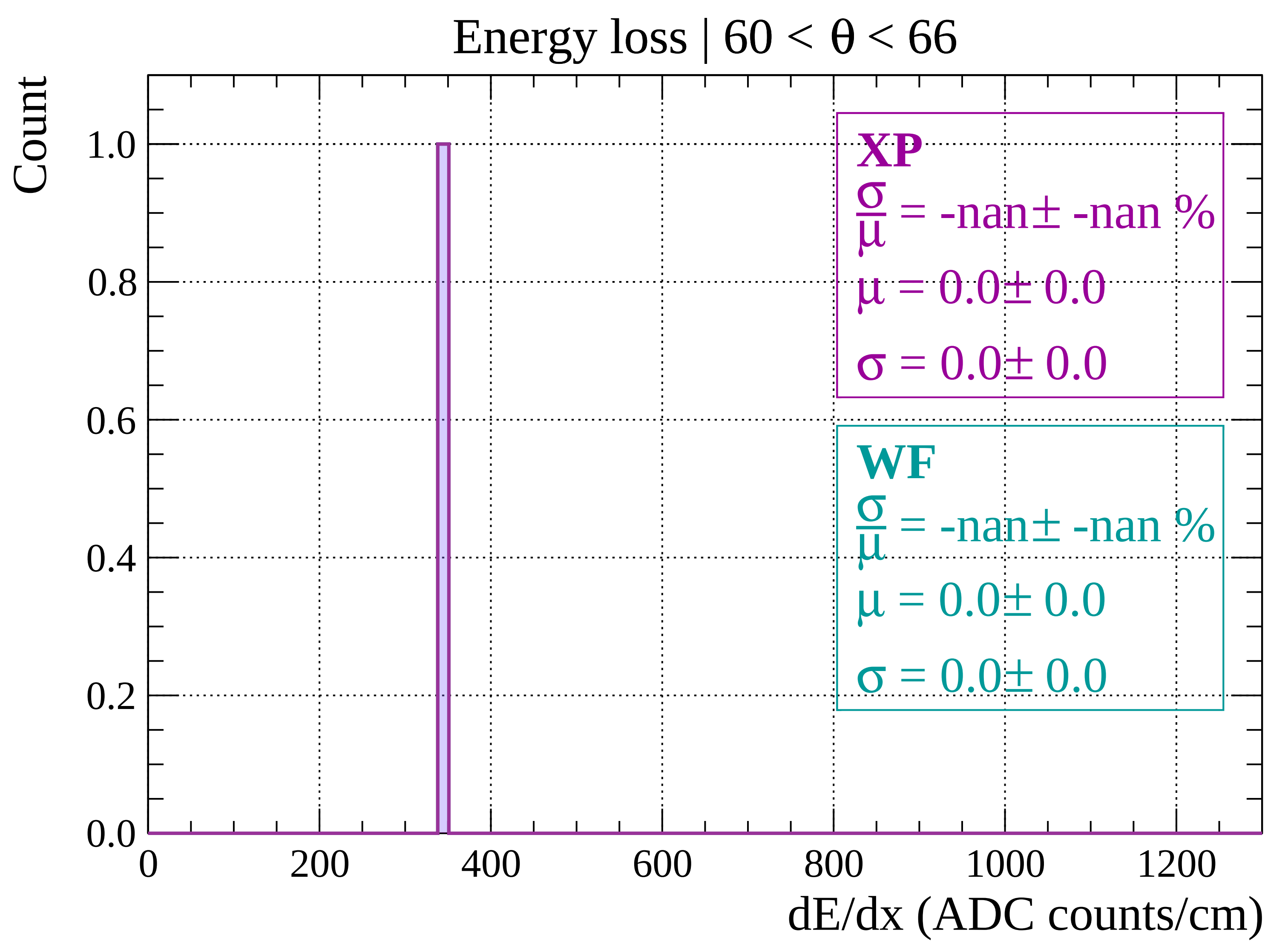
Energy loss | 42 < θ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
XP
$\frac{\sigma}{\mu}$ = 9.23± 1.51 %
μ = 421.7± 6.1
σ = 38.9± 5.8
WF
$\frac{\sigma}{\mu}$ = 10.75± 1.80 %
μ = 416.6± 7.4
σ = 44.8± 6.7



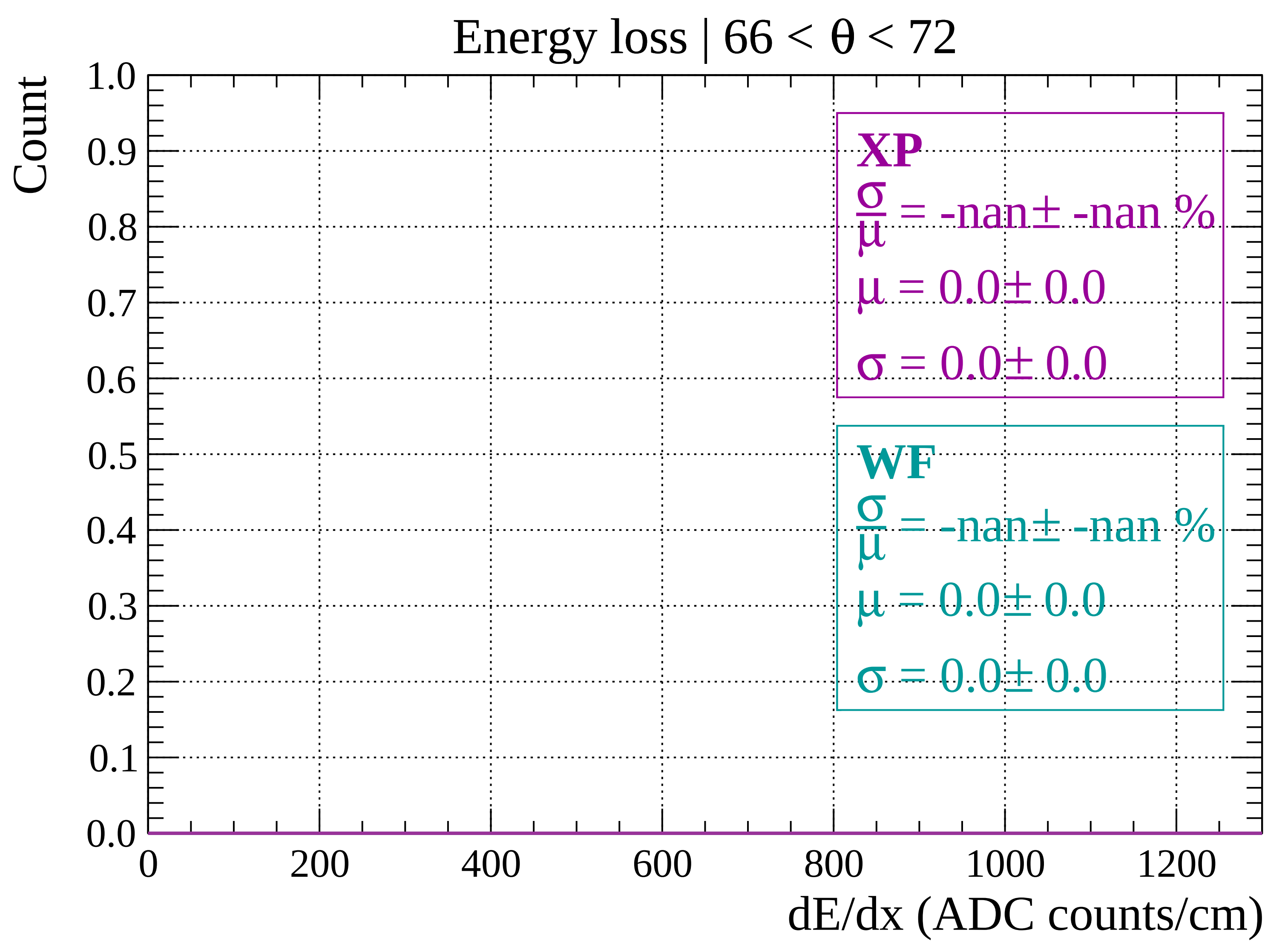
Energy loss | 48 < θ < 54
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.75± 3.42 %
μ = 410.2± 11.6
σ = 44.1± 12.8
WF
$\frac{\sigma}{\mu}$ = 12.52± 6.59 %
μ = 415.9± 17.0
σ = 52.1± 25.3



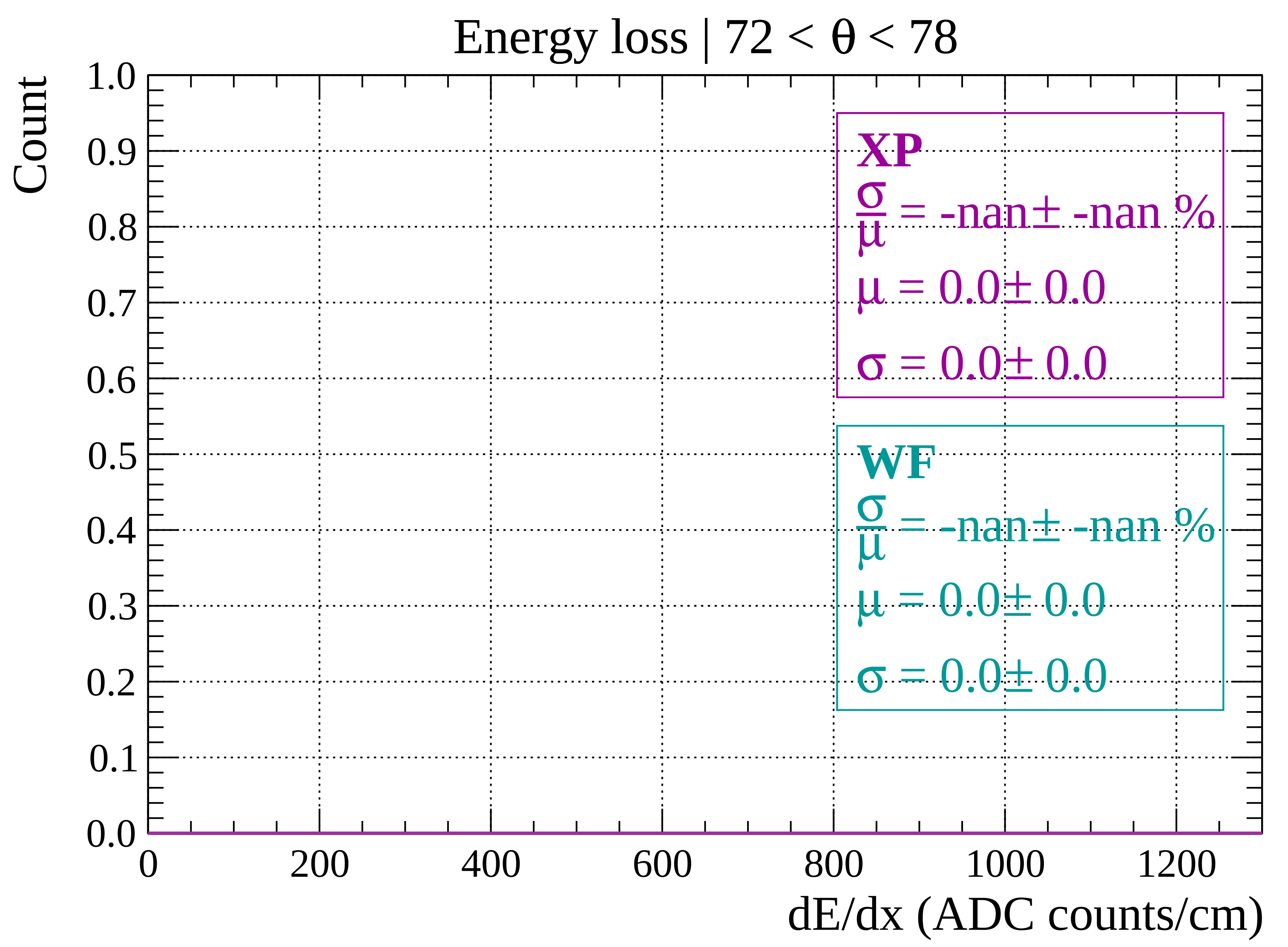
Energy loss | 54 < θ < 60
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 34.80± 101.69 %
μ = 439.1± 156.9
σ = 152.8± 391.9
WF
σ/μ = 38.39± 85.49 %
μ = 456.7± 267.8
σ = 175.3± 287.6

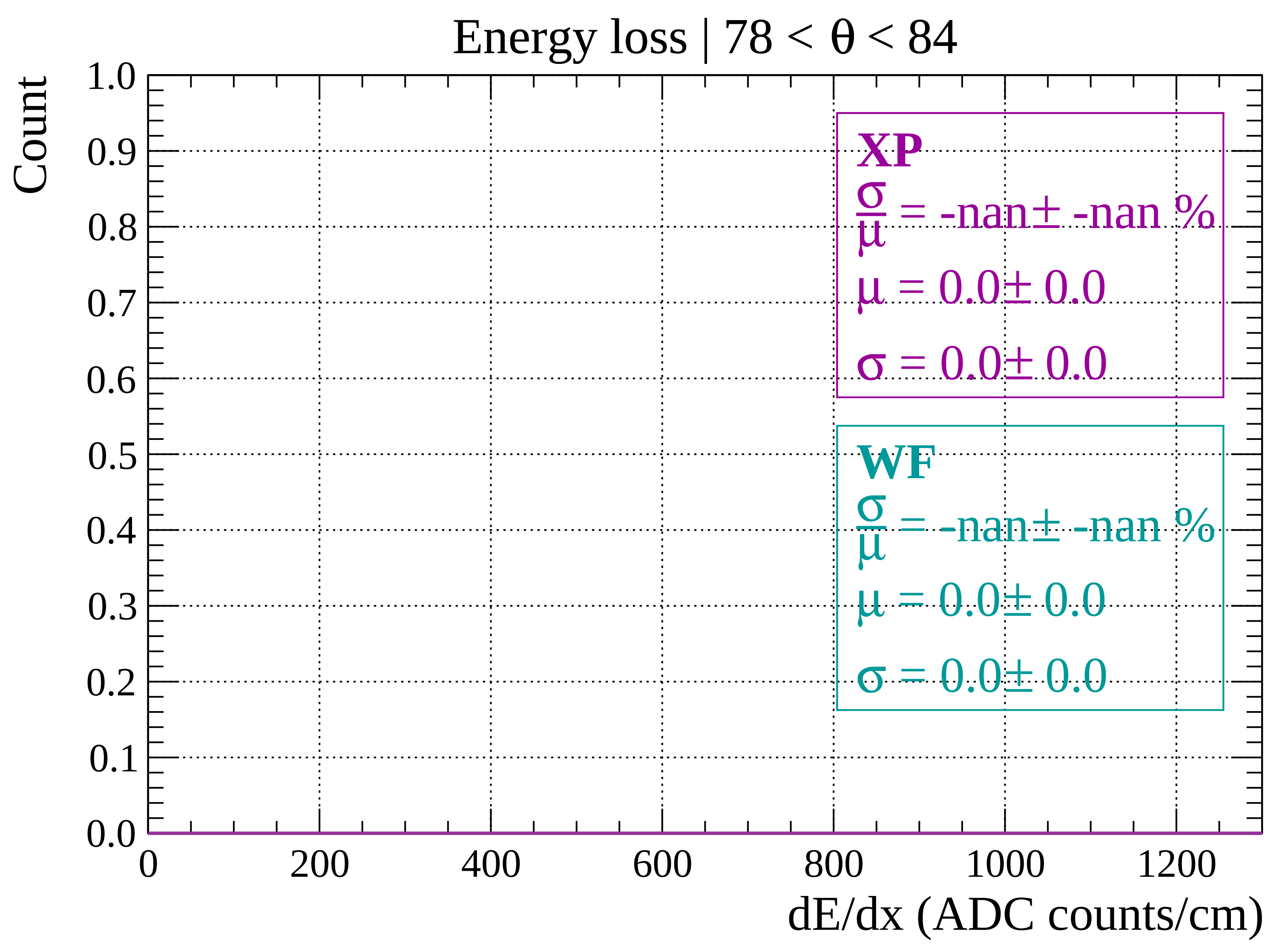


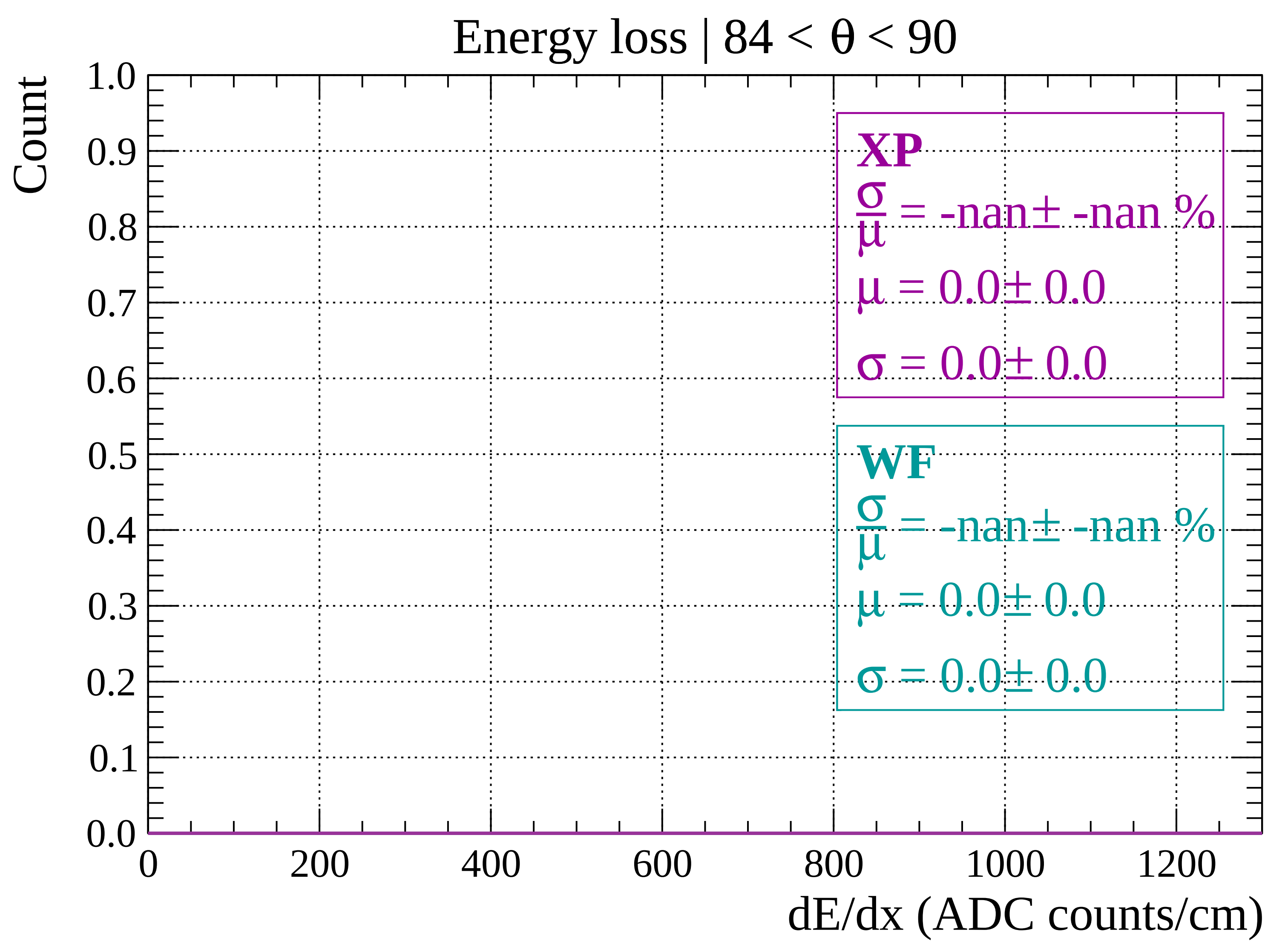
Energy loss | 60 < θ < 66
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

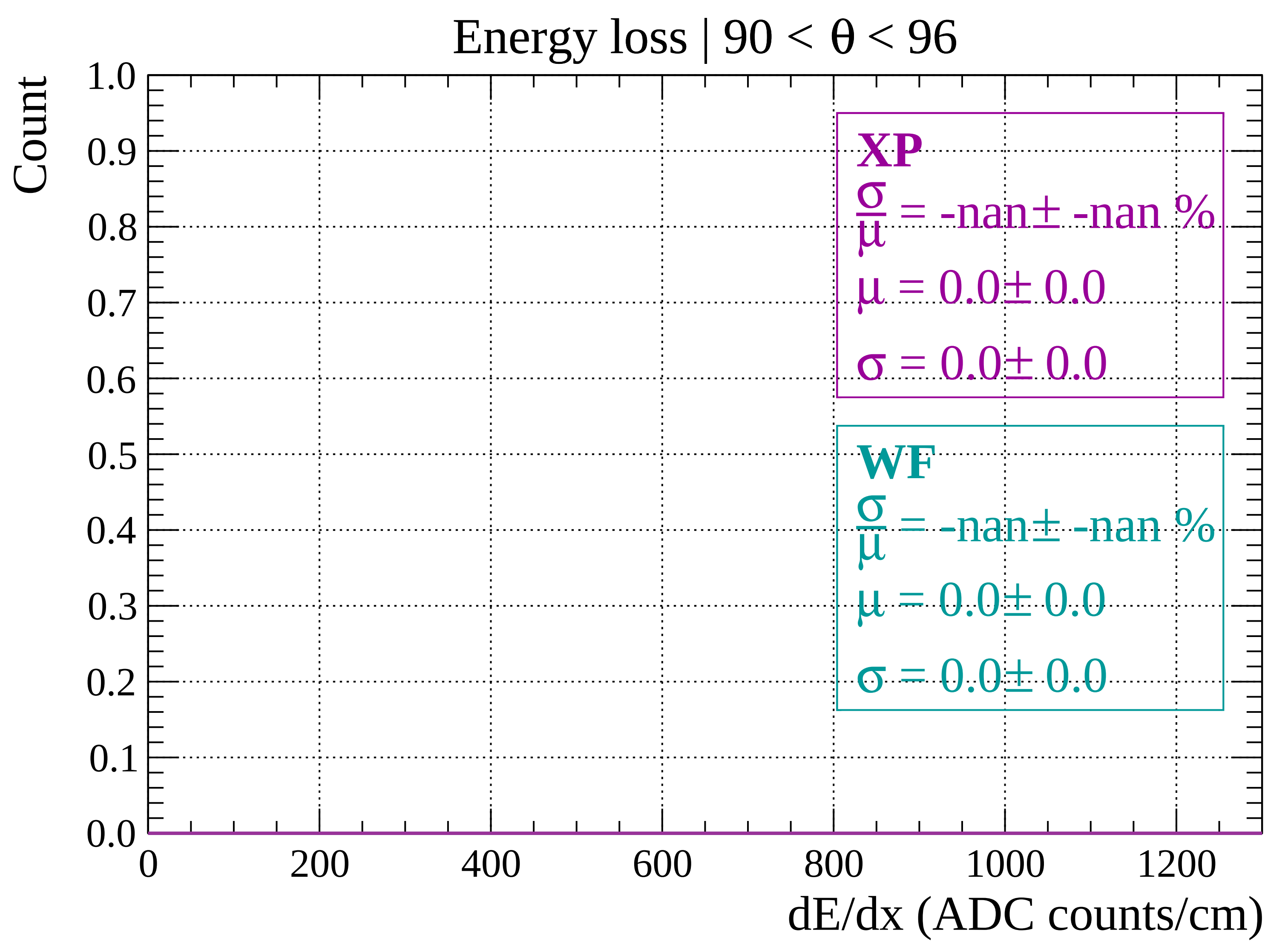


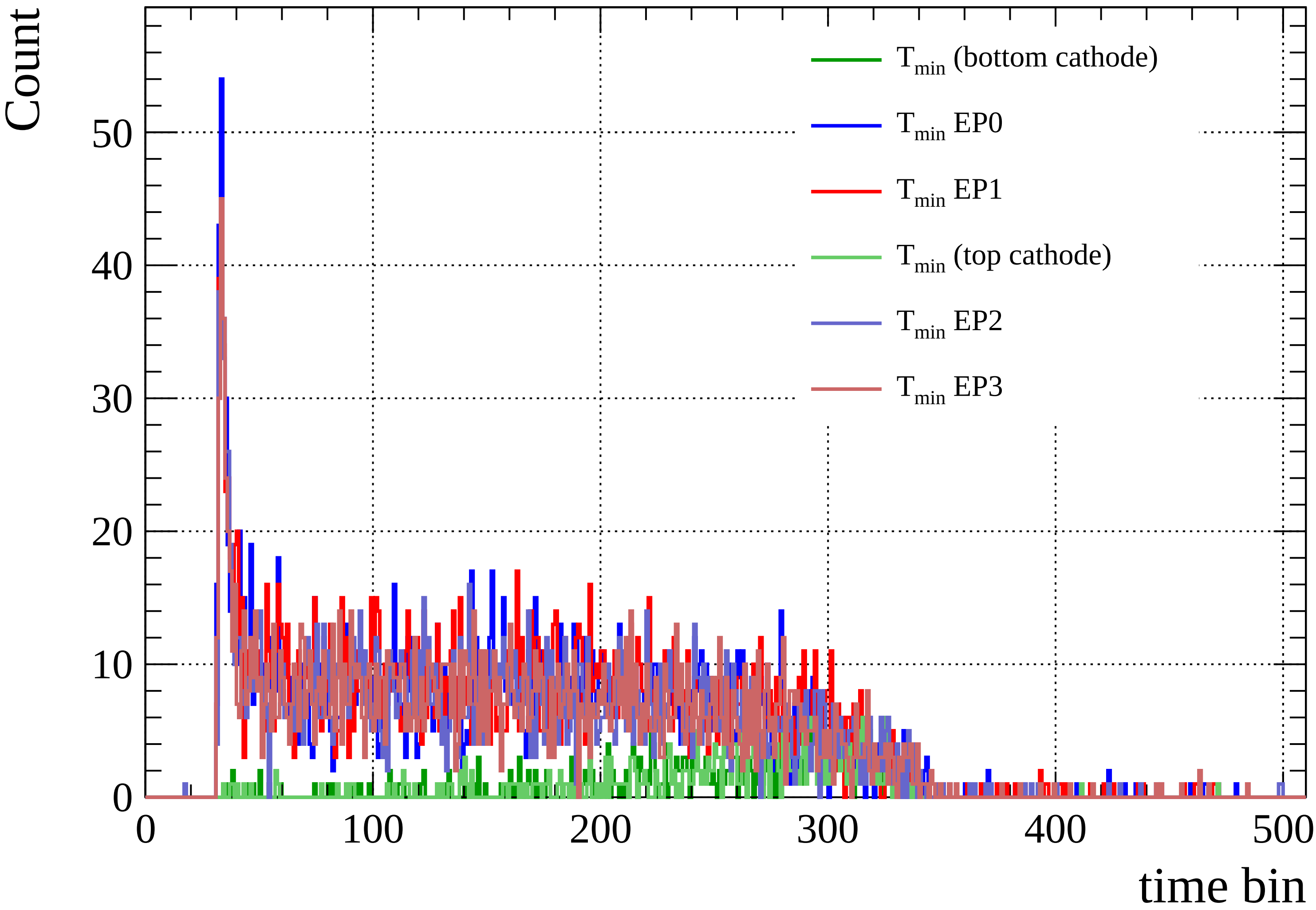
Energy loss | 66 < θ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 72 < θ < 78
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

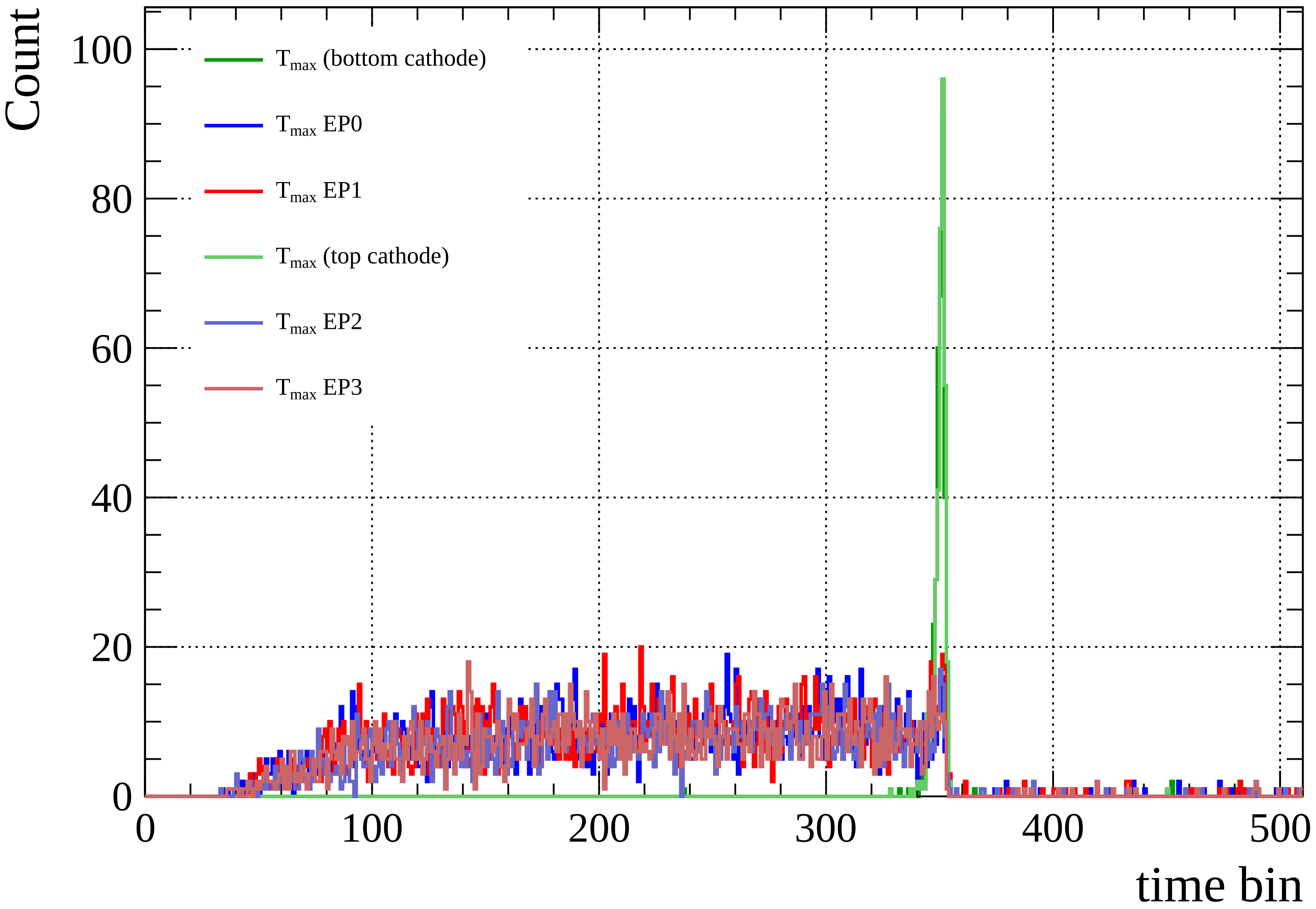

Energy loss | 78 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 84 < θ < 90
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 90 < θ < 96
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
0
10
20
30
40
50
0
100
200
300
400
500
time bin
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
20
40
60
80
100
0
100
200
300
400
500